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1. 体验式零售：Offering Memories to Sell Goods? Pricing and Welfare Implications of Experiential Retail (MSOM, 2023)

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Offering memories to sell goods? Pricing and welfare implications of experiential retail

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Offering Memories to Sell Goods? Pricing and Welfare Implications of Experiential Retail

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Abstract. ***Problem definition*:** In an environment where consumers' rising valuation of Instagrammable memories drives their spending from products to experiences, retailers offer experiences to attract consumers back to their stores. Yet, it is not obvious under which settings consumers can benefit from these experiences and raise retailers' profits. ***Methodology/results*:** We use a random utility model for consumer choice in both monopoly and duopoly settings. For the latter, we pose a game-theoretic model to analyze the equilibrium prices, profits, and consumer welfare for various problem cases. We show that medium-quality experiences can lower the product sales and store traffic below the level when no experience is offered. Sufficiently low- or high-quality experiences overcome this issue, making the consumers better off. Yet, the former presents the only profitable option when a single retailer in the market offers an experience. In contrast, when both retailers adopt experiences, low-quality experiences may not be profitable, and the retailers would need to adopt even higher-quality experiences, which lead to a "win-win-win" outcome for the two retailers and consumers. A fee structure for the experience elevates retailer profits but can turn stores into outlets where consumers visit to purchase experiences and not products. When experiential retailing is the common practice in the market, it enables "win-win-win" outcomes when free experiences fail. ***Managerial implications*:** Being the first to study this new retail format, our results contribute to the ongoing debate on the settings under which experiential offerings are beneficial for the retailers and consumers and highlight that there is no one-size-fits-all strategy. Our results show that experiences affect main product sales in nonobvious ways, especially in competitive markets. In a (post-)pandemic world where retailers try to attract consumers back to stores, we offer insights that can guide retailers in the process.

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Keywords: complementary offerings • crosscategory choice • multivariate logit model • MVL model

1. Introduction

Dick's Sporting Goods opened their third House of Sport store in Minnetonka, Minnesota in June 2022 after debuting this new store concept in 2021 first in Rochester, New York and then in Knoxville, Tennessee. These new stores are different than the traditional Dick's stores; in addition to products, they feature a rock-climbing wall, a batting cage with simulation technology, and golf sessions that the customers can enjoy for free. The company states that the stores have been in the works since 2018.

Dick's is not the only company to adopt this new store concept. Several Lululemon stores that opened before the pandemic continue to serve not only as sale locations for yoga wear but also, as studios where consumers can practice yoga/meditation for free (Well+Good LLC 2017). Similarly, Whole Foods has not been only a grocery retailer, but it has also been featuring wine bars, cooking classes, and consultations with nutritionists in its stores (Hanley Investment Group 2016).

The trend for these "experiential" stores started before the pandemic when the spending on experiences grew tremendously compared with that on goods. In 2017, Barclaycard in the United Kingdom reported that the spending in pubs had a 20% year-to-year increase. Although theaters and restaurants also enjoyed a sales increase (by 13% and 16%, respectively), the department store, vehicle, and household appliance sales suffered losses (by 1%, 11%, and 2.5%, respectively). *The Guardian* (2017) confirmed that "[w]e are continuing to spend less money on buying things, and more on doing things."

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Although the pandemic forced consumers to stay indoors, consumers are now more eager than ever for experiences outside of their homes. Hence, retailers such as Dick's aim to capitalize on that and introduce experiences to entice those consumers back to their stores to gain a competitive advantage.

The concept of *experiential retailing* dates back to 1990 s. In their *Harvard Business Review* article, Pine and Gilmore (1998) argue that in retail, "experiences are a distinct economic offering, as different from services as services from goods." Hence, they can be considered as a unique category of store offerings, which can be enjoyed independently of the consumption of the main products. Experiences may have several benefits. First, they may increase the traffic in stores by attracting those who value this new offering. Second, beyond their intrinsic value, experiences can also facilitate the sales of main products via crosscategory complementarities. Consumers may incur a travel cost for visiting the stores. As LSG Marketing Solutions states, "[g]ive people a reason to walk through your door besides shopping. Be it a charity event, an activity or something educational" (Dor Technologies, Inc 2019). Hence, attracting the consumers to the store for experiences would also encourage a main product purchase because of travel cost synergy in the same store visit. Another type of complementarity arises because of the brand synergy between the two offerings. In practice, retailers typically offer experiences that synergize well with the products of their brand (e.g., via yoga classes offered by the yoga apparel firm Lululemon, skating lessons offered by the skating gear firm Vans, playgrounds at Toys "R" Us stores, and cooking classes at Whole Foods stores). As such, retailers do not just sell products, but using their brand as a vehicle, they sell a "memory" facilitated via synergistic experiences (NPD Group 2020). Hence, experiences can act as complementary sales mechanisms for the main products, as the main products can act as souvenirs of the memorable time spent at the retailer because of the underlying brand synergy. Consequently, the advantages of offering experiences in stores may go beyond the well-known benefits of traditional brick-and-mortar stores (e.g., enabling in-store product trials).

Yet, despite the potential benefits, not all retailers become experiential retailers. As Pine and Gilmore (1998) highlight, experiences are a distinct economic offering, which may not be attractive to all consumers. For example, on one extreme, although yoga-savvy consumers can be eager to join the free yoga sessions at Lululemon stores, these sessions may not be appealing to those who like to wear a yoga outfit but do not practice yoga. On the other extreme, there may be a substantial number of people who visit an experiential retailer to solely enjoy the free experience while making a purchase elsewhere as "it's not completely out of the ordinary for people to visit these stores just for the Instagram pictures" (Lightspeed 2019). This type of free-riding behavior can be costly to retailers. As a result, whether to offer an experience is not an easy task, as it is not straight forward to decide on its quality level and the pricing policy such that the additional traffic generated in stores via the experience is exploited in the most profitable way. This requires considering the complementarity benefits of experiences in facilitating product sales via those consumers who value this new offering and the free riders who may arise to potentially dominate the store traffic in a competitive market.

Consequently, different strategies are observed in the way retailers offer experiences in practice. For example, Nike allows visitors to use its basketball court and get their performance analyzed for free in its Soho store in Manhattan (Business Wire, Inc. 2019). Vans and Lululemon follow the same lead. However, "[r]etailers that sell physical goods need to make sales. The risk is that customers may browse, have a glass of wine, try on a few things, and walk out the door and never order anything" (Practical Ecommerce 2017). To counter such risks, Sephora charges customers a fee for a makeup experience with a beautician, which is waived if the customer makes a purchase. Similarly, Livraria Lello (2023), touted as one of the most beautiful bookshops in the world, charges consumers a store entrance fee, which is deducted from a book purchase. In this paper, our goal is to study the impact of experiential offerings both on retailers and on consumers, considering their spillover effects in competitive markets. Specifically, we focus on the following research questions.

1. What is the impact of offering experiences on the main product prices and demand?

2. When are experiential offerings beneficial for the retailers and consumers?

3. How do profits and social welfare change under the two most common mechanisms: "free for all" and "conditionally free on purchase"?

To disentangle the primary trade-offs, we first address these questions in a monopoly where the experience is offered for free. We then analyze three duopoly scenarios, where none, one, or both of the retailers provide(s) experience(s). In our modeling framework, we adopt a random utility consumer choice model where consumers can consume neither of the offerings or consume them individually or jointly (at either of the retailers), depending on their heterogeneous valuation of the two offerings. We also study the policy where the retailer charges consumers a fee for the experience that is waived with a main product purchase. We identify the cases in which an experiential offering is beneficial for the consumers and the retailers and derive several structural results as well as managerial insights.

Being the first study of this new retail format, our paper offers several new insights on how retailers can use experiential offers to their advantage while also

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considering their implications for consumer welfare. In the following, we discuss our main insights and contributions.

1.1. Insights

In a monopoly setting, we find that although offering experiences raises the main product price, higher-quality experiences do not always lead to higher prices despite their higher costs. Counterintuitively, medium-quality experiences can severely hurt the main product demand and push the store traffic below the level that is observed without an experience. These free good-quality experiences become strong competitors to the highly priced main products. As a result, reduced sales combined with a high number of free riders lowers the profit and potentially, consumer welfare. In contrast, sufficiently low- or high-quality experiences boost the product sales via complementarities and make the consumers better off. Yet, the former makes the only winning proposition because of lower costs.

Experiences can help a retailer to differentiate itself in a competitive market. Surprisingly, our analysis of a duopoly competition with a single experiential retailer shows that this differentiation can backfire on the experiential retailer by benefiting solely the competitor. With medium-quality experiences, the experiential retailer may end up serving mainly the experiential demand, whereas the competitor enjoys most of the product sales, creating a fragmented market. Yet, we find that differentiation is beneficial for the experiential retailer when kept in moderation. That is, low-quality experiences benefit the retailer by helping to raise the main product demand with the right mix of store visitors (i.e., a high number of main product buyers with a low number of free riders). On the other extreme, high-quality experiences are guaranteed to benefit no one but the consumers.

The problem dynamics become more intricate when both retailers offer experiences, leading to new insights. As opposed to the earlier cases, low-quality experiences do not guarantee higher profits when competing experiences exist. Surprisingly, higher-quality experiences can make the retailers better off in this setting, highlighting that the conditions for the profitability go beyond the quality and cost dimensions, and they highly depend on the portfolio of the offerings and the market structure. With such experiences, both retailers benefit from the enlarged pie of demand with a balanced store traffic, which is also beneficial for the consumers. Hence, this setting can lead to a desirable "win-win-win" outcome for both retailers and consumers, which is not possible in other competitive settings.

Our findings show that the free-riding consumers can significantly hurt the profitability of the experiential offers. The retailers may attempt to resolve this issue by implementing a fee for the experience that is waived with a product purchase (as implemented by, for example, Sephora and Livraria Lello). Counterintuitively, when a single retailer offers an experience, this strategy leads to higher main product prices (compared with the "free for all" policy), which in turn, hurts demand, potentially making experiences the main source of revenue. Yet, the revenue generated through experiences also contributes to the retailer's profit, making this strategy more profitable compared with the "free for all" policy. On the downside, the store may lose its function as a product sale location and can turn into an outlet where consumers visit for experiences. As a result, the competitive market may operate like a setting with two monopolists, each serving a different type of demand, which benefits neither the experiential retailer nor the consumers compared with the case when no experiences are offered. On the bright side, when both retailers offer experiences, this fee mechanism enables "win-win-win" outcomes when a "free for all" policy fails to do so.

1.2. Contributions

Although the idea of using experience-rich retail environments to stimulate purchases goes back to the 1990 s, the earlier literature on this topic has been mostly conceptual (e.g., Schmitt 1999, Kim 2001, Healy et al. 2007). More recent works employ survey-based approaches to study the effect of various store design elements on consumer experience (e.g., Ballantine et al. 2010, Foster and McLelland 2015). However, as highlighted by Practical Ecommerce (2017) and Lightspeed (2019), the profitability of offering experiences is still an open question, especially under competition, and there are no academic studies (neither analytical nor empirical) on this operational aspect. Our study addresses this research gap by studying the implications of such offerings for the retailer profits and consumer welfare, considering two primary benefits of experiences: (1) increasing the traffic in stores and (2) facilitating the sales of the main products through crosscategory complementarities. As we highlight, these two aspects interact differently at different quality levels of the experiences under different competitive settings, even when the other potential benefits of experiences (e.g., resolving consumers' value uncertainty of products, learning about consumer tastes for product design, or inducing positive network externalities) are ignored. Despite the technically challenging nature of the problem, using a novel modeling approach that employs a multivariate logit (MVL) choice model, we are able to derive several analytical results that are also independent technical contributions to the price competition literature with multiple offerings. From a managerial perspective, our results highlight that there is no one-size-fits-all solution to offering profitable experiences and provide some counterintuitive insights as highlighted above. Hence, we contribute to the ongoing debate on the profitability of experiential retailing in different settings by explaining some important

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trade-offs that have not yet been well understood. In a postpandemic world, where the physical stores are slowly opening up, we provide valuable guidelines on this new retail format.

2. Literature Review

Managing experiences in retail is relevant to the research on ancillary services, price and service competition, and crosscategory choice modeling, which we briefly review below.

Experiential offerings relate to the ancillary (add-on) services literature, where retailers offer additional services that some of their consumers value in addition to their main business. Some examples include airlines' entertainment or Wi-Fi services on their flights. Building onto the early bundling literature (see Adams and Yellen 1976, Schmalensee 1984, McAfee et al. 1989, and more recently, Phillips and Simsek 2020), the ancillary service literature studies price discrimination and subscription mechanisms for these services while potentially facing boundedly rational consumers (see Ellison 2005, Gabaix and Laibson 2006, Shulman and Geng 2013, Geng and Shulman 2015, Cui et al. 2018, Wang et al. 2019). In our paper, we also study an additional offering with respect to the retailers' core business, which may increase the attractiveness of visiting the retailer. However, in contrast to ancillary services, which can only be consumed on the condition of a main product purchase from the firm (e.g., the Wi-Fi service can only be used with a flight ticket), experiences can be enjoyed, often for free, by anyone who is interested in this additional offer, resulting in experiences to compete even with the retailer's own product. In addition, although the main question in the ancillary service literature is to bundle or offer them as add-ons to the main service (as ancillary services are profit generators in both settings), in this paper the fundamental questions are whether to offer experiences at all, at what quality level, and under what type of market conditions.

The price and service competition literature studies problems where firms compete with multiple service attributes, potentially including price. These problems have been studied in different contexts. For example, So (2000) and Allon and Federgruen (2007, 2008) study competition in price and service waiting time. Debo and Veeraraghavan (2014) model service quality by the service rate as a second attribute of the product in addition to service value in a queuing setting (see also Anand et al. 2011, Cui and Veeraraghavan 2016). In a recent paper, Yuan et al. (2021) study co-opetition within a cluster of retailers that share the cost of managing common spaces so as to decrease consumers' disutility of waiting in shopping malls. Different from these works, which treat waiting time as the primary service attribute that creates a net disutility for consumers, an experience is the main service offering in our paper. That is, whereas some consumers may value this economic offering highly, others may not (e.g., not all consumers who purchase yoga clothes may practice yoga in real life). Hence, experiences are not simple utility additions to the product utilities, as they are not mere product quality attributes. Rather, consumers may solely obtain the main product even in the presence of an experiential offer. Alternatively, they may only obtain the experience, or they may consume it together with the main product that is sold at a price. This results in complex interactions between the competing retailers that offer experiences, leading to new insights. For example, different from So (2000), which shows that the firms need to exploit their firm characteristics to differentiate their services in the market, we find that firms may suffer from differentiating themselves in the market via offering experiences, as experiential offerings may become strong competitors to the main products, lowering their demand. These aspects of our problem relate to the modeling of crosscategory choice.

Although it is common for consumers to obtain multiple offerings in a shopping trip, the majority of research on choice modeling has focused on single item choice (see Train 2009 for an overview of these models, which are largely derived from the random utility-based logit model). More recently, crosscategory choice models have been developed to study the decision of selecting multiple items, each corresponding to a specific category, at a given decision moment. A key feature of these problems concerns the interdependencies across the items in different categories. By taking such correlations into account, these models are used to derive the choice probabilities for any possible basket. The first stream of models in this area stems from the spatial statistics literature. One of the most widely used models in this area includes the random utility-based MVL model. The MVL model was first proposed by Cox (1972) in the statistics literature for analyzing multivariate binary data. Russell and Petersen (2000) have later shown that this model can be used to estimate multicategory purchases with interdependencies across them (e.g., a main product and an experiential offering). Specifically, they have demonstrated that by modeling the conditional utility of obtaining a category (given the decisions regarding all other categories) as a logit model, it can be shown that the joint probability distribution for all possible baskets follows the model proposed by Cox (1972). Variations of this model have been used in marketing and operations management studies to study basket purchases by parsimoniously extending the standard logit model to incorporate the complementarity/substitution effects across categories (see, for example, Hruschka et al. 1999, Kopalle et al. 1999, Singh et al. 2005, Bel and Paap 2014, Kwak et al. 2015, Tulabandhula et al. 2020).

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The second stream of research in crosscategory choice modeling is based on the economic principals of utility maximization across baskets with multiple items. As such, Song and Chintagunta (2006), Niraj et al. (2008), Sinitsyn (2012), and Ma et al. (2021) adopt the traditional multinomial logit (MNL) framework where the random utility for each basket includes the summation of the mean utilities for all the items in the basket (e.g., none, one, or both of the experiential offering and main product), along with additional terms indicating the interaction between the items included in the basket and a Gumbel error term that is independent and identically distributed (i.i.d.) across the available baskets. Remarkably, the researchers unanimously confirm that the estimated probabilities via such models are equivalent to those obtained by the MVL model that is derived from the spatial statistics ideas (see the discussions in Russell and Petersen 2000, Song and Chintagunta 2006, Kwak et al. 2015, and Kamakura and Kwak 2020). Relying on the consistency of the two approaches and the wide acceptance of the MVL model by academics and practitioners, we employ the MVL model to study consumer choice where consumers face the choice of obtaining neither, one, or both of the main product and experiential offering while considering the interactions between the two offerings in both monopoly and competitive settings. We also discuss the equivalence and interpretation of this model within a basket utility maximization framework. We thus contribute to the this line of research by analytically studying the pricing problem with potentially complementary multicategory offerings in different competitive settings.

3. Experiential Offering: Free for All

In our base case, we analyze one of the most widely adopted strategies in practice, where an experience is offered for free. Note that despite being free, not all consumers may enjoy the experiential offer. For example, although Lululemon's free yoga sessions can be appealing to yoga-savvy consumers, there may be a considerable number of consumers who find the yoga practice not enjoyable despite wearing yoga clothes. Similarly, Whole Foods' free consultation with a diet coach is mainly targeted for those who want to improve their eating habits. Because those who value the experiential offer may obtain it for free with or without making a product purchase, free riding may occur. Hence, although the retailers' revenue is generated only via product purchases, the costs are also incurred because of experiences, which can be consumed by anyone with a high valuation for free, resulting in nontrivial interactions.

3.1. Monopoly Retailer's Problem

We follow the MVL model to formulate consumer choice in a monopoly setting where the retailer sells a main product and offers a free experience. That is, the retailer provides two offerings of different categories, with potential crosscategory interactions. Let indices x and m denote the experience and the main product categories, respectively. Consumers have the option to obtain neither of the categories or obtain one, potentially complemented with the other, depending on their valuation. In line with the MVL model, we start by formulating the conditional utility of obtaining a specific category, $k \in \{x,m\}$, given the decision regarding the other by accounting for the randomness in consumer taste. By formulating a series of local (conditional) logit choice models, a global choice model is then constructed, taking crosscategory interactions into account (see Cox 1972, Russell and Petersen 2000). Let $Z^k, k \in \{x,m\}$ denote a binary variable with realization z^k, which equals one if the consumer obtains category k and equals zero otherwise. Then, the conditional utility of obtaining (not obtaining) category $k \in \{x,m\}$, given the decision regarding the remaining category $l \neq k, l \in \{x,m\}$, is formulated by setting $z^k = 1$ ($z^k = 0$) in the equation:

$$U(Z^k = z^k | Z^l = z^l, l \neq k, l \in \{x,m\}) = v^k z^k + \psi^{kl} z^k z^l + \epsilon^k_{z^k}, \quad \forall k \in \{x,m\}, \tag{1}$$

where in line with the logit specification, $\epsilon^k_{z^k}, z^k \in \{0,1\}$, $k \in \{x,m\}$ are i.i.d. Gumbel-distributed random variables with zero mean and a variance of $\pi^2/6$, representing the heterogeneity in consumer tastes with respect to the alternatives. Equation (1) indicates that the conditional utility of obtaining category $k \in \{x,m\}$ depends on two elements. First, it depends on the baseline utility derived from the offering, which is denoted by v^k. As such, if consumers visit the retailer to purely enjoy the free experience, they obtain a utility of $v^x \equiv u^x - t$, where u^x denotes the nominal utility associated with the experience and $t > 0$ denotes the travel cost incurred for visiting the retailer. Similarly, consumers obtain a utility of $v^m \equiv u^m - t - p$ by consuming the main product alone at the retailer, where u^m denotes the nominal utility associated with the main product and p denotes the price charged for the product. In addition to the baseline utility, the conditional utility of obtaining a category depends on the interaction between the two offerings. This is represented by the interaction parameter, ψ^{xm}, which is assumed to be symmetric in the category pairs for logical consistency (i.e., $\psi \equiv \psi^{xm} = \psi^{mx}$) (Russell and Petersen 2000). Intuitively, the model dictates that this interaction exists only if the experience and the main product are obtained together (i.e., $z^x = z^m = 1$).

Obtaining the experience and the main product together leads to two types of interactions. First, visiting the retailer for both offerings leads to a natural *travel cost synergy*, which saves consumers an additional travel cost t when both offerings are obtained together. Second,

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retailers can implement experiences that synergize well with the products of their brand (e.g., via yoga classes at Lululemon stores, playgrounds at Toys "R" Us stores, and cooking classes at Whole Foods stores). That is, the retailers do not just sell products, but using their brand as a vehicle, they attempt to sell a "memory" facilitated via experiences. Hence, the marginal value obtained from the main product may increase with the consumption of the experience and vice versa. We let function $\beta(u^x)$ represent this *brand synergy* between the two offerings when they are consumed together. We assume that $\beta(u^x)$ is positively valued and concave increasing in u^x to reflect the effort put forward by the retailer to enhance the synergy level via higher-quality experiences and the diminishing returns to reach the state of a "memorable event" as a result of providing a sufficient-quality experiential offering. Specifically, we assume the functional form of $\beta(u^x) = s(1 - \exp(-\lambda u^x))$, where parameter λ denotes the sensitivity of the synergy level to the quality of the experience provided by the retailer and parameter s denotes the utility obtained from reaching a memorable event (all our analytical results hold when $\beta(u^x)$ is a constant or a piecewise linear function as well). Hence, the interaction term is defined as $\psi \equiv t + \beta(u^x)$, which allows us to explicitly model how the choice for a given offering affects the total utility obtained from the offerings, as formulated in Equation (1). Using this formulation, the conditional probability of obtaining category $k \in \{x, m\}$ can be expressed in logit form:

$$Pr(Z^k = z^k | Z^l = z^l, l \neq k, l \in \{x, m\}) = \frac{\exp(v^k z^k + \psi^{kl} z^k z^l)}{\exp(v^k + \psi^{kl} z^l) + 1}, \quad \forall k \in \{x, m\}. \tag{2}$$

By using the conditional choice probabilities in Equation (2) and the Besag (1974) characterization theorem, the joint probabilities for the full set of four distinct choices can be derived. As such, the probability of choosing option $z^x \in \{0, 1\}$ for the experience and $z^m \in \{0, 1\}$ for the main product can be formulated as follows (see Russell and Petersen 2000 for the details in derivation):

$$q_{z^x z^m} \equiv Pr(Z^x = z^x, Z^m = z^m) = \frac{\exp(\mu_{z^x z^m})}{\sum_{j,i \in \{0,1\}} \exp(\mu_{ji})}, \quad \text{where} \quad \mu_{z^x z^m} \equiv v^x z^x + v^m z^m + \psi z^x z^m, \tag{3}$$

with $v^x = u^x - t, v^m = u^m - t - p$, and $\psi = t + \beta(u^x)$. Note that Equation (3) provides a convenient form that reduces to the standard logit model when considering a single category. A key point to note is that, rather than a basket utility maximization approach, the proposed model starts with a logit assumption on the conditional utility of obtaining a category given the decision on the other. Remarkably, however, the final probabilities obtained from such an approach essentially follow an MNL structure following basket utility maximization, where each basket is represented by the sum of the baseline utility of each item included in the basket corrected by the pairwise interactions between them and a Gumbel error term that is i.i.d. across all the possible baskets. Hence, the validity of Equation (3) in formulating the crosscategory choice probabilities is supported not only by the spatial statistics principals but also, via a basket utility maximization argument.

Given the choice probabilities in Equation (3), the retailer's objective is to maximize profit, which equals the revenue generated through main product sales less the procurement cost of the main product and the cost associated with the experiential offering. We let c^m denote the per unit cost of the main product, whereas $c(u^x)$ represents the cost of offering an experience of quality u^x. We assume that $c(u^x)$ is a convex increasing function in u^x (i.e., the retailer incurs increasing cost per unit increase in the quality of experience). (Note that all our main findings hold under a linear cost function as well.) In the remainder of analysis, we assume that, without loss of generality, the cost function follows the form $c(u^x) = \eta \cdot (u^x)^2$, where parameter $\eta > 0$ captures the sensitivity of the cost function to u^x. Then, the retailer's profit function is given by

$$\pi(p) = (q_{01} + q_{11})(p - c^m) - (q_{10} + q_{11})c(u^x). \tag{4}$$

The first term in Equation (4) denotes the revenue less the procurement cost associated with main product sales, which are composed of two types of consumers who visit the retailer: those who consume the experience along with the main product (q_{11}) and those who do not (q_{01}). The second term in Equation (4) is the cost associated with the experiential offering, which is utilized not only by the consumers who make a main product purchase (q_{11}) but also, by those who take advantage of it for free without making a purchase (q_{10}). We denote the optimal solution to this problem with *. If the retailer does not offer an experience, then the profit function is simply

$$\pi(p) = q_1(p - c^m), \tag{5}$$

where q_1 is the portion of consumers who purchase the main product, and it can be derived within a traditional MNL framework (i.e., $q_1 \equiv \frac{\exp(v^m)}{\exp(v^m)+1}$, which is essentially a simplification of Equation (3) for a single-category problem). We denote the optimal solution to this problem with ^{*N}.

We start with the following result. We rely on several lemmas to derive our main results. We have relegated these lemmas to the online appendix to keep the main body of our paper concise.

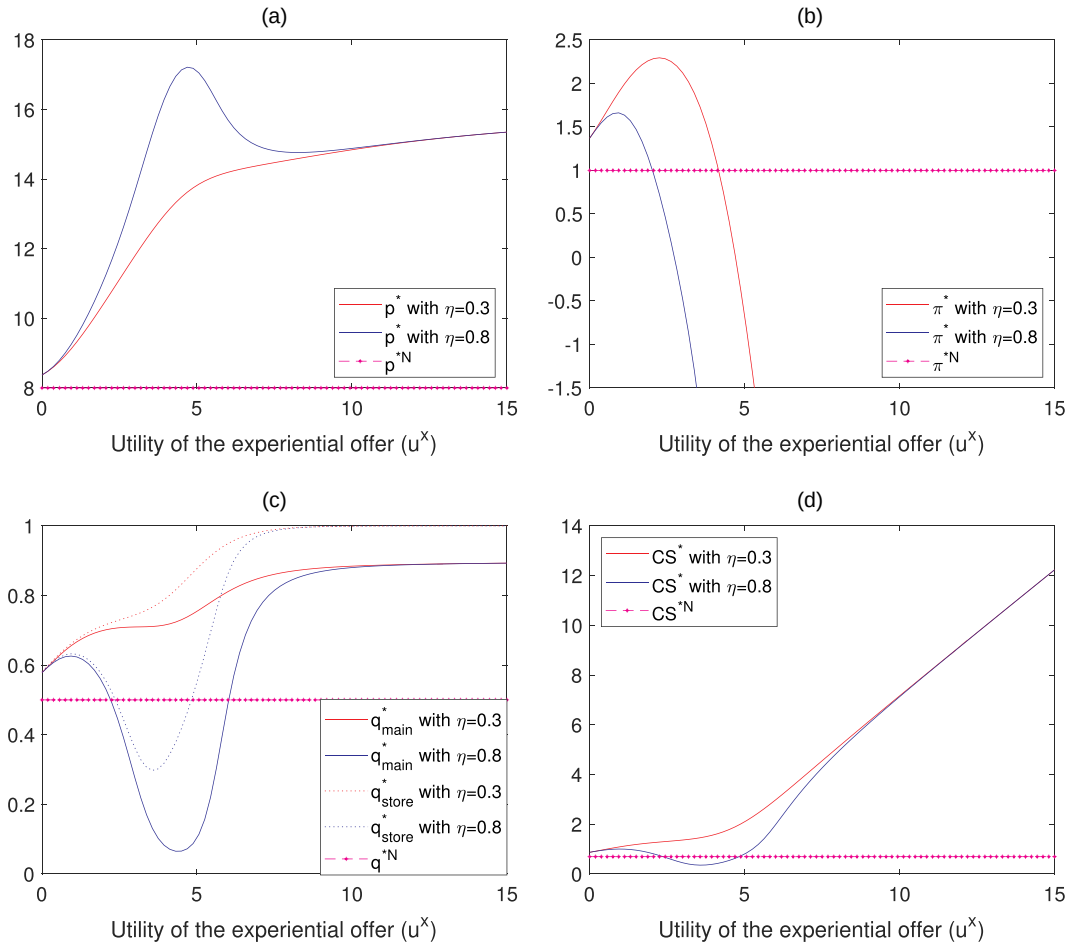
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Theorem 1. *It holds that* $p^* > p^{*N}$, $\forall u^x \geq 0$, *with* $\frac{\partial \pi^*}{\partial u^x}\big|_{u^x=0} > 0$, *and if* p^* *is decreasing in* u^x, *then* π^* *is also decreasing in* u^x, *with* $\lim_{u^x \to \infty} \pi^* = -\infty < \pi^{*N} > 0$. *Furthermore,* p^* *increases in* η, *whereas* π^* *decreases in* η, *and* p^* *and* π^* *both increase in* λ.

Theorem 1 shows that although offering experiences raises the main product prices, the increasing price behavior may not be monotone in the quality of experiences. This is surprising, as higher-quality experiences come at higher costs, which are expected to further increase the prices. As Dor Technologies, Inc (2019) indicates, experiences, especially when offered for free, can help to attract additional consumers to the store. Higher store traffic naturally translates to higher product sales at first because of the complementarity benefits in the store, which enable the retailer to charge a higher price for the product initially. If the cost of experiences can be counteracted by these complementarity benefits (i.e., low η), higher-quality experiences keep raising the main product price (see the red line in Figure 1(a)).

Otherwise, experiences can turn into a double-edged sword on sales when offered at higher quality levels. That is, if the experiences are costly (i.e., high η), higher-quality experiences start hurting the main product sales because of high prices, and potentially, they could lower the sales below the level when no experience is offered, as shown in Figure 1(c) (solid blue line starting at $u^x = 2.25$). This is because higher-quality free experiences paired with highly priced main products eventually deter product purchases while increasing the proportion of free riders (i.e., those who visit the store to purely enjoy the free good-quality experiences). These consumers potentially dominate the store traffic, which can drop below the level that is observed when no experience is offered because of the loss in sales for the main product. This is illustrated in Figure 1(c) (blue lines), which shows a case where the total store traffic and hence, the sales are $q^{*N} = 0.5$ in the absence of an experience. When an experience of quality $u^x = 4.5$ is offered, however, the sales drop to $q^*_{main} = 0.07$, also resulting in lower store traffic of $q^*_{store} = 0.42$, which is dominated by the free

Figure 1. (Color online) Price, Profit, Demand, and Consumer Surplus in a Monopoly Setting



Notes. Parameter values are $u^m = 13, t = 5, c^m = 6, s = 5$, and $\lambda = 0.15$. Also, $q^*_{main} \equiv q^*_{01} + q^*_{11}, q^*_{store} \equiv q^*_{main} + q^*_{10}, q^{*N} \equiv q^*_1$. (a) Price. (b) Profit. (c) Demand. (d) Consumer surplus.

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riders $q^*_{10} = 0.35$. Counterintuitively, the retailer would then be forced to lower the product prices with higher-quality experiences. Although this strategy helps to start regaining the product sales (see Figure 1(c), solid blue line at $u^x = 5$), it does not improve the profits; as the gap between the quality of the experience and the price charged for the product grows wider, profit losses become inevitable despite the rise in sales, as stated in Theorem 1. Given these findings, Theorem 2 sheds light on the settings where it is beneficial to offer experiences.

Theorem 2. *In a monopoly setting, there exists a threshold $\hat{u}^x > 0$ such that offering an experience with utility $u^x \le \hat{u}^x$ improves both the retailer's profit and consumer welfare. Otherwise, an experiential offering can make the retailer and/or consumers worse off.*

Theorem 2 indicates that low-quality experiences are a win-win for the retailer and the consumers. Such a setup allows the retailer to keep the main product price increase relatively low, leading to a good mix of the store visitors. By avoiding an excessive number of free riders, such a strategy boosts sales via complementarities and raises the profit while also improving the consumer welfare (see Figure 1(d) between $u^x = 0$ and $u^x = 2.4$). This is contrary to medium-quality experiences that may result in a tremendous price increase, which can undermine the complementarity benefits. Hence, the profit is hurt with potentially fewer people visiting the store. This may lower the consumer welfare as well. On the other hand, high-quality experiences help to gain back the sales and improve consumer welfare, unfortunately at the cost of retailer profits, as stated in Theorem 2.

3.2. Duopoly Competition

In this section, we study the impact of offering experiences in a setting where two retailers (a and b) engage in a price competition for the main product. We examine three cases. (1) In Case N, neither retailer offers an experience. (2) In Case A, only one of the retailers offers an experience (without loss of generality, retailer a is the experiential retailer). (3) In Case AB, both retailers a and b offer experiences. In all cases, consumers have three options to choose from when it comes to the main product purchase (i.e., obtaining it at retailer a or b or not obtaining it at all). Experiences can also be obtained from multiple sources when they are offered by both retailers (i.e., in Case AB).

Bel and Paap (2014) show that a straight-forward extension of the MVL model proposed by Cox (1972) and Russell and Petersen (2000) can be used for cross-category purchases where each category has more than two options. As such, we let Z^k denote a random variable representing the choice for category $k \in \{x, m\}$, with realization $z^k \in J^k$, where $J^m \equiv \{0, a, b\}$ and $J^x \equiv \{0, a\}$ for Case A and $J^x \equiv \{0, a, b\}$ for Case AB. Unless otherwise noted, we let choice 0 denote the option to not obtain the category (i.e., the only option for the experience in Case N, which is represented by $J^x \equiv \{0\}$ for completeness). Then, the conditional utility of choosing option z^k, for category $k \in \{x, m\}$, given the decision regarding the remaining category $l \neq k, l \in \{x, m\}$, can be formulated as

$$U(Z^k = z^k | Z^l = z^l, l \neq k, l \in \{x, m\}) = v^k_{z^k} + \psi^{kl}_{z^k z^l} + \epsilon^k_{z^k}, \tag{6}$$

where $\epsilon^k_{z^k}, z^k \in J^k, k \in \{x, m\}$ are i.i.d. Gumbel-distributed random variables with zero mean and a variance of $\pi^2/6$. Also, analogous to the monopoly setting, $v^m_i \equiv u^m_i - t_i - p_i, v^x_i \equiv u^x_i - t_i, i \in \{a, b\}, v^k_0 \equiv 0, k \in \{x, m\}$. Further, $\psi^{xm}_{z^x z^m} = \psi^{mx}_{z^m z^x}$, with $\psi^{xm}_{z^x z^m} \equiv (t_{z^x} + \beta_{z^x}(u^x_{z^x}))\mathbb{1}\{z^x = z^m\}, z^x \in J^x \setminus 0, z^m \in J^m \setminus 0$, and $\psi^{xm}_{z^x z^m} = 0, z^k = 0, k \in \{x, m\}$, where $\beta_{z^x}(u^x_{z^x}) \equiv s_{z^x}(1 - \exp(-\lambda_{z^x} u^x_{z^x}))$ and $\mathbb{1}\{\cdot\}$ is an indicator function that equals one if the argument is true and equals zero otherwise. Hence, the properties of the interaction terms are analogous to the monopoly setting such that (1) the interactions between the options across categories are symmetric and they only exist when both categories are obtained at a retailer and (2) the interaction terms take into account both the travel cost and brand synergy between the two offerings at a retailer.

Then, using Equation (6), the conditional probability of choosing option z^k for category $k \in \{x, m\}$, given the choice for the remaining category, is formulated as

$$Pr(Z^k = z^k | Z^l = z^l, l \neq k, l \in \{x, m\}) = \frac{\exp(v^k_{z^k} + \psi^{kl}_{z^k z^l})}{\sum_{z^k \in J^k} \exp(v^k_{z^k} + \psi^{kl}_{z^k z^l})}, \tag{7}$$

from which the following joint probabilities follow

$$q_{z^x z^m} \equiv Pr(Z^x = z^x, Z^m = z^m) = \frac{\exp(\mu_{z^x z^m})}{\sum_{j \in J^x, i \in J^m} \exp(\mu_{ji})}, \quad \text{where} \quad \mu_{z^x z^m} \equiv v^x_{z^x} + v^m_{z^m} + \psi^{xm}_{z^x z^m}. \tag{8}$$

Given the choice probabilities in Equation (8), the profit function of retailer $i \in \{a, b\}$ is

$$\pi_i(p_i) = \sum_{z^x \in J^x} q_{z^x i}(p_i - c^m_i) - \sum_{z^m \in J^m} q_{i z^m} C_i, \tag{9}$$

where c^m_i denotes the cost of the main product at retailer i and C_i denotes the cost incurred for offering the experience by retailer $i \in \{a, b\}$. Similar to the monopoly setting, C_i is defined as the following. (1) $C_i \equiv 0, i \in \{a, b\}$ for Case N; (2) $C_a \equiv c_a(u^x_a)$ and $C_b \equiv 0$ for Case A; and (3) $C_i \equiv c_i(u^x_i)$ for Case AB, where $c_i(u^x_i) \equiv \eta_i (u^x_i)^2, i \in \{a, b\}$.

Given Equation (9), we first study the case when only one of the retailers offers an experience (Case A) in relation to the case when neither of the retailers offer an experience (Case N).

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3.2.1. Only One of the Retailers Offers an Experience. This setting enables consumers to enjoy the experiential offering at retailer a with a main product purchase. Yet, it also makes it possible for consumers to exploit retailer a by consuming its experiential offer for free while making their product purchase at retailer b. This is an interesting case to study as it demonstrates how such dynamics affect the equilibrium outcomes when one of the retailers differentiates itself in the market by adding another potentially complementary category to its main offering. We start with the following result. The equilibrium outcomes are denoted with $^\circ$.

Theorem 3.

1. *There exists a unique Nash equilibrium when neither retailer offers an experience, and at equilibrium, the following relationships always hold*: $\pi_i^\circ = p_i^\circ - c_i^m - 1, i \in \{a, b\}$.

2. *There exists a unique Nash equilibrium when only retailer a offers an experience, and at equilibrium, the following relationships always hold*: $\pi_a^\circ = p_a^\circ - 1 - c_a^m - \frac{e^{u_a^x+\beta_a(u_a^x)}}{e^{u_a^x+\beta_a(u_a^x)}+1} c_a(u_a^x)$ *and* $\pi_b^\circ = p_b^\circ - c_b^m - 1$.

Theorem 3 enables us to provide analytical results on the equilibrium market outcomes on prices, profits, and consumer welfare when retailer a differentiates itself in the market by expanding its offerings via an experience (Case A) compared with the case when the retailers in the market only sell products (Case N), denoted by $^{\circ A}$ and $^{\circ N}$, respectively.

In the following, we present our results on the behavior of the equilibrium outcomes in the quality of the experience u_a^x; see Online Appendix C for additional results on the other problem parameters.

Theorem 4. *When only retailer a offers an experience, the equilibrium price, market share, and profit of retailer a (retailer b) increase (decrease) in* u_a^x *if* $\eta_a = 0$. *Further, in this setting,* $p_a^{\circ N} < p_a^{\circ A}$, $\pi_a^{\circ N} < \pi_a^{\circ A}$, *and* $CS^{\circ N} < CS^{\circ A}$, *whereas* $p_b^{\circ N} > p_b^{\circ A}$ *and* $\pi_b^{\circ N} > \pi_b^{\circ A}$.

As η_a *increases, however, the relationships may reverse because the equilibrium prices of both retailers as well as the market share and profit of retailer b increase in* η_a *for any* u_a^x. *When* $\eta_a > 0$, *it holds that* $\lim_{u_a^x \to \infty} \pi_a^{\circ A} = -\infty$ *and* $\lim_{u_a^x \to \infty} CS^{\circ A} = \infty$.

Theorem 4 helps to develop some intuition on the impact of introducing experiences in a competitive setting. It shows that if retailer a's experiential offering is costless ($\eta_a = 0$), then higher-quality experiences are always better for the retailer. Because such an offering improves the main product demand without an additional cost, a higher-quality experience always raises retailer a's price, market share, and profit while hurting those of the competitor. Consumer welfare also improves with an increase in the quality of the experience, as the moderate rise in price is compensated for by the additional utility obtained from the experience. As a result, in such a setting, retailer a and consumers are always better off with an experience, whereas retailer b is worse off. However, as the cost of experience increases ($\eta_a > 0$), retailer a charges consumers a higher price. This is beneficial for the competitor, as she gains additional market share and profit despite the price that she sets. Surprisingly, very high-quality experiences eventually make both retailers suffer in this setting, and the only winner is the consumer, as we elaborate below.

Theorem 5. *When only retailer a offers an experience, the following results hold.*

1. *There exists a threshold* $\breve{u}_a^x > 0$ *such that if the utility of the experience satisfies* $u_a^x \geq \breve{u}_a^x$, *then both retailers are worse off, whereas consumers are better off with an experiential offer.*

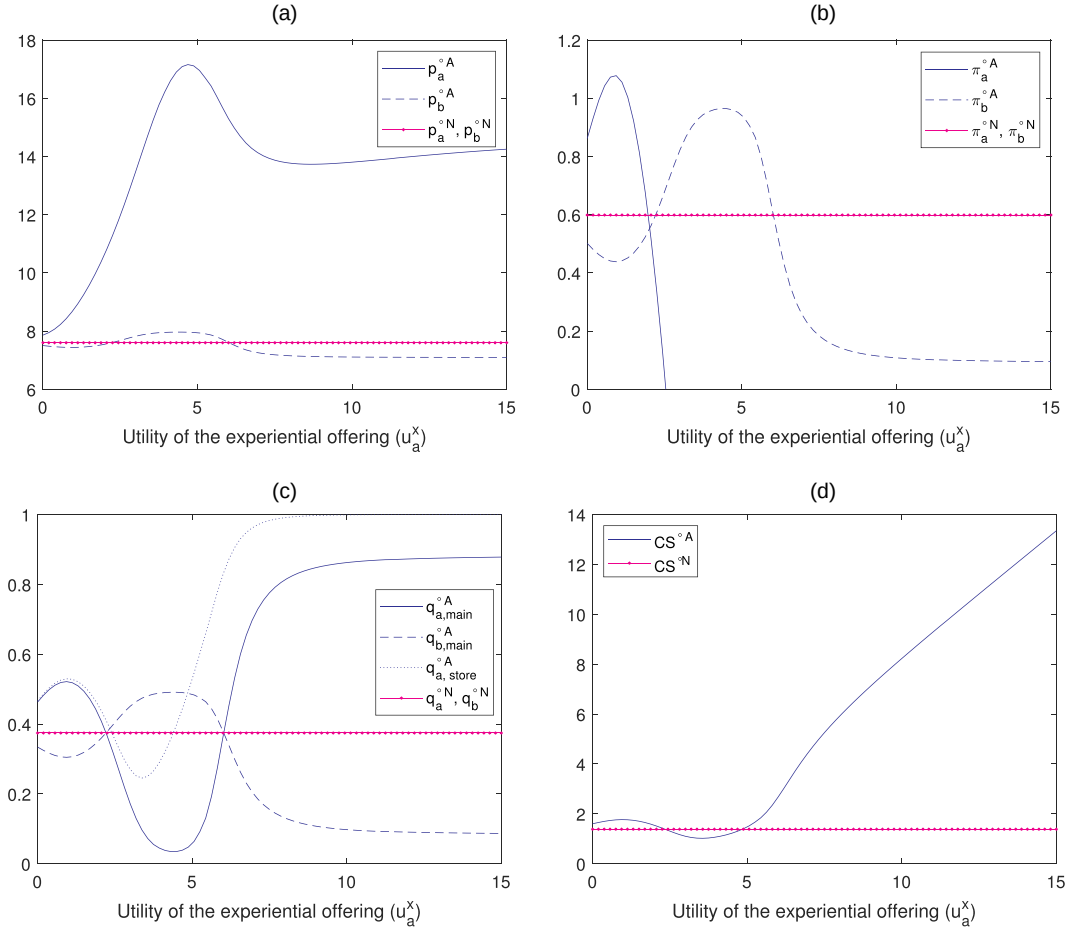
2. *There exists a threshold* $\tilde{u}_a^x > 0$ *such that if the utility of the experience satisfies* $u_a^x \leq \tilde{u}_a^x$, *then retailer a and consumers are better off, whereas retailer b is worse off with an experiential offer.*

Theorem 5(1) indicates that when a high-quality experience is offered by retailer a, no one benefits but the consumer. As a result, both retailers suffer from the differentiation in the market with respect to the retailers' offerings. The high-quality experiences attract many consumers to the store of retailer a. Although this raises the sales, the high cost that comes with such experiences makes the profit loss unavoidable for retailer a. Retailer b is also worse off because of the loss of market share to retailer a, resulting in a "lose-lose" outcome for the retailers. This finding is consistent with what practitioners predict; Kambiz Hemati, Vice President of global retail design at Foot Locker, says that there is a much greater cost to doing nothing at all than trying experiences that fail (Retail Dive 2019).

On the other extreme, if the quality of the experiential offering is kept sufficiently low, then it is guaranteed that both the consumers and the experiential retailer benefit from it, whereas the competitor is hurt (i.e., a "win-lose" outcome is realized for the retailers) as indicated in Theorem 5(2). As a result, retailer a benefits from differentiating itself in the market when done in moderation. In such a case, offering an experience results in a mild price increase (see Figure 2(a)), which allows retailer a to gain additional market share for the product while observing a relatively low number of free riders, as depicted in Figure 2(c) between $u_a^x = 0$ and $u_a^x = 1.9$ (Note that $q_{a,main}^{\circ A}$ represents only the store visitors who buy the main product, whereas $q_{a,store}^{\circ A}$ represents all the store visitors, including those who buy the main product as well as those who are free riders). A good mix of store visitors (with a high number of main product buyers and a low number of free riders) yields higher profits with a slightly higher main product price, which does not lead to a reduction in the overall consumer welfare, as depicted in Figure 2(d). On the other

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Figure 2. (Color online) Price, Profit, Market Share, and Consumer Welfare in a Duopoly Setting When One of the Retailers Offers an Experience Compared with When Neither of the Retailers Offer an Experience



Notes. Parameter values are $u_i^m = 13, c_i^m = 6, t_i = 5, i \in \{a, b\}, \eta_a = 0.8, s_a = 5$, and $\lambda_a = 0.15$. (a) Price. (b) Profit. (c) Market share. (d) Consumer surplus.

hand, the competitor fails to attract main product demand, despite lower price that she implements.

Most interestingly, we find that medium-quality experiences can backfire on the retailer by benefiting solely the competitor (i.e., a "lose-win" outcome for the retailers) (see Figure 2(b) between $u_a^x = 2.2$ and $u_a^x = 6$). Although low-quality experiences help attract additional sales via a mild price jump, higher-quality experiences lead to a much greater price increase. Hence, the free high-quality experiential offer becomes a strong competitor to the highly priced main product. Although there are complementarity benefits of obtaining both offerings together, these are undermined by high prices, which deter customers from making main product purchases while increasing the demand for the experience alone. This hurts the main product sales while encouraging free riders at retailer a, leading to profit losses. In contrast, the competitor benefits by capturing a part of the demand that is lost by retailer a. As shown in Figure 2(c), these dynamics lead to a fragmentation in the market where the competitor enjoys a greater market share for the main product than the experiential retailer, which in turn, attracts mainly the demand for the experiential offering at the expense of the main product sales (compare the solid line with the dotted and dashed lines in Figure 2(c)). This benefits neither the experiential retailer nor the consumers (because of high product prices in the market), and the only winner is the competitor (see the dashed line in Figure 2(b)).

Next, we turn our attention to the case when both retailers offer an experience.

3.2.2. Both Retailers Offer Experiences. In this setting, both retailers compete for the attention of the consumers via experiential offers along with the main product. That is, consumers can obtain both the main product and the experiential offering during their visit to one of the retailers and take advantage of complementarities between them. Alternatively, consumers can opt to visit different retailers for the main product and experience. This makes the competition for the main product demand more fierce

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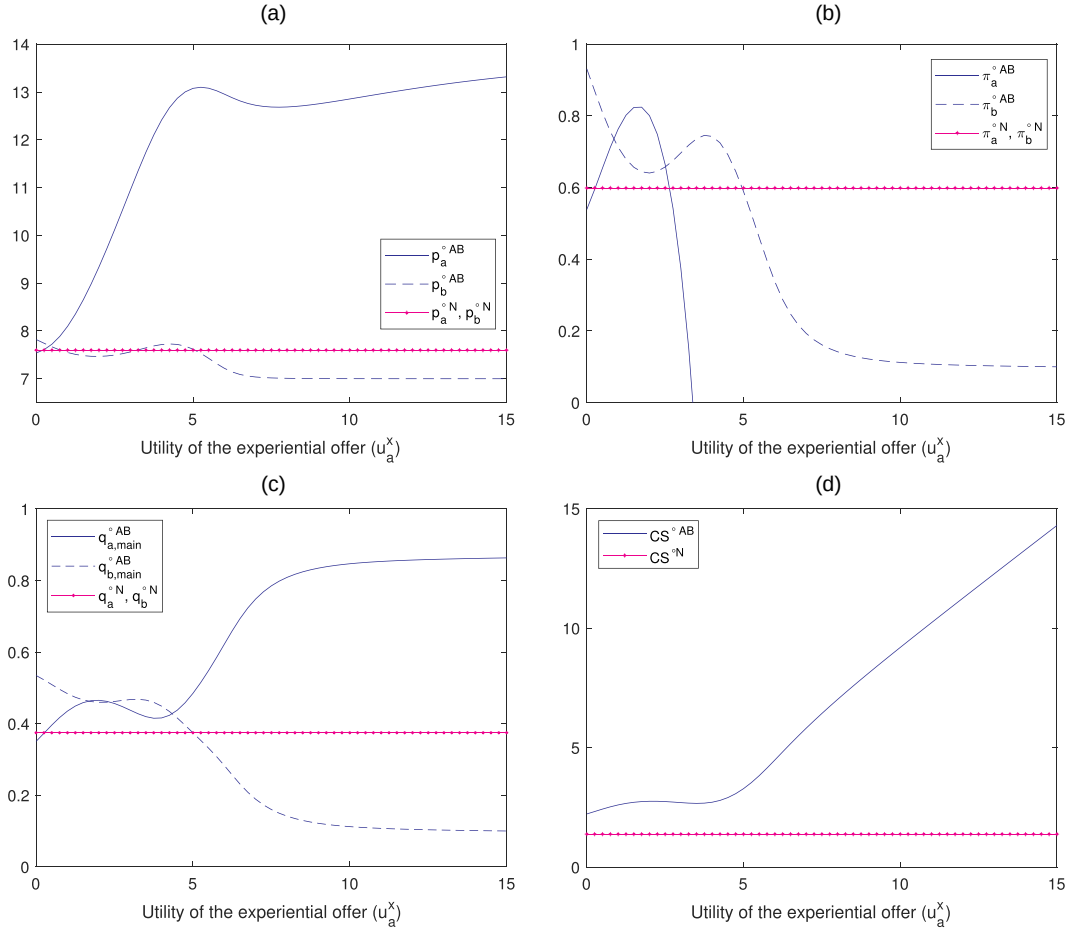
and the trade-offs more intricate, which result in an analytically challenging problem. Despite this challenge, we are able to show that there exists a unique Nash equilibrium under mild conditions, shown in Online Appendix B.

Theorem 6. *When both retailers offer experiences, there always exists a Nash equilibrium, and its uniqueness is guaranteed under mild conditions. At equilibrium, the following relationship always holds*: $\pi_i^\circ = p_i^\circ - 1 - c_i^m - \frac{e^{u_i^x+\beta_i(u_i^x)}}{e^{u_i^x+\beta_i(u_i^x)}+e^{u_j^x-t_j}+1} c_i(u_i^x)$, $i,j \in \{a,b\}, i \neq j$.

In all the earlier cases of monopoly and duopoly settings, we have shown that low-quality experiences are always beneficial for the experiential retailer and the consumers (see Theorems 2 and 5). Surprisingly, Figure 3(b) shows that this result does not hold when both retailers offer experiences. That is, low-quality experiences do not guarantee higher profits when competing experiences exist. This is because a highly attractive competing experience would make the retention of main product demand extremely hard for the retailer if sufficient complementarities are not provided. Hence, low-quality experiences may not be sufficient to compete with the other retailer, which would hold the majority of the market share. The retailer would need to adopt higher-quality experiences to gain back the market share in this setting.

These higher-quality experiences lead to an improved market share for both retailers by increasing the size of the total pie. As shown in Figure 3(c), higher-quality experiences enable the retailers to grow their market share and benefit from the enlarged pie of demand through the right mix of store visitors. This also benefits the consumers as the product prices in the market are kept sufficiently low. Hence, experiences can lead to a desirable "win-win-win" outcome for the retailers and consumers when both retailers offer them (see Figure 3(b) and (d) between $u_a^x = 1$ and $u_a^x = 2$). This is in contrast to the earlier cases, where our extensive numerical

Figure 3. (Color online) Price, Profit, Market Share, and Consumer Welfare in a Duopoly Setting When Both Retailers Offer Experiences Compared with When Neither of the Retailers Offer an Experience



Notes. Parameter values are $u_i^m = 12, c_i^m = 6, t_i = 4, \lambda_i = 0.15, i \in \{a,b\}, \eta_a = 0.4, \eta_b = 0.1, u_b^x = 1$, and $s_a = 5, s_b = 1$. (a) Price. (b) Profit. (c) Market share. (d) Consumer surplus.

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analyses show that at most, one retailer benefits from experiences. Hence, it is a stark difference from the earlier settings where the low-quality experiences are guaranteed to make the retailer better off, highlighting that the profitability of experiences goes beyond their cost and quality dimensions, and it is highly dependent on the market structure. Hence, some intuitive results (i.e., low-quality experiences being always profitable) may not hold in all competitive settings.

4. An Alternative Mechanism: Conditionally Free on Purchase

Noting the effect of free-riding consumers on profitability, we study a mechanism that discourages such behavior in this section. Specifically, we explore the policy under which the retailer charges experience-seeking consumers a fee that is counted toward a main product purchase. This setup is analogous to the business model adopted by Sephora, where consumers can enjoy makeup sessions for a fee that is waived with a product purchase. Similarly, Livraria Lello, a bookshop located in Portugal, charges consumers a store entrance fee that is redeemed on a book purchase.

In the following, we study the impact of adopting such a mechanism on the optimal price, profit, and consumer welfare. We assume a general function for the fee, $f(u^x)$, that is linear in the utility of the experience that is being offered (i.e., $f(u^x) = \alpha u^x$, with $0 < \alpha \leq 1$). We also use indices *FFA* and *CFP* to denote the strategies "free for all" and "conditionally free on purchase," respectively.

4.1. Monopoly Retailer's Problem

The MVL model that we employ for the *FFA* policy provides a general and flexible framework that can be easily adapted to the policy that we explore in this section. Hence, Equation (3) also applies to the *CFP* policy, albeit with slight adjustments to the parameters v^x and ψ. As such, the following are defined for the *CFP* policy: $v^x \equiv u^x - t - f, v^m \equiv u^m - t - p$, and $\psi \equiv t + \beta(u^x) + f$. The modified parameters v^x and ψ indicate that, compared with the *FFA* policy, the baseline utility obtained from the experience (v^x) is reduced by the fee that is charged for the experience, $f \equiv f(u^x)$. This fee, however, is dropped with a main product purchase (i.e., the complementarity between the two offerings is now greater by f, which is now also included in the interaction term ψ). Then, the following insights hold.

Theorem 7. *Optimal price and profit under the CFP policy are higher than those under the FFA policy, whereas the consumer welfare is lower*: $p^{*CFP} > p^{*FFA}, \pi^{*CFP} > \pi^{*FFA}$, *and* $CS^{*CFP} < CS^{*FFA}$. *Furthermore, under the CFP policy, there exists a threshold* $\hat{u}^{x,CFP} > 0$ *such that offering an experience with utility* $u^x \leq \hat{u}^{x,CFP}$ *improves both the retailer's profit and consumer welfare. Otherwise, an experiential offering can make the retailer and/or consumers worse off.*

Interestingly, Theorem 7 shows that the main product price increases when the retailer starts charging free-riding consumers a fee for the use of experiences. When the fee structure is implemented, the experiences also become a source of revenue for the retailer, enabling the retailer to raise the main product pricing. Although this comes at the expense of the demand for the main product, the revenue that is raised via the experience-purchasing consumers also contributes to the retailer's profit, making this strategy more profitable compared with the "free for all" policy. On the downside, Figure 5 in Online Appendix D shows that the demand drop would be extreme with medium-quality experiences, where the price increase is at its peak. As a result, these experiences lead to a setting where the store may lose its function as a sale location for the product and simply turns into an outlet where consumers visit to purchase experiences. Hence, even though the profit recovers slightly, it still makes the retailer and the consumers worse off compared with the policy of not offering experiences.

Yet, Theorem 7 indicates that there still exist "win-win" outcomes where both the retailer and consumers are better off with experiences under the *CFP* policy. Similar to the *FFA* policy, if the quality of the experience is sufficiently low, then offering an experience with a fee charged on free-riding customers is more profitable than not offering an experience at all. Although consumer welfare drops slightly with the implementation of a fee structure, it still poses a better outcome than the setting in which the retailer does not offer an experience at all. As a result, the strategy that gives a higher profit (i.e., implementing a fee for the experience) also improves consumer welfare compared with the setting where an experience is not offered.

4.2. Duopoly Competition

Similar to the monopoly setting, we can adapt the model we used for the *FFA* policy to study the impact of implementing a fee structure in a duopoly setting. That is, Equation (8) applies with some adjustments to parameters v_i^x and $\psi_{z^x z^m}^{xm}$, which are now defined as $v_i^x \equiv u_i^x - t_i - f_i$, $i \in \{a, b\}$, with $f_i \equiv f_i(u_i^x) = \alpha_i u_i^x$ with $0 < \alpha_i \leq 1, i \in \{a, b\}$, and $\psi_{z^x z^m}^{xm} \equiv (t_{z^x} + f_{z^x} + \beta_{z^x}(u_{z^x}^x))\mathbb{1}\{z^x = z^m\}, z^x \in J^x \setminus 0, z^m \in J^m \setminus 0$. In this setting, the following result holds.

Theorem 8. *When only retailer a offers an experience, there exists a unique Nash equilibrium under the CFP policy, and at equilibrium, the following results hold.*

1. *There exists a threshold* $\breve{u}_a^{x,CFP} > 0$ *such that if the utility of the experience satisfies* $u_a^x \geq \breve{u}_a^{x,CFP}$, *then both retailers are worse off, whereas consumers are better off with an experiential offer.*

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2. *There exists a threshold* $\tilde{u}_a^{x,CFP} > 0$ *such that if the utility of the experience satisfies* $u_a^x \le \tilde{u}_a^{x,CFP}$, *then retailer a and consumers are better off, whereas retailer b is worse off with an experiential offer*.

Theorem 8 indicates that the earlier findings with respect to the "win-win" and "lose-lose" outcomes in this setting are robust considering the different types of offer policies. In addition, our numerical results show that the fee structure implemented by retailer *a* leads to improvements in the profits of both retailers; see Figure 6 in Online Appendix D. Yet, similar to the monopoly setting, the fee structure leads to a drop in the main product demand. Hence, with medium-quality experiences, there is a greater fragmentation in the market, where the experiential retailer serves solely the consumers who seek the experience, whereas the competitor serves all the consumers who seek the main product. As a result, the market essentially operates like a setting with two monopolists, each serving a different type of demand. This benefits neither the experiential retailer nor the consumers, and the only winner is the competitor, which is able to attract greater main product demand with higher prices.

Our numerical results show that when both retailers offer experiences, the fee structure elevates the prices and profits of both retailers as well. As a result, a fee structure has the potential to enable a "win-win" outcome for retailers under certain conditions when a "free for all" policy fails to do so. Although consumers are worse off with a fee structure, they are still better off compared with the case where retailers do not offer experiences at all. Also, when both retailers compete with experiences in addition to price, the distribution of demand for the two types of offerings is more balanced across the retailers. Therefore, offering experiences "conditionally free on purchase" has the potential to raise retailers' profits beyond the level that is achieved by offering them "free for all" while also improving consumer welfare compared with the setting where experiences are not offered at all.

5. Conclusion

As consumer spending slowly shifts from goods toward experiences, retailers are offering both in their stores to attract consumers back to stores and take advantage of the complementarities provided by experiences to convince consumers to make product purchases. Although experiences can have a variety of complementarity benefits in consumption with the main products, they can also be consumed individually, leading to potential competition for the store traffic because of consumer heterogeneity. In this paper, we study the pricing and consumer welfare implications of these dynamics in monopoly and duopoly settings. We derive several structural results along with managerial insights on how experiences should be managed to ensure profitability while also improving consumer welfare.

Being the first study on this new retail format, we consider two primary benefits of experiences in our paper: (1) increasing the traffic in stores and (2) facilitating the sales of the main products through crosscategory complementarities. Although we do not consider any additional potential benefits of experiences (e.g., resolving consumers' value uncertainty of products, learning about consumer tastes for the design of products, or inducing positive network externalities), we show that the two aspects we consider have an intricate interplay at different quality levels of the experiences in different competitive settings. Hence, through a well-supported model, our results explain some aspects of this new retail format that have not yet been well understood. We highlight that there is no one-size-fits-all solution to offering profitable experiences and provide some valuable insights into the ways in which retailers can use experiences to their advantage based on the retail landscape in which they operate while considering the consumer welfare implications. Broader implications of our study pertain to online businesses that can potentially benefit from having an offline footprint. Through new value creation via complementarities induced by experiences, online retailers can leverage stores to attract certain consumers who highly value memories, which may be harder to create in online retail.

Future research directions include studying other aspects of experiences. In our model, we assume that the consumers know the precise utilities of the main product and experience before they decide whether to visit the store. This may lead to limitations for certain types of experiences implemented in practice. For example, it may be possible that consumers are not aware of these utilities or the fact that an experiential offer even exists at the retailer before visiting the store. On the other extreme, some retailers offer experiences that the consumers cannot avoid (e.g., Starbucks Reserve Roasteries are designed in a way that the consumers have to take an elevator or stairs that go around the roaster in the store). Alternatively, some companies offer very high-quality experiences to resolve consumers' value uncertainty on new products. In such settings, other kinds of pricing mechanisms that offer additional discounts on products can be adopted. Our model does not consider these aspects but can still serve a starting point for studying them.

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推荐理由：本文研究换电需求随机到达、且电池会衰减并需要更换时的换电站运营决策。作者用有限期马尔可夫决策过程刻画随机调度、分配与库存补充，确定各时段电池充电、放电和更换的数量。论文证明了价值函数的单调性及特殊情形下最优策略的单调结构，并为缓解维数灾难，提出利用回归初始化价值函数的单调近似动态规划方法。计算实验验证了该方法相较精确算法及其他单调近似方法的表现，并提炼无人机换电站的运营启示。

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A Monotone Approximate Dynamic Programming Approach for the Stochastic Scheduling, Allocation, and Inventory Replenishment Problem: Applications to Drone and Electric Vehicle Battery Swap Stations

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Abstract. There is a growing interest in using electric vehicles (EVs) and drones for many applications. However, battery-oriented issues, including range anxiety and battery degradation, impede adoption. Battery swap stations are one alternative to reduce these concerns that allow the swap of depleted for full batteries in minutes. We consider the problem of deriving actions at a battery swap station when explicitly considering the uncertain arrival of swap demand, battery degradation, and replacement. We model the operations at a battery swap station using a finite horizon Markov decision process model for the stochastic scheduling, allocation, and inventory replenishment problem (SAIRP), which determines when and how many batteries are charged, discharged, and replaced over time. We present theoretical proofs for the monotonicity of the value function and monotone structure of an optimal policy for special SAIRP cases. Because of the curses of dimensionality, we develop a new monotone approximate dynamic programming (ADP) method, which intelligently initializes a value function approximation using regression. In computational tests, we demonstrate the superior performance of the new regression-based monotone ADP method compared with exact methods and other monotone ADP methods. Furthermore, with the tests, we deduce policy insights for drone swap stations.

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Keywords: electric vehicles and drones • battery swap station • Markov decision processes • battery degradation • monotone policy and value function • regression-based initialization • approximate dynamic programming

1. Introduction

Electric vehicles (EVs) and drones hold great promise for revolutionizing transportation and supply chains. The U.S. Department of Energy (2014) reports that EVs can reduce oil dependence and carbon emissions, but vehicle adoption is hindered by range anxiety, purchase price, recharge times, and battery degradation (Rezvani, Jansson, and Bodin 2015, Saxena et al. 2015). Drone applications have increased in recent years; many organizations are using drones or undergoing testing to use drones for different purposes, including, but not limited to, delivery (CBS NEWS 2013, Wallace 2015, DHL 2016, Weise 2017), transportation (Mutzabaugh 2017), and agriculture (Jensen 2019). However, drone use is restricted by short flight times, long battery recharge times, and battery degradation (Drones Etc. 2015, Park, Zhang, and Chakraborty 2017, James 2020). An option to overcome these barriers for EVs and drones is a battery swap station. A battery swap station is a physical location that enables the automated or manual exchange of depleted batteries for full batteries in a matter of seconds to a few minutes.

Swap stations have many benefits, including their ability to help reduce battery degradation. Battery degradation, or, more specifically, battery capacity degradation, is the act of the battery capacity decreasing over time with use. Each recharge and use of a battery cause a battery to degrade. Degraded battery capacity means EVs and drones have shorter maximum flight times and ranges. Thus, an interesting aspect of managing battery swap stations is that both battery charge and battery capacity are needed; however, the recharging and use of battery charge is the *exact cause* of battery capacity degradation. Thus, this presents a unique problem where recharging batteries, which enables the system to operate in the short

term, is harmful for long-term operation. Although all recharging causes degradation, the regular-rate charging used at swap stations reduces the speed in which batteries degrade compared with fast-charging (Lacey et al. 2013, Shirk and Wishart 2015). This increases battery lifespans and causes less environmental waste from disposal. Despite this benefit, swap stations still must determine when to recharge and replace batteries.

The benefits of battery swap stations are not restricted to decreasing battery degradation. Swap stations also alleviate range anxiety by allowing users to swap their batteries in a couple of minutes. Furthermore, battery swap stations are projected to be a key component within a smart grid through the use of battery-to-grid (B2G) and vehicle-to-grid (V2G). B2G and V2G enable a charged battery to *discharge* the stored energy back to the power grid (Dunn, Kamath, and Tarascon 2011). In practice, several companies such as Toyota Tsusho and Chubu Electric Power in Japan (Nuvve 2021a, b), NIO and State Grid Corporation in China (Zhang 2020), and Nissan and E.ON in the United Kingdom (Inside EVs 2020) have installed or plan to install V2G technology. Swap stations can also reduce the purchase price barrier through a business plan where the swap station owns and leases the high-cost batteries (Mak, Rong, and Shen 2013). For the many organizations seeking to use drones, a set of continuously operating drones is often vital. However, continuous operation is difficult because the realized flight time of a drone is often less than the recharge time (Drones Etc. 2015, James 2020). Thus, automated drone battery swap stations are a promising option because no downtime for recharging is necessary. Given the benefits and applications of swap stations, we examine the problem of optimally managing a battery swap station when considering battery degradation.

We model the operations at a battery swap station using the new class of stochastic scheduling, allocation, and inventory replenishment problems (SAIRPs). In SAIRPs, we decide on the number of batteries to recharge, discharge, and replace over time when faced with time-varying recharging prices, time-varying discharge revenue, uncertain nonstationary demand over time, and capacity-dependent swap revenue. SAIRPs consider the *key interaction* between battery charge and battery capacity and link the use and replenishment (recharging) actions of charge inventory with the degradation and replenishment needs of battery capacity inventory. Battery charge and capacity are linked because each recharge and discharge of a battery causes the battery capacity to degrade, and the level of battery capacity directly limits the maximum amount of battery charge. To replenish battery capacity, we must determine when and how many batteries to replace over time. For SAIRPs, the combination of battery charge and battery capacity is necessary to satisfy nonstationary, stochastic demand over time.

We model the problem as a finite horizon Markov decision process (MDP) model allowing us to capture the nonstationary elements of battery swap stations over time, including mean battery swap demand, recharging price, and discharging revenue (Puterman 2005). The MDP's state space is two dimensional, indicating the total number of fully charged batteries and the average capacity of all batteries at the station. The action of the model is two dimensional. The first dimension indicates the total number of batteries to recharge or discharge. The second dimension indicates the total number of batteries to replace. The selected action results in an immediate reward, equal to profit, comprised of capacity-dependent revenue from battery swaps, revenue from discharging batteries back to the power grid, cost from recharging batteries, and cost from replacing batteries. The system transitions to a new state according to a discrete probability distribution representing battery swap demand over time, the current state, and the selected action. For our MDP model of the stochastic SAIRP, we seek to determine an optimal policy that maximizes the expected total reward, which is equal to the station's expected total profit. A standard solution method for solving MDPs is backward induction (BI) (Puterman 2005). We solve a set of modest SAIRPs using BI to provide a baseline for comparing the approximate solution methods; however, as Asadi and Nurre Pinkley (2021) showed, BI is not effective for deriving optimal policies for realistic-sized SAIRPs.

Stochastic SAIRPs suffer from the curses of dimensionality; thus, we investigate the theoretical properties of the problem to inform more efficient solution methods. We prove that the stochastic SAIRP has a monotone nondecreasing value function in the first, second, and both dimensions of the state. We also prove that general SAIRPs violate the sufficient conditions for the existence of a monotone optimal policy in the second dimension of the state. However, if the number of battery replacements at each decision epoch is constrained to be less than a constant upper bound, we prove there exists a monotone optimal policy for the second dimension of the state in the stochastic SAIRP.

To overcome the curses of dimensionality, we exploit these theoretical results and investigate efficient solution methods. We investigate methods that exploit our proven monotone structure, including monotone backward induction (MBI) (Puterman 2005) and monotone approximate dynamic programming (ADP) algorithms. First, we examine the monotone approximate dynamic programming (MADP) algorithm

of Jiang and Powell (2015) that exploits the monotonicity of the value function. Next, we propose a new regression-based monotone ADP algorithm, which we denote MADP-RB. In our MADP-RB, we build on the foundation of MADP and introduce a regression-based approach to intelligently initialize the value function approximation.

We design a comprehensive set of experiments using Latin hypercube sampling (LHS). We compare the performance of ADP methods with the BI and MBI for the LHS's generated scenarios of a modest size. Experimentally, we show our regression-based ADP generates near-optimal solutions for modest SAIRPs. Besides, using the same LHS scenarios, we solve large-scale SAIRPs with our proposed ADP algorithms. We demonstrate that our proposed ADP approaches can overcome the inherent curses of dimensionality of SAIRPs that BI and MBI failed to succeed.

1.1. Main Contributions

The main contributions of this work are as follows: (i) we demonstrate that stochastic SAIRPs violate the sufficient conditions for the optimality of a monotone policy in the second dimension of the state and prove the existence of a monotone optimal policy for the second dimension of state when an upper bound is placed on the number of batteries replaced at each decision epoch; (ii) we prove the monotone structure for the MDP value function; (iii) we propose a regression-based monotone ADP method by using the theoretical structure of the MDP optimal value function to intelligently approximate the initial value function and make updates in each iteration; and (iv) we computationally demonstrate the superior performance of our regression-based monotone ADP algorithm and deduce managerial insights about managing battery swap stations.

The remainder of the paper is organized as follows. In Section 2, we outline literature relevant to our modeling approach, solution approaches, and EV and drone applications. In Section 3, we formally define the stochastic scheduling, allocation, and inventory replenishment problem as a two-dimensional Markov decision process. In Section 4, we present theoretical results for the stochastic SAIRP. In Section 5, we present solution methods and outline the monotone ADP algorithm with regression-based initialization to solve stochastic SAIRP instances. In Section 6, we present results and insights from computational tests of the solution methods and realistic-sized instances of the stochastic SAIRP. We summarize the contributions in Section 7 and provide opportunities for future work.

2. Literature Review

There is growing interest surrounding EVs and drones in industry and academia. We proceed by discussing the relevant literature pertaining to (i) the EV and drone swap station application; (ii) the background knowledge for the proposed approach using aspects of optimal timing and reliability, inventory management, and equipment replacement problems; (iii) the scientific works that explain the lithium-ion battery degradation process; and (iv) ADP approaches that address the curses of dimensionality. To the best of our knowledge, no past research has derived the structure of the optimal policy and value function for the scheduling, allocation, and inventory replenishment problem nor solved the realistic-sized instances of SAIRPs to derive insights for managing the operations at a battery swap station faced with battery degradation.

Swap stations were initially introduced for EVs and thus have a more extensive research base. However, there is growing interest surrounding drone battery swap stations. We first examine the work on managing the internal operations of a battery swap station that are most similar to the model presented in this paper and Asadi and Nurre Pinkley (2021). Widrick, Nurre, and Robbins (2018) develop an inventory control MDP for a swap station that only considers the number of batteries to recharge and discharge over time but excludes battery capacity levels, degradation, and replacement. They prove the existence of a monotone optimal policy only when the demand is governed by a nonincreasing discrete distribution (e.g., geometric). Nurre et al. (2014) also consider determining the optimal charging, discharging, and swapping at a swap station using a deterministic integer program that excludes uncertainty. Worley and Klabjan (2011) examine an EV swap station with uncertainty and seek to determine the number of batteries to purchase and recharge over time. Purchasing batteries is fundamentally different from battery replacement in SAIRPs. Worley and Klabjan (2011) examine the one-time purchase of batteries to open a swap station and do not consider purchasing decisions over time. Contrarily, we assume the initial number of batteries at the swap station is previously determined and instead consider replacing batteries over time. Sun, Tan, and Tsang (2014) propose a constrained MDP model for determining an optimal charging policy at a single battery swap station and examine the tradeoffs between quality of service for customers and energy consumption costs.

Other research considers a mix of long-term strategic and short-term operational swap station decisions. Schneider, Thonemann, and Klabjan (2018) consider a network of swap stations that seeks to determine the long-term number of charging bays and batteries to locate at each station and the short-term number of batteries to recharge over time. Schneider, Thonemann, and Klabjan (2018) do consider charging *capacity*; however, their use of capacity indicates the number of batteries that can be recharged at one time in

the station and do not model *battery capacity*. Kang et al. (2016) propose the EV charging management problem, which determines the optimal locations for a network of swap stations and further determines the charging policy for each location. Their definition of charging policy only considers charging and excludes discharging or replacement. Excluding the explicit charging actions over time, Zhang et al. (2014) determine the number of batteries that are necessary for swapping over time. For further studies in the area of EV operations management, we refer the reader to a review by Shen et al. (2019).

A common limitation of the aforementioned research is that it fails to account for battery degradation. To the best of our knowledge, there are very few articles that consider battery degradation. Asadi and Nurre Pinkley (2021) are the first to introduce stochastic SAIRPs for managing battery swap stations with degradation. However, they do not theoretically analyze this problem class, do not introduce intelligent approximate dynamic solution methods that exploit the theoretical results, and do not provide insights from solving realistic-sized SAIRPs.

Others have examined battery degradation in a deterministic setting without any uncertainty (Park, Zhang, and Chakraborty 2017, Kwizera and Nurre 2018, Tan et al. 2019). Sarker et al. (2015) consider the problem of determining the next day operation plan for a battery swap station under uncertainty. They do consider battery degradation; however, they solely penalize battery degradation with a cost in the objective and do not link it to a reduction in operational capabilities.

Others have examined battery swap stations from different perspectives. Researchers have studied how to find the optimal number and location of swap stations in a system (Kim, Song, and Morrison 2013, Hong, Kuby, and Murray 2018, Shavarani et al. 2018). Extending this idea further, Yang and Sun (2015) look to locate swap stations and route vehicles through the swap stations. Others have examined how to locate and/or operate swap stations that are coordinated with green power resources (Pan et al. 2010), stabilize uncertainties from wind power (Gao, Zhao, and Wang 2012), or coordinate with the power grid (Dai et al. 2014).

Our research is related to optimal timing and reliability problems. There is rich literature on finding the optimal timing of decisions to maximize systems' lifespan and reliability. For instance, researchers maximize the expected quality-adjusted life years by finding the optimal timing of living-donor liver transplantation (Alagoz et al. 2004), biopsy test (Chhatwal, Alagoz, and Burnside 2010, Zhang et al. 2012), and replacement of an implantable cardioverter defibrillator generator (Khojandi et al. 2014). There are two options for the actions in these works (e.g., transplant/wait, take/skip the biopsy test, replace/not replace). However, our action determines the *number* of batteries to recharge/discharge and replace at each epoch because it is not a single battery that enables the station to operate. Instead, it is a set of batteries that enables operation, which creates a significantly larger action space depending on the number of batteries at the station. Similar to our work is that of Bloch-Mercier (2001), which determines the optimal timing and duration of a degrading repairable system. There is extensive research in the nexus of optimization, reliability, and systems maintenance. We refer the reader to the recent review paper by de Jonge and Scarf (2020) for further study.

Our research can be placed under the umbrella of inventory management and equipment replacement problems with stochastic elements. There is a large research base examining these types of problems under different characteristics. We proceed by reviewing a small sample of this body of knowledge by focusing on foundational work and research most similar to the scope of this paper. Researchers have extensively studied inventory problems, including those with stochastic demand (Porteus 2002), two- and multiechelon supply chains (Clark and Scarf 1960, Clark 1972), and multiple products (DeCroix and Arreola-Risa 1998). A desirable feature of the solutions to inventory problems is that the optimal policy has a simple structure. A classic example of such an optimal policy is the (s, S) policy that indicates to order up to S units when the inventory level drops below s (Scarf 1960). Others have examined more sophisticated inventory problems that include scheduling production (Laan and Salomon 1997, ElHafsi 2009, Maity 2011, Golari, Fan, and Jin 2017), performing maintenance or replacement (Horenbeek et al. 2013), and ordering spare parts for maintenance (Elwany and Gebraeel 2008, Rausch and Liao 2010). Additionally, researchers have examined perishable inventory that degrades over time (Nahmias 1982) or inventory that can be recycled or remanufactured in a closed-loop supply chain (Toktay, Wein, and Zenios 2000, Zhou, Tao, and Chao 2011, Govindan, Soleimani, and Kannan 2015).

The proposed work is distinct from this previous literature as it links the actions of recharging batteries to the actions that must be taken for replacing battery capacity. No prior work includes the counter-intuitive property that the act of maintaining the system in the short-term (e.g., through recharging batteries which can be analogous to short-term maintenance or short-term inventory replenishment) is harmful for long-term performance (e.g., future need to replace equipment or replenish other types of inventory).

A novel component of our work is the consideration of battery degradation within the decision-making process. Battery degradation is most traditionally measured based on calendar life or cycles, where a cycle consists of one use and one recharge (Lacey et al. 2013, Ribbernick, Darcovich, and Pincet 2015). Using physical

experiments, simulation, and mathematical modeling, researchers aim to capture the rate of battery degradation for different batteries and conditions such as temperature and depth of discharge (Dubarry et al. 2011, Plett 2011, Abe et al. 2012, Hussein 2015, Ribbernick, Darcovich, and Pincet 2015). We approximate battery degradation using a linear degradation factor derived from the work of Lacey et al. (2013) and Ribbernick, Darcovich, and Pincet (2015), as is consistent with other research using a linear forecast (Wood, Alexander, and Bradley 2011, Abdollahi et al. 2015, Xu et al. 2018).

Our MDP model suffers from the curses of dimensionality because of the very large size of all MDP elements together, including state and action spaces, transition probability, and reward. ADP is a method that has had great success in determining near-optimal policies for large-scale MDPs (Powell 2011). Researchers have used ADP methods to solve problems in energy, healthcare, transportation, resource allocation, and inventory management (Bertsimas and Demir 2002, Powell and Topaloglu 2006, Simao et al. 2009, Erdelyi and Topaloglu 2010, Maxwell et al. 2010, Çimen and Kirkbride 2013, Çimen and Kirkbride 2017, Meissner and Senicheva 2018, Nasrollahzadeh, Khademi, and Mayorga 2018). Jiang and Powell (2015) propose a monotone ADP algorithm that is specifically designed for problems with monotone value functions. In this paper, we prove that the value function of the stochastic SAIRP has a monotone nondecreasing structure. Hence, we use the monotone ADP algorithm of Jiang and Powell (2015) and enhance it by adding a regression-based initialization.

3. Problem Statement

In this section, we present and model the MDP model of the SAIRP that considers stochastic demand for swaps over time, nonstationary costs for recharging depleted batteries, nonstationary revenue from discharging, and capacity-dependent swap revenue. The MDP model captures the dynamic average battery capacity over time, the associated replacement policies, and the interaction between battery charge and battery capacity at a battery swap station. This model was originally presented in Asadi and Nurre Pinkley (2021); however, we believe it is necessary to provide the reader with the formal problem definition to enable understanding of the main theoretical and algorithmic contributions that follow. We use a finite horizon MDP to capture the high variability of data over time, including the mean demand for battery swaps, the price for recharging batteries, and the revenue earned from discharging batteries back to the power grid. The uncertainty in the system is the stochastic demand for battery swaps (i.e., exchange of a depleted battery for a fully charged battery). We model this uncertainty (stochastic demand) using the random variable, D_t, for each time period t. These random variables are explicitly used to calculate the transition probabilities. The objective is to maximize the expected total reward of the swap station and determine optimal policies which denote how many batteries to recharge, discharge, and replace over time. For our model, the expected total reward equals the expected total profit calculated as the revenue from satisfying demand and discharging batteries to the power grid minus the costs from recharging and replacing batteries.

We formulate our MDP model with the following elements. We define T as the finite set of decision epochs, which are the discrete periods in time in which decisions are made. By defining N as the terminal epoch, $T = \{1, \ldots, N-1\}$, $N < \infty$.

We denote the two-dimensional state of the system at epoch t, $s_t = (s_t^1, s_t^2) \in S = (S^1 \times S^2)$, as the total number of fully charged batteries, $s_t^1 \in S^1$, and the average capacity of all batteries, $s_t^2 \in S^2$, at the swap station. In the design of s_t^1, we only consider that batteries are either fully charged or depleted. The number of full batteries at epoch t, s_t^1, is an integral value between zero and M, where M is the total number of batteries in the station; thus, $S^1 = \{0, 1, 2, \ldots, M\}$.

We use an aggregated MDP in which we track the discretized average battery capacity rather than a disaggregated MDP, which tracks each battery capacity individually, to reduce the curses of dimensionality from the second dimension of the state. The disaggregated MDP severely suffers from the curse of dimensionality as the state space's size grows exponentially as the number of batteries increases. We discretize the average battery capacity where $S^2 = \{0, \theta, \theta + \varepsilon, \ \theta + 2\varepsilon, \ldots, 1\}$, in which θ equals the lowest acceptable average battery capacity and ε in the discretized capacity increment. State zero in S^2 is an absorbing state representing that the average battery capacity dropped below θ. To discourage the station from allowing the battery capacity to drop below θ thereby resulting in lower quality batteries at the station, we disallow charging, discharging, swapping, and replacement when in this absorbing state. Hence, the set of feasible actions when in an absorbing state, $s_t^2 < \theta$ or $s_t^2 = 0$, only includes no recharge/discharge and no replacement. We note, with this aggregated modeling proposed by Asadi and Nurre Pinkley (2021), the problem size and complexity are reduced, which is not always necessary when using approximate solution methods. However, the aggregated model allows us to benchmark the performance of new and existing approximate solution methods and analyze larger SAIRP instances. Furthermore, we previously showed that the results do not significantly change with aggregation (Asadi and Nurre Pinkley 2021).

We denote the two-dimensional action to represent the number of batteries to recharge/discharge, a_t^1, and the number of batteries to replace, a_t^2, at epoch t. In our aggregated MDP model, there is no known difference between the capacity of batteries as we only track the average capacity of all batteries. In reality, swap stations, applying the aggregated MDP, may track/not track the capacity of each battery. If, consistent with the model, the swap station does not track individual battery capacity values, we assume the specific batteries that are selected to be recharged/replaced or discharged/swapped are *arbitrarily* selected from the set of empty and fully charged batteries, respectively. However, if the swap station does track the individual battery capacity value, we assume that the station selects to recharge/discharge and swap batteries with the highest capacity values and selects to replace batteries with the lowest capacity values. With this selection mechanism, individual battery capacity values will be closer to the average battery capacity of the system and, thus, further emphasizes the aggregated modeling decision. Regarding the first dimension of the action, a_t^1, we attribute a positive value to the number of batteries to recharge, and a negative value to the number of batteries to discharge. To clarify the distinction between recharging and discharging actions, we define positive recharging, a_t^{1+}, and discharging actions, a_t^{1-}, with Equations (1) and (2). We note that only dealing with the positive number of batteries that are recharged or discharged using Equations (1) and (2) is helpful to clarify the forthcoming state transition, probability transitions, and reward calculations:

$$a_t^{1+} = \begin{cases} a_t^1 & \text{if } \ a_t^1 \geq 0, \\ 0 & \text{otherwise,} \end{cases} \tag{1}$$

$$a_t^{1-} = \begin{cases} |a_t^1| & \text{if } \ a_t^1 < 0, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

The action a_t^1 represents both the number of batteries that are recharged, when a_t^1 is positive, and the number of batteries that are discharged, when a_t^1 is negative. We designed the action in this way as it is not beneficial to recharge and discharge at the same epoch, as they will cancel each other out and cause the capacity to degrade. Thus, we select one value for a_t^1 for each epoch t, and state, s_t. Depending on whether the selected action is positive or negative indicates whether recharging or discharging will occur. We denote the number of plug-ins in the station as Φ. We assume all plug-ins are capable of supplying energy from the grid to recharge batteries and receiving energy from batteries discharged using Battery to Grid (Dunn, Kamath, and Tarascon 2011). We define the first dimension of action as $a_t^1 \in A_t^1 = \{\max(-s_t^1, -\Phi), \ldots, 0, \ldots, \min(M - s_t^1 - a_t^2, \Phi)\}$, which limits the number of discharged batteries by the minimum of the number of plug-ins and the number of full batteries ($-\min(s_t^1, \Phi) = \max(-s_t^1, -\Phi)$) and limits the number of recharged batteries by the minimum of the number of plug-ins and the number of depleted batteries that were not replaced. In the second dimension of the action space, $a_t^2 \in A_t^2 = \{0, \ldots, M - s_t^1\}$, we only allow depleted batteries to be replaced at each epoch t which arrive at epoch $t+1$ with full charge and capacity. We define $A_{s_t} = (A_{s_t}^1 \times A_{s_t}^2) \subseteq (A_t^1 \times A_t^2)$ as the set of feasible actions for the state s_t at epoch t. In our model, the set of feasible actions when in an absorbing state, $s_t^2 < \theta$ or $s_t^2 = 0$, only includes no recharge/discharge and no replacement; that is, $A_{(s_t^1,0)} = \{(0,0)\}$.

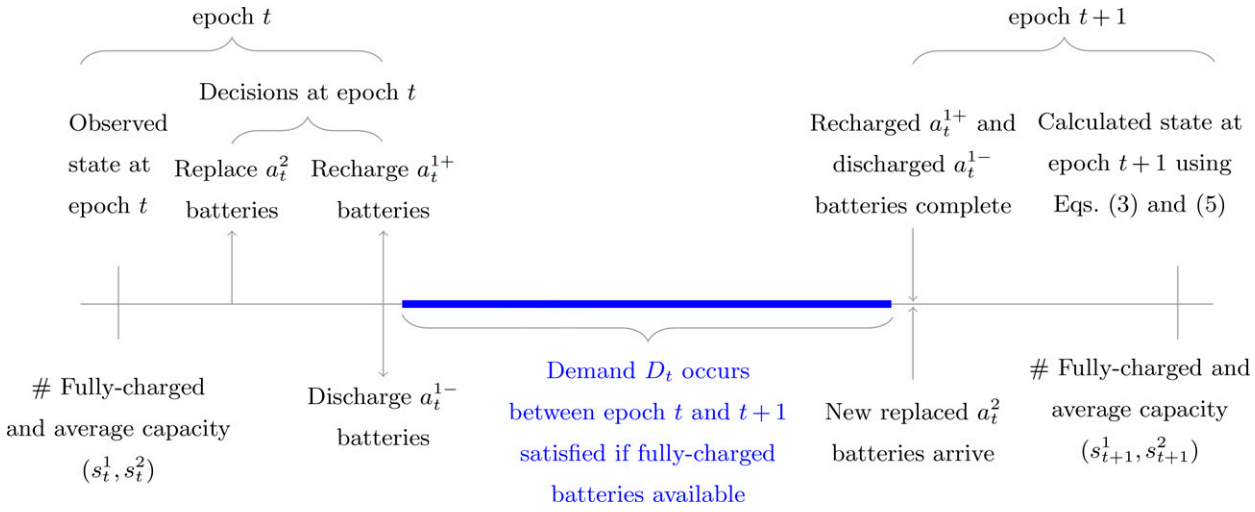
In Figure 1, we display the timing of the events at the swap station. The decisions for actions/operations, including recharging, discharging, and replacing, are made at decision epochs. Between two consecutive decision epochs is a period in which demand is realized and batteries are swapped (as permitted). We assume that the time period between two consecutive epochs is sufficient to recharge or discharge a battery completely. Our model does not allow interruption of recharging, discharging, and replacing actions between two consecutive decision epochs. Therefore, depleted (full) batteries selected for recharging (discharging) at epoch t are fully charged (depleted) at epoch $t+1$. When stochastic demand for a battery swap arrives between epochs t and $t+1$, we can swap up to the number of fully charged batteries in our inventory, which equals the number of fully charged batteries at epoch t minus the number of discharged batteries. We subtract the fully charged batteries assigned to be discharged as they are unavailable for swapping until the next decision epoch.

Transition probabilities indicate the likelihood of transitioning between states when considering the uncertainty of the system. In our MDP model, the uncertainty in the system is the stochastic demand for battery swaps (i.e., exchange of a depleted battery for a fully charged battery) between decision epochs t and $t+1$, D_t. The amount of satisfied swap demand between epochs t and $t+1$ equals $\min\{D_t, s_t^1 - a_t^{1-}\}$, wherein the second term indicates the number of full batteries that are not already assigned to discharging at t. We outline the state transition for the first dimension of the state in Equation (3) that determines the number of full batteries at epoch $t+1$ based on the number of swapped batteries between t and $t+1$ and the number of full, recharged, discharged, and replaced at epoch t:

$$s_{t+1}^1 = s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} - \min\{D_t, s_t^1 - a_t^{1-}\}. \tag{3}$$

The second state transitions according to Equation (4), which determines the future average capacity at $t+1$ based on the current average capacity and the number of full, recharged, discharged, and replaced batteries at epoch t, and Equation (5). We assume that all batteries swapped between epochs t and $t+1$ have a

Figure 1. (Color online) Diagram Outlining the Timing of Events for the SAIRP Model



capacity equal to the average capacity of the batteries at the swap station. We justify the assumption with the following logic. Batteries previously swapped at epoch $t_1 < t$, which are in use outside of the station between t_1 and t and need to be swapped again between epochs t and $t+1$, have a capacity similar to the average station capacity at t when the swap station is used regularly (i.e., $t - t_1$ is small). In other words, if the EVs and drones in the system use swap station frequently, the out-of-control change in the capacity of batteries outside the swap station is negligible, which justifies that the average capacity of batteries inside and outside swap stations are approximately equal. If the swap station is not used frequently, then we leave this case for future work to consider the impact of swapping batteries on the capacity of batteries. We define δ^C to represent the amount of battery capacity degradation from one battery cycle. We adopt the cycle-based degradation measure (Lacey et al. 2013, Abdollahi et al. 2015) and assume that batteries do not degrade when not in use. Furthermore, without loss of generality, we attribute the degradation from a full cycle to the recharge/discharge portion of the cycle. We use *round*() to represent that Equation (4) returns values in the discretized state space, S^2, with ε precision:

$$g^2(s_t^1, s_t^2, a_t^1, a_t^2) = round\left(\frac{(s_t^2 - \delta^C)(a_t^{1+} + a_t^{1-}) + a_t^2 + s_t^2(M - a_t^{1+} - a_t^{1-} - a_t^2)}{M}\right). \tag{4}$$

In the first term in the numerator of Equation (4), we multiply the summation of the number of recharged ($a_t^{1+} > 0$) and discharged ($a_t^{1-} > 0$) batteries by the reduced average capacity ($s_t^2 - \delta^C$) because of the recharging/discharging actions. The second term adds the a_t^2 replaced batteries with 100% capacity. The third term maintains the same capacity for batteries not recharged, discharged, and replaced. These terms are all averaged over the M batteries in the swap station. The system enters the absorbing state $0 \in S^2$ when the average capacity is less than θ. To discourage entrance into this absorbing state, no recharging, discharging, swapping, or replacement is allowed. This setting ensures that swap stations should take appropriate actions *before* allowing the average capacity to drop below θ. Thus, the transition of the second dimension of the state (average capacity of batteries) is precisely defined with Equation (5). If the output of $g^2()$ for taking a specific action is a number $j^2 \geq \theta$, the output of the transition function for the second state of the system (average capacity of batteries), $f^2()$, is j^2. Otherwise, if $j^2 < \theta$ the system transfers to the absorbing state. Hence, the output of function $f^2()$ equals zero:

$$f^2(s_t^1, s_t^2, a_t^1, a_t^2) = s_{t+1}^2 = \begin{cases} g^2(s_t^1, s_t^2, a_t^1, a_t^2) & \text{if } g^2(s_t^1, s_t^2, a_t^1, a_t^2) \geq \theta, \\ 0 & \text{otherwise.} \end{cases} \tag{5}$$

In Equation (6), we define the probability of transitioning from state $s_t = (s_t^1, s_t^2)$ at epoch t to the state $j = (j^1, j^2)$ at epoch $t+1$ when action $a_t = (a_t^1, a_t^2)$ is taken:

$$p(j^1, j^2 | s_t^1, s_t^2, a_t^{1+}, a_t^{1-}, a_t^2) = \begin{cases} p_{s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} - j^1} & \text{if } a_t^2 + a_t^{1+} < j^1 \leq s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} \text{ and} \\ & j^2 = f^2(s_t^1, s_t^2, a_t^1, a_t^2), \\ q_{s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} - j^1} & \text{if } j^1 = a_t^2 + a_t^{1+} \text{ and } j^2 = f^2(s_t^1, s_t^2, a_t^1, a_t^2), \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

We define $p_j = P(D_t = j)$ and $q_u = \sum_{j=u}^{\infty} p_j = P(D_t \geq u)$. Each probability in Equation (6) depends on the number

of batteries swapped, that is, $s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} - j^1$ (see Equation (3)). When no batteries are swapped at epoch t, the station still has $s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-}$ fully charged batteries at epoch $t+1$. Instead, if all available fully charged batteries at epoch t are swapped, the station will have $a_t^2 + a_t^{1+}$ fully charged batteries at epoch $t+1$, which are the result of the a_t^2 replaced and a_t^{1+} recharged batteries at epoch t.

In Equation (6), the probability of transitioning to another state is nonzero only when Equation (5) is satisfied. When the transition probability is $P(D_t = s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} - j^1)$, the demand for swaps is less than or equal to the number of full batteries available for swapping (as in condition 1 of Equation (6)). Alternatively, when the demand for swaps is greater than the number of available full batteries, the state transitions according to the cumulative probability $P(D_t \geq s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} - j^1)$. If Equation (5) is not satisfied, j^1 is lower than the total number of batteries recharged and replaced ($a_t^2 + a_t^{1+}$), or j^1 exceeds than the maximum number of fully charged batteries, $s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-}$, the probability of transition is zero.

To clarify the transition probability function, we illustrate using an example. Consider the case when at epoch t, the swap station has 80 full batteries and 20 depleted batteries in inventory (i.e., $M = 100, s_t^1 = 80$), the average battery capacity equals 0.85, and we take the action to recharge 10 batteries (i.e., $a_t^1 = a_t^{1+} = 10$) and replace 5 batteries (i.e., $a_t^2 = 5$). For this example, we assume recharging or discharging for one time period results in a capacity degradation equal to 0.01 (i.e., $\delta^C = 0.01$) and the discretized capacity increment is also 0.01 (i.e., $\varepsilon = 0.01$). If there is no demand for battery swaps (i.e., $D_t = 0$), at epoch $t+1$ the station will have 95 full batteries with a discretized average capacity equal to 0.86. Thus, the probability of transitioning to a state with more than 95 full batteries or an average capacity not equal to 0.86 is zero. Contrarily, if the demand for swaps is 80 or more (i.e., $D_t \geq 80$), then all full batteries in inventory will be swapped and the number of full batteries at epoch $t+1$ equals $a_t^2 + a_t^{1+} = 10 + 5 = 15$. Thus, the probability of transitioning to a state with less than 15 full batteries is zero. Furthermore, the probability of transitioning to a state with exactly 15 full batteries and average capacity equal to 0.86 indicates that demand for swaps met or exceeded $s_t^1 - a_t^{1-} = 80$. Last, consider the case that we transition to a state with 30 full batteries and average capacity equal to 0.86. The 30 full batteries is between the minimum, $a_t^2 + a_t^{1+} = 15$, and maximum, $s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} = 95$ number of full batteries; thus, we know that $a_t^2 + a_t^{1+} = 15$ batteries arrive at the end of t indicating that we swapped $s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} - j^1 = 80 + 5 + 10 - 0 - 30 = 65$ batteries at epoch t. As follows, the probability of transitioning to this state equals the probably that demand for swaps equals 65, that is, $P(D_t = 65)$.

The actions taken seek to maximize the expected total reward. The expected total reward depends on the immediate reward earned at each epoch. Specifically, the immediate reward is the profit earned. In our setting, swap stations earn revenue from swapping and/or discharging fully charged batteries and incur costs to recharge and/or replace depleted batteries. We calculate the immediate reward at epoch t according to the state of the system $s_t = (s_t^1, s_t^2)$, the taken action $a_t = (a_t^1, a_t^2)$, and the future state $s_{t+1} = (s_{t+1}^1, s_{t+1}^2)$. Specifically, the immediate reward is calculated according to Equation (7):

$$r_t(s_t, a_t, s_{t+1}) = \rho_{s_t^2}(s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} - s_{t+1}^1) - K_t a_t^{1+} + J_t a_t^{1-} - L_t a_t^2, \tag{7}$$

where $s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} - s_{t+1}^1$ equals the number of batteries swapped and the time-dependent recharging cost, discharging revenue, and replacement cost are defined as K_t, J_t, and L_t, respectively. SAIRPs consider two aspects of a battery: charge and capacity. In this model, the fully charged/empty batteries are not necessarily full capacity because they might already be degraded because of the previous recharge/discharge actions. Thus, the average capacity of batteries can take a value less than 100%. We assume the realized swap revenue depends on the current average capacity. Thus, we define $\rho_{s_t^2}$ to be the capacity-dependent revenue per battery swapped in Equation (8):

$$\rho_{s_t^2} = \beta\left(1 + \frac{s_t^2 - \theta}{1 - \theta}\right) = \frac{\beta(1 + s_t^2 - 2\theta)}{1 - \theta}. \tag{8}$$

We set $\beta \geq \max_{t \in T} K_t$ to ensure the swap station is profitable with each battery swapped (i.e., the swap revenue is no less than the maximum recharging cost). We use the average capacity of batteries as the indicator of the quality of batteries in the station when developing the revenue per swap function. Revenue per battery swap is a linear function of the average capacity of batteries in the station. This setting ensures that the stations can gain higher revenue when the average capacity is higher. It also provides an incentive for swap stations to replace batteries for higher revenue and benefits customers by receiving higher quality batteries. In the design of Equation (8), when the average capacity is at the lowest operational value ($s_t^2 = \theta$), the revenue per swap $\rho_{s_t^2}$ equals β, which is at least equal to the maximum price paid for recharging batteries. When the swap station has an average

battery capacity equal to one, $s_t^2 = 1$, then $\rho_{s_t^2} = 2\beta$, which equates to a higher revenue earned due to higher customer satisfaction from swapping a higher quality battery. Hence, in our design, the revenue per swap has a value between $[\beta, 2\beta]$ depending on the average capacity of batteries in the station at epoch t. We calculate the terminal reward in Equation (9) as the potential revenue from swapping all remaining fully charged batteries provided the average battery capacity is at least θ:

$$r_N(s_N) = \begin{cases} \rho_{s_N^2} s_N^1 & \text{if } s_N^2 \geq \theta, \\ 0 & \text{otherwise.} \end{cases} \tag{9}$$

Using the probability transition function and the immediate reward, we define the immediate expected reward in Equation (10):

$$r_t(s_t, a_t) = \sum_{s_{t+1} \in S} \left[p_t(s_{t+1} \mid s_t, a_t)(\rho_{s_t^2}(s_t^1 + a_t^2 + a_t^{1+} - a_t^{1-} - s_{t+1}^1)) \right] - K_t a_t^{1+} + J_t a_t^{1-} - L_t a_t^2. \tag{10}$$

We define the decision rules, $d_t(s_t) : s_t \rightarrow A_{s_t}$, as a function of the current state and time. Our decision rules determine the selected action $a_t \in A_{s_t}$ when the system is in s_t at decision epoch $t \in T$. In our problem setting, we use deterministic Markovian decision rules because we choose which action to take if we know the current state (Puterman 2005). A policy π consists of a sequence of decision rules $(d_1^\pi(s_1), d_2^\pi(s_2), \ldots, d_{N-1}^\pi(s_{N-1})$ for all decision epochs. The expected total reward of policy π, denoted $v_N^\pi(s_1)$ when the system starts in state s_1 at epoch $t = 1$ is calculated according to Equation (11):

$$v_N^\pi(s_1) = \mathbb{E}_{s_1}^\pi \left[\sum_{t=1}^{N-1} r_t(s_t, a_t) + r_N(s_N) \right]. \tag{11}$$

In Section 5, we describe our solution methodology to find optimal/near-optimal solutions to maximize the expected total reward of the stochastic SAIRPs.

4. Theoretical Results

In this section, we prove theoretical properties regarding the structure of the optimal SAIRP policy and value function. First, we show that the stochastic SAIRPs violate the sufficient conditions for the optimality of a monotone policy in the second dimension of the state (average capacity of batteries). Second, we prove the existence of a monotone optimal policy for the second dimension of the state in a special case of the SAIRP. Last, we prove the monotonicity of the value function when considering the first, second, and both dimensions of the state. In the remainder of this section, we present the main theorems and point the reader to the appendices for the formal mathematical proofs.

4.1. Monotone Policy

Our investigation in proving the structure of an optimal policy for the SAIRP is motivated by the desire to exploit efficient algorithms that require less computational effort to find optimal policies and increase the ability to solve larger problem instances (Puterman 2005). Widrick, Nurre, and Robbins (2018) examined the problem of managing a battery swap station when only considering battery charge, or equivalently the first dimension of our MDP model. They proved the existence of a monotone optimal policy when demand is governed by a nonincreasing discrete distribution. In our investigation of the second dimension, our intuition was that monotonicity would be preserved for the stochastic SAIRP. Informally, this equates to the optimal policy indicating to replace more batteries when the average capacity is lower. However, in Lemma 1, we prove a counter-intuitive result that, in general, the sufficient conditions for the optimality of a monotone policy in the second dimension of the state do not exist for the stochastic SAIRPs. Instead, we can prove the existence of a monotone optimal policy for the second dimension of the state when an upper bound is placed on the number of batteries replaced at the swap station at each decision epoch.

First, we formally define a monotone optimal policy. A monotone nonincreasing policy π has the property that for any $s_i, s_j \in S$ with $s_i \leq s_j$ (for multidimensional states, please see the partial ordering definition in Definition 2 of Online Appendix B), there exist decision rules $d_t^\pi(s_i) \geq d_t^\pi(s_j)$ for each $t = 1, \ldots, N-1$ (Puterman 2005). The sufficient conditions for the existence of a monotone optimal policy in the second dimension of the state are as follows (Puterman 2005).

This list needs to be appeared this way as it is in Puterman 2005 and Widrick et al. (2018).

1. $r_t(s_t^2, a_t^2)$ is nondecreasing in s_t^2 for all $a_t^2 \in A'$.
2. $q_t(k \mid s_t^2, a_t^2)$ is nondecreasing in s_t^2 for all $k \in S^2$ and $a \in A'$.
3. $r_t(s_t^2, a_t^2)$ is a subadditive function on $S^2 \times A'$.
4. $q_t(k \mid s_t^2, a_t^2)$ is subadditive on $S^2 \times A'$ for every $k \in S^2$.
5. $r_N(s_N)$ is nondecreasing in s_N^2.

Where A' includes all possible actions for the second dimension of the action space. Specifically, $A' = \{\cup_{s_t \in S} A_{s_t}^2\}$. We note that $q_t(k \mid s, a) = \sum_{j=k}^{\infty} p_t(j \mid s, a)$, which is the sum of the probabilities from k to ∞, in general. For the second dimension, we have $q_t(k \mid s_t^2, a_t^2) = \sum_{j^2=k}^{\infty} p_t(j^2 \mid s_t^2, a_t^2)$.

In Lemma 1, we prove that one of the aforementioned conditions is not satisfied for stochastic SAIRPs. In Theorem 1, we can to prove that a monotone optimal policy in the second dimension of the

state (average capacity of batteries) does exist when there is an upper bound on the number of batteries replaced at each decision epoch. We refer the reader to Online Appendix A for full details of the proof of Lemma 1 and Theorem 1.

Lemma 1. *The stochastic SAIRPs violate the sufficient conditions for the optimality of a monotone policy in the second dimension of the state (average capacity of batteries).*

Theorem 1. *There exist optimal decision rules $d_t^* : S \rightarrow A_{s_t}$ for the stochastic SAIRP which are monotone nonincreasing in the second dimension of the state (average capacity of batteries) for $t = 1, \ldots, N-1$ if there is an upper-bound U on the number of batteries replaced at each decision epoch where $U = M\varepsilon/2(1-s_t^2)$, such that M is the number of batteries at the swap station and ε is the discretized increment in capacity.*

We provide an example to explain the optimality of the monotone policy in the second dimension of the state. Consider a swap station with $M = 100$ batteries, a discretized capacity increment $\varepsilon = 0.01$, and a replacement threshold $\theta = 0.8$. The monotone policy is optimal when the maximum number of batteries replaced per epoch, U, is between 2 and 50. The specific value between 2 and 50 depends on the value of the average capacity. If the average capacity $s_t^2 = 0.8$, then $U = 100(0.01)/2(1-0.8) = 2$, whereas if $s_t^2 = 0.99$, then $U = 100(0.01)/2(1-0.99) = 50$. We note that when $s_t^2 = 1$, then $U = 100(0.01)/2(1-1) = \infty$, meaning there is no limit on the number of replaced batteries. However, as the average capacity is already at the highest value of one, it is not advantageous for swap stations to incur the cost for replacing a full capacity battery after which the average capacity will remain at one. Although there is a restriction on the number replaced at each epoch, there are no restrictions on the consistent replacement of batteries over multiple consecutive decision epochs.

4.2. Monotone Value Function

We now investigate the structure of the value function for stochastic SAIRPs. Although proving that a value function has a monotone structure is a weaker result than proving the structure of an optimal policy, it enables the application of computationally efficient solution methods. We prove that the MDP value function for the stochastic SAIRP is monotone nondecreasing in the first, second, and both dimensions. These results directly motivate our selection of efficient approximate dynamic programming algorithms.

A value function $V(s)$ is monotone nondecreasing in state s, if for any $s_i, s_j \in S$ with $s_i \leq s_j$, we have $V(s_i) \leq V(s_j)$ for any given action in any decision epoch t (Papadakia and Powell 2007, Jiang and Powell 2015). The MDP value function for the stochastic SAIRP is given in Equation (12), which is comprised of the immediate expected reward (as given by Equation (7)) and the transition probabilities (as given by Equation (6)). In Theorem 2, we show that the value function is monotone in s_t^2. This means for any given action, at each decision epoch t, as the average capacity increases, the MDP value function will not decrease.

Theorem 2. *The MDP value function of the stochastic SAIRP is monotone nondecreasing in s_t^2.*

In Theorem 3, we prove that the value function is monotone in s_t^1. This means for any given action at each decision epoch t, as the number of fully charged batteries increases, the MDP value function will not decrease. If the demand for the MDP model is governed by a nonincreasing discrete distribution, this result is implied from the result of Widrick, Nurre, and Robbins (2018). However, we strengthen the result as we *do not* require a nonincreasing discrete distribution in Theorem 3.

Theorem 3. *The MDP value function of the stochastic SAIRP is monotone nondecreasing in s_t^1.*

When considering both dimensions simultaneously, in Theorem 4, we prove that the value function is monotone in (s_t^1, s_t^2).

Theorem 4. *The MDP value function of the stochastic SAIRP is monotone nondecreasing in (s_t^1, s_t^2).*

In a multidimensional setting, we need to define the concepts of partial ordering and partially nondecreasing function, which are given by Definitions 2 and 3 in Online Appendix B. We also refer the reader to Online Appendix B for full details of the proofs of Theorems 2–4.

5. Solution Methodology

This section presents the solution methods used to solve the stochastic SAIRP. First, we briefly describe the dynamic programming solution methods with the BI approach to provide exact solutions when the problem is not large-scale. Next, we present the approximate dynamic programming methods to overcome the curses of dimensionality and yield high-quality solutions for the stochastic SAIRPs.

5.1. Exact Solution Method: Dynamic Programming

Backward Induction (BI) is an exact solution method to find optimal policies for the MDP problems (Puterman 2005). Our goal is to find the optimal policy π^* that maximizes the expected total reward given by Equation (11). We attribute the optimal value

function, $V_t(s_t)$, to the optimal policy. We calculate the optimal value based on cumulative values of taking the best actions onward from decision epoch t to N when in state s_t at epoch t (see Equation (12)). We use Bellman equations as presented in Equation (12) to find optimal policies and corresponding optimal value functions for $t = 1, \ldots, N-1$ and $s_t \in S$:

$$V_t(s_t) = \max_{a_t \in A_{s_t}} \left\{ r_t(s_t, a_t) + \sum_{j \in S} p_t(j \mid s_t, a_t) V_{t+1}(j) \right\}. \quad (12)$$

The BI algorithm starts from $t = N$ and sets $V_N(s_N) = r_N(S_N)$ according to Equation (9). Then, it finds the actions that maximize $V_t(s_t)$ for every state s_t moving backward in time $(t = N-1, \ldots, 1)$ using Equation (13). The optimal expected total reward over the time horizon is $V_1^*(s_1)$ where s_1 is the state of the system at the first decision epoch:

$$a^*_{s_t,t} = \arg\max_{a_t \in A_{s_t}} \left\{ r_t(s_t, a_t) + \sum_{j \in S} p_t(j \mid s_t, a_t) V_{t+1}(j) \right\}. \quad (13)$$

Now, we clarify the terms used in the remainder of this paper. A *sample path of demand* is the collection of realized demand (uncertainty element) during the time period between two consecutive epochs generated from a given probability distribution. A *sample path of state* is comprised of the collection of consecutive visited states, one per time period. To calculate the visited states, we need the decision rule returned by a solution method for all states or visited states over time, the *sample path of demand*, and the present state. Using this information, we use the state transition functions (Equations (3) and (5)) to calculate the *sample path of state*. A *sample path of policy* is the set of consecutive decision rules of the visited states of the system. We use the term *instance* to refer to an example of stochastic SAIRP, specifically when we discuss the size of stochastic SAIRPs. The term *scenario* is used to refer to examples within our space-filling designed experiment that include different values for parameters. These values are generated such that to cover the designed experiment space (see Section 6.3).

5.2. Approximate Dynamic Programming Solution Methods

In this section, we outline our monotone approximate dynamic programming algorithm with regression-based initialization (MADP-RB). Approximate dynamic programming is a proven solution method that overcomes the curses of dimensionality (Powell 2011). Using the foundation of the monotone approximate dynamic programming algorithm proposed by Jiang and Powell (2015), we make enhancements by exploiting our theoretical results to intelligently approximate the initial value function approximation and update the approximation with each algorithmic iteration.

The stochastic SAIRP suffers from the curses of dimensionality considering the size of all MDP elements together. Asadi and Nurre Pinkley (2021) showed that the size of the state space, the action space, the transition probability function, and the optimal policy are $O(M(1-\theta)/\varepsilon)$, $O(M^2)$, $O(M^4N(1-\theta/\varepsilon)^2)$, and $O(MN1-\theta/\varepsilon)$, respectively, where M, N, ε, and θ are the number of batteries, the time horizon, the capacity increment, and the replacement threshold, respectively. For instance, the size of the transition probability function is $O(10^{15})$ for a realistic-sized problem with $M = 100$ batteries, planning over a one month time horizon in one hour increments $n = 744$ with discretized battery capacity in increments of $\varepsilon = 0.001$. Because of these large sizes, standard MDP solution methods, such as BI, were ineffective in solving realistic-sized instances of the stochastic SAIRP (Asadi and Nurre Pinkley 2021).

5.2.1. Monotone ADP. Here, we discuss our choice of monotone ADP as the foundation of our proposed approximate solution method. Although there are many different ADP algorithms and approaches (Powell 2011), there is no standard method to link the best algorithm to solve any particular problem. However, using the problem structure is good practice when developing efficient and effective algorithms. As we proved in Theorems 2–4, the value function is monotone nondecreasing in both dimensions: s_t^1 and s_t^2. Hence, it is reasonable to use and enhance the monotone approximate dynamic programming algorithm proposed by Jiang and Powell (2015). This algorithm has already shown promising performance for several application areas (Jiang and Powell 2015). We proceed by outlining the core steps of the MADP algorithm of Jiang and Powell (2015) while highlighting our additions and changes to create the monotone approximate dynamic programming algorithm with regression-based initialization (MADP-RB). To aid with the explanation, in Algorithm 1, we outline the MADP and underline the enhancements for our MADP-RB. First, we introduce the notation necessary for the ADP algorithms in Table 1.

5.2.2. Monotone ADP with Regression-Based Initialization. In this section, we describe the core steps of the MADP algorithm proposed by Jiang and Powell (2015) and our enhancement using regression-based initialization in Algorithm 1. We display our enhancements in Algorithm 1 with underlines to make it more

Table 1. Notation Used in the ADP Algorithms

Notation	Description
$maxIteration + 1$	The maximum number of regression-based initialization iterations
$\overline{M}$	The starting number of batteries used for the small SAIRPs solved using BI
$\overline{T}$	The time horizon in the small SAIRP
$u_t^{iter}(s_t)$	The optimal value of being in state s_t at iteration *iter* and epoch t
τ	The maximum number of core ADP iterations
$V_t^n(s_t)$	The optimal value of being in state s_t at epoch t for iteration n
$\overline{V}_t^n(s_t)$	The approximate value of being in state s_t at epoch t for iteration n
$\hat{v}_t^n(s_t^n)$	The observed value of state s_t^n at epoch t for iteration n
$z_t^n(s_t^n)$	The smoothed value of being in state s_t at epoch t for iteration n

clear for the reader. We proceed by explaining the implementation of the algorithm.

The monotone ADP with regression-based initialization (MADP-RB) has two main steps. In the first step, we intelligently initialize the value function approximation using a linear regression function. The coefficients of the regression function are derived from feeding the optimal solutions of small SAIRPs. The second step is the core MADP algorithm that consists of updating the approximated values of visited and nonvisited states over time through an iterative process. The states are visited over time at each iteration using the information of the present state, realized uncertainty, and taken action. The approximated value of the visited state is updated based on the observed value and previous approximated value. At each iteration, the monotonicity operator updates the approximated value of the nonvisited states over time. We proceed by explaining each step in detail.

5.2.2.1. Regression-Based Initialization. In this section, we explain our regression-based method to intelligently initialize the MADP-RB. The first step of the MADP algorithm is to initialize the value function approximation for all decision epochs such that the monotonicity of the value function is preserved. Commonly, this is done by assigning a constant value, for example, zero, to $\overline{V}_t^0(s_t)$ for all $s_t \in S$ and $t = 1, \ldots N-1$. However, using 0 or any constant value fails to exploit how the monotone value function changes based on state and time. Thus, our enhancement to the MADP algorithm is to intelligently approximate the initial value function approximation by exploiting the monotonicity of the value function and considering the relationship between the value function and model's parameters. In empirical experiments of the small SAIRPs (see Section 6.2), we observe that the value function of the optimal policies $V_t(s_t)$ is a linear function of the decision epoch (t), the state of the system (s_t), and the number of batteries in the station (M). As we need a fast and simple but intelligent initialization, we use linear regression, given by Equation (14), to approximate the initial value function for larger problem instances. To do so, for each scenario, which is a set of input parameters or factors (see three input parameters of our problem in Section 6.3), we iteratively calculate the optimal value function for small but increasing problem instances and store the optimal values for every t, s_t, and M. Then, we fit a linear regression on the aggregated results of small SAIRPs to derive the coefficients' values for h_0, h_1, h_2, h_3, and h_4 for each scenario. We note that the different values of input parameters of scenarios may lead to different BI solutions. As a result, the multipliers' value of the fitted regression may vary for different scenarios of SAIRPs. To further explain this enhancement, we outline how these steps can be applied to the stochastic SAIRP:

$$\overline{V}_t^0(s_t) = h_0 + h_1 M + h_2 s_t^1 + h_3 s_t^2 + h_4 t. \tag{14}$$

In the stochastic SAIRP, as M and T increase, the problem suffers from the curses of dimensionality. For instance, the transition probability is $O(10^{15})$ for a realistic-sized problem when $M = 100$, $n = 744$, and $\varepsilon = 0.001$. Thus, for every set of input parameters (scenario) of a stochastic SAIRP, we first optimally solve small instances with small numbers of batteries $\overline{M} \ll M$ and time horizons $\overline{T} \ll T$. We repeat this step by slowly increasing the number of batteries by one until we reach a user-defined maximum number of iterations (or until it is computationally infeasible to optimally solve small instances with BI). For each instance, we determine the optimal value function and the associated trends based on changes in the state, time, and number of batteries. From the value functions for smaller instances of the scenario, we use linear regression on the decision epoch (t), the state of the system (s_t), and the number of batteries in the station (M) to approximate the initial value function approximation for larger problem instances. See lines 1–6 in Algorithm 1 that describe this regression-based enhancement that is new and distinct from Jiang and Powell (2015). We complete the initialization phase in line 7, where we set the approximate value for the terminal epoch for all states and all core iterations $n = 1, \ldots, \tau$ to the terminal reward (see Equation (9)).

Algorithm 1 (Monotone Approximate Dynamic Program with Regression-Based Initialization)

1. Initialize $\overline{M}$ batteries and $\overline{T}$ time horizons, where $\overline{M} \ll M$ and $\overline{T} \ll T$, and input the parameters of the scenario.
2. Set iteration counter $iter = 0$.
3. **for** $iter \leq MaxIterations$ **do**
4. $u_t^{iter}(s_t) \leftarrow$ *Use backward induction to find the value function for the problem with* $\overline{M} + iter$ *batteries and* $\overline{T}$ *time horizons, where* $\overline{M} \ll M$ *and* $\overline{T} \ll T$.
5. **end for**
6. Initialize $\overline{V}_t^0(s_t)$ using linear regression on the combined $u_t^{iter}(s_t)$ for all states at $t = 1, \ldots, N-1$.
7. Set $\overline{V}_N^n(s) = r_N(s)$ for $s \in S$ and $n = 1, \ldots, \tau$
8. Set $n = 1$
9. **while** $n \leq \tau$ **do**
10. Select initial state S_1^n
11. **for** $t = 1, \ldots, N-1$ **do**
12. Sample an observation of the uncertainty, D_t, determine optimal action a_t^n and future value $\hat{v}_t^n(s_t^n)$
13. Smooth the new observation with the previous value,
$$z_t^n(s_t^n) = (1-\alpha_n)\overline{V}_t^{n-1}(s_t^n) + \alpha_n \hat{v}_t^n(s_t^n) \quad (15)$$
14. Perform value function monotonicity projection operator as in Jiang and Powell (2015)
15. Determine next state, S_{t+1}^n
16. **end for**
17. Increment $n = n + 1$
18. **end while**

5.2.2.2. Core Steps of MADP-RB (MADP). The algorithm iteratively proceeds in accordance with Jiang and Powell (2015) in lines 8–12. With each iteration, we select the best action starting from a random initial state (line 10) and move forward in time (line 11) using a sample observation of uncertainty and the approximate value of the future state, $\overline{V}_{t+1}^{n-1}(j)$ (line 12). Specifically, we pick the best action (line 12) using Equation (16) and store the current observed value using Equation (17). In Equation (17), we use the previous approximation of the future states, $\overline{V}_{t+1}^{n-1}(s_{t+1})$, as the approximation of $\mathbb{E}(V_{t+1}(s_{t+1}) \mid s_t, a_t)$:

$$a_{s_t,t}^n = \arg\max_{a_t \in A_{s_t}} \left\{ r_t(s_t, a_t) + \overline{V}_{t+1}^{n-1}(s_{t+1}) \right\}. \quad (16)$$

$$\hat{v}_t^n(s_t^n) = \max_{a_t \in A_{s_t}} \{ r_t(s_t, a_t) + \mathbb{E}(V_{t+1}(s_{t+1}) \mid s_t, a_t) \}. \quad (17)$$

In line 13, we use a combination of the previous value function approximation, $\overline{V}_t^{n-1}(s_t^n)$, and the current observed value, $\hat{v}_t^n(s_t^n)$ to calculate the smoothed value of being in state s_t, $z_t^n(s_t^n)$. This combination is weighted based on a stepsize function, α_n. Traditional stepsize functions, including $1/n$ (Powell 2011) and harmonic (Powell 2011, Rettke, Robbins, and Lunday 2016, Meissner and Senicheva 2018), usually smooth the value function using pure observations for early iterations and gradually put less weight on the observation and put more on the approximations as the number of iterations increases.

Next, we apply the monotonicity projection operator as defined in Jiang and Powell (2015) (line 14). We note that the algorithm stores the value of all the states, and the monotonicity operator adjusts the value of nonvisited states according to the value of the visited state. It ensures that no approximated value violates the monotonicity of the value function. For instance, consider the visited state s_t and an arbitrary state $\widetilde{s_t}$ such that $s_t \leq \widetilde{s_t}$ and $\overline{V}_t(s_t) > \overline{V}_t(\widetilde{s_t})$. The monotonicity operator increases the approximation for $\widetilde{s_t}$ up to $\overline{V}_t(s_t)$ and preserves the monotonicity property of the value function. Similarly, we decrease the approximation for lower states (i.e., $\widetilde{s_t} \leq s_t$) with higher value approximations than the visited state. Lastly, the algorithm moves forward in time to the next decision epoch until the last decision epoch, wherein a new iteration begins. Every new iteration starts from an arbitrary state and steps forward in time until the input number of iterations are completed.

5.2.3. Stepsize Function. Finding a good stepsize is a problem-dependent procedure that requires empirical experiments (Powell 2011). A stepsize function, α_n is used to scale the current observed value and $(1-\alpha_n)$ is used to scale the current value function approximation. Because the value function approximation is often initially set to constant values and therefore is not informative, many stepsizes start with higher α_n values that emphasize the observed value. Then, as more iterations are conducted, α_n is decreased to place a greater emphasis on the value function approximation. We note that a stepsize function needs to satisfy the three basic conditions given by Powell (2011) to guarantee convergence. A basic example of such stepsize function is $1/n$. Our preliminary experiments show that using the $1/n$ stepsize function is not appropriate as the rate of decreasing α_n over iterations is too fast for SAIRPs. Instead, we use the harmonic stepsize function (Powell 2011) that uses a user-defined parameter, w, to control the rate of decrease over iterations. In Equation (18), we provide the formal definition of the harmonic stepsize function:

$$\alpha_n = \frac{w}{w + n - 1} \quad (18)$$

Additionally, we use the search-then-converge (STC) stepsize rule that can control the rate of decrease in α_n by appropriately setting the parameter values (Powell 2011). Hence, the STC function is suitable for cases

like ours that need an extended learning phase (Powell 2011). The STC rule was initially proposed by Darken and Moody (1992); however, we use the generalized STC formula given by George and Powell (2006) presented in Equation (19):

$$\alpha_n = \alpha_0 \frac{\left(\frac{\mu_2}{n} + \mu_1\right)}{\left(\frac{\mu_2}{n} + \mu_1 + n^{\zeta} - 1\right)}. \tag{19}$$

The harmonic and STC stepsize functions follow the common process of weighting the observation higher earlier and decreasing this weighting with each iteration. The harmonic and STC stepsize functions are classified as deterministic as their values do not change based on the observations. In contrast, adaptive stepsize functions are sensitive to changes in the observations. To broaden our investigation, we also use the adaptive stepsize first introduced in George and Powell (2006) in Section 6.

6. Computational Results

In this section, we present the results and insights from computational experiments on different SAIRP instances. First, we present the data used to solve modest and realistic-sized SAIRP instances. We clarify how we distinguish modest, realistic-sized, and small SAIRP instances. We denote SAIRP instances as modest if they are optimally solvable using backward induction (BI) (i.e., seven batteries total and only a one-week time horizon). We denote SAIRP instances as realistic sized if they include larger numbers of batteries and the system is considered for longer time horizons (i.e., 100 batteries over a one-month time horizon). Because of the curses of dimensionality, the realistic-sized instances are not optimally solvable using BI. Within our monotone approximate dynamic programming with regression-based initialization solution procedure, we initially solve small SAIRP instances that are optimally solvable using BI in a matter of minutes (i.e., two, three, or four batteries over a one-week time horizon). After explaining the data for these instances, we proceed by explaining in detail the regression-based initialization used in MADP-RB, the LHS designed experiments, solution method comparison, and solutions/insights for both the modest and realistic-sized SAIRPs.

6.1. Explanation of Data

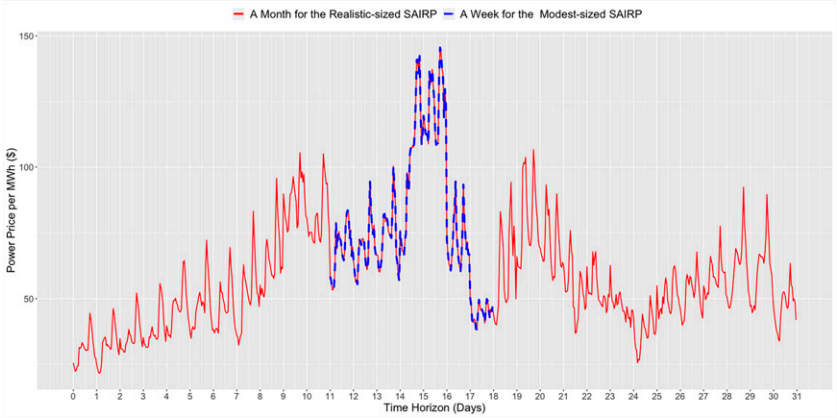
For the computational results, we use realistic data representing the costs to recharge, discharge, and replace a battery, the demand, and the battery degradation rate. To avoid redundancy in presenting similar insights for both EVs and drones, we focus on drones. First, we present the associated data, and then we show the results of a comprehensive set of experiments in Sections 6.3 and 6.4.

First, we set parameters associated with drone battery costs. We set the cost to recharge a battery using the historical power prices from the Capital Region, New York area in 2016 (National Grid 2016). We use the time frame with the highest total power price for the modest and realistic-sized instances with the time horizon of a week and a month, respectively. Hence, as displayed in Figure 2, we select December 12–18 and the month of December for the modest and realistic-sized problems, using dashed and solid lines, respectively. We multiply these historical time-varying power prices by the maximum capacity of a battery to calculate the nonstationary, time-varying costs to recharge a battery. We assume a drone battery has a maximum capacity equal to 400 Wh as in the DJI Spreading Wing S1000 battery (DJI 2019). We assume the revenue earned from discharging a battery back to the grid is equal to the charge price. Consistent with level 2 or 3 battery charging (Morrow, Karner, and Francfort 2008, Ribbernick, Darcovich, and Pincet 2015, Tesla 2017), we assume a depleted (full) battery takes one hour (i.e., the time between two consecutive decision epochs) of recharging (discharging) to become full (depleted). We use the purchase price for batteries to calculate the cost of battery replacement. Assuming the price per kilowatt hour of a battery is approximately $235 (Romm 2017), we set the replacement cost of a drone battery to be $100. This calculated replacement cost serves as the baseline price, which is used and varied in the Latin hypercube designed experiments in Section 6.3.

In the absence of real data representing the number of customers demanding swaps at the station over time, we use the methodology of Asadi and Nurre Pinkley (2021), Widrick, Nurre, and Robbins (2018), and Nurre et al. (2014) to derive the mean demand at the swap station over time. We assume the mean demand, λ_t, is equivalent to the historical arrival of customers at Chevron gas stations (Nexant, Inc. et al. 2008). Using λ_t, we assume the demand follows a Poisson distribution where t is the hour of the day. We scale λ_t to be in line with the number of batteries in the problem instance. Let λ'_t be the scaled demand for M' number of batteries. Because λ_t is originally used for $M = 7$, we calculate λ'_t values by multiplying $M'/7$ by λ_t. In Figure 3, we display the mean demand by hour over a one-week time horizon for the modest instances with $M = 7$. For longer time horizons, we assume the mean arrival of demand repeats every week.

We use existing studies to calculate the battery degradation rate per cycle. Although there are several factors that influence the battery degradation process, research states that the capacity fading has a linear behavior, especially in the first 500 cycles (Dubarry et al. 2011, Lacey et al. 2013, Hussein 2015, Ribbernick, Darcovich, and Pincet 2015). We note that the

Figure 2. (Color online) Power Price Fluctuations over December (Realistic-Sized SAIRP) and the Week of December 12–18 (Modest Size SAIRP), 2016, in the Capital Region, New York



standard number of cycles for a drone lithium-ion battery is 300–500 ($\delta^C \approx 0.1\%$) (Battery University 2017). We select a higher value for the degradation rate to account for the elements that accelerate the degradation process, such as temperature from continuous use and recharging in swap stations. Thus, we select $\delta^C = 2\%$ as the baseline degradation rate and vary this value in our computational experiments to capture how changes in the degradation rate impact the policy and performance of the system. The model is robust in that future experiments can be conducted for different baseline δ^C values to represent different degradation characteristics. In general, the industry-accepted battery end-of-life value is 80% of capacity. As follows, we set the replacement threshold $\theta = 80\%$ (Wood, Alexander, and Bradley 2011, Debnath, Ahmad, and Habibi 2014). We note that $s_t^2 < 80\%$ is equivalent to the absorbing state of the system $s_t^2 = 0$ wherein the feasible action set is $A_{(s_t^1,0)} = \{(0,0)\}$. Hence, we can only replace batteries when the average capacity of batteries is not less than 80%.

6.2. Data Used for Regression-Based Initialization

In this section, we explain the data we used for our regression-based method to initialize the MADP-RB intelligently. We examine the empirical experiments of small SAIRPs and detect that the value function of the optimal policies $V_t(s_t)$ is a function of the decision epoch (t), the state of the system (s_t), and the number of batteries in the station (M). To clarify, we display the optimal values $V_t(s_t)$ of scenario 6 from the Latin hypercube designed experiments presented in Section 6.3. We display two consecutive decision epochs in Figure 4. In this example, we compute and show the optimal $V_t(s_t)$ when we solve small SAIRPs with $M = 2, 3, 4$.

As shown in Figure 4, the horizontal axis denotes the states, and the vertical axis shows the corresponding values. Among various techniques to estimate or forecast $V_t^0(s_t)$, we want a fast and simple method to generate and assign the initial approximations. Therefore, we propose using the linear regression function presented in Equation (14) to initialize the approximated value function, $\overline{V}_t^0(s_t)$.

Having solved the small SAIRPs, we find the appropriate values for h_0, h_1, h_2, h_3, and h_4. In Sections 6.3 and 6.4, we demonstrate how the intelligent initial value function approximation leads to superior results.

6.3. Latin Hypercube Designed Experiments for Modest SAIRPs

In this section, we perform a Latin hypercube sampling designed experiment that is used to assess the quality of the solution methods and to deduce insights

Figure 3. (Color online) Mean Demand for Swaps over Time

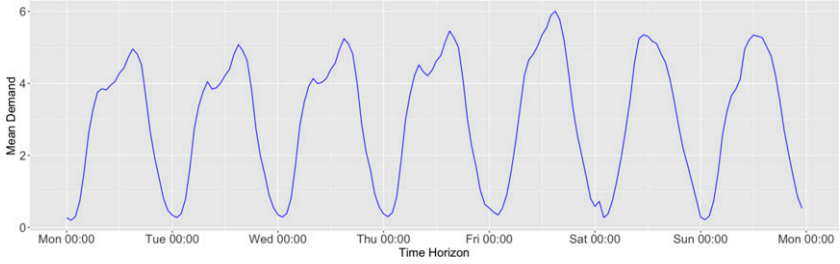
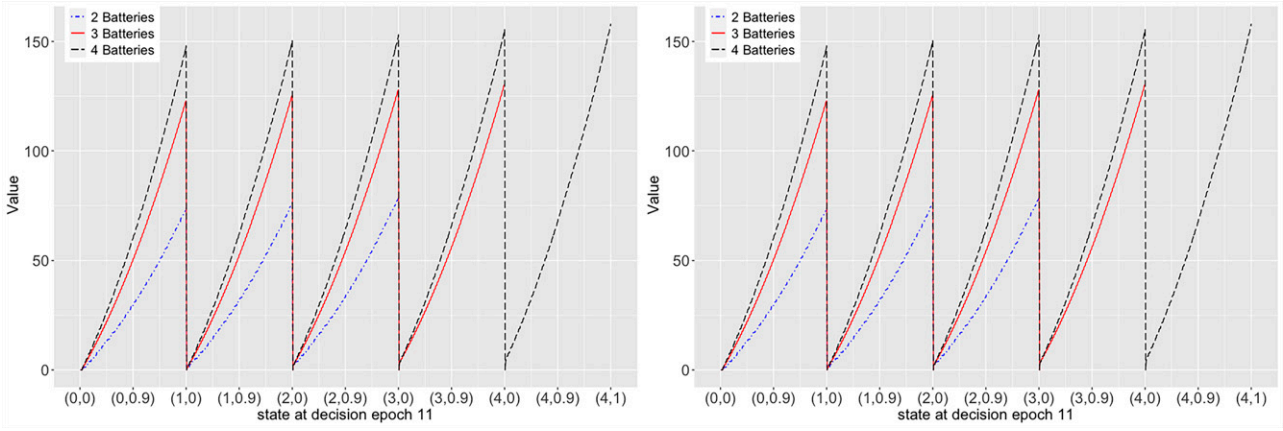


Figure 4. (Color online) Instance of the Optimal Values over States at Two Consecutive Decision Epochs



for the battery swap station application. An LHS designed experiment is a space-filling design with broad application in computer simulation (Montgomery 2008). Using the LHS configuration of Asadi and Nurre Pinkley (2021), with the set of 40 scenarios generated to cover the design space of parameters, we first quantify the performance of MBI, MADP, and MADP-RB.

In prior work, Asadi and Nurre Pinkley (2021) were able to optimally solve 40 modest LHS scenarios using BI. We run all computational tests using a high-performance computer with four shared memory quad Xeon octa-core 2.4-GHz E5-4640 processors and 768 GB of memory. Even on a high-performance computer, the largest scenario Asadi and Nurre Pinkley (2021) were able to optimally solve using BI is with $M = 7$, $\varepsilon = 0.001$, and $n = 168$, which represents a full week of operations where each decision epoch is one hour. Using these modest scenarios, herein, we are specifically concerned with quantifying the speed and performance of MBI, MADP, and MADP-RB against a known optimal policy and optimal expected total reward. The factors and their associated lower and upper bounds are defined as in Table 3 (we refer the reader to Asadi and Nurre Pinkley (2021) for a complete justification for how these low and high values are calculated).

6.3.1. Memory Intensive Processing vs. Compute Intensive Processing. We can implement BI and MBI in two ways. In the first way, which we denote *Memory Intensive*, we calculate and store all of the transition probability values, so they are readily available throughout the execution of the algorithm. In the second way, which we denote *Compute Intensive*, we do not store any probability values and instead calculate each probability value when needed for calculations through the execution of the algorithm. As is indicated in their names, the *Memory Intensive* way requires more memory, and the *Compute Intensive* way requires more time as it needs to calculate probability values numerous times. We implement and use both ways to solve the modest SAIRP instances on a high-performance computer (HPC). We note, on this HPC, we have different options. If we want to run the algorithm for longer, we have up to 72 hours of computational time available per run with access to only 768 GB of memory using four shared memory quad Xeon octa-core 2.4-GHz E5-4640 processors. However, if we want to use more memory, we have access to 3 TB of memory (four shared memory Intel(R) Xeon(R) CPU E7-4860 v2 2.60-GHz processors) but only six hours of computational time. Thus, our results reflect these limitations.

In Table 2, we report the memory used and computation times for BI and MBI by instance and method. We highlight the instances where we either exceeded the memory or time available and thus do not find a solution. As is evident by the results, BI and MBI are not tractable solution methods for even modest SAIRPs. MBI does outperform BI; however, the computation time is significant even for $M = 10$. Although not shown in Table 2, when $M \geq 12$, MBI exceeds the 72-hour time limit. We note that our approximate solution methods do not run into memory issues. Hence, we move forward with the faster processing method, *Memory Intensive*, to solve SAIRPs hereafter.

We computationally test MBI for the general case when no limits are placed on how many batteries are replaced per decision epoch and the MADP and MADP-RB methods with the harmonic, STC, and adaptive stepsize functions. For the core ADP procedure, we run the scenarios for $\tau = 500{,}000$ iterations. The number of iterations is large enough to ensure that the tolerance of the converged values is less than or equal to 0.1%. For these stepsize functions, based on extensive computational experiments, we present the most favorable results. For the harmonic stepsize function, Powell (2011) recommends that $\alpha_n < 0.05$ in the last iteration ($\tau = 500{,}000$). Thus, we set $\alpha_\tau = 0.05$

Table 2. Time and Memory Used for Different Size of SAIRPs Using Memory Intensive and Compute Intensive Methods

	Memory intensive			Compute intensive		
Size	Memory used (GB)	BI computation time (h)	MBI computation time (h)	Memory used (GB)	BI computation time (h)	MBI computation time (h)
$M = 7$	830	3.8	3.6	0.2	29.2	7.5
$M = 8$	1,120	**>6**	**>6**	0.2	52.6	11.3
$M = 9$	2,030	**>6**	**>6**	0.2	**>72**	16.3
$M = 10$	**>3,072**	**>6**	**>6**	0.2	**>72**	24.8

Note. The bold font indicates instances where we either exceeded the memory or time available and do not find a solution.

and $w = 25{,}000$. We note, our tests show that the average optimality gap increases when w is below 25,000, thus we use $w = 25{,}000$. For setting the STC parameters, we let $\alpha_0 = 1$ to put the total weight on the observation at the beginning of the procedure. This results in $\alpha_1 = 1$. We test six values for $\mu_1 = 10$, 100, 500, 600, 1,000, and 10,000, five values for $\mu_2 = 10$, 100, 1,000, 10,000, and 100,000, and three values for $\zeta = 0.6$, 0.7, and 0.8 to adjust the tuple of parameters (μ_1, μ_2, ζ). Over these 90 experiments, we observe that $(\mu_1, \mu_2, \zeta) = (600, 1{,}000, 0.7)$ yields the best result in terms of the average optimality gap. Thus, we use this tuple of parameter values for the STC stepsize. For the adaptive stepsize function, we use the setting presented in Powell (2011). In the adaptive stepsize, we need to estimate the bias and its variance. For smoothing the observation and approximation of the bias and its variance, Powell (2011) recommends using a harmonic stepsize that tends to a value between 0.05 and 0.1 in the last iteration. Consistent with this recommendation, we use the harmonic stepsize function with $w = 25{,}000$.

For the MADP-RB, we set $\overline{M} = 2$, $\overline{T} = T = 168$ hours, MaxIterations = 2 (i.e., three iterations are used as the input for the linear regression and value of *iter* can be zero, one, or two), and use BI to optimally solve all of the 40 scenarios for two, three, and four batteries. Then, for each scenario, we plug its associated result into Equation (14) to set the initial approximation as in line 6 of Algorithm 1.

Later, we will show that adding the regression-based initialization to MADP significantly improves the quality of solutions. One can argue that initialization using any monotone value function leads to the same result. We tested this argument using a set of arbitrary monotone value functions and report the best-founded function. We call the algorithm with this arbitrary monotone value function initialization MADP-M. We note that in MADP, the initial value function is a constant and equals to $\overline{V}_t^0(s_t) = 0 \ \forall s_t \in S$. In MADP-M, the monotone value function used for initialization is $\overline{V}_t^0(s_t) = (\frac{\beta(1+s_t^2-2\theta)}{1-\theta})(s_t^1) + k(N-t)$ where k is a constant. The first term is a monotone value function that equals the revenue generated from swapping all the full batteries when in state s_t at epoch t. The second term is a nonnegative term with an opposite relationship with time, which means the value of being in every state, s_t, is higher at earlier decision epochs. The logic behind adding the nonnegative term is as follows. The value of being in state s_t, $V_t(s_t)$, accumulates the reward from time t onward to the end of the time horizon, so we may expect to gain a higher reward (profit) over a longer time. Although $V_t(s_t)$ is not always decreasing in t, the general trend is observed in the optimal values of small SAIRPs. Our tests show that MADP-M with $k \leq 1$ outperforms MADP. Then, we test $k = 0.1, 0.2, \ldots, 1$ and observe that MADP-M with $k = 0.5$ yields the best result.

Table 3. Factors with Associated Low and High Values for Use in the Latin Hypercube Designed Experiment

Factor	Low	High
Basic revenue per swap (β)	1	3
Replacement cost L_t	2	100
Battery degradation factor (δ^C)	0.005	0.02

In Table 4, we summarize the 40 scenarios and the results comparing the approximate methods to backward induction. Specifically, we calculate an *average optimality gap over all scenarios*, comparing MADP, MADP-M, MADP-RB, and monotone backward induction (MBI) to the optimal expected total reward calculated using BI. In Equation (20) for each scenario of MADP, MADP-M, and MADP-RB, we input the expected total reward of the last iteration into *the expected reward of the approximated method*. Then, we can calculate the average and maximum optimality gaps over 40 scenarios given by the last two rows of Table 4. We report the values related to the average of 500 sample paths in the last iteration. This number of sample paths provides robust results even for a realistic-sized problem. Each sample path is generated independently from the nonhomogeneous Poisson distributions of the stochastic demands. Furthermore, the sample path generated for each iteration is independent of the sample paths of previous iterations. Hence, all the sample paths in the last iteration used for the value evaluation are independent of the

Table 4. Optimality Gap (%) of MADP, MADP-M, MADP-RB, and MBI

				Harmonic stepsize			STC stepsize			Adaptive stepsize			
Scenario	β	L_t	δ^c	MADP	MADP-M	MADP-RB	MADP	MADP-M	MADP-RB	MADP	MADP-M	MADP-RB	MBI
1	1.03	45	0.009	27.11	6.66	8.26	30.67	8.57	10.45	20.00	7.71	8.77	0.00
2	1.08	82	0.011	14.14	13.68	12.10	16.09	12.19	10.58	7.67	3.90	2.40	0.09
3	1.13	61	0.005	43.50	12.14	6.27	47.01	17.28	7.54	11.69	8.31	7.38	6.05
4	1.20	69	0.009	26.08	8.28	1.26	30.10	10.24	3.00	19.84	8.81	3.34	0.00
5	1.23	89	0.008	14.80	2.20	8.62	18.62	0.53	7.88	4.61	0.15	1.73	0.00
6	1.26	47	0.010	27.76	12.06	9.72	30.10	13.51	10.95	19.66	12.37	9.59	0.00
7	1.32	98	0.019	12.11	22.34	9.15	13.73	22.8	9.32	9.07	4.17	0.71	0.00
8	1.39	94	0.011	13.49	8.38	12.76	16.33	7.97	10.82	7.28	2.56	2.68	0.00
9	1.41	87	0.014	6.93	19.06	15.37	9.39	18.87	14.55	4.33	6.35	3.31	0.00
10	1.48	36	0.006	25.41	15.61	9.20	33.78	15.97	9.58	26.50	14.75	3.47	7.25
11	1.53	68	0.016	17.28	7.24	3.17	18.05	6.56	1.60	13.51	1.43	0.94	0.00
12	1.56	28	0.018	15.04	2.22	1.25	15.37	0.78	0.06	12.79	6.46	6.11	5.95
13	1.60	57	0.010	27.51	14.45	10.10	31.56	15.74	11.84	20.77	14.18	1.46	0.00
14	1.68	31	0.017	30.02	15.21	6.18	28.94	15.29	4.92	23.95	14.52	10.45	11.05
15	1.71	62	0.006	26.79	10.13	5.25	33.69	10.78	4.62	15.92	8.63	1.19	0.00
16	1.76	44	0.017	19.14	0.36	1.14	20.91	0.32	0.20	14.72	3.62	2.88	0.00
17	1.83	55	0.016	17.36	3.00	1.84	17.63	2.39	1.13	13.29	1.46	1.83	0.00
18	1.88	51	0.019	11.38	10.00	7.55	13.74	10.51	6.57	8.32	1.32	1.00	0.00
19	1.94	66	0.012	17.12	2.01	3.99	18.23	2.35	3.06	11.17	2.91	0.17	0.00
20	1.95	73	0.019	11.22	11.59	9.15	13.51	10.74	8.47	9.27	1.76	0.82	0.00
21	2.00	83	0.012	16.41	1.95	4.27	19.50	3.06	3.95	11.45	3.29	0.01	0.00
22	2.07	20	0.013	60.56	52.87	0.38	53.56	54.87	1.59	56.97	54.10	0.39	53.97
23	2.15	90	0.013	13.40	3.44	12.19	15.41	2.44	11.46	9.88	0.27	2.06	0.00
24	2.16	41	0.007	33.37	31.29	2.02	27.58	29.17	3.07	35.00	33.40	0.94	30.69
25	2.22	79	0.015	7.42	7.37	12.45	10.11	7.45	12.03	5.07	3.08	1.55	0.00
26	2.27	8	0.007	64.60	67.51	8.99	68.96	70.09	8.91	66.91	70.85	8.79	47.85
27	2.32	25	0.017	59.25	45.07	4.37	59.72	40.87	5.03	44.85	44.52	5.13	49.80
28	2.37	74	0.014	8.40	8.27	13.87	8.27	7.62	12.80	4.45	3.34	2.95	0.00
29	2.40	7	0.018	68.69	69.38	15.65	72.46	72.84	15.22	75.11	73.61	15.47	74.43
30	2.50	14	0.008	65.68	68.12	7.05	68.96	67.59	2.48	64.64	68.68	7.38	43.56
31	2.51	24	0.015	59.93	54.56	2.08	48.80	54.09	1.12	59.03	51.93	3.00	56.76
32	2.56	34	0.009	49.50	41.41	1.64	36.87	42.90	2.67	49.92	38.64	0.25	30.92
33	2.64	39	0.010	52.19	34.27	7.55	49.33	36.56	6.50	37.41	32.55	7.73	25.82
34	2.68	54	0.016	23.60	11.83	8.06	23.82	12.74	8.13	19.71	12.34	9.42	7.20
35	2.71	3	0.007	31.16	34.05	7.69	31.22	33.44	7.45	33.33	33.40	7.69	53.11
36	2.77	16	0.012	68.79	69.50	6.64	71.81	68.35	6.62	68.49	66.33	6.29	56.91
37	2.83	21	0.013	69.68	64.86	2.29	69.44	63.10	3.12	67.79	63.29	3.33	58.57
38	2.90	77	0.015	8.18	4.20	5.16	9.62	3.62	5.48	4.58	1.10	1.14	0.00
39	2.91	96	0.020	10.27	7.51	11.55	10.91	7.61	11.18	7.05	1.17	1.51	0.00
40	2.96	12	0.005	59.05	66.65	7.42	63.61	67.87	6.93	65.77	68.09	7.69	17.20
Average gap				30.86	23.52	7.09	31.94	23.74	6.82	26.54	21.23	4.07	15.93
Maximum gap				69.68	69.50	15.65	72.46	72.84	15.22	75.11	73.61	15.47	74.43

sample paths used to update the value functions in previous iterations:

$$\text{Optimality Gap} = \left| \frac{\text{Expected Reward BI} - \text{Expected Reward Approximated Method}}{\text{Expected Reward BI}} \right| \times 100\%. \quad (20)$$

Immediately evident from Table 4 is that MADP-RB led to significantly smaller average and maximum optimality gaps than the MADP, MADP-M, and MBI of Jiang and Powell (2015). Overall, we find that MADP-M outperforms MADP in regard to the average optimality gap. This demonstrates that initializing an ADP approach, even with an arbitrarily monotone value function, does result in benefit. However, we further find that when considering both the average and maximum optimality gaps, MADP-RB significantly outperforms MADP-M. This demonstrates the significant benefit of the regression-based initialization. We proceed by further analysis of MADP and MADP-RB.

MBI is a reasonable approximation for scenarios with high replacement costs, but it does not provide competitive optimality gaps when replacement cost is low. However, when we limit the number of batteries replaced at each epoch in accordance with Theorem 1, the optimality gap is 0.00%. MBI is smarter than BI and can save computational time when searching for the best policies. In other words, we do not need to

loop over all the actions when using the monotonicity property of the optimal policy. However, as we showed, BI and even MBI are not computationally tractable for realistic-sized instances of SAIRPs, which necessitates using approximate solution methods.

As is expected, ADP methods take significantly less time than BI and MBI, which take on average 3.8 and 3.6 hours, respectively. The computational time of MADP-RB includes three major operations in Algorithm 1: (i) obtaining the data for the regression function in steps 1–5; (ii) deriving the regression function in step 6; and (iii) calculating the monotone ADP results in steps 7–18. The average computational time for operations (i), (ii), and (iii) are 902 seconds (0.25 hours), less than 1 second, and 1,025 seconds (0.28 hours), respectively. We note that (iii) is equivalent to the average computational time of MADP without the regression-based initialization. In other words, if we execute the initialization steps offline, there is no difference between the computational time of MADP and MADP-RB as finding the coefficients of the regression function takes less than a second. As the problem instance of the SAIRP increases significantly, the required memory and time for BI and MBI also increase significantly. However, MADP and MADP-RB enable us to solve realistic-sized SAIRP instances, as we demonstrate in Section 6.4.

Next, to compare the two ADP methods, we quantify a measure of convergence. One way to evaluate an ADP method is to focus not only on the ending performance but also on the convergence with each iteration. Thus, we calculate *the average optimality gap over all iterations*, g_m^s, for each method m and scenario s. In this calculation, we average the optimality gap for all iterations (τ) as in Equation (21):

$$g_m^s = \frac{1}{\tau}\sum_{i=1}^{\tau}\frac{|\text{Exp. Tot. Rew. atIter. } i - \text{Opt. Exp. Tot. Rew.}|}{\text{Opt. Exp. Tot. Rew.}} \times 100\%. \tag{21}$$

In Figure 5, we show the expected total reward (value) as a function of the number of iterations for MADP and MADP-RB with the three different stepsizes. The rows of Figure 5 correspond to the harmonic, STC, and adaptive stepsizes, respectively. The columns correspond to the scenarios that result in the low, average, and high g_m^s in the LHS where m includes MADP with the three stepsizes. Within each subfigure, we display three lines showing the progression of the value by iteration for MADP and MADP-RB and the optimal value for the state s_1, which corresponds to the initial state of the swap station when all batteries are full and new with full capacity. From the subfigures,

Figure 5. (Color online) Expected Total Reward Convergence of MADP and MADP-RB Using Different Stepsize Functions vs. Optimal Expected Reward

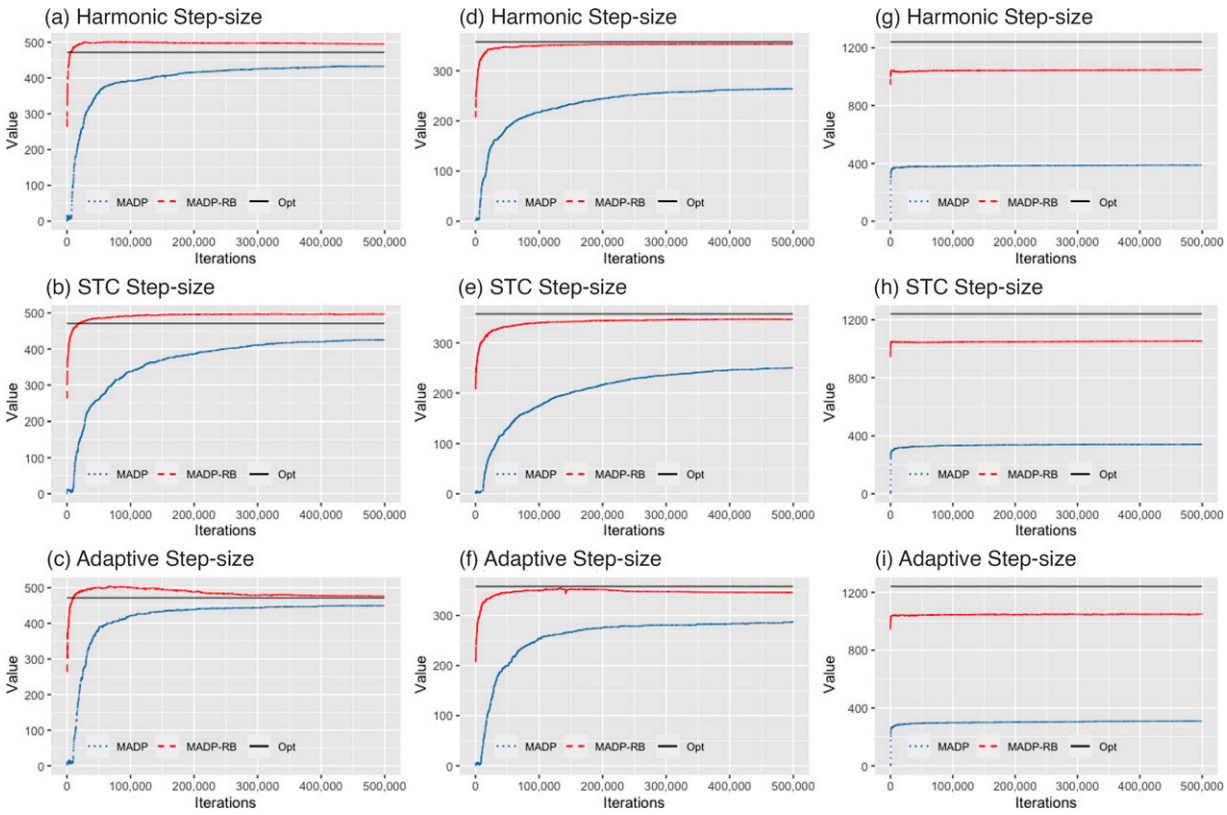


Table 5. Comparison Between Convergence over Iterations Using MADP and MADP-RB

Average Optimality Gap over MADP			Iterations over scenarios (%) MADP-RB		
Harmonic	STC	Adaptive	Harmonic	STC	Adaptive
36.72	40.84	32.58	6.89	8.53	5.90

we observe that MADP-RB consistently outperforms MADP in regard to convergence.

Now, we explain the possibility of convergence to a value greater than optimal (see Figure 5, (a)–(c)). The approximated value depends on two factors: (i) the previous approximation value and (ii) the present observation value. These values are smoothed using a stepsize function and set equal to the approximation value for the next iteration. Hence, as either value can be greater than the optimal value in any iteration, the smoothed value can take a value higher than the optimal value. For example, suppose the initial approximation is too high. In that case, the smoothed value can be higher than the expected value of being in any state.

We also calculate the *average optimality gap over iterations over scenarios* for each method m using Equation (22). In this equation, m = MADP and MADP-RB, and the total number of scenarios in our Latin hypercube designed experiments is 40:

$$\overline{g_m} = \frac{\sum_{s=1}^{40} g_m^s}{40}. \tag{22}$$

In Table 5, we display the *average optimality gap over iterations over scenarios* for MADP and MADP-RB using different stepsize functions given by Equation (22). As shown in Table 5, the average optimality gap associated with MADP-RB for all stepsizes is remarkably lower than MADP. This indicates that MADP-RB can more quickly converge to the optimal value compared with MADP.

The required computation effort for both ADP approaches depends on, in part, the number of iterations, τ. Thus, we analyze how the value function changes by iteration, which will inform our choice of τ for future experiments. In Figure 5, we observe the value function plateaus before τ = 500,000. Furthermore, we notice that both methods do not significantly change after τ = 100,000 in many scenarios. When examining all scenarios solved by MARP-RB, we calculate the percentage difference between the value at τ = 100,000 and τ = 500,000. For the harmonic stepsize, the average percentage difference is 1.03%, and the maximum percentage difference is 8.86%. We obtained similar values for the STC and adaptive stepsize and scenarios; however, to avoid redundancy, we proceed with an analysis of the harmonic stepsize.

Because the value does not always convey the operational changes in the policy that decision makers implement, we proceed by analyzing the changes in policy at iteration 100,000 and 500,000. We use Equation (23) to compute the percentage difference in the policies using at iteration 500,000 and 100,000 when independent 500 sample paths of realized demand generated. With this equation, we calculate the percentage difference between the summation of the recharging/discharging and replacement actions as we decrease the number of iterations for each scenario. We then average this value over all scenarios. Specifically, in Equation (23), TC_s^a and $\overline{TC_s^a}$ represent the total number of actions, recharging/discharging ($a = a^1$), and replacements ($a = a^2$), for scenario s using τ = 500,000 and τ = 100,000, respectively. That is, $TC_s^{a^1}$ and $\overline{TC_s^{a^1}}$ denote the total number of recharging/discharging actions, and $TC_s^{a^2}$ and $\overline{TC_s^{a^2}}$ are the total number of replacements for scenario s using 500,000 and 100,000 iterations, respectively. Our results show the average percentage difference in the number of recharged/discharged and replaced batteries over 40 scenarios are 6% and 5%, respectively:

$$\text{Average (\%) difference for action } a \text{ over all scenarios} = \frac{1}{40}\sum_{s=1}^{40} \frac{|\overline{TC_s^a} - TC_s^a|}{TC_s^a} \times 100\%. \tag{23}$$

Thus, as the changes in the value and policies are not significant, we use τ = 100,000 in our future computational experiments for solving realistic-sized SAIRPs in Section 6.4.

Exploiting the monotonicity property does significantly improve the quality of solutions and convergence of ADP methods. To demonstrate this result, we run two versions of standard approximate value iteration (AVI) without and with regression-based initialization, AVI and AVI-RB, respectively. Adding the monotonicity property to AVI and AVI-RB converts them to MADP and MADP-RB, respectively. In Table 6, we present a summary comparing AVI, MADP, AVI-RB, and MADP-RB using the harmonic stepsize function and 500,000 iterations. In this table, we present the average optimality gap of AVI, MADP, AVI-RB, and MADP-RB over all 40 scenarios of our Latin hypercube designed experiments (see Table 4 for full details of

Table 6. Comparison Between the Average Optimality Gap of Different Approximate Solution Methods

Approximate method	AVI	MADP	AVI-RB	MADP-RB
Average optimality gap (%)	41.17%	30.86%	14.75%	7.09%

these 40 scenarios). As is evident by these results, using the monotonicity property decreases the average optimality gap. However, we observe that the regression-based initialization is even more impactful than the monotonicity operator. By adding the monotonicity operator, the average optimality gap decreases by 10.31% (AVI to MADP) but by adding the regression-based initialization, the average optimality gap decreases by 26.42% (AVI to AVI-RB). The lowest average optimality gap occurs with MADP-RB that has both the monotonicity property and regression-based initialization.

Finally, we validate the robustness of our results, focusing on the regression-based initialization within MADP-RB. First, we explain why we do not need to define a separate training set and test set for our regression-based initialization process. To cover the design space, we generate 40 scenarios using LHS. The generated scenarios include different combinations of input parameters. When the combined result of BI for small sizes of *each scenario* becomes available, we fit a linear regression function *for each scenario*, to initialize MADP-RB for that specific scenario. In other words, after inputting the parameters of a scenario, we solve small sizes of that scenario (in regard to the number of batteries and time horizon), store the resulting values, combine them, and fit the regression using Equation (14). Hence, if any scenario outside the LHS is generated (possibly from another LHS), it needs to follow the same process. Therefore, a specific regression function would be fitted for this scenario. On the other hand, if we had used a single regression function for all the generated scenarios of the first LHS (could be interpreted as the training set), then we would have needed a test set (second LHS) to validate and measure the performance of the fitted regression function.

Table 7. Results of Realistic-Sized SAIRPs for the Latin Hypercube Designed Experiment

	MADP expected total reward ($)			MADP-RB expected total Reward ($)		
Scenario	Harmonic	STC	Adaptive	Harmonic	STC	Adaptive
1	3,468.3	2,791.7	3,639.3	6,964.3	7,303.6	**7,884.1**
2	5,404.1	4,900.6	4,773.9	7,073.1	**7,622.2**	7,549.9
3	9,324.8	4,219.4	8,788.3	15,443.0	15,840.8	**16,491.2**
4	5,397.1	3,890.3	5,836.5	8,684.0	9,645.6	**9,710.6**
5	4,783.1	4,923.2	4,986.2	10,308.5	10,597.4	**11,079.1**
6	4,824.0	2,363.0	5,443.4	7,545.4	8,123.5	**8,797.8**
7	3,779.6	4,415.5	4,422.3	**5,269.7**	5,176.2	5,212.2
8	4,479.5	3,401.0	3,521.3	8,128.5	9,336.6	**9,486.3**
9	2,087.0	2,546.6	2,730.5	6,848.6	7,446.5	**7,780.6**
10	8,461.4	8,736.6	7,297.4	**22,060.2**	19,707.0	20,229.4
11	3,684.1	3,385.4	3,749.2	6,581.5	7,463.9	**7,889.3**
12	2,888.5	2,842.8	3,357.9	5,903.6	**6,052.8**	5,900.4
13	6,248.1	4,061.9	5,994.1	11,620.6	12,228.4	**12,950.7**
14	3,253.2	2,578.2	3,426.1	7,176.3	7,258.9	**7,573.4**
15	9,983.3	9,573.4	8,814.8	16,484.4	16,326.9	**16,614.3**
16	4,446.9	3,631.8	4,474.7	7,177.7	**7,417.1**	7,268.3
17	4,789.0	3,875.1	5,372.4	7,804.2	8,601.2	**8,787.8**
18	3,958.3	3,634.7	4,291.3	6,900.5	6,998.8	**7,260.1**
19	5,855.0	4,561.5	6,722.5	10,514.8	10,892.3	**12,124.5**
20	4,090.8	3,676.9	2,051.8	7,415.2	7,501.5	**7,529.3**
21	7,545.6	5,562.7	7,116.9	10,685.1	12,764.6	**13,478.4**
22	5,454.8	5,539.8	5,699.1	**33,221.8**	31,557.5	32,379.8
23	5,629.3	5,707.7	7,111.5	11,848.4	12,005.5	**12,811.4**
24	10,593.2	10,698.7	11,037.8	**34,805.9**	32,215.2	32,399.8
25	4,963.0	5,349.2	7,186.8	10,505.9	10,074.7	**11,664.4**
26	51,874.6	48,577.1	50,855.3	84,745.6	83,125.1	**85,354.7**
27	5,224.7	5,512.7	4,699.4	16,876.2	16,771.0	**17,203.0**
28	6,366.9	5,006.7	6,514.1	11,356.8	12,461.2	**12,748.8**
29	39,244.6	34,101.7	37,832.0	**71,877.0**	70,920.1	71,396.5
30	25,188.0	24,640.2	26,046.0	79,376.8	77,325.9	**80,233.1**
31	5,373.0	5,971.0	6,832.8	**33,432.0**	31,564.1	32,693.5
32	9,216.7	10,299.8	10,055.9	**44,189.8**	42,772.4	43,355.7
33	9,929.9	9,783.1	10,293.0	**29,615.5**	28,922.0	28,786.7
34	5,920.8	4,409.0	6,452.9	11,341.8	12,115.3	**13,178.0**
35	1,00,038.0	98,060.7	98,254.7	1,22,397.0	1,22,683.0	**1,24,779.0**
36	18,743.7	17,610.1	16,362.2	**73,562.5**	72,612.0	73,060.5
37	8,067.0	9,426.7	12,138.2	**60,294.6**	58,483.0	59,488.7
38	7,726.4	5,789.2	8,285.0	12,137.7	14,070.3	**14,674.7**
39	5,699.1	5,683.8	5,168.2	11,029.9	10,982.9	**11,158.6**
40	68,536.0	63,309.2	62,951.7	**1,12,542.0**	1,11,153.0	1,12,073.0
Average CPU time (h)	8.5	8.5	8.6	8.6	8.6	8.7

Nevertheless, we tested the robustness of our result by generating a new set of LHS scenarios, finding the regression function's coefficients of Equation (14) for each scenario, applying MADP-RB, and comparing the results with the presented LHS outputs. We observe no significant difference between the results of the two LHSs as the percentage change in average optimality gap, average charging/recharging, and average replacements are 0.2%, 0.9%, and 0.3%, respectively.

6.4. Monotone ADP Results and Performance for Realistic-Sized SAIRPs

In this next set of computational experiments, we focus on the ability to solve and deduce insights from realistic-sized stochastic SAIRPs. We proceed by determining the parameters, summarizing the results, including the expected total reward and computational times, presenting sample paths of policies, and analyzing the relationship between the outputs and inputs of the experiments.

For the test instances, we solve all 40 scenarios of the designed experiment presented in Section 6.3 using the realistic data summarized in Section 6.1. Specifically, we consider 100 batteries, a one-month time horizon with each decision epoch representing one hour, and scaled mean demand $\lambda'_t = (\lambda_t)(M'/7)$ where $M' = 100$, and λ_t is the original mean arrival of demand at epoch t in line with the original modest problem with $M = 7$. Because of the curses of dimensionality, BI and MBI are not capable of solving these large problems; thus, we focus on the performance of MADP and MADP-RB with $\tau = 100{,}000$ iterations.

In Table 7, we present the expected total reward and required computational time for the MADP and MADP-RB methods with harmonic, STC, and adaptive stepsizes. The average computational time of MADP-RB is only eight minutes longer than MADP for all stepsize functions as a result of executing the regression-based initialization (see steps 1–6 in Algorithm 1). We highlight the highest expected reward for each row of Table 7. We acknowledge that the highest expected total reward does not always indicate the lowest optimality gap; however, we use this as a metric of comparison in the absence of being able to determine the optimal solution and value for these realistic-sized SAIRPs.

The results are very clear and consistent. The MADP-RB value is always greater than MADP for all stepsize functions and scenarios. Within MARP-RB, we observe that harmonic, STC, and adaptive stepsize generate the highest expected total reward in 11, 3, and 26 scenarios, respectively.

In addition to examining the expected total reward of the two ADP methods, we examine sample paths of the policies when 500 sample paths of realized demand are generated. Our results show that the average percentage of demand met levels off around when the number of sample paths is greater than 200. As a result and to be conservative, for all scenarios, we calculate the average of demand met for 500 sample paths. We observe that the average of demand met for 500 sample paths over all the scenarios under MADP-RB is 25%, 29%, and 19% more than MADP with harmonic, STC, and adaptive stepsize functions, respectively. The optimal policies are similar across these three stepsize functions; thus, we proceed by presenting further analysis for MADP-RB policy using the harmonic stepsize function.

From the sample paths, we observe three typical behaviors within all LHS scenarios. In Figure 6, we displayed examples of the fluctuations of the average capacity for these three categories when the realized demand equals mean demand. In type 1, the average capacity remains more than 95% throughout the time horizon (see Figure 6(a)). This is achieved with many replacement actions. The scenarios that exhibit type 1 behavior have low replacement costs and high revenue per swap (including scenarios 26, 29, 30, 35, 36, 40, 37, and 22). In type 2 (see Figure 6(b)), the average capacity does not stay as high as type 1 but is maintained above 85% for the entire time horizon except the very end of the month. For 500 sample paths of demand, on average, in scenarios with type 1 behavior,

Figure 6. (Color online) Sample Paths of Average Capacity over Time Horizon for Three Types of LHS Scenario When Realized Demand Equals Means Demand

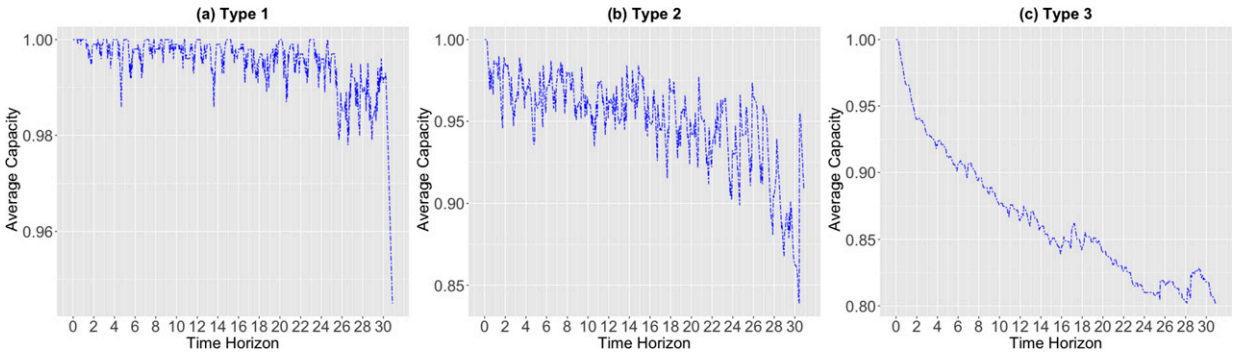
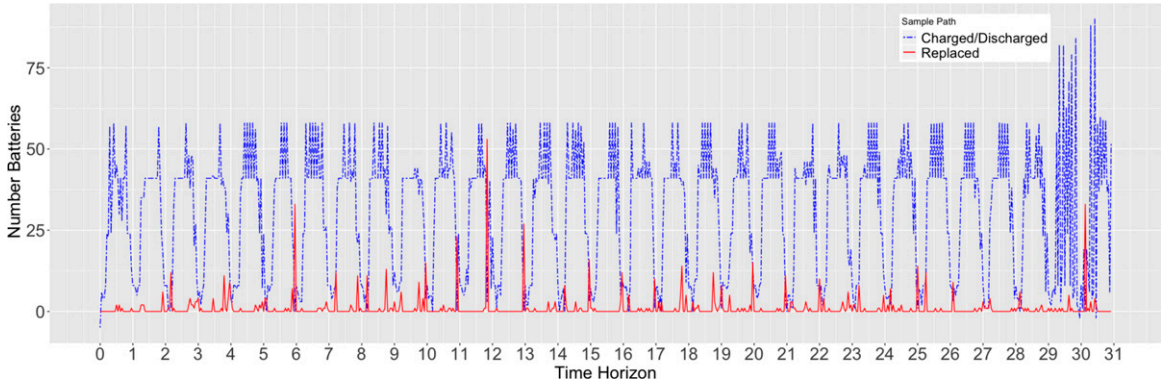


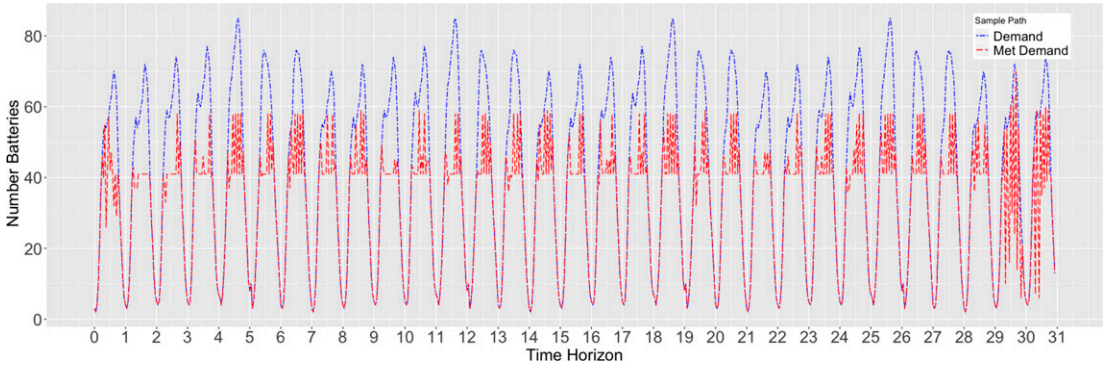
Figure 7. (Color online) Sample Path of the Policy for Scenario 15 When Realized Demand Equals Mean Demand



we observe replacement in 60% of the epochs, and when a replacement occurs, 15% of the batteries are replaced. In contrast, in scenarios with type 2 behavior, replacement occurs in 41% of the decision epochs, and 7% of the batteries are replaced when a replacement occurs. The scenarios that exhibit type 2 behavior have average replacement cost and revenue per swap values (includes scenarios 31, 27, 32, 24, 33, 10, and 15). In these scenarios, it is beneficial to recharge batteries to ensure demand is met, but the station allows battery capacity to degrade and does not replace batteries as frequently. Finally, in type 3 (see Figure 6(c)), the average battery capacity consistently decreases over the time horizon to the replacement threshold of 80%. Fewer batteries are replaced as, on average, we observe replacement in 11% of the decision epochs, and only 3% of batteries are replaced when a replacement occurs. This demonstrates a reduced priority for maintaining a high battery capacity. At epochs in which replacements occur, a few batteries are replaced at a time such that to ensure the minimum capacity is maintained. The scenarios that exhibit type 3 behavior have high replacement costs and low swapping revenue values.

Next, we dig into a specific scenario to compare and contrast the policies for the modest and realistic-sized instances. Our intent in this analysis is to determine whether similar actions are taken for the same scenario when considering more batteries over a long time horizon. We observed this is not the case, thereby justifying the need to solve realistic-sized instances. To demonstrate our observations, we present results for scenario 15 that has meaningful differences with regard to the replacement actions, charging actions, and amount of demand met between the modest and realistic-sized instances. In the realistic-sized problem, on average, we observe replacement in 27% of the decision epochs, and when replacement occurs 4% of the batteries are replaced. However, the derived policy in the modest problem consists of no replacement actions. With regard to charging, on average, we see charging actions in 97% and 40% of decision epochs for the realistic-sized and modest problems, respectively. With more frequent charging and replacement actions in the realistic-sized problem, the percentage of met demand is higher than the modest problem. The derived policy for the realistic-sized problem satisfies 76% of demand, which is significantly higher than 54% of the demand met for the modest problem. From this analysis, although we must solve a finite horizon MDP to capture the time-varying elements, it is necessary to be able to computationally solve instances with longer time horizons and a larger number of batteries that mimic reality to determine the general operating policies.

Figure 8. (Color online) Demand and Met Demand for Scenario 15 Based on the Sample Path When Realized Demand Equals Mean Demand



In Figure 7, we present a sample path of the policy when realized demand equals the mean demand for the realistic-sized SAIRP associated with scenario 15. We observe that more batteries are replaced before epochs with high power prices (e.g., 53% on day 12 before the high-power price days of December 12–18), which is consistent among type 2 scenarios. When this replacement occurs, the battery capacity is higher, resulting in higher swapping revenue to offset higher costs to recharge batteries. Overall, we observe a consistent trend for recharging batteries and meeting demand during the time horizon. The number of full batteries is kept between 40% and 60% of the total number of batteries during the middle of the day, and recharging is conducted to raise the number of full batteries to 90% overnight. We show in Figure 8 that the policy meets 100% of demand during off-peak epochs and more than 50% during peak epochs.

7. Conclusions

We examined a stochastic SAIRP for a battery swap station where there is a direct link between the inventory level and necessary recharging and replacement actions for battery charge and battery capacity. Specifically, the direct link is that the act of recharging a battery to enable short-term operation is the exact cause for long-term battery capacity degradation. This creates a unique problem where tradeoffs between recharging and replacing batteries must be analyzed. We used an MDP model for a battery swap station faced with battery degradation and uncertain arrival of demand for swaps. In the MDP model of the stochastic SAIRP, we determine the optimal policy for charging, discharging, and replacing batteries over time when faced with nonstationary charging prices, nonstationary discharging revenue, and capacity-dependent swap revenue. We prove theoretical properties for the MDP model, including the existence of an optimal monotone policy for the second dimension of the state (average capacity of batteries) when there is an upper bound placed on the number of batteries replaced at each decision epoch. Furthermore, we prove the monotonicity of the value function. Given these results, we solved the stochastic SAIRP using both BI and MBI. However, we run into the curses of dimensionality as we are unable to find an optimal policy for realistic-sized instances.

To overcome these curses, we propose a monotone ADP solution method with regression-based initialization (MADP-RB). The MADP-RB builds upon the monotone ADP (MADP) algorithm, which is fitting for problems with monotone value functions (Jiang and Powell 2015). In our MADP-RB, we initialize the value function based on an intelligent approximation using regression. We used a Latin hypercube designed experiment to test the performance of MBI, MADP, and MADP-RB. In the designed experiment, we examine both modest-sized problem instances that are optimally solvable using BI and realistic-sized instances. Overall, we observed that MADP-RB resulted in the greatest performance in terms of computational time, optimality gap, convergence, and expected total reward.

Our investigation into SAIRPs opens the avenue for many opportunities for future work. In terms of the model and solution method, researchers should examine state reduction/aggregation approaches and quantify how they impact the quality of the solutions and computational time. Future research should consider different mechanisms within a disaggregated MDP model that captures individual battery states and actions. Furthermore, it is valuable to solve the disaggregated MDP using approximate solution methods and compare the insights with the presented aggregated MDP. Moreover, it is interesting to study discretizing the state of charge of batteries and allowing partial recharging decisions. For the battery swap station application, future work should consider different types of charging options, multiple swap station locations, and multiple classes of demand dictated by how long the battery must operate to satisfy the demand (e.g., how far a drone must fly). Furthermore, researchers should examine the exact tasks that EVs and drones are doing to properly capture the capacity of batteries when seeking a swap at the station.

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3. 拍卖机制：**Choice Screen Auctions (AER, 2023)**

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关键词：拍卖；市场设计；搜索引擎；选择屏幕；竞争政策

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推荐理由：欧洲安卓平台曾通过选择屏幕拍卖，让用户在初始设置中选择搜索引擎。本文说明，按展示付费还是按安装付费这一设计差异，会显著改变中标者的构成、特征与预期市场份额。按安装付费的拍卖还会扭曲搜索引擎的激励，使其偏向提高单个安装用户的变现收入，即使这会减少用户数量；竞争加剧可能进一步放大这种扭曲。2020年安卓选择屏幕拍卖的数据与理论预测一致。

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Choice Screen Auctions[†]

By Michael Ostrovsky*

Choice screen auctions have been recently deployed in 31 European countries, allowing consumers to choose their preferred search engine on Google's Android platform instead of being automatically defaulted to Google's own search engine. I show that a seemingly minor detail in the design of these auctions—whether they are conducted on a "per appearance" or a "per install" basis—plays a major role in the mix and characteristics of auction winners and, consequently, in their expected market share. Furthermore, per install auctions distort search engines' incentives. Empirical evidence from Android choice screen auctions conducted in 2020 is consistent with my theoretical results. (*JEL* D44, D47, K21, L40, L86)

Optimal regulation of digital platforms is one of the thorniest issues in competition policy. A particularly challenging dimension for regulation is the fact that dominant platforms are often active in multiple distinct businesses and may leverage their position in one area into gaining an advantage in another. The net effect of such leverage on consumer welfare is often ambiguous and hard to determine. On one hand, a dominant platform's expertise and technological complementarities may make the adjacent product genuinely superior to the alternatives. On the other hand, such leverage may make it harder for other firms to successfully compete, even if their products, on their own, would be preferred by some consumers to that of the platform.

These linkages across product lines have led regulators to sometimes propose extreme measures to regulate large digital platforms, all the way to breaking them up and prohibiting them from entering certain lines of business. Notable examples in the United States include the Microsoft case of the late 1990s, in which the initial court decision was to break up the company,[1] and the recently concluded congressional investigation into the business practices of Amazon, Apple, Facebook, and Google, which proposes "structural separations and line of business restrictions"

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[1] *United States v. Microsoft Corp.*, 97 F. Supp. 2d 59 (D.D.C. 2000), https://law.justia.com/cases/federal/district-courts/FSupp2/97/59/2339529/.

as a solution for "restoring competition in the digital economy."[2] Regulators in the European Union and other parts of the world have often reached similar conclusions. Of course, the breakup of a company is a very heavy-handed solution, difficult to implement, rife with potential unintended consequences, and, unsurprisingly, adamantly opposed by the digital platforms.[3]

In light of these problems, platforms and regulators have, in some cases, adopted a more "lightweight" alternative as a compromise solution: choice screens. The logic of a choice screen is straightforward: instead of having the consumer use the dominant platform's product automatically and by default, the platform agrees to present the consumer with a menu of choices. This menu includes the platform's own product as one of the options and also includes several competing products as alternatives. Consumers can then choose whichever products they prefer, leveling the playing field between the dominant platform and its competitors.

Choice screens for web browsers on the Windows platform were first proposed by Microsoft in 1999 as a remedy in its negotiations with the US Department of Justice.[4] They were not adopted at that time but were subsequently accepted as a compromise solution between the European Commission and Microsoft in 2009 and were displayed to users in Europe from 2010 until 2014.[5] (Due to a technical error, the choice screen was not displayed on one of the versions of Windows from May 2011 to July 2012, affecting approximately 15 million users. Microsoft admitted its responsibility for this error and was subsequently fined €561 million.[6]) In 2017, Google reached a settlement with the competition authority in Russia to display choice screens for the default search engine on the Android platform there.[7] A similar agreement was reached between Google and the European Commission

[2] "To address this underlying conflict of interest, Subcommittee staff recommends that Congress consider legislation that draws on two mainstay tools of the antimonopoly toolkit: structural separation and line of business restrictions. Structural separations prohibit a dominant intermediary from operating in markets that place the intermediary in competition with the firms dependent on its infrastructure. Line of business restrictions, meanwhile, generally limit the markets in which a dominant firm can engage" (Section VI.A.1, https://web.archive.org/web/20201006214413/https://judiciary.house.gov/uploadedfiles/competition_in_digital_markets.pdf).

[3] To give just one recent example (out of many available ones): "A government effort to break up Facebook Inc. from Instagram and WhatsApp would defy established law, cost billions of dollars and harm consumers, according to a paper company lawyers have prepared in the wake of rising antitrust legal threats. […] In the paper, Facebook says unwinding the deals would be nearly impossible to achieve, forcing the company to spend billions of dollars maintaining separate systems, weakening security and harming users' experience" (https://www.wsj.com/articles/facebook-says-government-breakup-of-instagram-whatsapp-would-be-complete-nonstarter-11601803800).

[4] https://www.wired.com/2000/11/microsoft-7/

[5] "Under the commitments approved by the Commission, Microsoft will make available for five years in the European Economic Area […] a "Choice Screen" enabling users of Windows XP, Windows Vista and Windows 7 to choose which web browser(s) they want to install in addition to, or instead of, Microsoft's browser Internet Explorer. … The Commission's preliminary view was that competition was distorted by Microsoft tying Internet Explorer to Windows. This was because it offered Microsoft an artificial distribution advantage not related to the merits of its product on more than 90 per cent of personal computers. Furthermore, the Commission's preliminary view was that this tying hindered innovation in the market and created artificial incentives for software developers and content providers to design their products or websites primarily for Internet Explorer. The approved commitments address these concerns. PC users, by means of the Choice Screen, will have an effective and unbiased choice between Internet Explorer and competing web browsers. This should ensure competition on the merits and allow consumers to benefit from technical developments and innovation both on the web browser market and on related markets, such as web-based applications" (https://ec.europa.eu/commission/presscorner/detail/en/IP_09_1941).

[6] https://ec.europa.eu/commission/presscorner/detail/en/IP_13_196

[7] https://www.reuters.com/article/us-alphabet-google-russia-idUSKBN17J11C, https://yandex.com/blog/yacompany-com/choosing-yandex-search-on-android

following a €4.3 billion fine imposed on the company by the Commission in 2018,[8] and Google began displaying choice screens for both default search engines and web browsers to Android users in Europe in 2019.[9] An analogous solution is now being considered by the Australian Competition and Consumer Commission.[10]

Choice screen menus can be an effective and powerful tool. For instance, following the 2010 introduction of browser choice screen menus on the Windows platform in Europe, the number of downloads of Opera Software's web browser more than doubled.[11] Discussing Google's introduction of choice screen menus on Android, the European Commissioner for Competition, Margrethe Vestager, stated, "We've seen in the past that a choice screen can be an effective way to promote user choice."[12] However, from the viewpoint of a company that owns the platform, the initial implementations of choice screen menus suffered from one serious shortcoming: zero revenue. This may be a particularly salient issue in the case of search engines. First, being chosen by a consumer is extremely valuable to a search engine due to the advertising revenues it expects to receive when the consumer uses it. Second, the dominant company itself may be making large payments to *another* platform to have consumers use its search engine there.[13] In this case, it is logical for the company to argue that it should be allowed to charge others for the right to have their products be shown on its platform's choice screens—and a natural way to do so is via an auction.

That is the decision that Google announced in August 2019,[14] and the first "choice screen auctions" took place in early 2020 (see Figure 1 for an illustration of an Android choice screen), with the subsequent auctions run on a quarterly basis. The basic rules of Google's choice screen auctions are very simple.

> In each country auction, search providers will state the price that they are willing to pay each time a user selects them from the choice screen in the given country. The three highest bidders will appear in the choice screen

[8] "The Commission decision has concluded that Google has engaged in two instances of illegal tying: First, the tying of the Google Search app. As a result, Google has ensured that its Google Search app is pre-installed on practically all Android devices sold in the EEA. Search apps represent an important entry point for search queries on mobile devices. The Commission has found this tying conduct to be illegal as of 2011, which is the date Google became dominant in the market for app stores for the Android mobile operating system. Second, the tying of the Google Chrome browser. As a result, Google has ensured that its mobile browser is pre-installed on practically all Android devices sold in the EEA. Browsers also represent an important entry point for search queries on mobile devices and Google Search is the default search engine on Google Chrome. The Commission found this tying conduct to be illegal as of 2012, which is the date from which Google has included the Chrome browser in its app bundle" (https://ec.europa.eu/commission/presscorner/detail/en/IP_18_4581).

[9] https://www.blog.google/around-the-globe/google-europe/presenting-search-app-and-browser-options-android-users-europe/

[10] "The ACCC is seeking submissions from consumers and industry participants about choice screens, which give users a choice of internet search services on mobiles and tablets, rather than a pre-selected search service, and about the supply of web browsers in Australia" (https://www.accc.gov.au/media-release/feedback-sought-on-choice-and-competition-in-internet-search-and-web-browsers).

[11] https://press.opera.com/201%3/18/opera-more-than-doubles-download-numbers-in-europe-after-choice-screen-introduction/

[12] https://ec.europa.eu/commission/presscorner/detail/en/STATEMENT_19_1774

[13] While the companies do not directly disclose these numbers, analysts estimate that Google is paying Apple on the order of $8–$12 billion per year to have Google be the default search engine on Apple's Safari browser (https://www.npr.org/2020/10/22/926290942/google-paid-apple-billions-to-dominate-search-on-iphones-justice-department-says). It is worth noting that this agreement itself has also been a subject of recent regulatory scrutiny (https://www.justice.gov/opa/press-release/file/1328941/download). Similarly, Google is paying an estimated $400–$450 million per year to Mozilla to be the default search engine on the Firefox browser (https://www.zdnet.com/article/sources-mozilla-extends-its-google-search-deal/).

[14] https://www.blog.google/around-the-globe/google-europe/update-android-search-providers-europe/

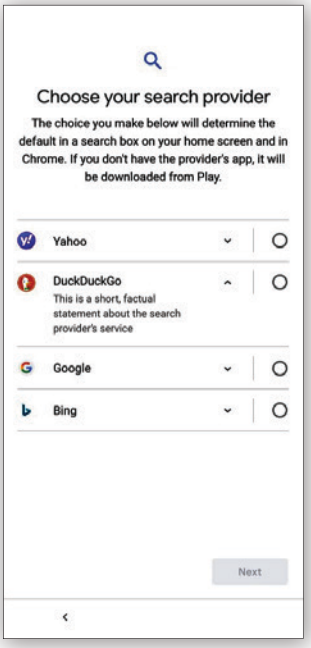


FIGURE 1. ANDROID CHOICE SCREEN

> for that country. The provider that is selected by the user will pay the amount of the fourth-highest bid.[15]

In the same document, Google explains why it chose to auction off slots in the choice screen this way:

> Q: Why does Google use an auction to determine the search providers that appear in the choice screen?
>
> A: An auction is a fair and objective method to determine which search providers are included in the choice screen. It allows search providers to decide what value they place on appearing in the choice screen and to bid accordingly.
>
> The auction revenues help us to continue to invest in developing and maintaining the Android platform.

In this paper, I show that a seemingly minor detail of the implementation of choice screen auctions plays a major role in their outcomes—and thus in the overall effectiveness of the antitrust remedy. Specifically, while the answer in the Q&A section of the document states that an auction "allows search providers to decide what value they place on appearing in the choice screen and to bid accordingly" (see https://

[15] https://web.archive.org/web/20200501202513/https://www.android.com/choicescreen/

web.archive.org/web/20200501202513/https://www.android.com/choicescreen/), the auction, as implemented, charges these providers not for *appearing* in the choice screen but for *being chosen by a user*.

While the difference may seem to be just a matter of language, it is not. To see the intuition for the difference, consider a version of the auction with just one available spot and two bidders. Bidder A gets revenue US$10 from each user who installs its search engine, and if it is shown as an option in the choice screen, then the probability that a user will choose it is 10 percent. Bidder B gets revenue US$20 from each user who installs its search engine, but the probability that a user will choose it (if it is shown as an option in the choice screen) is only 1 percent. The value that bidder A has for appearing on the screen is therefore US$1, and the value that bidder B has for appearing on the screen is US$0.20. Thus, if the auction is conducted on the "per appearance" basis, then bidder A will win, will pay US$0.20 per appearance, and will have its search engine chosen by users 10 percent of the time, while the dominant platform's own search engine will be chosen 90 percent of the time. If, instead, the auction is conducted as implemented, with bidding and payment on the "per install" basis, then bidder B will win and will pay US$10 every time its search engine is chosen (corresponding to US$0.10 per appearance). The winner's search engine will be chosen only 1 percent of the time, and the dominant platform's one will be chosen the remaining 99 percent of the time. Thus, relative to the per appearance auction, the per install auction results in a lower likelihood that an alternative search engine will be chosen by the user (making it correspondingly more attractive to the dominant platform) and gives advantage to search engines that generate higher revenue per user versus those that are more popular but generate less revenue on a per user basis. In Section I, I show that these conclusions hold more generally, in a basic model in which alternative search engines differ on two dimensions: revenue per user (i.e., how much revenue the search engine generates, on average, when a user chooses to install it) and popularity (i.e., the likelihood that the search engine will be chosen if it is shown to the user in the choice screen). Moreover, I show that the difference is exacerbated by competition. As the number of alternative search engines grows, under the per appearance auction, the expected popularity of the winner also grows, so the probability that the dominant platform's own search engine is chosen decreases. By contrast, under the per install auction, these measures are not affected by the number of bidders.

In the example above and in the model of Section I, a search engine's popularity and revenue per user (RPU) are fixed. In practice, a search engine has some ability to trade them off against each other. For instance, a search engine may choose to show more intrusive ads, increasing its revenue per user but decreasing its popularity. Conversely, a search engine may donate some of its proceeds to charity or implement very strict privacy rules, lowering its revenue per user but increasing the probability that a user will choose it. I introduce this possibility in Section II and show that the two auction formats result in very different incentives to the search engines regarding this trade-off. Under the per appearance auction, each bidder chooses the same point on the popularity-RPU frontier as it would choose if it were the only bidder (and were thus guaranteed the spot on the choice screen). In particular, this implies that just as in the model of Section I, as the number of competitors increases, the expected popularity of the winner also increases, and the expected probability that the dominant platform's search engine is chosen goes down. By

contrast, under the per install auction, each bidder has a strong incentive to distort the choice toward higher revenue per each user who chooses the product, at the expense of lowering the probability of actually being chosen. This distortion grows stronger as the number of bidders grows. In the limit, as that number approaches infinity, the distortion results in a "race to the bottom," with all bidders pushing to the extreme point on the popularity-RPU frontier: the highest possible RPU and the lowest possible popularity. As a result, the expected popularity of the winner goes in the opposite direction versus that in the case of the per appearance auction, minimizing the probability that an alternative search engine will be chosen.

In Section III, I present empirical evidence using publicly available data from the four sets of choice screen auctions (for the periods of March–June 2020, July–September 2020, October–December 2020, and January–March 2021) conducted by Google in 2020 in 31 European countries, as well as various statistics publicly available on Google Play Store on the numbers of downloads and ratings of various search engine apps. The evidence is consistent with my theoretical conclusions. In particular, to mention just one data point from the section, the search engine that was most successful in these auctions, winning a slot in every country and in every period, has been installed only approximately 100,000 times *worldwide*, has only 74 user reviews in Google's Play Store, and has one of the lowest ratings among the search engines participating in choice screen auctions. For the sake of comparison, Google's own search app has been installed more than 5 billion times, while alternative search engines Bing (produced by Microsoft) and DuckDuckGo (produced independently and focused on user privacy) have been installed more than 10 million times. (These numbers include all installs, including those that come from choice screens and those that do not.)

Subsequent to the circulation of the initial draft of this paper, Google has announced the decision to discard the use of choice screen auctions and instead determine the list of providers shown in the choice screen by their overall popularity.[16] In Section IV, I conclude by discussing this decision and other issues related to the design of choice screen auctions.

Related Literature.— Two recent surveys by Crémer, de Montjoye, and Schweitzer (2019) and Scott Morton et al. (2019) provide extensive discussions of the challenges of regulating digital platforms, potential remedies, and other related issues. Salinger (2020) reviews the economics of self-preferencing. On the specific issue of choice screens (without auctioning off the slots), Economides and Lianos (2011) provide a discussion of the 2009 Microsoft-EU agreement regarding the Windows platform and the Internet Explorer web browser.

On the issue of search engine monetization, see Edelman, Ostrovsky, and Schwarz (2007) and Varian (2007). These papers also contain discussions of search engines adjusting advertisers' "per click" bids by their estimated probabilities of being clicked, essentially transforming those "per click" auctions into "per appearance" ones. See also Varian (2008) for a discussion of reasons why search engines adjust "per click" bids in ad auctions by the click-through rates instead of ranking the ads solely by "per click" bids. This distinction is conceptually similar to the difference between the "per

[16] See https://blog.google/around-the-globe/google-europe/changes-android-choice-screen-europe/ (June 8, 2021).

install" and "per appearance" formats in choice screen auctions that I discuss in the current paper.[17] There are, however, several important differences. First, there is a matter of degree. In sponsored search auctions, the "per click" format, while imperfect, has worked successfully at the ad auction pioneer GoTo (subsequently renamed Overture and then acquired by Yahoo) for almost a decade, from 1998 until 2007.[18] By contrast, the failure of the "per install" choice screen auction format was so dramatic that the entire auction was scrapped after barely a year, despite initially getting the blessing of both Google and the European Commission. Second, the auctioneer's incentives are very different in the sponsored search setting versus the choice screen one: in the latter, the auctioneer places a high value on the outcome in which the winning bidder's listing is *not* chosen by the end user, which is not the case in the former setting. Finally, in the choice screen auction setting, I explore the important issue of search engines' endogenous choice of popularity versus revenue per user and show how the per install auction distorts this decision (with the distortion growing larger as the number of bidders grows), while the per appearance auction does not.

The "nondistortionary" property of the per appearance auction is not a coincidence. In the choice screen setting, the per appearance auction can be viewed as an implementation of the Vickrey-Clarke-Groves mechanism (VCG) (Vickrey 1961; Clarke 1971; Groves 1973), with the right to be shown on the choice screen being the object auctioned off. VCG is known to preserve various incentives (e.g., pre-auction investment or information acquisition), even in cases in which other auction formats may not (Rogerson 1992; Bergemann and Välimäki 2002; Arozamena and Cantillon 2004; Hatfield, Kojima, and Kominers 2014, 2018). As I show in Section IIA, the per appearance auction in the choice screen setting likewise has this nondistortionary property.

I. Basic Model: Exogenous Popularity and Revenue per User

A platform is auctioning off the right to be shown on the choice screen. There is one slot available, next to the platform's own product.[19] There are $n = 2$ bidders, $i \in \{1, 2\}$. Each bidder i has an exogenously determined *popularity* q_i, and if its product is chosen by a user, then bidder i receives revenue r_i from that. Variables q_i and r_i are private information of bidder i. Variables q_1, q_2, r_1, and r_2 are independently and identically distributed, and each is drawn from the uniform distribution on $[0, 1]$.[20]

If a product of popularity q is shown to a user, then it is chosen with probability q. With probability $1 - q$, the platform's own product is chosen instead, in which case the platform gets benefit $\pi > 1$.[21]

Under the "per appearance" auction, each bidder submits a bid for the right to be shown to users. The bidder with the highest bid wins, is shown on the choice

[17] See also the discussion of practical implementation of per appearance auctions, and the connections to sponsored search auctions, in Section IV.

[18] https://www.nytimes.com/2007/02/05/technology/05yahoo.html

[19] In Google's Android choice screen auctions, there are three slots next to the platform's own listing. I consider the case of only one alternative slot for simplicity; this assumption does not qualitatively change my conclusions.

[20] This assumption is made for the ease of exposition. In Appendix A.A, I extend the results of the current section to the case of more general distributions.

[21] That is, the platform's most preferred outcome is to have its own product chosen by a user; after all, if that was not the case, there would be no need for the antitrust remedy.

screen next to the platform's own product, and pays the amount equal to the bid of the second-highest bidder.

Under the "per install" auction, each bidder submits a bid. The bidder with the highest bid wins, is shown on the choice screen next to the platform's own product, and pays the amount equal to the bid of the second-highest bidder *if the user chooses its product*.

Note that both auction formats are incentive-compatible: it is a dominant strategy for each bidder to submit its valuation truthfully. That is, under the "per install" auction, each bidder i will bid r_i, while under the "per appearance" auction, each bidder i will bid $q_i r_i$.[22]

To characterize the distribution of outcomes in the per appearance auction, we need to perform some calculations.

First, observe that the unconditional distribution of each bid $b_i = q_i r_i$ is given by the CDF $G(x) = x - x\ln x$ and the corresponding density function $g(x) = -\ln x$ (for $x \in (0,1]$).[23] Thus, for a bidder with type (q,r), the probability of winning the auction is $(qr) - (qr)\ln(qr)$. For the population of bidders with type q and with types r distributed uniformly on $[0,1]$, the probability of winning is therefore $\int_0^1 \left[(qr) - (qr)\ln(qr)\right]dr = q\int_0^1 (r - r\ln q - r\ln r)dr = q\left(\frac{1}{2} - \frac{1}{2}\ln q + \frac{1}{4}\right) = \frac{3}{4}q - \frac{1}{2}q\ln q$.[24] We can now calculate the expected popularity of the winner of the auction, which is equal to $2\int_0^1 q\left(\frac{3}{4}q - \frac{1}{2}q\ln q\right)dq = \int_0^1 \frac{3}{2}q^2 dq - \int_0^1 q^2 \ln q dq = \frac{1}{2} + \frac{1}{9} = \frac{11}{18}$.[25]

The expected payment made by the winner of the auction is equal to $E\left[\min\{q_1 r_1, q_2 r_2\}\right]$. Given the distribution $G(\cdot)$ of each $q_i r_i$ derived above, the distribution of $\min\{q_1 r_1, q_2 r_2\}$ is given by $G(x)^2 + 2G(x)\left[1 - G(x)\right]$, with the corresponding density $2g(x) - 2g(x)G(x) = -2\ln x(1 - x + x\ln x)$. Thus, $E\left[\min\{q_1 r_1, q_2 r_2\}\right] = -2\int_0^1 \left[x\ln x(1 - x + x\ln x)\right]dx = \frac{7}{54}$, and the expected payoff of the platform is $\frac{7}{54} + \frac{7}{18}\pi$.

Characterizing the outcomes of the per install auction is straightforward. Under this format, the expected popularity of the winner is $1/2$—it is independent of the bids in the auction and plays no role in determining the winner. This is lower than the expected popularity of the winner in the per appearance auction $(11/18)$, and thus, the probability that an alternative product will be chosen is reduced and the probability that the dominant platform's product is chosen is increased.

The expected per install payment made by the winner is $1/3$ (and is independent of its popularity), and the expected payoff of the platform in the per install auction is therefore $1/6 + 1/2\pi$, which is higher than its expected payoff in the per appearance auction $(\frac{7}{54} + \frac{7}{18}\pi)$.

[22] I will ignore other equilibria of these auctions.

[23] For $x \in [0,1]$, the probability that $q_i r_i$ is less than or equal to x is equal to $G(x) = x + \int_x^1 \frac{x}{q}dq = x + x(\ln 1 - \ln x) = x - x\ln x$.

[24] This calculation uses the fact that $\int x\ln x = \frac{1}{2}x^2\ln x - \frac{x^2}{4} + c$. In the next step, we will also use the fact that $\int x^2\ln x = \frac{1}{3}x^3\ln x - \frac{x^3}{9} + c$.

[25] The logic behind this formula is that by symmetry, (the expected popularity of the winner of the auction) is equal to (the expected popularity of the winner of the auction conditional on that winner being bidder 1). The latter expression is equal to $\frac{\int_0^1 q_1 \Pr(\text{bidder 1 is the winner of the auction}\,|\,q_1)dq_1}{\Pr(\text{bidder 1 is the winner of the auction})}$, whose denominator is equal to $1/2$.

The contrast between the outcomes of the per appearance and per install auctions becomes even more striking if the number of bidders, n, becomes large; i.e., the number of potential alternative products grows. It is immediate that as n grows, the expected popularity of the winner of the per appearance auction converges to one: the highest possible popularity. By contrast, the expected popularity of the winner of the per install auction remains unchanged, at one-half; under that format, the increase in the competition has no impact on the popularity of the winner and thus on the probability that an alternative product will be chosen. Of course, if the platform's payoff from having a user choose its own product is higher than the revenue that an alternative product generates, this outcome is preferred by the platform, just as it was in the case of $n = 2$.

II. Extension: Endogenous Popularity and Revenue per User

It is clear from the results of Section I that the choice between per install and per appearance auction formats is a first-order issue. However, that only tells a part of the story. In this section, I consider the incentives of bidders to choose between making their product more attractive to users (at the expense of lower per user revenue) versus moving in the opposite direction. This is an important issue for the case of default search engines on a platform. Search engines may be able to increase their popularity by reducing the intrusiveness of ads, enhancing privacy protections, or donating their advertising revenue to charitable causes. These measures, while making the search engine more attractive to users, reduce the revenue it makes from each one of them. In this section, I show that the issues discussed in Section I are exacerbated once these incentives are taken into account; in some cases, dramatically so.

I maintain most of the assumptions of Section I and make only one change. Instead of assuming that each bidder i has an exogenously given popularity q_i and revenue per user r_i, I assume that each bidder has an exogenously given type t_i drawn independently from the uniform distribution $F(t)$ on $[0,1]$ (with the corresponding density function $f(t)$).[26] The bidder can then select its popularity q_i from $[0,t_i]$, and its revenue per user is then equal to $r_i = t_i - q_i$. After making this decision, the bidder submits its auction bid as before.

Bidding decisions are still straightforward under both rules: it is optimal to bid truthfully in both per appearance and per install auctions. However, each bidder now needs to decide, given its type t_i, how much of that type to allocate to popularity q_i and how much to allocate to revenue per user. I work out the resulting equilibria in the next two subsections.

A. *Per Appearance Auction*

Consider a per appearance auction with n bidders, and suppose bidder i has type $t_i \in [0,1]$. Fix other bidders' strategies, let $G(x)$ denote the distribution of the first-order statistic of those bidders' bids, and let $P(x)$ denote the expected payment that bidder i would make, conditional on winning the auction, if it submitted bid x

[26] In Appendix A.B, I extend the results of the current section to the case of more general distributions.

(note that $G(x)$ and $P(x)$ are purely functions of the other bidders' strategies and x). Bidder i has two decisions to make: popularity q_i and bid b_i. Its payoff as a function of these two decisions is given by

$$\Pi(q_i, b_i) = G(b_i) \times \left[q_i(t_i - q_i) - P(b_i)\right].$$

It is immediate that bidder i's optimal choice of popularity is to set

$$q_i = \frac{t_i}{2}. \tag{1}$$

This is an optimal strategy *regardless* of what other bidders' strategies are (or how many of those bidders there are). Thus, the strategy profile in which each bidder sets $q_i = t_i/2$ and then bids $(t_i/2)^2$ per appearance constitutes an equilibrium.

B. *Per Install Auction*

Equilibrium characterization in the case of the per install auction requires a more involved argument. Consider a symmetric equilibrium of the per install auction with n bidders, and suppose equilibrium strategies are given by functions $q(t)$ and $b(t)$, with the first one denoting the popularity chosen by a bidder with type t and the second one denoting its bid. We know that in equilibrium, function $b(t)$ will be truthful (given the choice of popularity $q(t)$ and the corresponding revenue per user $t - q(t)$); however, just as in the case of the per appearance auction, it is more convenient to not yet impose that restriction on function $b(t)$.

Take a bidder of type $t_i \in (0,1)$ and a real number Δ_q such that $q(t_i) + \Delta_q \in (0, t_i)$. Let $\Pi(\Delta_q; t_i)$ denote the expected payoff of bidder i whose type is t_i *if it chooses popularity* $q(t_i) + \Delta_q$ *but bids* $b(t_i)$, given that other bidders are bidding according to strategies $q(t)$ and $b(t)$. We then have

$$\Pi(\Delta_q; t_i) = F^{n-1}(t_i) \times \left[q(t_i) + \Delta_q\right] \times \Bigg\{ t_i - \left[q(t_i) + \Delta_q\right] - E\left[b\left(\max_{j \neq i}\{t_j\}\right) \,\middle|\, \max_{j \neq i}\{t_j\} \leq t_i\right] \Bigg\}.$$

Because we started with an equilibrium profile of strategies, the partial derivative of $\Pi(\Delta_q; t_i)$ with respect to Δ_q has to be equal to zero when evaluated at $(0; t_i)$. This implies the following equation for $q(t_i)$:

$$q(t_i) = \frac{t_i - E\left[b\left(\max_{j \neq i}\{t_j\}\right) \,\middle|\, \max_{j \neq i}\{t_j\} \leq t_i\right]}{2}. \tag{2}$$

Note that even without fully characterizing the equilibrium, from the comparison of equations (1) and (2), it is immediate that equilibrium popularity chosen by each type $t_i > 0$ will be strictly lower under the per install auction than under the per appearance auction (assuming, of course, that the equilibrium of the per install auction actually exists, which we will show below).

Recall that by incentive compatibility, we have $b(t_j) = t_j - q(t_j)$. We can then rewrite the expectation in equation (2) as

$$E\left[b\left(\max_{j\neq i}\{t_j\}\right)\middle|\max_{j\neq i}\{t_j\} \leq t_i\right] = \frac{\int_0^{t_i}\left[s - q(s)\right]dF^{n-1}(s)}{F^{n-1}(t_i)}$$

$$= \frac{(n-1)\int_0^{t_i}\left[s - q(s)\right]f(s)F^{n-2}(s)ds}{F^{n-1}(t_i)},$$

and subsequently rewrite equation (2) as

$$2q(t_i)F^{n-1}(t_i) = t_iF^{n-1}(t_i) - (n-1)\int_0^{t_i}\left[s - q(s)\right]f(s)F^{n-2}(s)ds. \tag{3}$$

The next step is to take a derivative of both sides of equation (3) with respect to t_i, which gives us

$$2q'(t_i)F^{n-1}(t_i) + 2(n-1)q(t_i)f(t_i)F^{n-2}(t_i)$$

$$= F^{n-1}(t_i) + (n-1)t_if(t_i)F^{n-2}(t_i) - (n-1)t_if(t_i)F^{n-2}(t_i)$$

$$+ (n-1)q(t_i)f(t_i)F^{n-2}(t_i),$$

which simplifies to

$$2q'(t_i) + (n-1)q(t_i)\frac{f(t_i)}{F(t_i)} = 1. \tag{4}$$

Equation (4) is a first-order linear differential equation, with the initial condition $q(0) = 0$. In our case, $F(t_i) = t_i$ and $f(t_i) = 1$, and so the equation becomes

$$2q'(t_i) + (n-1)q(t_i)\frac{1}{t_i} = 1,$$

with the solution

$$q(t_i) = \frac{t_i}{n+1}.$$

C. *Comparison*

The comparison between the two formats is immediate. Even with just two bidders, $n = 2$, a bidder of each type t_i chooses a much lower popularity under the per install auction than under the per appearance auction: $t_i/3$ versus $t_i/2$. Of course, this lower popularity on a per type basis immediately translates into a correspondingly lower overall probability that a product alternative to the dominant platform's one will be picked by users from the choice screen.

As the number of bidders grows, the expected popularity of the winner of the per appearance auction also grows: with n bidders, it is equal to $n/\left[2(n+1)\right]$, which

is increasing in n.[27] By contrast, the expected popularity of the winner of the per install auction moves in the *opposite* direction, falling as the number of bidders grows: with n bidders, it is equal to $n/(n+1)^2$, which is decreasing in n.[28]

In the limit, as $n \to \infty$, the difference becomes particularly dramatic. Under the per appearance auction format, the popularity of the winning bidder converges to $1/2$. By contrast, under the per install format, as $n \to \infty$, the popularity of the winner converges to 0; therefore, the share of installs that goes to the dominant platform's own product converges to 100 percent, completely undoing the choice screen auction's raison d'être.

III. Evidence from Android Choice Screen Auctions

I now turn to the empirical evidence on the outcomes of Google's choice screen auctions for default search engines on the Android platform conducted in 2020. These auctions took place in January 2020 (for the period from March to June), June 2020 (for the period from July to September), September 2020 (for the period from October to December), and December 2020 (for the period from January to March 2021). For each time period, 31 independent auctions were conducted (1 per country). These auctions were conducted on a per install basis, with those submitting top three bids being shown on the choice screen and paying the fourth-highest bid every time a user chose one of them from the choice screen. In the event of a tie, more than three bidders could win, with the ties broken randomly on a per device basis.

Historical data on the winners of these auctions are publicly available on the Internet Archive at https://web.archive.org/web/*/https://www.android.com/choicescreen-winners/. These results, on a country-by-country basis, are summarized in Table 1. In each search engine–country cell in the table, I list the periods in which that search engine won a slot on the choice screen menu in that country. Countries are sorted by population, while search engines are sorted by the average population they won across the four time periods. For example, the search engine DuckDuckGo won a slot in every country in the first period (total population 519.4 million) and also in the second one (same total population), but only 4 countries in period 3 (with the total population 11.4 million) and 1 country in period 4 (population 11.6 million). The average population won by DuckDuckGo across the four time periods is thus $(519.4 + 519.4 + 11.4 + 11.6)/4 = 265.5$ million, as listed in the last row of Table 1. Sorting the search engines by the total number of auctions won (next-to-last row in the table), i.e., giving equal weight to countries of different sizes, would result in an almost identical ordering.

Table 2 lists the search engines by the average population they won across the four periods, includes again the total number of auctions they won, and adds data collected from the Android Play Store (https://play.google.com/store/apps) on the popularity and quality of these search engines.[29] To be eligible to participate in a

[27] It is helpful to recall that with n independent draws from the uniform distribution, the expected value of their maximum is $n/(n+1)$.

[28] The derivative of function $f(x) = \frac{x}{(x+1)^2}$ is equal to $f'(x) = \frac{(x+1)^2 - 2(x+1)x}{(x+1)^4} = \frac{1-x}{(x+1)^3}$, which is less than zero for $x > 1$.

[29] I collected the data on March 1, 2021—exactly one year after the first day of the first choice screen auction period. The screenshots of Play Store pages from which these data were taken are available upon request.

Table 1—Winners of Android Choice Screen Auctions (by Country)

Country	Population	Info.com	PrivacyWall	DuckDuckGo	Bing	GMX	Yandex	Qwant	Seznam	Givero	Ecosia
Germany	83,990,646	1, 2, 3, 4	2, 3, 4	1, 2	3, 4	1					
United Kingdom	67,999,326	1, 2, 3, 4	2, 3, 4	1, 2	1, 3, 4						
France	65,297,182	1, 2, 3, 4	2, 3, 4	1, 2	3, 4			1			
Italy	60,457,546	1, 2, 3, 4	2, 3, 4	1, 2	3, 4			1			
Spain	46,789,532	1, 2, 3, 4	2, 3, 4	1, 2	3, 4			1			
Poland	37,847,219	1, 2, 3, 4	3, 4	1, 2		2, 3, 4	1				
Romania	19,196,044	1, 2, 3, 4	1, 3, 4	1, 2		2, 3, 4	1, 3, 4				
Netherlands	17,142,323	1, 2, 3, 4	2, 3, 4	1, 2	3, 4	1					
Belgium	11,608,284	1, 2, 3, 4	2, 3, 4	1, 2, 4	3			1			
Czech Republic	10,717,516	1, 2, 3, 4	3, 4	1, 2					1, 2, 3, 4		
Greece	10,415,204	1, 2, 3, 4	1, 3, 4	1, 2		2, 3, 4	1, 3, 4	1			
Portugal	10,190,296	1, 2, 3, 4	1, 4	1, 2		2, 3, 4	1, 3	1			
Sweden	10,114,623	1, 2, 3, 4	1, 2, 3, 4	1, 2	3, 4						
Hungary	9,657,366	1, 2, 3, 4	1, 3, 4	1, 2		2, 3, 4	1, 3, 4				
Austria	9,032,162	1, 2, 3, 4	2, 3, 4	1, 2	3	1, 4					
Bulgaria	6,938,828	1, 2, 3, 4	1, 3, 4	1, 2, 3		2, 3, 4	1, 3, 4				
Denmark	5,795,666	1, 2, 3, 4	1, 2, 3, 4	1, 2	3, 4		1	1		1	
Finland	5,543,674	1, 2, 3, 4	2, 3, 4	1, 2	3, 4		1				
Slovakia	5,461,816	1, 2, 3, 4	4	1, 2		3			1, 2, 3, 4		
Norway	5,428,345	1, 2, 3, 4	1, 2, 3, 4	1, 2	3, 4						
Ireland	4,960,177	1, 2, 3, 4	1, 2, 3, 4	1, 2	3, 4						
Croatia	4,099,199	1, 2, 3, 4	1, 4	1, 2, 3		2, 3, 4	1, 4				
Lithuania	2,710,479	1, 2, 3, 4	3, 4	1, 2		2, 3, 4	1				
Slovenia	2,079,635	1, 2, 3, 4	1	1, 2		2, 3, 4	1				3, 4
Latvia	1,881,006	1, 2, 3, 4	3, 4	1, 2		2, 4	1, 3, 4				
Estonia	1,328,929	1, 2, 3, 4		1, 2		2, 3, 4	1, 3, 4				
Cyprus	1,190,962	1, 2, 3, 4	1	1, 2		2, 3, 4	1, 3, 4				
Luxembourg	629,798	1, 2, 3, 4		1, 2		3, 4		1, 2, 3, 4			
Malta	514,564	1, 2, 3, 4	1, 3, 4	1, 2		2, 3, 4	1				
Iceland	341,834	1, 2, 3, 4	1, 4	1, 2, 3		2, 3, 4	1				
Liechtenstein	38,150	1, 2, 3, 4	4	1, 2, 3		2, 3, 4		1			
Auctions won		124	76	67	25	51	32	12	8	1	2
Average pop. (M)	519.4	519.4	373.3	265.5	208.9	112.3	58.8	53.3	16.2	1.4	1.0

Table 2—Popularity and Ratings of Android Search Engine Apps

Search engine	Av. pop.	Auct. won	Installs	Rating	Reviews
info.com	519.4	124	100,000+	4.1	74
PrivacyWall	373.3	76	100,000+	4.2	378
DuckDuckGo	265.5	67	10,000,000+	4.8	1,014,301
Bing	208.9	25	10,000,000+	4.4	187,314
GMX	112.3	51	10,000+	4.5	19
Yandex	58.8	32	100,000,000+	4.5	1,042,638
Qwant	53.3	12	1,000,000+	3.8	10,968
Seznam	16.2	8	1,000,000+	4.3	67,137
Givero	1.4	1	100+	n/a	n/a
Ecosia	1.0	2	5,000,000+	4.6	125,723

choice screen auction, a "search provider must have an app that is available for free in Google Play," and if a search engine is chosen by a user from the choice screen, "[that app] will be downloaded from Play" in addition to the search engine being set as the default in the Chrome browser on the user's device (https://www.android.com/choicescreen/). Of course, users can install these apps even without the choice screen (as they do, e.g., outside of Europe); moreover, the numbers listed on Play Store count the number of installs worldwide, not just in Europe. Thus, while the numbers of downloads resulting from choice screen auctions are not publicly disclosed by Google or the European Commission, the install numbers from Play Store

listed in Table 2 provide upper bounds on those numbers.[30] These install numbers, of course, also provide a measure of the overall popularity of these search engines by including the installs that were made independently of choice screen auctions.[31]

Table 2 also includes the ratings from Android users on the quality of these apps as well as the number of user reviews. Play Store requires a minimum number of installs, ratings, and reviews before it starts disclosing this information. One of the winners of choice screen auctions did not pass this bar and, thus, does not have average ratings reported by the Play Store platform (indicated as "n/a" in the table).

Consistent with the theoretical results from Section I, there is no meaningful correlation between the quality and popularity of the search engines participating in these auctions and the average population or the number of auctions they won in them.[32] Strikingly, of the top 5 winners, each of whom has on average won a slot in choice screen auctions covering a population of more than 100 million, 3 had on the order of only 100,000 installs or less (from all sources, worldwide, over their lifetimes) and had very small numbers of user reviews (74 for info.com, 378 for PrivacyWall, and 19 for GMX—compared with, e.g., 1 million reviews for DuckDuckGo or 187,000 reviews for Bing). The two largest winners, info.com and PrivacyWall, had some of the lowest user ratings among the search engines available (4.1 and 4.2, respectively, versus, e.g., 4.8 for DuckDuckGo or 4.6 for Ecosia—the two most highly rated search engines). These two biggest winners of choice screen auctions also do not show up in the aggregate market share statistics for search engines on mobile platforms in Europe, despite winning very large populations in choice screen auctions for an entire year (519 million people in the case of info.com and 373 million in the case of PrivacyWall).[33]

Regarding the theoretical results of Section II on endogenous quality choice, I do not have direct evidence on the impact of the incentives from the auction on the actual design choices that search engines make to trade off revenue they make per each user versus their popularity and quality. The time period of only a year may also be insufficient for these design changes to manifest themselves. However, these incentives are clearly understood (and lamented) by at least some of the participants in these auctions.

The most highly rated search engine, DuckDuckGo, won a slot in every country in periods 1 and 2, but then by period 3 only won slots in 4 countries, covering in total only 2 percent of the available population, and then won a slot in only 1 country in period 4. As explained by DuckDuckGo, "Despite DuckDuckGo being robustly

[30] Note that Play Store does not disclose the exact number of installs but instead shows a range, which is what Table 2 reports.

[31] One caveat on this measure is that the search engines' popularity may vary by geography. For example, the majority of users of the Yandex search engine are from Russia and countries of the former Soviet Union.

[32] Ideal data to check this correlation would include the actual choice probabilities from choice screens displayed to the users and the actual bids submitted by the search engines to the auctioneer. Unfortunately, these data are not available; they are proprietary and confidential. Instead, I use publicly available proxies for these measures: I use the average population and the number of auctions won by these various search engines as a proxy for their average value per user and the numbers of installs and reviews as a measure of popularity.

[33] Data on aggregate market share statistics for search engines on mobile platforms in Europe are available at https://gs.statcounter.com/search-engine-market-share/mobile/europe. In February 2021, the following search engines (of those participating in choice screen auctions) had nonnegligible market shares: Yandex (1.34 percent), Bing (0.51 percent), DuckDuckGo (0.38 percent), Ecosia (0.22 percent), Seznam (0.07 percent), and Qwant (0.04 percent).

profitable since 2014, we have been priced out of this auction because we choose to not maximize our profits by exploiting our users. In practical terms, this means our commitment to privacy and a cleaner search experience translates into less money per search. This means we must bid less relative to other, profit-maximizing companies."[34]

The second most highly rated search engine is Ecosia, which uses its profits to plant trees around the world (https://info.ecosia.org/). Despite its overall popularity with users (more than 5,000,000 installs and more than 100,000 user reviews), it only won coverage in the last 2 periods and in only 1 country (covering less than 0.4 percent of overall available population). Ecosia explained its decision not to participate in the initial auction (for the first time period) as follows: "Ecosia is a not-for-profit search engine. Taking part in Google's auction would force us to spend our income on an unnecessary bidding war with other (profit-oriented) search engines. We'd rather use it to plant trees on our endangered planet."[35]

IV. Concluding Remarks

On June 8, 2021, Google announced the decision to discontinue the use of choice screen auctions, saying, "Following further feedback from the Commission, we are now making some final changes to the Choice Screen including making participation free for eligible search providers."[36,37] While Google and the European Commission did not further elaborate on the details of the Commission's feedback and reasoning, the makeup of the winners of choice screen auctions, and the fact that some virtually unknown and unpopular search engines have won most of the positions in them, likely played an important role in them.

The results of my paper, however, show that abandoning the auction is not the only possible solution. The issues described in Section III arise not from the usage of auctions per se but rather from the specific implementation of these auctions. Replacing the "per install" bidding rule in choice screen auctions with a "per appearance" one might be sufficient to alleviate the shortcomings of Google's implementation of choice screen auctions or at least meaningfully reduce their impact.

A "per appearance" auction can be implemented in practice in a number of different ways. Search engines can bid directly per appearance, specifying how much they are willing to pay every time they are shown on the choice screen and paying the highest losing bid every time they are shown. To help them make bidding decisions, the platform could provide to each search engine historical data on its "conversion probability," i.e., the likelihood that the search engine will be chosen by a user conditional on being shown on the choice screen. Alternatively, the auction could still ask bidders to report their values "per install" but then rank bidders by the product

[34] https://spreadprivacy.com/search-preference-menu-duckduckgo-elimination/

[35] https://blog.ecosia.org/google-auction-choice-screen/.

[36] https://blog.google/around-the-globe/google-europe/changes-android-choice-screen-europe/

[37] The specifics of how Google decided to list the search engines on the choice screen starting September 2021 are as follows: (i) "The five most popular eligible general search services in each country according to StatCounter (including Google) will be displayed at the top, ordered randomly each time the choice screen is shown," and (ii) "Up to seven remaining eligible general search services will be shown below the initial five services, similarly ordered randomly. In the event there are more than seven remaining general search services in a given country, the seven services to be shown on the choice screen in that country will be selected at random each time the choice screen is displayed" (https://www.android.com/choicescreen/).

of those reported values and estimated conversion probabilities and charge each winning bidder the lowest bid they could have reported and still won the auction, similarly to how many search engines auction off ad slots (Edelman, Ostrovsky, and Schwarz 2007; Varian 2007). The platform can estimate these conversion probabilities by allocating a small fraction of choice screen impressions to random selections of all eligible search engines and then using the data from this fraction of traffic to estimate the probabilities. In summary, while deploying per appearance choice screen auctions in one of the above forms would require some additional work on the part of the auctioneer, they are sufficiently similar to the existing formats in other domains that their implementation would not be too problematic.

While per appearance choice screen auctions may provide a simple, transparent, and potentially effective tool in the regulatory toolbox, it is important to note that this solution is not necessarily optimal. To judge the optimality of this (or any other) solution, one would need, at a minimum, to explicitly specify the regulator's objective function. Moreover, even holding the overall auction format fixed, there are parameters of its implementation that will have an impact on its overall effectiveness and thus need to be chosen appropriately. For example, one would need to decide how many options to show on the choice screen,[38] how often to show the choice screen to the users, and how often to conduct these auctions. And going beyond just the design of a particular choice screen auction, there is a question of how common they should be. For example, many apps from Apple and Google come pre-installed on iOS and Android platforms, respectively, and in principle, each one could have a choice screen to give a level playing field to the competitors. But having to go through a choice screen for every single one of them may lead to a poor experience for end users, substantially reducing consumer utility, which gives rise to the question of how to determine the circumstances and cases to which choice screen auctions should be applied, without overwhelming the end users. I leave the analysis of these questions to future research.

Choice screen auctions can be a powerful tool for leveling the playing field in the now widespread settings in which a dominant platform offers a product that is competing with several others. More generally, the tools of market design can be useful for constructing elegant and effective regulatory solutions for the increasingly complex and interconnected digital economy. Having said that, when deploying such solutions, it is important to analyze the equilibrium properties of the resulting systems on a detailed level. Seemingly minor details can have major effects on the outcomes. And the case of Android choice screen auctions serves as a cautionary tale: getting these details wrong can undermine the entire mechanism.

Appendix A. Model of Choice Screen Auctions with General Distributions of Bidder Types

In Sections I and II, I restrict attention to uniform distributions of bidder types. In this Appendix, I consider analogous settings with more general distributions.

[38] As discussed above, along with abandoning auctions in the choice screen solution, Google and the European Commission have also increased the number of search engines displayed in the menu. See also https://www.auction-technologies.net/images/Papers/competitive-edge-the-google-auction-that-may-turn-into-a-trap.pdf for a discussion of whether the number of slots should be increased.

A. *Exogenous Popularity and Revenue per User*

First, consider the setting of Section I, with the following change. Instead of assuming that variables q_1, q_2, r_1, and r_2 are all distributed uniformly on $[0,1]$, I now make the following set of assumptions:

- Variables q_1 and q_2 are drawn from a continuous distribution F with full support on $[0,1]$.
- Variables r_1 and r_2 are drawn from a continuous distribution G with full support on $[0,1]$.
- Variables q_1, q_2, r_1, and r_2 are independent.

Let $\bar{q} = E[F]$. Just as in the uniform case, it is immediate that in the per install auction, the expected popularity of the winner is equal to $\bar{q}$. Let us now show that in the per appearance auction, the expected popularity of the winner (which we denote q^A) is strictly greater than $\bar{q}$.

For each $q \in [0,1]$, let $W(q)$ be the probability that $qr_1 > q_2 r_2$, when q_2 is drawn from F, while r_1 and r_2 are drawn from G, independently of each other and of q_2. As observed in footnote 25 in the main body of the paper, the expected popularity of the winner of the auction is equal to the expected popularity of the winner of the auction *conditional on that winner being bidder* 1 (simply by symmetry between the bidders). That latter quantity, in turn, is equal to the ratio

$$\frac{\int_0^1 q_1 \Pr(\text{bidder 1 is the winner of the auction} \,|\, q_1) dF(q_1)}{\Pr(\text{bidder 1 is the winner of the auction})},$$

the denominator of which (again simply by symmetry) is equal to $1/2$.

We thus have the following expression for q^A:

$$q^A = 2\int_0^1 qW(q)dF(q).$$

Note that both q and $W(q)$ are nonnegative, strictly increasing functions of q and that $\int_0^1 qdF(q) = \bar{q}$ and $\int_0^1 W(q)dF(q) = 1/2$. We thus have

$$q^A = 2\int_0^1 qW(q)dF(q) > 2\left[\int_0^1 qdF(q)\right]\left[\int_0^1 W(q)dF(q)\right] = 2 \cdot \bar{q} \cdot \frac{1}{2} = \bar{q},$$

concluding the proof.

Next, let us show that in the per appearance auction, the expected payoff of the platform is strictly lower than that in the per install auction. Consider two draws q_1 and q_2 from distribution F and two draws r_α and r_β from distribution G. Let $\Pi^A(q_1, q_2, r_\alpha, r_\beta)$ be the payoff to the platform from the per appearance auction when bidder 1's popularity is equal to q_1 and her revenue per user is equal to r_α, while bidder 2's popularity is equal to q_2 and his revenue per user is equal to r_β.[39] Define $\Pi^I(q_1, q_2, r_\alpha, r_\beta)$ analogously for the per install auction. We need to show that

$$E_{q_1,q_2,r_\alpha,r_\beta}\left[\Pi^I(q_1, q_2, r_\alpha, r_\beta) - \Pi^A(q_1, q_2, r_\alpha, r_\beta)\right] > 0. \tag{A1}$$

[39] For concreteness, whenever there are ties, one can assume that they are broken randomly with equal probabilities (though the specific tie-breaking rule will not matter since ties happen with probability zero).

Note that by symmetry,

$$E_{q_1,q_2,r_\alpha,r_\beta}\left[\Pi^I(q_1,q_2,r_\alpha,r_\beta) - \Pi^A(q_1,q_2,r_\alpha,r_\beta)\right]$$
$$= E_{q_1,q_2,r_\alpha,r_\beta}\left[\Pi^I(q_1,q_2,r_\beta,r_\alpha) - \Pi^A(q_1,q_2,r_\beta,r_\alpha)\right],$$

where we swapped r_α and r_β under Π^I and Π^A, and so equation (A1) is equivalent to

$$\text{(A2)}\qquad E_{q_1,q_2,r_\alpha,r_\beta}\left[\Pi^I(q_1,q_2,r_\alpha,r_\beta) + \Pi^I(q_1,q_2,r_\beta,r_\alpha) - \Pi^A(q_1,q_2,r_\alpha,r_\beta) - \Pi^A(q_1,q_2,r_\beta,r_\alpha)\right] > 0.$$

To prove equation (A2), it is sufficient to show that for any q_1, q_2, r_α, and r_β such that $r_\alpha \neq r_\beta$, $q_1 r_\alpha \neq q_2 r_\beta$, and $q_2 r_\alpha \neq q_1 r_\beta$ (so that ties do not occur),

$$\text{(A3)}\qquad \Pi^I(q_1,q_2,r_\alpha,r_\beta) + \Pi^I(q_1,q_2,r_\beta,r_\alpha) - \Pi^A(q_1,q_2,r_\alpha,r_\beta) - \Pi^A(q_1,q_2,r_\beta,r_\alpha) \geq 0,$$

with the inequality being strict for a positive mass of tuples $(q_1,q_2,r_\alpha,r_\beta)$.

Without loss of generality, assume that $r_\alpha > r_\beta$, and recall that we denote by $\pi > 1$ the payoff of the platform when its own search engine is chosen from the choice screen. Then $\Pi^I(q_1,q_2,r_\alpha,r_\beta) = (1-q_1)\pi + q_1 r_\beta$ and $\Pi^I(q_1,q_2,r_\beta,r_\alpha) = (1-q_2)\pi + q_2 r_\beta$. Next, without loss of generality, suppose that $q_1 \geq q_2$. Then $q_1 r_\alpha > q_2 r_\beta$, and $\Pi^A(q_1,q_2,r_\alpha,r_\beta) = (1-q_1)\pi + q_2 r_\beta$. Finally, we need to consider two cases: $q_2 r_\alpha > q_1 r_\beta$ and $q_2 r_\alpha < q_1 r_\beta$.

Case 1: $q_2 r_\alpha > q_1 r_\beta$. In this case, $\Pi^A(q_1,q_2,r_\beta,r_\alpha) = (1-q_2)\pi + q_1 r_\beta$, and so $\Pi^I(q_1,q_2,r_\alpha,r_\beta) + \Pi^I(q_1,q_2,r_\beta,r_\alpha) - \Pi^A(q_1,q_2,r_\alpha,r_\beta) - \Pi^A(q_1,q_2,r_\beta,r_\alpha) = (1-q_1)\pi + q_1 r_\beta + (1-q_2)\pi + q_2 r_\beta - (1-q_1)\pi - q_2 r_\beta - (1-q_2)\pi - q_1 r_\beta = 0$.

Case 2: $q_2 r_\alpha < q_1 r_\beta$. In this case, $\Pi^A(q_1,q_2,r_\beta,r_\alpha) = (1-q_1)\pi + q_2 r_\alpha$, and so $\Pi^I(q_1,q_2,r_\alpha,r_\beta) + \Pi^I(q_1,q_2,r_\beta,r_\alpha) - \Pi^A(q_1,q_2,r_\alpha,r_\beta) - \Pi^A(q_1,q_2,r_\beta,r_\alpha) = (1-q_1)\pi + q_1 r_\beta + (1-q_2)\pi + q_2 r_\beta - (1-q_1)\pi - q_2 r_\beta - (1-q_1)\pi - q_2 r_\alpha = (q_1-q_2)\pi + (q_1 r_\beta - q_2 r_\alpha)$. By our assumptions, $(q_1-q_2)\pi \geq 0$ and $q_1 r_\beta - q_2 r_\alpha > 0$, and so in this case, the expression in equation (A3) is strictly positive.

Note that Case 2 happens with positive probability, and thus, the expression in equation (A2) is also strictly positive, concluding the proof.

For the limit case of the number of bidders n approaching infinity (last paragraph of Section I), the conclusions and the arguments are identical to those for the uniform case because for those arguments the only characteristics of the distributions that matter are the upper bounds of their supports.

Finally, the proof above also shows that for the case with two bidders, the search engine's expected payoff from the per install auction exceeds that from the per appearance auction even if its own payoff π from getting a user is less than one, as long as $\pi \geq 0$. For the limit case of large n, the comparison does rely on the assumption $\pi > 1$.

B. *Endogenous Popularity and Revenue per User*

I now consider an extension of the model in Section II, in which instead of assuming that distribution F of types t is uniform on $[0,1]$, I make the following set of assumptions:

- Types t_i are drawn independently from distribution $F(t)$ with full support on $[0,1]$.
- Distribution $F(t)$ has a continuous positive density function $f(t)$ on $[0,1]$.

The analysis of the per appearance auction (Section IIA) remains unchanged.

The analysis of the per install auction also remains unchanged up to the differential equation (4). Thus, in particular, as in Section IIB, we can immediately conclude that if the per install auction has a symmetric equilibrium characterized by some function $q(t)$, the equilibrium popularity chosen by each type $t_i > 0$ will be strictly lower under the per install auction than under the per appearance auction.

Next, dropping index i in equation (4) for convenience, we have

$$2q'(t) + (n-1)q(t)\frac{f(t)}{F(t)} = 1.$$

To solve this differential equation, first multiply both sides by $F(t)^{\frac{n-1}{2}}/2$ to get

$$q'(t)F(t)^{\frac{n-1}{2}} + \frac{n-1}{2}q(t)f(t)F(t)^{\frac{n-3}{2}} = \frac{F(t)^{\frac{n-1}{2}}}{2}.$$

This equation, in turn, can be rewritten as

$$\left[q(t)F(t)^{\frac{n-1}{2}}\right]' = \frac{F(t)^{\frac{n-1}{2}}}{2},$$

and thus,

$$q(t)F(t)^{\frac{n-1}{2}} = \int_0^t \frac{F(x)^{\frac{n-1}{2}}}{2}dx + C$$

for some constant C.

Since $q(0) = 0$ and $F(0) = 0$, it has to be that $C = 0$, and so

$$q(t) = \frac{\int_0^t F(x)^{\frac{n-1}{2}}dx}{2F(t)^{\frac{n-1}{2}}}.$$

For example, for the case of the uniform distribution considered in Section II, we get $q(t) = \left(\frac{2}{n+1}t^{\frac{n+1}{2}}\right)/\left(2t^{\frac{n-1}{2}}\right) = \frac{t}{n+1}$. Note also that the search engine's bid, equal to its revenue per user, is increasing in its type:

$$r(t) = t - q(t) = t - \frac{\int_0^t F(x)^{\frac{n-1}{2}}dx}{2F(t)^{\frac{n-1}{2}}},$$

and so

$$r'(t) = 1 - \frac{2F(t)^{n-1} - (n-1)f(t)F(t)^{\frac{n-3}{2}}\int_0^t F(x)^{\frac{n-1}{2}}dx}{4F(t)^{n-1}} \geq \frac{1}{2} > 0.$$

Finally, since $F(x)$ is a strictly increasing function, for any $t > 0$ (including the highest possible type $t = 1$), we have $\lim_{n\to\infty}\frac{\int_0^t F(x)^{\frac{n-1}{2}}dx}{2F(t)^{\frac{n-1}{2}}} = 0$, and so just like in the uniform case, as the number of bidders becomes large, the popularity of the winner of the per install auction converges to zero.

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4. 随机维修员路径：The k-Traveling Repairman Problem with Stochastic Service Request Times (TS, 2026)

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关键词：旅行维修员问题；随机优化；先验路径；情景采样；分支定价

推荐人：张芳菲

推荐理由：旅行维修员问题以降低顾客提交服务请求后的等待损失为目标。本文在经典模型中加入未来请求时刻的概率信息，并采用更灵活的等待不效用度量。作者设计先验路径及执行时的补救规则，以兼顾已知与未来请求，最小化总预期等待损失。问题被表述为基于路径的随机优化模型，通过分支定价算法求解，并利用采样情景估计期望等待。论文通过析因实验比较不同路径策略，并将方法用于大规模真实共享单车场景。

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The *k*-Traveling Repairman Problem with Stochastic Service Request Times

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Abstract. The traveling repairman problem focuses on minimizing total latency, defined as the waiting time experienced by customers after submitting a service request. This problem naturally arises in settings where customer experience is a key objective, particularly in time-sensitive service operations. We address a generalized traveling repairman problem that extends the classical setting by incorporating available probabilistic information on the timing of future requests as well as a more flexible method for measuring the impact of service latency. To account for potential future requests in addition to those known at the time of route planning, we develop a priori routes combined with recourse rules for their execution, aiming to minimize the total expected disutility caused by service latency across all customers. We formulate the problem as a stochastic, path-based traveling repairman problem and solve it using a branch-and-price algorithm, where expected latency is estimated through sample-scenario planning. We then evaluate the performance of our a priori routing approach through a factorial experiment; compare it with alternative routing strategies; and apply these methods to a large-scale, real-world bike-sharing setting.

Supplemental Material: The online appendix is available at https://doi.org/10.1287/trsc.2023.0478.

Keywords: traveling repairman problem • stochastic optimization • a priori optimization • sample-scenario planning

1. Introduction

In service operations management, effective allocation and routing of technicians to time-sensitive service requests are crucial for maintaining high-quality service delivery. Minimizing delays is essential, whether repairing household appliances (Pham and Klesmüller 2024) or performing maintenance on public infrastructure (Muritiba et al. 2021). Prolonged wait times between a service request and its fulfillment can lead to significant damages or profit losses (Ahmadi et al. 2025). Service operations are often plagued by uncertainty, particularly when it comes to the stochastic nature of service request times and locations (Bertsimas and Van Ryzin 1991). However, the abundance of historical data or real-time data made available through advances in sensor technologies and the Internet of Things coupled with virtual replicas of physical systems (often referred to as digital twins) allow for accurate predictions of critical events, such as service requests and maintenance times, in time-sensitive industries (Bhowmik 2019, Zhao et al. 2021, Mihai et al. 2022, Priyanka et al. 2022, Yan, Wang, and Wu 2022). This paper proposes a solution method to utilize this information and address the challenging interplay of variability and optimization in technician routing problems.

We explore the complex landscape of technician routing in the face of uncertain service request times, focusing on methodologies and practical implications of managing such stochastic dynamics. Specifically, we investigate the problem of designing a set of a priori routes to service a combination of *deterministic* customers with *known* service requests at the beginning of an operational period and *stochastic* customers with *potential* service requests at random times during the operational period. We assume that the utility of a customer following a service request declines with time according to a customer-specific utility decay (disutility) function. The objective of our problem concerns the minimization of expected disutility for customers who request service *before* and *during* a target operational period. There are different reasons why a priori routes that include both known and potential requests may be preferred over strategies that rely on route reoptimization (for example, when a stochastic customer's request is revealed) or a simple insertion of new orders to existing routes. Although the former strategy may require high computational efforts during the operational period, causing potential delays in subsequent services, the latter can result in arbitrarily poor performances. We investigate this trade-off in computational experiments.

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We believe that our approach is broadly applicable to a range of problems, particularly in the context of technician routing for corrective maintenance services. We summarize the salient characteristics of this class of problems as follows. (1) The planner manages a set of customers with known or approximately known locations who may request service at various times during the operational period. (2) The planner has access to information or beliefs that can be used to predict future requests and seeks to leverage this knowledge to design a priori routes—serving both customers with current requests and those likely to request service during the targeted operational period. (3) The planner's objective is to minimize a function of the expected customer latency defined as the time between the placement of a request and the start of service. Some applications that motivate our study are the bike-share system of a major metropolitan city and the water and sewage system of a city.

One application is the bike-share system of a major metropolitan city. These systems allow users to rent bikes without worrying about maintenance or security between rides. Bikes are docked at stations throughout the city, and customers can pick up and return bikes at any station depending on dock availability and their destination. The operator maintains the bikes, docks, and stations, generating revenue through subscriptions or pay-per-minute rentals. Inoperable bikes and infrastructure lead to lost revenue. Using historical data, the operator can form beliefs about which stations (or clusters of stations) may require service during the operational period beyond those with reported issues. The operator may use these beliefs to plan technician routes that minimize the total expected downtime across bikes, stations, and docks. Intuitively, the goal is to schedule technician activities along a priori routes that both address known issues and position technicians advantageously in time and space to respond to potential service requests.

Another application is the water and sewage system of a city. In a sewage "lift station," a series of pumps directs wastewater flow to treatment plants. Because of weather events, wear and tear, or foreign objects entering the system, these pumps may become faulty and require service. The scheduler forms a belief based on sensor data triggered by float switches of a potential pump failure. A failed pump can cause sewage to leak into nearby streams, making the time between failure and repair critical to maintain service and protect public health. To mitigate this risk, stations are often equipped with multiple pumps, allowing continued—but reduced—operation when one fails. Thus, restoring full capacity quickly remains essential. The system scheduler aims to address issues as soon as possible after they are observed. Total failure of a station is rare; even in the Mobile, Alabama area—a low-lying, rain-prone zone—only 17 such events were recorded across 450+ stations in 12 months, with just 5 because of pump issues (MAWSS 2025). Such severe failures require emergency dispatch and therefore, are handled separately from a priori scheduled routes.

In this paper, we develop a versatile solution framework for a priori routing in settings where a planner must schedule a mix of known and potential service requests. Our approach accommodates beliefs about potential request times given either as probability distributions or as discretized sampled scenarios derived from historical data. We develop novel recourse rules (strategic idling of service agents) and novel dominance conditions in our route generation procedure. We demonstrate the effectiveness of our approach through a series of full-factorial experiments using adapted benchmark instances from the literature and a case study of the New York City bike-sharing system. We compare the performance of our approach with the performance of several other approaches. We show the superiority of our solutions against methods that use less distributional information, use different recourse rules, or utilize reoptimization as the routes are operated (not a priori). Finally, we discuss promising future research directions that may extend this work.

1.1. Literature Review

Routing problems with stochastic service requests underlie many operational challenges in logistics and supply chain management Psaraftis et al. (2015). These challenges are characterized by the need to design delivery routes for uncapacitated vehicles to meet customer service requests arriving randomly over a given geographical area and time horizon. For example, package express firms (e.g., local couriers and United Parcel Service) often begin a working day with a set of known service requests and may dynamically adjust vehicle routes to accommodate additional calls arriving throughout the day (Hvattum, Løkketangen, and Laporte 2006). Similarly, service technicians (e.g., roadside assistance and utilities employees) may periodically adjust preliminary schedules to accommodate requests placed after the start of daily business (Jaillet 1988). Likewise, less-than-truckload logistics can be approximated via the routing of an uncapacitated vehicle (Thomas 2007). In each of these examples, past customer transaction data can be used to derive probability distributions on the timing and location of potential customer requests, thus opening the door to the possibility of adjusting vehicle routes in anticipation of future requests.

The deterministic version of our problem is known as the traveling repairman problem (TRP), in which the latency of a customer on a route is measured against the starting time of all routes (all service requests are known before the repairmen start their routes). The TRP is also known as the minimum latency problem,

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the deliveryman problem, and the traveling salesman problem with cumulative costs (Muritiba et al. 2021). The TRP focuses on minimizing the tour's total waiting time for all service clients. As a result, the TRP is more "customer centric" than the related TSP (Ngueveu, Prins, and Calvo 2010).

In the machine scheduling context, the TRP was first introduced in the literature by Conway, Maxwell, and Miller (1967), where the total completion time of jobs processing on a single machine is minimized. Picard and Queyranne (1978) and Domschke, Scholl, and Voβ (1997) also studied different versions of the TRP in a scheduling context. Sahni and Gonzalez (1976) established the TRP as NP-hard for general metrics. Research conducted to develop approximations includes Fakcharoenphol, Harrelson, and Rao (2007), Archer, Levin, and Williamson (2008), and Nagarajan and Ravi (2008). Recently, the integer programming formulations in Fischetti, Laporte, and Martello (1993), van Eijl (1995), Bigras, Gamache, and Savard (2008), Méndez-Díaz, Zabala, and Lucena (2008), and Godinho, Gouveia, and Pesneau (2014) present enumerative solution strategies in the TRP domain.

Different applications of the TRP can be found in disk-head scheduling (Blum et al. 1994), in the routing of automated guided vehicles through n cells in a flexible manufacturing system (Simchi-Levi and Berman 1991), in finding information stored in computer networks (Ausiello, Leonardi, and Marchetti-Spaccamela 2000), in minimizing the message latency in binary n cubes with virtual channels (Sarbazi-Azad 2003), in minimizing passenger waiting times at transitional nodes for public bus transportation (Saharidis, Dimitropoulos, and Skordilis 2014), and in the high-level synthesis of integrated circuits (Aras and Yurdakul 2016). Fischetti, Laporte, and Martello (1993) studied a pizza delivery TRP, minimizing the average arrival time to customers by clustering different orders.

In this paper, we study the problem of assigning k service agents to routes known as the kTRP. The kTRP is a special case of the weighted k traveling repairman problem (wkTRP) (Muritiba et al. 2021) in that equal weights represent the urgency of service for each client; however, in this work, we generalize the idea of weights to a *disutility* function of service time latency. Therefore, the wkTRP is a special case of the problem focus in this paper, which we call the generalized traveling repairman problem (gTRP). Our work is closely related to Luo, Qin, and Lim (2014), Bulhões, Sadykov, and Uchoa (2018), and Muritiba et al. (2021). In Luo, Qin, and Lim (2014), a kTRP problem with distance constraints is considered, and a branch-price-and-cut algorithm is tailored and implemented. A novel bounded bidirectional label-setting algorithm is proposed for the pricing subproblem. Bulhões, Sadykov, and Uchoa (2018) use a branch-and-price algorithm to solve the TRP. They show that the subproblem of the column generation procedure corresponds to an elementary resource-constrained shortest-path problem (ERCSPP), usually solved by a dynamic programming-based labeling algorithm. However, ERCSPP is NP hard (Dror 1994). For that reason, several path relaxation approaches have been proposed. k-cycle relaxation (Irnich and Villeneuve 2006) reduces the ERCSPP to a k-cycle resource-constrained shortest-path problem (RCSPP), in which cycles with length less than k are not allowed, and RCSPP is solvable in pseudopolynomial time. Baldacci, Mingozzi, and Roberti (2011) introduced ng-path relaxation in the context of the vehicle routing problem (VRP), and it is implemented in the TRP setting by Roberti and Mingozzi (2014). Nucamendi-Guillén et al. (2016) introduces a branch-price-and-cut approach and an effective metaheuristic. Bulhões, Sadykov, and Uchoa (2018) created enhanced dominance rules and generalized ng sets while decreasing the ng-set sizes. Finally, Muritiba et al. (2021) consider a deterministic and static wkTRP of a speed camera network. Their work assumes that all cameras to be serviced are known in advance and will be serviced during the planning horizon. k routes are constructed to minimize the total weighted latency of service. The authors introduce an integer programming model formulation and then, solve the problem with branch-and-cut and iterated local search approaches. Although Muritiba et al. (2021) showed that maximum route duration limits are never "tight" in their results, we utilize the maximum duration constraint to construct the maximum "wait-to" time for a service agent to wait for a customer to request service.

Bertsimas and Van Ryzin (1991) introduce a stochastic variant of the TRP called the dynamic traveling repairman problem, in which the probability of a customer requiring service is known before the operational period. They draw from queueing theory and focus on the performance of varying routing policies, including using first come, first serve; forming traveling salesman problem (TSP) tours; and using space-filling curves and a nearest neighbor approach. As stated in Bertsimas and Van Ryzin (1991), their stochastic variant of the TRP is related to so-called a priori optimization introduced in Jaillet (1988) and Bertsimas, Jaillet, and Odoni (1990). A priori routing methods consider n known customers, of which only a random subset must be visited in each operational period. The a priori route plans visits to all n customers in a specified order, skipping customers who do not require a visit. The optimal solution minimizes the expected cost over all possible (2^n) subsets given the probabilities of each stochastic customer's visit. Recently, van Ee and Sitters (2018) and Navidi, Gortz, and Nagarajan (2020) investigate the a priori TRP. van Ee and Sitters (2018) provides a constant-factor approximation in the uniform model of

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the problem. Navidi, Gortz, and Nagarajan (2020) achieve a constant-factor approximation under the nonuniform model (the probability of each stochastic customer's visit requirement is independent of other customers).

The motivation of the above a priori TRP work and our work is to provide solutions for each operational period to reduce or eliminate reoptimization during the operation of the routes. Under our considered problem setting, at the beginning of the operational period, customers are either deterministic or stochastic request. Deterministic request customers are visited if scheduled. As in Bertsimas and Van Ryzin (1991), stochastic customers may or may not request service during the operational period. In contrast to Bertsimas and Van Ryzin (1991), in our problem, a visit to stochastic customer c is dependent on the relative time at which customer c requests service relative to the position of customer c in the a priori route and the realizations of previous stochastic customers on the route. Our approach is to generate a priori routes to schedule potential visits to all customers and provide recourse rules in the case that a stochastic customer is not ready for or does not require a service when its turn has arrived. Specifically, we propose that the service agent might wait for the stochastic customer to request service for a period of time that maintains feasibility of the a priori route. For example, if sufficient evidence suggests that a nearby customer may request service soon, it may be beneficial for a technician to wait rather than move immediately. Although waiting may not be optimal in every setting and the acceptable amount of waiting may vary, our framework is flexible and does not assume a fixed limit on allowable waiting. However, we do propose and implement one specific waiting policy as part of our evaluation. The proposed waiting period alters the expectations of outcomes for customers on an a priori route (i.e., if and when a customer will be visited), and we show that it can reduce the total expected disutility of all customers. In summary, we consider the distribution of service request times and a recourse rule that depends on the time at which a customer can be serviced. Therefore, in contrast to previous work, the probability of a customer visit varies with the schedule sequence chosen.

1.2. Our Contributions

The contribution of this paper can be summarized as follows.

- We introduce and formulate a generalized version of the traveling repairman problem with stochastic service request times and a generalized disutility function of latency. We formulate the problem as a set-partitioning problem, which relies on the expected disutility of a priori routes. We are not aware of any other work that considers route-dependent probabilities of customer visits.
- We present an approach that uses all known distributional belief information on the service request times of customers to calculate the expected disutility of the scheduled customers. We mathematically characterize the expected disutility of customers and the relationship of a specific recourse rule (waiting for a customer to request service until a dynamic, route-dependent time) to this calculation. Further, this paper is the first to our knowledge to present a generalized objective (disutility) that is a function of the latency of each customer.
- We develop a solution approach that relies on the framework of branch and price, where the pricing problem (PP) takes the form of an elementary shortest-path problem with stochastic service request times. We introduce an extension of the dynamic programming-based labeling algorithm coupled with a sample-scenario planning (SPP)) approach to solve the shortest-path problem and to estimate the expected disutility of each partial path. We develop forward- and backward-labeling algorithms with appropriate dominance rules designed for the stochastic pricing problem.
- We show using adopted benchmarks from the literature on TRP and a case study inspired by real-world experiments that our approach generates routes with lower expected disutility than approaches that use less information, and we quantify the trade-off between solution time and quality.

1.3. Paper Organization

The remainder of the paper is organized as follows. In Section 2, we formally describe the gTRP. We introduce our mathematical formulation for the gTRP in Section 3, and in Section 4, we introduce a branch-and-price solution approach using all distributional information along with a description of our labeling algorithms. Section 5 describes alternative approaches to generating a priori routes. In Section 6, we present a computational experiment that compares solution approaches on the basis of competitive ratios. Finally, in Section 7, we summarize the contributions of the paper and the insights gained.

2. Generalized Traveling Repairman Problem

We consider the problem of a service provider providing on-demand services to n customers in $\bar{\mathcal{C}} = \{1, \ldots, n\}$ over a time horizon $\mathcal{H}$ from a single depot 0. The time horizon consists of a sequence of equal-length operational periods $h \in \mathcal{H}$, each with a maximum time span T (e.g., business hours during a day). In the remainder of this paper, we focus on the decision making associated with a single operational period h, and for the sake of simplicity of notation, we do not label every notation with an indicator h. Consider a graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, where

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$\mathcal{N} = \bar{\mathcal{C}} \cup \{0, 0'\}$ is the set of vertices with 0 and 0′ being the start and end (possibly coinciding) depots and $\mathcal{E}$ is the set of edges that connect vertices in $\mathcal{N}$. Associated with each edge $(i,j) \in \mathcal{E}$, there is a travel time $t_{i,j}$ that is proportional to the length of the edge.

At the beginning of a given operational period, a subset of *deterministic customers* denoted by $\mathcal{D} \subseteq \bar{\mathcal{C}}$ is known to need service. Given the on-demand nature of the service, we also consider a second subset of *stochastic customers* denoted by $\mathcal{S} \subseteq \bar{\mathcal{C}} \setminus \mathcal{D}$ who have not submitted a service request before the beginning of operational period h. The service provider has sufficient evidence to believe that such customers *might* request service during the operational period h. For all customers who have not already requested service, the service request time follows a probability distribution on the domain of times during and after the operational period h. Specifically, we consider a probability density function $f_c(\cdot)$ of service request times τ_c for each customer $c \in \mathcal{S}$. Further, each customer $c \in \mathcal{S}$ has cumulative service request distribution $F_c(\cdot)$.

Following a service request by customer c, the customer's utility begins to decline—that is, disutility increases—until a technician arrives to provide the service. Technicians follow routes, each comprising a sequence of customer visits. Accordingly, the time at which service is rendered depends on the route to which the customer is assigned. We then define a customer's *latency* as the time elapsed between the later of the start of the day and the customer's request time, and the time the service is rendered. Let $l_{c,r}$ denote the latency experienced by customer c when served on technician route r. The corresponding *disutility* is given by $g_c(l_{c,r})$, where $g_c(\cdot)$ is a (strictly) increasing function referred to as the *customer utility decay function*.

At or just before the beginning of an operational period h, the planner determines a set of service routes for a given fleet of technicians to serve a subset of customers from the set $\mathcal{C} = \mathcal{D} \cup \mathcal{S}$. Note that $\mathcal{C}$ does not necessarily equal $\bar{\mathcal{C}}$ because not all customers $\bar{\mathcal{C}} \setminus \mathcal{D}$ need be considered in the route planning (i.e., the operator does not need to consider stochastic request customers who have very little chance of requesting service). The objective is to minimize the total expected customer disutility across all served customers.

In our problem, we consider a finite set of k homogeneous agents who can service the chosen routes during the operational period h. We propose that service agents operate under the following policy; along a route, a deterministic customer is served immediately when their turn arrives. In contrast, for a stochastic customer, the agent may wait for the customer to request service as long as the service can begin on or before a predetermined, route-dependent time referred to as the *wait-to time*. Of course, if the stochastic request customer has requested service when their turn arrives, they are departed for immediately. The rule that determines the amount of waiting time (including the option of a no-waiting policy) (see Section 6) is specified by the user. If no request has been received from a stochastic customer by its wait-to time, that customer is skipped.

If a customer $c \in \mathcal{C}$ is not scheduled, we assume that a penalty γ_c is incurred. This penalty aims to approximate the expected disutility experienced by the customer under the assumption that their request will be served at some point in the future—potentially as early as the next operational period. In our implementation, we define ξ_c as the earliest (possibly optimistic) time at which customer c can be served during operational period $h+1$. The penalty γ_c is then defined as the expected latency-induced disutility experienced by customer c, assuming service at time ξ_c. For deterministic customers, latency is measured from the beginning of period h, whereas for stochastic customers, the expected latency is measured from their expected request time $\mathbb{E}[\tau_c]$ based on the distribution f_c.

3. A Path-Based Formulation

Let $\mathcal{R}$ be the set of routes that a service agent can take. Let the binary variable $y_r = 1$ iff route r is selected and zero otherwise. Let the binary variable $\beta_c = 1$ iff customer c is left unscheduled and zero otherwise. Let binary parameter $\alpha_{rc} := 1$ iff customer c is served along route r and zero otherwise:

$$\min \quad \sum_{r \in \mathcal{R}} \tilde{u}_r y_r + \sum_{c \in \mathcal{C}} \gamma_c \beta_c \tag{1a}$$

$$\text{s.t.} \quad \sum_{r \in \mathcal{R}} \alpha_{rc} y_r + \beta_c = 1 \qquad \forall c \in \mathcal{C}, \tag{1b}$$

$$\sum_{r \in \mathcal{R}} y_r \le k, \tag{1c}$$

$$y_r \in \{0,1\} \qquad \forall r \in \mathcal{R}, \tag{1d}$$

$$\beta_c \in \{0,1\} \qquad \forall c \in \mathcal{C}. \tag{1e}$$

In Model (1), the objective function (1a) minimizes the total expected customer disutility calculated as the sum of expected latency-induced disutilities, where $\tilde{u}_r := \sum_{c \in r} g_c(l_{c,r})$ and γ_c denotes the penalty for not scheduling customer c. Constraints (1b) ensure that each customer is either scheduled or not, whereas Constraint (1c) sets the cap on the number of available service agents, k. Constraints (1d) and (1e) give the type and domain of the decision variables, respectively.

3.1. Wait-to Time

If a service request from a stochastic customer is received after that customer has been skipped, the service will need to be rescheduled for a later operational period, which may result in significant delays. We

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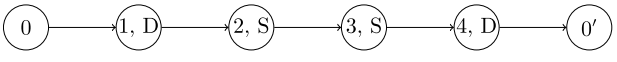
hypothesize that incorporating strategically timed waiting before visiting stochastic customers in the a priori route could reduce total expected customer disutility by allowing more time for a stochastic customer to submit a service request and therefore, increasing the likelihood of addressing the request within the current route. In the remainder of the paper, we examine the planning requirements for integrating such waiting times, and in Section 6.3.2, we quantify the potential benefits of this approach by evaluating its impact on total expected customer disutility.

Given a waiting policy, the latest start of service at a stochastic customer is referred to as the wait-to time. Different strategies to set the wait-to times can be proposed. In this paper, we propose a waiting rule that uses the *maximum wait-to time*, $\overline{\eta}_c$, which corresponds to the latest possible time for the start of the service at a stochastic customer c while keeping the route time feasible. Consider a route $r = [0 - r(1) - r(2) - \cdots - r(i) - \cdots - r(|r|-1) - r(|r|) - 0']$ starting and ending at the depot, where $r(i)$ is a pointer to the ith customer of r. The maximum wait-to time, $\overline{\eta}_{r(i)}$, is determined such that the time feasibility of the rest of the a priori route r is maintained (i.e., $\overline{\eta}_{r(i)} := T - \sum_{j=i}^{|r|-1} t_{r(j),r(j+1)} - t_{r(|r|),0'}$).

3.2. Route Realizations

For each stochastic customer on route r, there are at least two possible visit realizations: one where the customer is visited and one where they are skipped. In general, for a route with n_r^s stochastic customers, there are $2^{n_r^s}$ potential visit realizations corresponding to the various combinations of visiting or skipping each stochastic customer. Let $\mathcal{V}_r$ be the set of all possible visit realizations of a priori route r. For example, given the route $r = [0-1-2-3-4-0']$ in Figure 1, where customers 1 and 4 are deterministic request and customers 2 and 3 are stochastic request, four visit realizations $v \in \mathcal{V}_r$ exist: $[0-1-2-3-4-0']$, $[0-1-\ \ -\ \ -4-0']$, $[0-1-2-\ \ -4-0']$, and $[0-1-\ \ -3-4-0']$, where a circled customer is one who is scheduled in the a priori route but is skipped in a given visit realization. Note that each visit realization $v \in \mathcal{V}_r$ of a priori route r is associated with multiple *request realizations* concerning the service request and completion times of customers on the a priori route. Let $\mathcal{Q}_v$ be the set of request realizations associated with visit realization v. Suppose $v(i)$ is a pointer to the ith customer on visit realization $v \in \mathcal{V}_r$. Let $\tau_{v(i)}$ be the time that customer $v(i)$ requests a service, $\theta_{v(i)}$ be the service agent release time of customer $v(i)$, and $\eta_{v(i)}$ be the wait-to time set by the decision maker for customer $v(i)$ on visit realization v. Note that because of the stochastic nature of a service request time by a stochastic customer, the completion time of service for any other customer scheduled after them on an a priori route would be stochastic too.

Figure 1. Zero Indicates Depot, S Indicates Stochastic Customer, and D Indicates Deterministic Customer



The wait-to times of stochastic customers on a route play an important role in the calculation of the expected latency of that route. Let $\kappa_v(v(i))$ be a function that returns the last visited customer, immediately preceding $v(i)$, according to visit realization v. Let $\phi_v(v(i)) := \eta_{v(i)} - t_{\kappa_v(v(i)),v(i)}$ be a function that returns the latest time that the service agent can depart from customer $\kappa_v(v(i))$ such that the start of service at customer $v(i)$ is no later than its wait-to time, $\eta_{v(i)}$. In that sense, $\phi_v(v(i))$ represents the latest time that customer c can request a service and still be visited along the a priori route that it was scheduled on. Therefore, the probability of customer $v(i)$ being serviced is $F_{v(i)}(\phi_v(v(i)))$. Likewise, the probability that customer $v(i)$ is skipped and not serviced is $1 - F_{v(i)}(\phi_v(v(i)))$.

Associated with each stochastic customer $v(i)$, two request realizations are possible with respect to $\tau_{v(i)}$. (1) If customer $v(i)$ requests service before the release of the service agent from customer $\kappa_v(v(i))$ (i.e., $\tau_{v(i)} \le \theta_{\kappa_v(v(i))}$), then $v(i)$ is visited at time $\theta_{\kappa_v(v(i))} + t_{\kappa_v(v(i)),v(i)}$, and its latency is $\theta_{\kappa_v(v(i))} + t_{\kappa_v(v(i)),v(i)} - \tau_{v(i)}$. (2) If customer $v(i)$ requests service after time $\theta_{\kappa_v(v(i))}$ and before $\eta_{v(i)} - t_{\kappa_v(v(i)),v(i)}$, the latest departure time possible from $\kappa_v(v(i))$ such that the arrival time to $\eta_{v(i)}$ is on or before its wait-to time (i.e., $\theta_{\kappa_v(v(i))} < \tau_{v(i)} \le \phi_v(v(i))$), then $v(i)$ is visited at time $\tau_{v(i)} + t_{\kappa_v(v(i)),v(i)}$ (the technician stays at $\kappa_v(v(i))$ until a request from $v(i)$ is received before heading there), and its latency is $t_{\kappa_v(v(i)),v(i)}$. Note that a service request time $\tau_{v(i)}$ after $\phi_v(v(i))$ would result in skipping the customer, and therefore, it would not fall under the visit realization v.

The analytical calculations of the expected latency of different request realizations of the route described above are provided in Online Appendix EC.1. In the following section, we discuss our approach to solving the gTRP as formulated in Model (1) considering the disutility of all possible routes.

4. Solution Approach (SSP)

In this section, we introduce our algorithm for solving the path-based integer Model (1). The large number of paths in $\mathcal{R}$ makes it impractical to solve the model directly using an mixed-integer programming (MIP) solver. Therefore, the usual solution method involves a branch-and-price algorithm. This algorithm is based on a branch-and-bound approach, where lower bounds are computed using column generation (for a complete survey of column generation methods, see Lübbecke and Desrosiers 2005). Column generation is often

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successful when the associated integer programs are set-partitioning (or set-covering) problems. At each iteration of the column generation, a restricted linear master problem (RLMP) is solved, where the RLMP contains a subset of feasible columns already priced out by the pricing problem, which are in compliance with the branching decisions. The search tree starts at the root node and is initialized with enough columns to obtain a feasible solution. At each node, if the solution is integer, it is a valid solution to the original master problem, and so, it is compared with the incumbent solution. Otherwise, branching occurs to eliminate the current fractional solution. A valid branching scheme balances the search tree, eliminates the current fractional solution, and maintains the problem's structure. The optimal solution for the original problem is the best integer solution found at the end of the search process.

4.1. Restricted Linear Master Problem

In Model (1), binary variable y_r takes on the value of one if feasible route r is selected and zero otherwise. Consider a subset of all possible routes $\bar{\mathcal{R}} \subseteq \mathcal{R}$. The RLMP, which determines a linear combination of routes in $\bar{\mathcal{R}}$ generated by the pricing problem, takes the following form:

$$[RLMP] \quad \min \quad \sum_{r\in\bar{\mathcal{R}}} \tilde{u}_r y_r + \sum_{c\in\mathcal{C}} \gamma_c \beta_c \tag{2a}$$

$$\text{s.t.} \quad \sum_{r\in\bar{\mathcal{R}}} \alpha_r^c y_r + \beta_c \geq 1 \qquad \forall c \in \mathcal{C}, \tag{2b}$$

$$\sum_{r\in\bar{\mathcal{R}}} y_r \leq k, \tag{2c}$$

$$y_r \geq 0 \qquad \forall r \in \bar{\mathcal{R}}, \tag{2d}$$

$$\beta_c \geq 0 \qquad \forall c \in \mathcal{C}. \tag{2e}$$

Notice that Model (2) differs from Model (1) in three ways. (i) It restricts the set of routes to $\bar{\mathcal{R}} \subseteq \mathcal{R}$ (hence, the term "restricted" in the model's name), (ii) it employs a set-covering formulation rather than a set-partitioning formulation (see Constraints (2b)), and (iii) it relaxes the integrality requirements of the decision variables.

4.2. Pricing Problem

In a column generation method, where the master problem objective function is a minimization, the pricing problem aims to find negative reduced cost columns (variables) among those that are not yet considered in the RLMP or to prove that no such columns exist. In the former case, columns are added to the RLMP before starting a new iteration. In the latter one, the algorithm stops.

For Model (2), the variables $y_r, r \in \mathcal{R}$ are generated by column generation as needed using a pricing subproblem. The description of the subproblem requires the dual variables of the master problem. We denote the dual variables of the constraints of Model (2) as $\rho_c \geq 0 \ \forall c \in \mathcal{C}$ for Constraints (2b) and $\pi \leq 0$ for Constraint (2c). The reduced cost of a route r, denoted by $\overline{u}(r)$, would then be calculated as $\overline{u}(r) = \tilde{u}_r - \sum_{c\in\mathcal{C}} \alpha_r^c \rho_c + \pi$. Therefore, the pricing problem to generate routes based on the RLMP solution is

$$[PP] \qquad \min_{r\in\mathcal{R}} \overline{u}(r). \tag{3}$$

The PP takes the form of an elementary shortest-path problem with stochastic service request times. This problem is an extension of the elementary shortest-path problem with resource constraints (ESPPRC) (Feillet et al. 2004, Irnich and Desaulniers 2005), which is known to be strongly NP hard (Dror 1994). The typical algorithm to solve the ESPPRC is referred to as a labeling algorithm, which is an extension of the Ford–Bellman algorithm, taking resource constraints into account.

The algorithm starts with an initial label associated with depot (0), which has a resource usage of zero for all resources. The algorithm then extends labels along a path using extension functions that govern the resource usage along the path. Each node receives labels throughout the algorithm that stand for partial paths from the source node (depot (0)) and indicate the cost and resource consumption of these paths. Nodes are iteratively treated until no new labels are created. When a node is treated, all of its new labels are extended toward every possible successor node (customer or depot (0′)). To efficiently handle the resource constraints, the labeling algorithm uses a technique called dominance pruning, which allows it to discard labels that are dominated by other labels. A label is dominated if there exists another label that will provably perform better in all common extensions of the two labels. The labeling algorithm is a powerful and efficient method for solving the ESPPRC, and it has been applied in several applications, such as transportation and communication networks, logistics and supply chain management, and project scheduling. We propose two labeling mechanisms, forward labeling and backward labeling, which are discussed next.

4.3. Forward Labeling

In our forward-labeling implementation, we begin with an initial label that represents a partial path p beginning

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at the depot (0). Each label is structured based on the following resources:

- $n(p)$: customer node where the label of partial path p resides;
- $w(p)$: total travel time of partial path p;
- $\Pi(p)$: forbidden extensions of partial path p; and
- $\overline{u}(p)$: reduced cost of partial path p.

4.3.1. Extension Functions. Let p be a partial path at node $n(p) = c$ that is about to be extended to customer c' to create a new partial path $p' = p \oplus c'$. The feasibility steps of our labeling algorithm determine if an extension exceeds the maximum duration of a route, T, and whether c' is a member of the forbidden set $\Pi(p)$. The extension functions that update a label associated with a partial path p to a new partial path $p' = p \oplus c'$ can be defined as follows.

4.3.1.1. Vertex Extension Function. $n(p') := c'$. The vertex of p' is the extension customer c'.

4.3.1.2. Travel Time Resource Extension Function. $w(p') := w(p) + t_{c,c'}$. The travel time resource extension function adds the travel time from the current node of a label c to the extension customer c'.

4.3.1.3. Forbidden Set Extension Function. $\Pi(p') := \Pi(p) \cup \{c\}$. The forbidden set extension function adds the current node c to the forbidden set of the new label p'. The forbidden set is used to ensure that a customer is not visited more than once in a path.

4.3.1.4. Reduced Cost Extension Function.

$$\overline{u}(p') := \overline{u}(p \oplus c').$$

We propose a sample-scenario planning approach to calculate the reduced cost of a partial path, incorporating the expected latency-induced disutilities associated with scheduled visits on a partial path as described in Algorithm 1. Note that if the scenarios are taken randomly from the service request distributions, the approach is equivalent to a Monte Carlo approach. The cost extension function operates primarily by averaging the disutilities of the new route over $|\Omega|$ sampled scenarios, where each scenario ω represents a sampling from the request time distributions of the stochastic customers.

Let τ_c^ω denote the service request time of customer c under scenario $\omega \in \Omega$. The function $V(p, \omega)$ returns the visit realization v^ω corresponding to partial path p given the request times in scenario ω. As shown in Algorithm 1, a given partial path p is evaluated across all scenarios $\omega \in \Omega$, with the latency-induced disutility computed separately for each scenario. The expected disutility is then obtained by averaging the scenario-specific disutilities.

Algorithm 1 (SSP-Based Expected Latency-Induced Disutility Calculation for Partial Path $\boldsymbol{p}$)

Inputs:
- $\mathcal{G}, \mathcal{D}, \mathcal{S}$ ▷ graph, set of deterministic and stochastic customers
- Ω ▷ set of scenarios
- p ▷ a partial path

Output:
- $\tilde{u}(p)$ ▷ the expected latency of partial path p

Algorithm *Expected_Latency_Extension_Function* $(\Omega, p, (\rho_c)_{c\in\mathcal{C}}, \pi)$
- $\tilde{u}(p) := 0$
- **for each** $\omega \in \Omega$ **do**
  - $v^\omega \leftarrow V(p, \omega)$ ▷ visit realization according to scenario ω
  - $\theta \leftarrow 0$ ▷ service agent release time
  - $m \leftarrow 0$ ▷ the last visited customer—initially set to depot
  - $i \leftarrow 1$
  - **while** $i \le |p|$ **do**
    - **if** $p(i) \in \mathcal{D}$ **then**
      - $\theta \leftarrow \theta + t_{m,p(i)}$
      - $\tilde{u}(p) \leftarrow \tilde{u}(p) + g_{p(i)}(\theta)$
      - $m \leftarrow p(i)$
    - **else if** $p(i) \in \mathcal{S}$ **then**
      - $\overline{\eta}_{p(i)} = T - \sum_{j=i}^{|p|-1} t_{j,j+1} - t_{|p|,0'}$ ▷ the maximum wait-to time of $p(i)$
      - $\phi_{v^\omega}(p(i)) = \overline{\eta}_{p(i)} - t_{m,p(i)}$ ▷ the latest departure from $m = \kappa_{v^\omega}(p(i))$
      - **if** $\tau^\omega_{p(i)} \le \theta$ **then**
        - $\theta \leftarrow \theta + t_{m,p(i)}$
        - $\tilde{u}(p) \leftarrow \tilde{u}(p) + g_{p(i)}(\theta - \tau^\omega_{p(i)})$
        - $m \leftarrow p(i)$
      - **else if** $\tau^\omega_{p(i)} \le \phi_{v^\omega}(p(i))$ **then**
        - $\theta \leftarrow \tau^\omega_{p(i)} + t_{m,p(i)}$
        - $\tilde{u}(p) \leftarrow \tilde{u}(p) + g_{p(i)}(t_{m,p(i)})$
        - $m \leftarrow p(i)$
      - **else**
        - $\theta \leftarrow \phi_{v^\omega}(p(i))$
        - $\tilde{u}(p) \leftarrow \tilde{u}(p) + g_{p(i)}(\xi_{p(i)} - \tau^\omega_{p(i)})$
      - **end**
    - $i \leftarrow i + 1$
  - **end**
- **end**
- **return** $\tilde{u}(p) \leftarrow \frac{\tilde{u}(p)}{|\Omega|}$

Remark 1. Note that in our implementation, following each extension $p \oplus c'$ to obtain p', we recalculate the reduced cost $\overline{u}(p')$ following the steps of Algorithm 1 and adjusting for the associated dual variables. Although one might be able to restore part of the calculation done to compute $\overline{u}(p)$ while computing $\overline{u}(p')$, our experiments show that the amount of required memory can become prohibitively large. This is because an extension can result in changes to the wait-to

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times of the stochastic customers along the partial path, invalidating the calculations performed before the extension.

The calculation of the expected latency-induced disutility of a partial path is done by calculating the disutility of a path according to a scenario and averaging scenario-dependent disutilities in the final step. A scenario-dependent disutility is calculated starting from the beginning of the path and processing the scheduled customers one by one by calculating their latency and disutility accordingly. The calculation of the latency at a deterministic customer is straightforward as latency is calculated as the service agent release time from the previous customer visited plus the travel time to reach the customer, which can then be translated into the measure of disutility using the respective customer's utility decay function (lines 9–12 in Algorithm 1). The calculation of latency at a stochastic request customer contains three distinct possible outcomes that depend on the scenario at hand. (1) The customer requests service before the service agent release time of the previously visited customer m (lines 16–19 in Algorithm 1), (2) the customer requests service after the service agent release time of the previously visited customer but before its latest departure time (lines 20–23 in Algorithm 1), or (3) the customer requests service after its latest departure time (lines 24–26 in Algorithm 1). In the first case, customer p_i is visited at time $\theta + t_{m,p(i)}$, and the latency is calculated by subtracting the scenario-specific request time $\tau^{\omega}_{p(i)}$. In the second case, customer p_i is visited at time $\tau^{\omega}_{p(i)} + t_{m,p(i)}$ (i.e., the service agent begins travel to customer $p(i)$ as soon as the request is made), and the latency is simply $t_{m,p(i)}$ (the best-case latency for this visit realization). In the third case, the latency is calculated as the maximum of zero or the difference between the time that the customer is assumed to be visited in the next planning period $\xi_{p(i)}$ and the scenario-specific request time $\tau^{\omega}_{p(i)}$. The obtained latency is then converted to the measure of disutility using the customer's utility decay function.

4.3.2. Dominance Pruning.

Consider two partial paths p and p' such that

$$n(p') = n(p) \tag{4a}$$

$$w(p') \leq w(p) \tag{4b}$$

$$\Pi(p') \subseteq \Pi(p) \tag{4c}$$

$$\overline{u}(p') \leq \overline{u}(p). \tag{4d}$$

Conditions (4) formulate the ESPPRC traditional necessary rules (Desaulneirs, Desrosiers, and Solomon 2005) for p''s dominance over p. Dominance rules are meant to guarantee that for any common extension of the two labels, the one identified as the dominating label (p') will continue to outperform the dominant one (p).

Proposition 1. *Dominance Conditions* (4) *are not sufficient conditions for the dominance of p' over p in the presence of stochastic service request times.*

Proof. We show the insufficiency of Conditions (4) in the presence of stochastic service request times through a counterexample detailed in Appendix A. □

Proposition 1 demonstrates the insufficiency of the classical dominance Conditions (4). Next, we propose additional necessary conditions to enforce dominance.

Proposition 2. *Label p' dominates label p if*

i. *Conditions* (4a), (4b), *and* (4c) *are met and*
ii. $\overline{u}(p' \oplus e) \leq \overline{u}(p \oplus e)$,

where $e \in \mathcal{E}_p$ is a feasible extension to p and $\mathcal{E}_p$ is the set of all possible feasible extensions of label p.

Proof. The proof follows the necessary conditions for dominance (Irnich and Desaulniers 2005). □

4.3.2.1. Approximated Forward Dominance.

Enumerating $\mathcal{E}_p$ could be impractical. Therefore, we propose an approximated dominance rule, which relies on substituting condition (ii) in Proposition 2 with (ii') $\overline{u}(p' \oplus e_\delta) \leq \overline{u}(p \oplus e_\delta)$, where e_δ is a real or fictive extension to p, resulting in a shift δ to the maximum wait-to times for all $\delta \in [\delta^{\min}, \delta^{\min} + \iota, \delta^{\min} + 2\iota, \ldots, \delta^{\max}]$ when

- $\delta^{\min} = \min_{\hat{c} \in \mathcal{C} \setminus \Pi(p)} \{t_{n(p),\hat{c}} + t_{\hat{c},0'} - t_{n(p),0'}\}$,
- $\delta^{\max} = T - w(p)$, and
- $\iota = \min_{\substack{c_1, c_2 \in \mathcal{C} \setminus \{\Pi(p) \cup \{\hat{c}\}\} \\ c_1 \neq c_2}} \{t_{c_1,c_2} + t_{c_2,0'} - t_{c_1,0'}\}$.

We provide the reasoning behind such an approximation in Appendix B. We assess the quality of solutions obtained with a simplified approximation in Section 6 and compare these with a nonapproximated solution found using the labeling algorithm described next.

4.4. Backward Labeling

The dominance complications of the above forward-labeling procedure lead us to question whether the maximum wait-to times of service agents can be handled more computationally efficiently. Therefore, we consider a backward-labeling alternative. In a backward-labeling procedure, labels begin at the depot $0'$ and are extended "backward" while maintaining feasibility. *In our problem, the backward procedure has the advantage of having stable maximum wait-to times during dominance pruning.*

Backward labeling (see, for example, Righini and Salani 2006), as with forward labeling, starts with an initial label p at the depot $0'$. The label is propagated in feasible backward directions as extension functions record information pertinent to the label's resource consumptions. Only the backward reduced cost extension function requires additional explanation, although the mechanics of the expected latency-induced disutility

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function are identical to those discussed in Section 4.3.1 (see Algorithm 1) for forward labeling. In forward labeling, an extension of partial path p to vertex j implies that j is "scheduled" after vertex $n(p)$. Therefore, the earliest visit time to j is $w(p) + t_{n(p),j}$ (whereas the latest visit time, if it occurs, must take into account the most recently scheduled stochastic request customer, including j). On the other hand, in backward labeling, we find that j is scheduled before any customer currently on the partial path p, and therefore, the earliest visit time is $t_{0,j}$ (and the latest visit time, if it occurs, is only dependent on the stochastic nature of j).

Consider two labels p and p' encoding two different backward partial paths with current vertex $n(p) = n(p')$ as shown in Figure 2. Suppose p' meets the traditional conditions for dominance of p (i.e., Conditions (4a)–(4d)). The expected disutility of a backward partial path p and consequently, its reduced cost $\overline{u}(p)$ depend on the completion partial path p_0 that will be joined with p to form a complete path. This dependency arises from how quickly the technician can reach customer $n(p)$ after completing service at the last customer of p_0 in the event that a request from $n(p)$ is received in time to be serviced (otherwise, some other customer scheduled later than $n(p)$ may be the realized vertex). This consideration is critical when $n(p) \in \mathcal{S}$. That is, there may exist a common backward completion p_0 for p and p' that could reverse previously valid dominance conditions. Therefore, the dominance Conditions (4a)–(4d) must be augmented to account for such possibilities. More detail is provided in the proof of Proposition 3 in Appendix C.

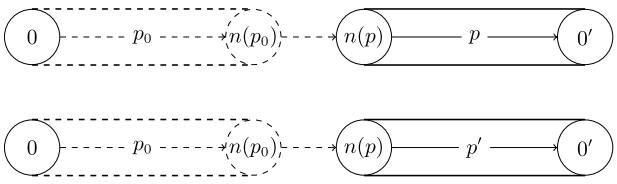
Suppose p_0^* is the completion partial path that when joined with p, minimizes the expected disutility and thus, the reduced cost $\overline{u}(p)$. Although identifying p_0^* without completing the labeling process is challenging, one can compute a valid lower bound on $\overline{u}(p)$, denoted by $\underline{U}(p)$, without exhaustively evaluating all possible completion partial paths. Given this lower bound, Proposition 3 establishes a sufficient condition for applying dominance rules.

Proposition 3. *In the backward-labeling procedure, the traditional conditions for dominance (i.e., Conditions (4a)–(4d)) must be augmented by the following additional condition:*

$$\overline{u}(p') \leq \underline{U}(p), \tag{5}$$

where $\underline{U}(p)$ corresponds to a lower bound on the expected disutility of partial path p.

Figure 2. Backward Labeling: Two Backward Labels p and p' on Node $n(p)$ with One Common Completion Partial Path p_0



Proof. The proof is provided in Appendix C. □

Notice that the looser the lower bound $\underline{U}(p)$, the more stringent the dominance Condition (5) becomes, resulting in fewer labels being identified as dominated. Therefore, a tighter lower bound $\underline{U}(p)$ is desirable to improve pruning effectiveness, although a trade-off may occur with computational time.

Without considering the specifics of p_0, the best-case scenario for a partial path p occurs when the service agent is available at a location nearest to the *realized* vertex of p. We use the term *realized* to refer to the case where $n(p) \in \mathcal{S}$. In this case, all stochastic request customers who appear at the beginning of the partial backward path p (up to the first deterministic customer if one exists) are potential first-visited customers—provided that all of their stochastic predecessors are skipped. To determine the best-case position from which to serve p, we must consider all of these customers and identify the most favorable unvisited location.

The true minimum expected disutility for p can be calculated by evaluating all possible remaining vertices that $n(p_0)$ can take and simulating the expected disutility associated with each one, assuming that it is the most recently visited customer (i.e., setting $m := n(p_0)$ in line 6 in Algorithm 1). The minimum among these values yields the true minimum expected disutility for p.

Alternatively, a lower bound on this true minimum expected disutility can be estimated by evaluating the best case for p within each scenario $\omega \in \Omega$. In this approach, let c_p^ω be the nearest customer to the realized vertex of p according to ω. Then, set $m := c_p^\omega$ in line 6 in Algorithm 1. This scenario-specific nearest-customer approach provides a potentially loose lower bound because it may rely on an unrealistic realization if c_p^ω is not consistent across all scenarios. However, this method avoids simulating over all remaining vertices and therefore, can yield a computational speedup. The trade-off is in bound quality; a looser bound weakens the effectiveness of dominance rules.

Both of the above bounds have limitations. The first approach yields the true minimum expected disutility for p but at a high computational cost. The speedup gained by the second approach in computing the lower bound may be offset by a less efficient labeling procedure because of the weakened effectiveness of the dominance rules.

Although bidirectional labeling proved to be much more efficient than monodirectional labeling for the VRP and the vehicle routing problem with time windows (VRPTW) (Righini and Salani 2006), in our setting, because of the deficiencies of the forward labeling

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(see Section 6.3.3), we solely utilize our proposed backward labeling in our computational analyses.

4.5. Acceleration Techniques

4.5.1. Dominance Tiers. The proof of dominance (i.e., applying Proposition 3) is more computationally expensive. Therefore, when solving the pricing problem, we first exhaust the search space for labels using the weaker dominance rules and then, apply Equation (5).

4.5.2. Local Neighborhood Search. Inspired by the heuristic column generation of Dayarian et al. (2015a), we employ local neighborhood search methods at each node of the branch-and-bound tree before the labeling algorithm is applied. We employ insertion, deletion, and customer pair swap operations on the columns in the current solution. All feasible operations are applied, new columns with the negative reduced costs are added to the RLMP, and the RLMP is resolved. This procedure is repeated until no new column with negative reduced cost can be generated.

4.5.3. *ng*-Route Decremental State-Space Relaxation. The forbidden set is the set of vertices that a label may not be extended to in order to maintain elementarity. The shortest-path problem with resource constraints is an NP-hard problem, and the elementarity condition adds extra complexity. The most effective labeling approaches to solve the ESPPRC follow two main strategies. The decremental state-space relaxation method (Boland, Dethridge, and Dumitrescu 2006, Righini and Salani 2009) initially relaxes the elementarity conditions and gradually adds restrictions to prevent cycles. Alternatively, the *ng*-path relaxation (Baldacci, Mingozzi, and Roberti 2011) partitions paths by neighborhood and mapping functions to impose partial elementarity. We impliment an efficient *ng*-path pricing in which a DSSR technique is embedded into the ng-path relaxation (Dayarian et al. 2015b). This technique effectively computes lower bounds for routing problems, like the VRP.

The *ng*-path relaxation can be described as follows. Suppose that $\mathcal{C}_p$ represents the set of customers visited by partial path p. Moreover, for each customer $c \in \mathcal{C}$, let $G_c \subseteq \mathcal{C}$, the so-called original neighborhood of customer c, represent a set (with an a priori fixed size) of customers selected according to a neighborhood criterion for customer c. For partial path $p = (0, p(1), \cdots, p(|\mathcal{C}_p|))$, we define a set $\Pi(p) \subseteq \mathcal{C}_p$ containing all prohibited extensions from customer $p(|\mathcal{C}_p|)$. The extension function that updates the forbidden extension set following an extension to c' is

$$\Pi(p') := \{p(i) | p(i) \in \bigcap_{k=i+1}^{|\mathcal{C}_p|} G_{p(k)}, \quad i = 1, \cdots, |\mathcal{C}_p| - 1\} \cup \{c'\}.$$

When no columns with negative reduced cost can be found, we set $G_c = \mathcal{C}$ and rerun the labeling procedure until no new columns with negative reduced cost can be found.

4.5.4. Subset-Row Cuts. Let σ be a triplet of customers in $\mathcal{C}$, and let Σ be the set of all customer triplets. Subset-row inequalities (Jepsen et al. 2008, Baldacci, Mingozzi, and Roberti 2011) are cuts of the nature $\sum_{r \in \mathcal{R}(\sigma)} y_r \leq 1 \ \forall \ \sigma \in \Sigma$, where $\mathcal{R}(\sigma)$ is a set of routes that visit at least two customers in the cut σ. Cuts of this type can help to improve the lower bounds formed at each node in the branch-and-bound tree but complicate the structure of the pricing problem. Therefore, we employ what Baldacci, Mingozzi, and Roberti (2011) call weak subset row inequalities such that $\mathcal{R}(\sigma)$ is made up of only routes that visit at least one arc $(c_1, c_2) \in \sigma$, which are robust (Costa, Contardo, and Desaulniers 2019). Each of the $|\Sigma|$ cuts is associated with a dual variable ζ that is by nature nonpositive. These dual variables are used in the pricing problem to "punish" violations of the cut and therefore, are subtracted from the expected latency of a route r.

5. Alternative Approaches

The described solution approach in Section 4 considers the full distribution of service request times for each customer. In contrast to that proposed approach, we analyze three alternative approaches, namely the point estimate (PE), greedy (G), and no wait approaches.

5.1. Point Estimate Approach

One hypothesis that we would like to test is that using the full distributional information of stochastic requests provides expected disutility or computational advantages to using a point estimate approach. The point estimate approach follows from Section 3 except that Ω contains one scenario—determined using only the expected values of request times for stochastic customers $c \in \mathcal{S}$. Note that this approach allows a less computationally expensive calculation of disutility because no integrals need to be evaluated. If the expected service request time $\mathbb{E}[\tau_c]$ is less than the scheduled service time, the customer will be visited at that time θ_c with associated latency $\theta_c - \mathbb{E}[\tau_c]$.

5.2. Greedy Approach

Another hypothesis that we would like to test is whether the proposed solution approach will provide better routes than dynamically adjusting the routes during operation. The greedy approach does not include stochastic customers in the initial set of routes. Routes are generated using branch and price to minimize the disutility of deterministic customers. As the routes are operated, if a stochastic customer $c \in \mathcal{S}$ requests service during the operational period h, the customer c is inserted into the remainder of an ongoing route in the lowest total disutility position with respect to all customers. The generation of deterministic routes

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follows the branch-and-price procedure described in Section 4 without the components dealing with stochastic service request times. In summary, when a service request is received from a stochastic customer c, the effect of the insertion of c in any route r on disutility will be deterministic as by the time of the insertion, the request time is known to the decision maker.

5.3. No Wait Approach

Given that we seek to test the hypothesis that waiting is beneficial to the solution of a stochastic TRP, we also propose an approach that does not allow any waiting for stochastic request customers. The approach follows the proposed solution approach (SSP) in every way except that in the execution of Algorithm 1, the latest request time of customer $p(i)$ on partial path p is set equal to the service agent release time from the previous customer on the partial path $(p(i-1))$ (i.e., $\phi_{v^\omega}(p(i)) = \theta$ in line 16 in Algorithm 1).

6. Computational Results

We analyze the effectiveness of our proposed approach in two studies. The first study analyzes the proposed approach by (i) the sensitivity of the results to the number of scenarios considered in the proposed SSP, (ii) comparing the effectiveness and performance of the proposed forward- and backward-labeling approaches and their respective dominance rules, (iii) the benefit of waiting before stochastic customers along the a priori route, and (iv) the trade-off between computational time and total expected latency in comparison with the alternative solution approaches described in Section 5. The second study analyzes the effectiveness of our approach using real-world data from the CitiBike bike-share service of New York City in the United States. Our method was coded in Julia, and the instances were solved on identical compute nodes each with a single hyperthread of an M3 processor. The linear programming models were solved using Gurobi 10.0. In what follows, we first describe our instance generation procedure for the comparison study followed by the detailed results of the comparison study and the bike-sharing case study.

6.1. Instance Generation

We draw our benchmark instances from the recent TRP literature, specifically drawing from a set of instances solved in Luo, Qin, and Lim (2014), Nucamendi-Guillén et al. (2016), and Muritiba et al. (2021) that were originally drawn from TSPLIB instances of the following names: "brd1405," "d15112," "d18512," "fnl4461," "nrw1379," and "pr1002." As in those works, we define $t_{max} = \max_{c \in \mathcal{C}} t_{0,c} + t_{c,0}$, set $T = 2t_{max}$, and set $k = \lceil |\mathcal{C}|/5 \rceil$. The instances generated have 30, 40, and 50 vertices. Each instance class (e.g., a row in Table E.1 in Appendix E) consists of 10 instances. Solutions to the benchmark instances all assume the classical "weighting" of latency in the TRP literature (i.e., $g(\cdot) = \cdot$), and the expected disutility is directly measured by the expected latency.

We analyze the effect of differing the number of stochastic request customers (varied from 25% to 50% of the number of customers; selected randomly). We set the next time of service of an unvisited customer c as $2T + t_{0,c}$. Therefore the parameter γ_c is equal to the expected time between this and the request of customer c. The service request times of stochastic customers are assumed to follow a truncated lognormal distribution, the parameters of which are set to place the expected value of the service request time between zero and T. There is a vast literature on the suitability of the lognormal distribution to model breakdown times in the contexts of reliability and failure of electronic materials and devices as well as solid materials when exposed to long-term mechanical comminution (Chang 2015, Sutton 2015, Neikov and Yefimov 2019).

To validate our solution approach, we first solve the deterministic instances using our approach and compare the results with the available best-known solutions from the literature in terms of latency. Please note that several of the components of our approach are designed to tackle the potential stochasticity of request times, and therefore, we do not aim to be competitive in terms of solution time when compared against the approaches specifically designed for deterministic setting. These results are reported in Table D.1 in Appendix D.

6.2. Sensitivity Analysis on the Number of Scenarios

To generate a scenario, we randomly sample service request times from each stochastic customer service request distribution to form scenarios. The number of scenarios has an impact on the computation required to calculate the expected latencies of paths and partial paths. As the number of scenarios increases, the law of large numbers dictates that the sample more closely approximates the underlying distribution.

To evaluate the impact of the number of scenarios on the quality of the optimal solution approximation obtained, we take a sample of instances consisting of one instance of each TSPLIB instance type. Therefore, we solve six instances for each experimental setting, resulting in a total of 36 instances. The sampled instances include those with $\mathcal{C} \in \{29, 39, 49\}$. Each instance in this set is solved using 25, 50, 75, 100, 250, or 500 scenarios generated using the stochastic customer service request distributions.

In Table 1, we report the percentage change in expected latencies from the expected latency of solutions solved with 500 sampled scenarios in the row "Disutility gap (%)." All expected latencies are calculated using 10,000 scenarios that were not used to solve

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Table 1. Out-of-Sample Results with Different Numbers of Scenarios

	Number of scenarios				
Metric	25	50	75	100	250
Disutility gap (%)	1.02	0.45	0.26	0.18	0.08
Time gap (%)	−48.13	−51.36	−46.01	−32.04	−24.9
Solved (/36)	34.0	34.0	33.0	35.0	33.0

Note. Time limit = 7,200 seconds.

the model (i.e., an out-of-sample test set of size 10,000). We also report the average required computational time for each experiment with a given number of scenarios as well as the total number of solved instances within the two-hour time limit. Based on the whole of our preliminary evidence, we proceed with 100 scenarios in the ensuing experiments, but note that careful experimentation is needed to understand the effect of the number of scenarios on out-of-sample performance.

6.3. Comparison Study of the Proposed Approaches

To assess the quality of different algorithmic features of our proposed approach, we compare the performance of our approach with that of several modified approaches using the same sample of TSPLIB instances discussed above. To ensure a fair comparisons, the routing solutions of the SSP, no wait, PE, G, and approximated forward dominance (AFD) approaches are tested against a holdout set of 10,000 scenarios. These results are presented in Table 2. The values reported in the rows Disutility gap (%) and Time gap (%) in Table 2 are averages calculated relative to the out-of-sample expected disutility of solutions obtained using 100 scenarios across all experimental settings. We set a time limit of two hours; if an experiment reaches this limit, the time limit is recorded as its run time, and the best upper bound found is reported as its disutility.

6.3.1. PE and G. We are interested in the trade-off between solution times as an indication of the complexity of a solution approach and its resulting expected latency. We report the results of a comparison study between the proposed approach (SSP) and the alternative approaches discussed in Section 5 in Table 2. The percentage increases in the expected latency and computational times in the G and PE approaches are reported under the headers Greedy and PE, respectively, in Table 2 (with a negative value representing a reduction). The objective values of the PE and G approaches do not use all of the information that we have on the distribution of expected service request times. The PE approach uses only the expected value of the service request time for each customer calculated as an average over the same 100 scenarios used by SSP. We find that the PE approach is outperformed by SSP in all settings. Although we do not report the detailed results in Table 2, we find that the disadvantage is greater as the proportion of stochastic customers increases. Further, the PE approach is computationally less efficient, although the individual simulations of expected disutility are less complicated.

Table 2. Comparison Table

	Comparison approach			
Metric	PE	Greedy	No wait	AFD
Disutility gap (%)	10.95	91.09	64.96	0.18
Time gap (%)	37.39	−94.27	357.81	9.62
Solved (/36)	32.00	36.00	26.00	35.00

Notes. Out-of-sample performances of alternative approaches in comparison with SSP are shown. Time limit = 7,200 seconds.

In the G approach, the obtained a priori routes include only deterministic customers. Stochastic request customers are inserted as described in Section 5.2 for each scenario. This allows the G approach a flexibility not afforded by the maximum wait-to time rule that we have implemented for the SSP and PE approaches. We do not find that this flexibility is beneficial to create solutions that minimize expected disutility, although the computation time is significantly less because of the consideration of fewer customers and no distributional information.

In summary, we find that the proposed SSP approach outperforms the PE approach and that the PE approach outperforms the G approach on average as shown in Table 2. We posit that the expected latency performance gap is because of the SSP approach's skill in scheduling deterministic customers to provide waiting time for stochastic requests. Specifically, the SSP and PE approaches are able to plan for the expected requests of stochastic customers and therefore, may balance the expected latency of deterministic and stochastic request customers.

6.3.2. The Benefit of Waiting. A hypothesis of this work is that allowing a service agent to wait for stochastic request customers to request service after being released from service at the preceding customer on their a priori route can be beneficial to the total expected latency of customers in the system. To test this hypothesis, we solve the gTRP problem with no waiting time allowed using the same instance scenarios as discussed in Section 6.2. If no waiting time is allowed, a stochastic request customer scheduled on an a priori route is visited only if a request has been received by the time that the service agent is released from the preceding customer on the route (i.e., $\phi_{v^\omega}(p(i)) = \theta$ in line 16 in Algorithm 1). Results reported under the headers "No wait" in Table 2 report the average change in expected latency and required computational time compared with SSP

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across experimental settings. The results of this experiment show that waiting time leads to a substantial reduction in the total expected latency across all settings. We find that SSP needs, on average, less computation time than the no waiting strategy, and therefore, it outperforms this strategy in objective and expense.

6.3.3. Forward vs. Backward Labeling. We also compare the backward-labeling procedure with the forward labeling employing our proposed AFD in Table 2 using the sampled scenarios from TSPLIB. Results reported under the header "AFD" in Table 2 report the increase in expected latency and required computational time compared with SSP using $\iota = \frac{\delta_{max} - \delta_{min}}{2}$. We find that the approximation scheme of the forward approach produces expected latencies higher than the backward approach, on average, because of the insufficiency of the approximation in dominating labels. Further, the backward-labeling approach is generally more efficient because of its simpler yet stronger dominance rules.

6.4. Factorial Experiment Results

We perform a factorial experiment over all of the TSPLIB instances that differ from each other in terms of the number and locations of customers, $\mathcal{C}$, and the size of the subset of stochastic customers $\mathcal{S}$. The summary results are shown in Table 3, whereas the detailed results are reported in Table E.1 in Appendix E.

On average, we find that the most significant factor to the change in CPU time is the number of customers followed by the ratio of stochastic request customers. The result unique to this work is that an increase in the proportion of stochastic request customers increases the solution time by a factor of at least two and sometimes greater. This follows from the difficulties discussed with respect to dominance in Section 4.4; an increase in stochastic requests customers increases the need to apply the computationally expensive dominance criteria. However, we also note that the distribution of the stochastic request customers may play a significant role. If the request time of a stochastic customer is known with relative certainty, it may be easier to identify labels that do not visit the customer at an advantageous time. On the other hand, as the request time distribution tends toward uniformity, parity is introduced among the possible routes.

Table 3. SSP Factorial Experiment Results with Disutility Calculated from the Out-of-Sample Test Set

$\lvert\mathcal{C}\rvert$	$\lvert\mathcal{S}\rvert$	Disutility	Time	Solved
29	7	126,727.83	55.42	60
29	15	145,718.60	133.35	60
39	10	162,211.73	223.39	58
39	20	195,831.16	901.68	60
49	12	202,883.54	916.80	58
49	25	240,476.91	2,064.61	51

Note. Time limit = 7,200 seconds.

6.5. Large Instance Results

In Tables E.1 and E.2 in Appendix E, we report the results of our approach on a set of instances adapted from the capacitated vehicle routing problem (CVRP) literature in Nucamendi-Guillén et al. (2016) and Muritiba et al. (2021). The instances do not have a specified T, and ours is set using the method described above. Despite the significantly higher complexity of our problem setting, we adopt a two-hour time limit to maintain consistency with previous work. These instances allow us to test our approach on larger problem sizes and to analyze the impact of the technician fleet size better. The computational effort required to solve the pricing problem using the labeling algorithm increases exponentially with problem size. Specifically, larger instances incur significant computational expense because of the additional simulation steps and dominance conditions required by our approach. However, this challenge is not unique to our problem or solution method; a similar trend is observed when these instances are solved as the deterministic TRP as reported by Muritiba et al. (2021), even without the added complexity of stochastic requests.

The results also provide the opportunity to examine the effect of the number of service agents k and the ratio of k to $|\mathcal{C}|$ on computational time. In the benchmark instances discussed above, the TRP literature has adopted a $|\mathcal{C}| : k$ ratio of five to one. Larger ratios of customers to service agents might require longer routes. Our approach requires resimulation of disutility for both extension and dominance verification, and therefore, we expect it to be tested more than the approaches of the deterministic TRP literature. In Tables E.1 and E.2 in Appendix E, we report the results of our approach on instances with ratios up to ~ 10 : 1, and we find that larger ratios have the expected negative effect on computational performance. For example, when the E instance with $|\mathcal{C}| = 76$ was solved with $k = 10$ and $k = 14$, we observed that the problem with the larger number of technicians was solved easily within the time limit, whereas the one with a smaller fleet size hit the time limit.

6.6. A Case Study: Bike-Share Station Maintenance

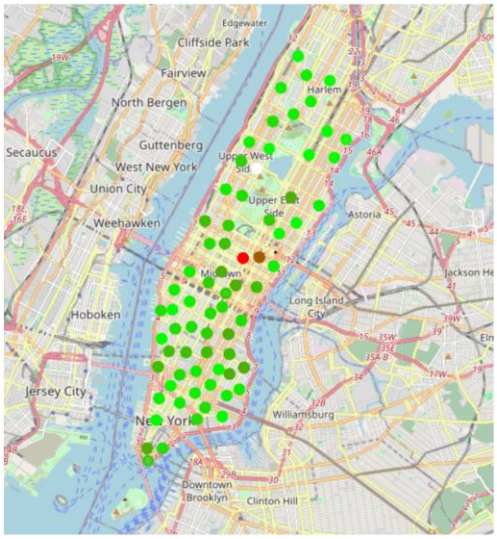
As mentioned in Section 1, we believe that the service operation for bike-share system maintenance is a gTRP. As with traffic camera maintenance (Muritiba et al. 2021), every moment that a bike is not available, the system loses some expected amount of revenue. We, therefore, use one of the largest bike-share systems in the world, the CitiBike system of New York City, as a case study. According to the September 2024 operating report, "There were 2,112 active Citi Bike stations at the

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end of September. Technicians made a total of 4,770 station visits to address dock and station field maintenance orders … Docking point repairs were the most common ticket types" (Citi Bike 2024).

We present an analysis of service agent routing for the stations operating out of a single depot. We focus on the most heavily trafficked part of the city from lower Manhattan to Central Park, which covers 419 bike-share stations. As shown in Figure 3, we split these into 70 clusters of stations of on average nearly six stations each using the *k*-means clustering algorithm (each dot represents a cluster). Each cluster is treated as a customer, matching the terminology used in our paper. We assume that the depot is located at the farthest south station on the island (the closest point to the area considered to be the central depot of CitiBike). For simplicity, we assume that travel times between stations are proportional to the Manhattan distance between stations at an average speed of nine miles per hour (the high end of the average speed range according to https://www.nyc.gov/). Travel times are calculated based on the geometric center of each cluster. Conservatively, we assume that the service time at a station is 30 minutes. We assume that the operational period starts at 6 a.m. so that any requests can be serviced before and during the morning rush hour and that the operational period is eight hours long. We assume that an unvisited customer with a known demand by the end of the current shift will be visited as soon as the next shift starts, taking into account the travel time from the depot to the customer. The parameter γ_c is then calculated as the expected difference between this time and the customer c request.

Figure 3. (Color online) Clustered Bike Stations with Scaled Stock-out Probabilities from Light Green (Lowest) to Red (Highest)



Data are available in real time on the number of disabled docks at each station. A report of a disabled dock at a given time is interpreted as a service request. We collected these data once per minute from September 1st through October7th, 2024. From the data, we estimate the time of the service at a station by measuring for any decrease in the number of disabled docks. We do not have access to the actual number of service teams used; therefore, we estimate the number of service teams required to make these service trips. To do so, we formulate a vehicle routing problem with time windows, where the nodes correspond to the set of stations visited by the recorded operations of CitiBike in a given day (hereafter referred to as the incumbent solution). To enforce the visit times, we allocate a short time window to each visited station (e.g., five-minute length), with a start time coinciding with the repair time observed from the actual operations. We set a large fixed cost for each vehicle employed and use Google OR-Tools (Perron and Furnon 2023) to solve this VRPTW. The solution to this problem enables the identification of the minimum fleet size required to serve all requests at their observed return-to-service times. We call this the incumbent approach.

We generate 10 instances (operational periods), each corresponding to a day within the last 10 days of the data collection period. For each obtained operational period, we fit Gaussian kernel density estimators to each of these cluster's request times using data up to one hour before an operational period begins. Inspired by the approach of Hvattum, Løkketangen, and Laporte (2006), all customers who have requested service by one hour before the operational period are treated as deterministic. All clusters that were visited by the incumbent solution (the actual events observed from the data) but have not requested service by one hour before the operational period are treated as stochastic. For each stochastic customer, we use these distributions to generate 100 service request time scenarios, consistent with the number of scenarios used in the previous section computational results.

6.7. Nonweighted Latency

Our goal is to determine if with the same or lower level of resources used in the incumbent solution, we can provide a better solution in terms of actual latency. In the first set of experiments, we focus on the classical "weighting" of latency in the TRP literature (i.e., $g(\cdot) = \cdot$), and the expected disutility is directly measured by the expected latency. We refer to these experiments as "nonweighted" case.

For planning purposes, we use 100 scenarios generated from the fitted request time distributions described above. However, we perform an ex post evaluation of the routing plans—measured in terms of total latency—using the actual request times from the data set.

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In Table 4, we report results for 10 days of operations (10 operational periods) starting from September 28, 2024. For each day, we report $|\mathcal{D}|$, $|\mathcal{S}|$, k(VRPTW), the number of deterministic and stochastic requests, and technicians used based on the solution of the incumbent approach, respectively. The column "Time" in Table 4 reports the CPU time to solve the instance. The column "NW latency" in Table 4 reports the latency of the solution of SSP with a nonweighted disutility function. In Table 4, the column "Incumbent latency" reports the latency according to the incumbent solution, whereas the column "Latency gap (%)" represents the SSP solution relative to the incumbent solution. We find that SSP outperforms the incumbent solution in all instances. We note that the incumbent solution visits the customers in their observed repair windows with as few vehicles as possible. The incumbent solution is based on an unknown approach by the CitiBike operating team that can respond dynamically to requests but also, may service customers outside of the area that we focus on. Therefore, a fair comparison would be to only compare our results using the same resources as the incumbent. The a priori routes that SSP produces demonstrate effective routes that consistently service customers with the same number of resources as the incumbent solution. We also note that our predictions could be quite primitive. In practice, a subject matter expert would have access to a higher-fidelity distribution of request times. Even given this possible shortcoming, the results indicate that the SSP approach leads to an average reduction in latency of 32.97% compared with the incumbent solution.

6.8. Weighted Latency

As discussed in Section 2, our method may be used to minimize the expected disutility of a set of a priori routes, where the customer utility decay function, $g_c(l_c)$, transforms the latency (l_c) of customer c. The results presented in the computational section up to this point have assumed $g_c(l_c) = l_c$. If the latency with respect to some customer is more important to the operations of the system, the system operator may want to reflect that by treating its latency differently in the objective of their model. For example, one may decide to prioritize station clusters with higher demands. As a proxy for demand, we use the stock-out probability of station clusters. As shown in Figure 3, some bike station clusters have a greater stock-out probability (calculated empirically by counting the operational days during which at least at one point, there was no bike available) than others. This stock-out frequency may be because of many factors, including the endogenous and exogenous bike rebalancing operations of the system or demand on a cluster of stations. As such, stock-out probability is a metric of the economic importance of a cluster of stations. As seen in Figure 3, the highest stock-out probability occurs at a station just south of Central Park in New York City, a known tourist destination.

Let the stock-out probability of cluster c be s_c. In Table 5, we compare the SSP approach with the incumbent approach on our bike-share problem instances altered such that $g_c(l_c) = s_c l_c$. The SSP is solved with the stated utility decay function, whereas there is no knowledge of any function used by the incumbent approach. Therefore, we are not surprised to find that SSP outperforms the incumbent approach on the weighted objective by 38.26% on average. For a more direct comparison of how g affects the solution approaches, we present Tables 6 and 7. In Table 6, we present a comparison of the latencies obtained using the nonweighted approach ($g_c(l_c) = l_c$) and the weighted approach ($g_c(l_c) = s_c l_c$), showing that the weighted approach sacrifices latency of around 9.79% on average. However, Table 7 shows that this sacrifice reduces the disutility (weighted latency) by 20.84% on average. Taken together, these results indicate a significant difference in the solutions obtained and an effective solution approach to the weighted problem.

7. Discussion

We introduce a novel approach to the routing of service agents in the service of stochastic, time-sensitive service

Table 4. Case Study Results

Day	$\lvert\mathcal{C}\rvert$	k (VRPTW)	$\lvert\mathcal{S}\rvert$	Time	NW latency	Incumbent latency	Latency gap (%)
28	26	9	2	38.72	2,392.75	3,410.37	−29.84
29	16	4	7	2.56	2,222.83	3,870.21	−42.57
30	28	7	5	33.96	3,089.87	4,128.28	−25.15
1	22	8	3	5.85	1,959.83	2,909.51	−32.64
2	25	10	4	4.55	2,366.7	3,664.87	−35.42
3	19	6	4	0.82	2,126.82	2,798.59	−24.0
4	22	7	2	2.94	2,105.56	2,794.28	−24.65
5	21	5	6	4.68	2,754.72	4,688.74	−41.25
6	21	4	7	58.61	2,875.05	3,943.92	−27.1
7	16	7	4	0.75	1,693.51	3,201.65	−47.1

Note. NW latency reports the latency of the solution of SSP with a nonweighted disutility function.

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Table 5. Weighted Case Study Results

Day	$\|\mathcal{C}\|$	k (VRPTW)	$\|\mathcal{S}\|$	Time	NW disutility	Incumbent disutiliy	Disutility gap (%)
28	26	9	2	18.53	150.14	230.79	−34.95
29	16	4	7	2.62	154.44	267.04	−42.16
30	28	7	5	3,602.93	201.1	259.85	−22.61
1	22	8	3	2.2	128.99	202.41	−36.27
2	25	10	4	4.66	186.15	281.47	−33.86
3	19	6	4	1.14	172.81	245.99	−29.75
4	22	7	2	20.94	144.59	278.25	−48.04
5	21	5	6	41.17	219.11	423.44	−48.25
6	21	4	7	257.3	184.21	296.5	−37.87
7	16	7	4	0.24	117.33	229.4	−48.85

Note. NW latency reports the latency of the solution of SSP with a nonweighted disutility function.

requests. The k traveling repairman problem with stochastic service requests is important to businesses and organizations that wish to minimize the latency-induced disutility of customers who have and have not requested service when the operational period begins. The problem that we study, as opposed to previous work, considers the timing of service requests and how a service agent route may be designed to take that timing into account. Previous approaches have considered the binomial probability of a customer requesting service. The benefit of our approach is that it may be applied to settings in which service requests are modeled as a distribution of times or enough scenarios from potential request times are available. As such, we provide a way to model the expected latency of customers in the system in relation to their predicted service request time.

We present a study to compare differing models of service request times as well as different routing approaches. Specifically, we model service requests as a full distribution of times and as a point estimate (mean). Our results show a reduction in the expected latency attributed to generating routes using the full distribution of service request times. Further, we show that the use of a service request time model has a significant benefit in relation to an approach that reacts to service request times by altering routes during operation.

We present a simple recourse rule for agents operating a route; if the agent will not be able to complete the planned route without skipping a customer, the customer should be skipped. We also propose and assess the effectiveness of a novel waiting strategy, incorporating wait times before stochastic customers and aiming to increase the chances of receiving their requests, whereas a service can still be provided without violating route feasibility. The waiting occurs either at the depot or at the customer who was last serviced (i.e., the current location of the agent). To solve the problem, we formulated it using a set-partitioning formulation and proposed a branch-and-price algorithm to solve it. Similar to most routing problems solved using column generation, the subproblem takes the form of an elementary shortest-path problem, which is solved using a dynamic programming-based labeling algorithm. We propose forward- and backward-labeling algorithms and demonstrate the advantages of the backward approach in the presence of waiting rules. One novel aspect of our approach is the incorporation of a sample-scenario planning framework within the labeling algorithm to estimate the expected latency of a partial path with respect to different request realizations of its stochastic customers.

Some computational complexities require further attention.

Table 6. Comparison of Nonweighted vs. Weighted Performance with Respect to Expected Latency Objective

Day	$\|\mathcal{C}\|$	k (VRPTW)	$\|\mathcal{S}\|$	NW latency	W latency	NW time	W time	Latency gap (%)
28	26	9	2	2,392.75	2,680.28	38.72	18.53	−10.73
29	16	4	7	2,222.83	2,509.16	2.56	2.62	−11.41
30	28	7	5	3,089.87	3,861.33	33.96	3,602.93	−19.98
1	22	8	3	1,959.83	2,271.06	5.85	2.2	−13.7
2	25	10	4	2,366.7	2,590.1	4.55	4.66	−8.63
3	19	6	4	2,126.82	2,222.2	0.82	1.14	−4.29
4	22	7	2	2,105.56	2,362.29	2.94	20.94	−10.87
5	21	5	6	2,754.72	3,016.17	4.68	41.17	−8.67
6	21	4	7	2,875.05	3,021.35	58.61	257.3	−4.84
7	16	7	4	1,693.51	1,778.32	0.75	0.24	−4.77

Notes. Latency gap shows decreases in the objective for the nonweighted (NW) approach. W, weighted.

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Table 7. Comparison of Nonweighted vs. Weighted Performance with Respect to Disutility

Day	$\|\mathcal{C}\|$	k (VRPTW)	$\|\mathcal{S}\|$	NW disutility	W disutility	Disutility gap (%)
28	26	9	2	204.03	150.14	35.9
29	16	4	7	185.56	154.44	20.15
30	28	7	5	272.28	201.1	35.39
1	22	8	3	169.15	128.99	31.14
2	25	10	4	210.76	186.15	13.22
3	19	6	4	185.09	172.81	7.11
4	22	7	2	182.48	144.59	26.2
5	21	5	6	263.99	219.11	20.48
6	21	4	7	210.44	184.21	14.24
7	16	7	4	122.7	117.33	4.58

Notes. Disutility gap shows increases in the objective for the nonweighted (NW) approach. W, weighted.

7.1. The Impact of Fleet Size k and Shift Length T

First, we note that little research has considered the effect of the number of service agents k or the length of the operational period T. Both parameters may indirectly impact the length of potential paths to be considered. As k decreases, more customers may need to be scheduled on each route. Further, as T increases, more feasible extensions must be considered in the labeling algorithm. Longer routes lead to more computational time to recalculate expected disutility.

7.2. The Choice of Stochastic Customers' Subset $\mathcal{S}$

The choice of which customers to include in the set $\mathcal{S}$, addressed in our case study, is a significant decision. Although our method is flexible enough to consider all possible stochastic request customers $\mathcal{C} \setminus \mathcal{D}$ (and either leave some unscheduled or schedule them with low priority), increasing the number of customers considered comes with a computational cost because these customers will still need to be considered in each labeling algorithm iteration. On the other hand, not including a customer means that the customer will certainly not be scheduled.

7.3. The Choice of Sampled Scenarios Ω

Although we address the impact of $|\Omega|$ on our approach, we believe that future work should consider the impact of the types of distributions considered. Further, we believe that some work should address the use of closed-form solution methods on known distributions to eliminate the need for a sampling approach altogether.

7.4. Directions for Future Work

Despite all of its innovative aspects, our study has limitations that warrant further investigation. The main limitation is related to the nature of the labeling algorithm and its combination with the proposed SSP mechanism, creating vulnerability because of the curse of dimensionality. Although we are capable of solving many of the existing benchmarks as well as the instances designed based on the bike-sharing data set in a relatively short amount of time, we anticipate that our methodology will suffer if used for much larger instances. Therefore, future research can focus on developing or adopting accelerating mechanisms, such as the inclusion of cutting planes to our branch-and-price framework or developing heuristic methods inspired by the proposed ideas in this paper.

Further, our proposed waiting rule may present an opportunity for future research on more refined waiting durations or may be replaced by another recourse rule. For example, an agent could wait for customers in a location different from the depot or its last serviced customer to improve the total expected latency outcome. Also, in the dominance conditions of our labeling algorithms, we ensure that no shift in *wait-to* times will shift the balance to dominated labels. From this, we note that the maximum wait-to times themselves can be optimized after the execution of our labeling algorithm. From a managerial perspective, we note that the optimal maximum wait-to times would need to be communicated *during* the operation of routes as they are dependent on the realized service request and completion times. We encourage future research to incorporate optimal wait-to times in the route-generating subproblem.

We consider a simplified setting, in which only one service request is considered for each operational period. An opportunity for further research may be to consider multiple potential requests at the same customer location such that the volume and timing of requests should be considered by the route planner.

We note here that the disutility function g scales latency in a way that affects the dominance procedures of the forward- and backward-labeling algorithms differently. For example, applying the forward rules, the function g may be computed many more times depending on the size of $\mathcal{E}_p$. g may be some abstraction of the true cost to the operating firm, and therefore, a trade-off may exist between the quality of this abstraction and the computational expense required to compute it. We believe that future research could investigate the approximation versus computation expense trade-off between different specifications of g.

Finally, we believe that future research should consider the possibility of scheduling unvisited customers who have requested service before the end of the operational period but after the completion of an a priori route. If an agent finishes their route with some time remaining, some unvisited customers may be serviced before the next operational period. Specifically, we believe that an interesting direct extension to our work

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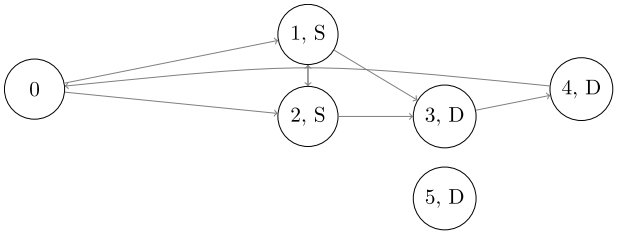
would allow for the scheduling of stochastic request customers in multiple positions in a route such that they may be revisited if they have not requested service when addressed by a service agent.

Appendix A. Proof of Proposition 1

We demonstrate the insufficiency of the traditional dominance rules (4) in forward labeling within the context of our gTRP through a counterexample. Let $g(\cdot) = \cdot$. Let the idling rule be maximum wait-to time. Consider five customers 1, 2, 3, 4, and 5, where customers 1 and 2 are stochastic customers and customers 3, 4, and 5 are deterministic customers, with (x, y) coordinate locations $(x_1, y_1) = (10, 2), (x_2, y_2) = (10, -1), (x_3, y_3) = (15, -2), (x_4, y_4) = (20, 0)$, and $(x_5, y_5) = (15.1, -2)$ served from depot $(x_0, y_0) = (0, 0)$. Let $p = [0 - 2 - 1 - 3]$ and $p' = [0 - 1 - 2 - 3]$ be two partial paths (labels) residing at customer 3. Let the service request time distribution of one be normal with small variance: $\tau_1 \sim N(10, 1)$; let the service request time distribution of two be uniform: $\tau_2 \sim U(0, T)$, where $T = 50$. Then, (4a), (4d), and (4b) are true (i.e., $n(p') = n(p) = 3$, $w(p') = 18.29 < 19.45 = w(p)$, $\Pi(p') = \Pi(p) = \{1, 2, 3\}$, and $\tilde{u}(p') = 90.23 < 95.48 = \tilde{u}(p)$; calculated using Algorithm 1). A graph of this problem is shown in Figure A.1. The visit realizations of p are $[0 - \ - \ - 3 - 0']$, $[0 - 2 - \ - 3 - 0']$, $[0 - \ - 1 - 3 - 0']$, and $[0 - 2 - 1 - 3 - 0']$. The visit realizations of p' are $[0 - \ - \ - 3 - 0']$, $[0 - \ - 2 - 3 - 0']$, $[0 - 1 - \ - 3 - 0']$, and $[0 - 1 - 2 - 3 - 0']$.

Next, consider the possible extension to customer 4 as shown in Figure A.1. Given this extension to the two paths p and p', the increase in total expected latency (having additional customers on a path always increases total expected latency) may be greater for p' than for p. Note that the addition of customer 4 reduces the maximum wait-to time for customer 1 on both p and p', making the visit of customer 1 less likely on both routes. The maximum wait-to time for customer 2 is also reduced, but because the service request distribution for two is uniform, the probability of visiting customer 2 is reduced less than the probability of visiting customer 1. The service agent operating p is, therefore, more likely to be waiting at customer 2 or the depot when time is equal to the last request time for customer 1, and the same agent operating p' is more likely to be waiting at the depot. Then, the service agent on p is more likely than the service agent on p' to visit customer 2 *and* to reach customer 3 at earlier times. This is reflected in the expected latency of each partial path (now $p \oplus 4 = [0 - 2 - 1 - 3 - 4]$ and $p' \oplus 4 = [0 - 1 - 2 - 3 - 4]$). In fact, in our example, the expected latency of p is now 213.68 and less than 215.42 for p'. Further, in Figure A.2(a), we show the evolution of the expected latency of the two partial paths p and p' following three possible extensions [5], [6], and [5, 6], illustrating that the advantage in expected latency is reverted back to the service agent on p' if the labels are extended to customer 5. Therefore, although based on (4a) – (4b), p' dominates p; this example shows that those conditions are not sufficient to ensure the dominance of a label p' over p.

Figure A.1. Zero Indicates Depot, S Indicates Stochastic Customer, and D Indicates Deterministic Customer



Note. Arcs do not represent travel time.

Remark A.1. The impact of a shift δ in a customer's wait-to time is contingent on the customer's current wait-to time position within their service request time distribution. Figure A.2(b) illustrates this relationship for a normally distributed request time. When the current wait-to time is near the expected request time (or the distribution's mode), a reduction of δ time units in the wait-to time may yield a more pronounced decrease in the cumulative probability of receiving a service request by that time compared with when the current wait-to time is farther from the expected request time (compare the blue area against the orange area in Figure A.2(b)).

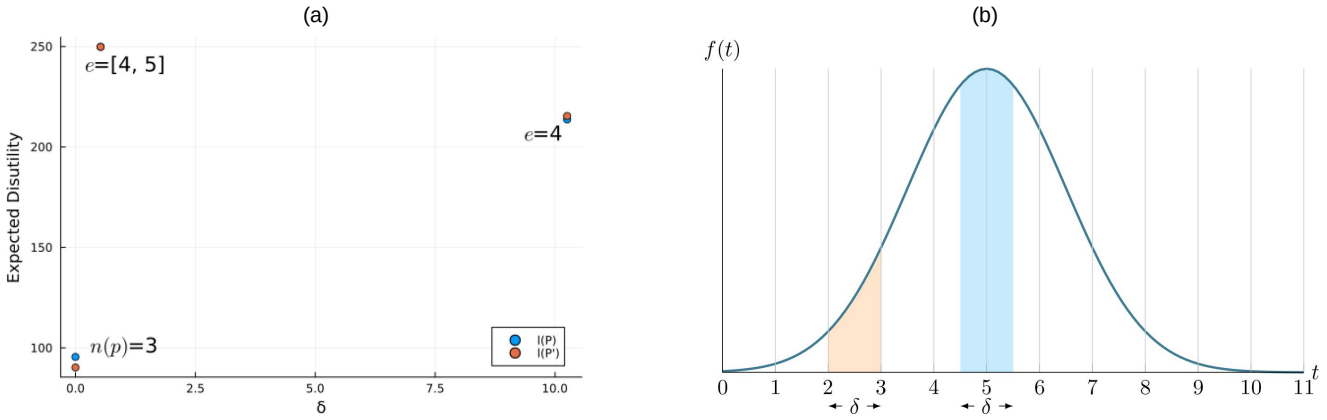
Appendix B. Rational Behind AFD

A valid rule for dominance of a partial path p' over p ensures that any common completion e of the two partial paths results in $\tilde{u}(p' \oplus e) \leq \tilde{u}(p \oplus e)$. We ensure that any feasible extension for p is also feasible for p' by Conditions (4a), (4b), and (4c). We note that for any stochastic customers visited on p and p', an extension e alters their maximum wait-to times. Specifically, the effect of an extension to any partial path p is a subtraction from all maximum wait-to times $\overline{\eta}_c$ for all stochastic customers $c \in p$ and all visit realizations v of p (the service agent has less time to wait because at least one additional customer is scheduled for later in the operational period).

Let δ_e be the amount of shift in the maximum wait-to times of stochastic customers on p and p' as a result of an extension e (a common extension from $n(p) = n(p')$ will result in the same shift from the current maximum wait-to times of stochastic customers on both partial paths). Because partial paths p and p' may have visited different sets of stochastic customers with different service request time distributions, a shift δ_e in their maximum wait-to times can result in different increases in their expected latency (recall the example). It is worth mentioning that even if the sets of stochastic customers of the two partial paths coincide based on Remark A.1 because such customers may be scheduled at different times in the two paths, a common shift δ_e may result in different increases in their expected latency. We refer to this behavior as nonmonotonic resource consumption. Therefore, for p' to dominate p for any common feasible extension e, we must have that any possible shift δ_e in the maximum wait-to times of customers currently on either label preserves the current superiority of p' over p in terms of the reduced cost (i.e., $\overline{u}(p' \oplus e) \leq \overline{u}(p \oplus e)$).

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Figure A.2. (Color online) The Effect of a Shift δ in the Wait-to Time



Notes. (a) Comparison of latencies for labels p and p' for three possible extensions e [$\in$] {[4], [5], [4, 5]}. (b) The effect of a shift δ in the wait-to time of a customer depending on their current wait-to time.

Consider an extension $e = [e(1), \cdots, e(n_e)]$. Extending partial paths p and p' along e will result in a shift

$$\delta_e = t_{n(p),e(1)} + \sum_{k=1}^{n_e-1} t_{e(k),e(k+1)} + t_{e(n_e),0'} - t_{n(p),0'}$$

to the maximum wait-to times of the stochastic customers of partial paths p and p'. The minimum shift among all possible extensions e is, therefore, the shortest path to the depot $0'$ from the current vertex through a reachable customer $\hat{c}$: that is, $\delta^{\min} = \min_{\hat{c}\in\mathcal{C}\setminus\Pi(p)}\{t_{n(p),\hat{c}} + t_{\hat{c},0'} - t_{n(p),0'}\}$. The next smallest shift to maximum wait-to times following the extending to $\hat{c}$ is calculated similarly as

$$\iota = \min_{c_1\neq c_2 c_1, c_2\in\mathcal{C}\setminus\{\Pi(p)\cup\{\hat{c}\}\}} \{t_{c_1,c_2} + t_{c_2,0'} - t_{c_1,0'}\}.$$

An upper bound on the shift to current wait-to times is given by the time remaining through the end of the operational period for the longer partial path (i.e., $\delta^{\max} = T - w(p)$). We can approximate the effect of δ_e by considering a sampling of potential shifts over the interval $[\delta^{\min}, \delta^{\max}]$ with a step size ι (i.e., $\delta \in [\delta^{\min}, \delta^{\min} + \iota, \delta^{\min} + 2\iota, \cdots, \delta^{\max}]$ to measure the difference $\overline{u}(p \oplus e_\delta) - \overline{u}(p' \oplus e_\delta)$, where e is a real or fictive extension resulting in a shift δ to the maximum wait-to times). If there exists any δ for which the difference is negative, p' cannot dominate p. Additionally, we must have that the expected arrival time to $e(1)$ is at least as early for p' as for p for any of their visit realizations. To approximate this, we must consider the expected arrival time to each possible extension $e(1) \in \mathcal{C}\setminus\Pi(p)$ for each of $\delta \in [\delta^{\min}, \delta^{\min} + \iota, \delta^{\min} + 2\iota, \cdots, \delta_{\max}]$.

Appendix C. Proof of Proposition 3

In the general setup of Figure 2, a valid rule for dominance of a partial backward path p' over p ensures that any common completion of these paths (i.e., a forward path $p_0 = [0 - \cdots - n(p_0)]$) results in $\overline{u}(p' \oplus p_0) \leq \overline{u}(p \oplus p_0)$. Given Conditions (4a), (4b), and (4c), we have that any feasible extension to p is also feasible to p'. Given that we are only considering common extensions, the change with respect to dual variables is the same for each.

The current advantage of p' over p has been calculated assuming that $n(p) = n(p')$ is traveled to from the depot (zero) at time 0. We must consider how this assumption is violated and what the consequences for that violation are. We have four general cases to consider. (1) The customers on p_0 and p are all deterministic. (2) Some customer on p_0 is stochastic request, and all customers on p are deterministic. (3) All customers on p_0 are deterministic, and some customer on p' is stochastic request. (4) Some customer on p_0 is stochastic request, and some customer on p' is stochastic request.

Under case (1), when all customers are deterministic, the departure point and service agent release time from p_0 are identical for an extension to both p and p'. Under case (2), given some stochastic request customer on p_0 and the potentially different wait-to times that result from the different resource usages of p and p' ($w(p)$ and $w(p')$, respectively), we have that the service agent may be released at different times and from different points along p_0. Specifically, we must note that the wait-to times of the stochastic request customers on p_0 are shifted earlier by $w(p) + t_{\hat{n}(p_0),n(p)}$ and $w(p') + t_{\hat{n}'(p_0),n(p')}$, respectively, where the former is by definition at least as great a shift (note that $n(p) = n(p')$). In scenarios where those stochastic customers will be skipped, this leads to an earlier release time from p_0 to p than to p' as seen in the example. This may lead to an advantage for p in some scenarios. However, in the worst case for p', this shift is the difference in the current resource usage $w(p) - w(p')$ for each of the customers who are common to both p and p', with any customers $p \setminus p'$ still having positive latency. If $n(p_0) \in \mathcal{D}$, then the departure point from p_0 and the travel time to $n(p)$ and $n(p')$ are the same. However, if $n(p_0) \in \mathcal{S}$, then $n(p_0)$ may or may not be serviced when p_0 is extended to p or p'; therefore, the travel times may be different, but because of the maximum wait-to time rule, the arrival time of the service agent to $n(p) = n(p')$ is the same.

Under cases (3) and (4), some stochastic request customer exists on p'. In case (3), the service agent release time from $n(p_0)$ is the same on an extension to both p and p'. The wait-to time of any stochastic request customer on

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p or p' is unchanged, and therefore, the probability of visiting any customer on p or p' is unchanged. This must mean that the probability of any visit realization (see Online Appendix EC.1) is unchanged. The earliest arrival time of each customer on each visit realization for each partial backward path has increased equally by $w(p_0)$. This may become an advantage for p. To see this, consider the subcase such that $n(p) \in \mathcal{S}$ with a distribution of service request times tightly distributed around some point in time later than $t_{0,n(p)}$. Suppose also that a large proportion of scenarios results in $n(p)$ being skipped and that $n(p_0)$ is near to the customer scheduled after $n(p)$ and far away from the customer scheduled after $n(p')$. Then, the agent will be able to quickly visit the next customer on p but not the next customer on p'. The agent then has perfectly used the time available, and the next customer on p may actually be visited earlier than before the extension to p_0 (i.e., an empty $p_0 = [0]$). The savings in latency because of this earlier visit time is passed on to all customers on p. The same potential improvement of p over p' follows for case (4).

Now, given the average disutility over all scenarios for p, each calculated with the best case for p, as $\underline{U}(p)$, we say that p' dominates p only if $\overline{u}(p') \leq \underline{U}(p)$.

Appendix D. Results on Instances from Luo, Qin, and Lim (2014)

In each row of Table D.1, we report the total latency (objective value), computational time, and number of instances (of 10) that we solve with our implementation of branch and price. In all instances, we find the optimal solution. This analysis serves as validation of our approach.

Table D.1. Replicated Objective Values and Reported Solution Times Using SSP

Instance	$\lvert\mathcal{C}\rvert$	Latency	Time	Luo, Qin, and Lim (2014)	Nucamendi-Guillén et al. (2016)	Muritiba et al. (2021)
brd14051	29	85,093.54	6.30	8.80	4.66	9.41
d15112	29	229,130.79	7.77	25.20	9.58	3.67
d18512	29	84,818.95	7.57	11.60	6.85	2.64
fnl4461	29	48,287.50	7.76	6.30	2.83	3.65
nrw1379	29	29,994.97	17.56	21.80	5.83	9.90
pr1002	29	164,763.90	9.34	12.20	8.09	4.76
brd14051	39	106,145.48	79.28	83.20	30.37	43.49
d15112	39	290,748.64	23.21	27.80	24.96	4.88
d18512	39	107,813.36	28.24	24.20	11.74	8.21
fnl4461	39	62,672.49	164.46	73.40	17.21	21.92
nrw1379	39	36,379.77	46.86	66.60	20.66	12.54
pr1002	39	217,071.57	164.87	209.60	52.12	41.26
brd14051	49	127,480.58	320.96	259.10	51.54	73.29
d15112	49	356,433.93	1,343.02	78.70	53.75	33.37
d18512	49	132,115.39	192.71	213.13	75.64	49.13
fnl4461	49	76,642.75	53.73	53.20	32.11	20.14
nrw1379	49	45,169.23	1,212.37	1,184.20	79.66	189.44
pr1002	49	268,605.26	170.09	94.50	60.70	36.73

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Appendix E. Detailed Results of the Factorial Experiments

Table E.1. SSP Factorial Experiment Results with Disutility Calculated from the Out-of-Sample Test Set

Instance	$\lvert\mathcal{C}\rvert$	$\lvert\mathcal{S}\rvert$	Latency	Time	Solved
brd14051	29	7	96,793.15	72.75	10
brd14051	29	15	110,473.32	281.71	10
brd14051	39	10	122,555.96	397.30	9
brd14051	39	20	159,915.43	1,155.69	10
brd14051	49	12	151,991.44	753.06	10
brd14051	49	25	179,128.34	1,566.85	8
d15112	29	7	281,194.67	43.10	10
d15112	29	15	301,543.56	80.31	10
d15112	39	10	330,320.97	236.53	10
d15112	39	20	426,861.65	785.23	10
d15112	49	12	462,029.68	740.82	9
d15112	49	25	506,025.10	2,010.13	10
d18512	29	7	93,416.10	61.58	10
d18512	29	15	119,559.64	114.46	10
d18512	39	10	124,699.33	248.52	10
d18512	39	20	153,191.03	801.02	10
d18512	49	12	156,748.07	1,224.98	10
d18512	49	25	177,730.34	1,783.80	9
fnl4461	29	7	57,605.13	89.02	10
fnl4461	29	15	67,923.33	65.60	10
fnl4461	39	10	75,175.63	168.10	9
fnl4461	39	20	87,939.30	1,424.66	10
fnl4461	49	12	96,773.17	906.72	9
fnl4461	49	25	105,715.11	1,731.04	8
nrw1379	29	7	36,829.83	32.25	10
nrw1379	29	15	48,993.82	116.04	10
nrw1379	39	10	43,904.85	147.65	10
nrw1379	39	20	54,718.94	784.25	10
nrw1379	49	12	53,378.81	775.98	10
nrw1379	49	25	73,792.59	3,509.34	8
pr1002	29	7	194,528.09	33.79	10
pr1002	29	15	225,817.90	141.99	10
pr1002	39	10	263,944.45	154.07	10
pr1002	39	20	292,360.64	459.24	10
pr1002	49	12	311,683.65	1,080.64	10
pr1002	49	25	341,926.24	1,835.25	8

Note. Time limit = 7,200 seconds.

Table E.2. SSP Large Instance Experiment Results with Disutility Calculated from the Out-of-Sample Test Set

Instance	$\lvert\mathcal{C}\rvert$	k	$\lvert\mathcal{S}\rvert$	Latency	Time
E-n22-k4	22	4	5	995.05	9.3
E-n23-k3	23	3	5	1,691.03	102.06
E-n30-k3	30	3	7	2,166.76	time-lim
E-n33-k4	33	4	8	3,341.59	time-lim
E-n51-k5	51	5	12	2,598.75	time-lim
E-n76-k7	76	7	19	3,484.63	time-lim
E-n76-k8	76	8	19	3,318.16	time-lim
E-n76-k10	76	10	19	2,790.47	time-lim
E-n76-k14	76	14	19	2,424.21	2,200.53
E-n101-k14	101	14	25	4,009.6	time-lim
P-n16-k8	16	8	4	815.03	7.16
P-n19-k2	19	2	4	813.66	710.36
P-n20-k2	20	2	5	976.6	1,143.73
P-n21-k2	21	2	5	1,149.59	time-lim

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Table E.2. (Continued)

Instance	$\lvert\mathcal{C}\rvert$	k	$\lvert\mathcal{S}\rvert$	Latency	Time
P-n22-k2	22	2	5	972.89	time-lim
P-n22-k8	22	8	5	770.55	1.0
P-n22-k8	22	8	5	770.55	1.0
P-n23-k8	23	8	5	632.19	1.18
P-n40-k5	40	5	10	1,596.74	time-lim
P-n45-k5	45	5	11	2,189.67	time-lim
P-n50-k7	50	7	12	1,763.48	time-lim
P-n50-k8	50	8	12	1,427.84	2,005.59
P-n50-k10	50	10	12	1,710.85	741.51
P-n51-k10	51	10	12	1,648.98	461.1
P-n55-k7	55	7	13	1,719.96	time-lim
P-n55-k8	55	8	13	1,661.88	time-lim
P-n55-k10	55	10	13	1,735.51	4,063.72
P-n55-k15	55	15	13	1,551.99	80.28
P-n60-k10	60	10	15	2,355.14	time-lim
P-n60-k15	60	15	15	1,504.47	88.73
P-n65-k10	65	10	16	2,398.84	time-lim
P-n70-k10	70	10	17	2,292.31	time-lim
P-n76-k4	76	4	19	6,687.59	time-lim
P-n76-k5	76	5	19	4,629.35	time-lim

Note. Time limit (time-lim) = 7,200 seconds.

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5. 销售人员激励：Optimal Incentives for Salespeople with Learning Potential (MS, 2023)

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Optimal Incentives for Salespeople with Learning Potential

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Abstract. We study a compensation problem for salespeople with learning potential. In our model, both the firm and sales agent are risk neutral and forward-looking; the agent can privately observe his skill, exert effort, and learn from experience; the firm can learn from the agent's choice and revise sales targets over time. The problem entails a dynamic tradeoff between exploiting learning, screening information, and maximizing efficiency. We find the optimal compensation plan differs substantially from the existing ones: it sets aggressive targets for expediting skill development, and pays the information rent for neutralizing the agent's misbehaving temptation over the entire relationship. We find learning drives the long-run outcomes; ignoring it can mislead compensation design and inflict substantial losses. Our results shed light on when and why firms distort sales, favor incumbents, and prefer long-term plans. By highlighting the critical role of learning in long-run performance, this study advances our understanding of salesforce theory and practice.

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1. Introduction

When it comes to selling, skill matters as much as effort. A rookie may be highly motivated, but he often lacks the skill to match the productivity of experienced colleagues. To catch up, he must hone his skills, for example, to extract information (Chen 2005), negotiate price (Mishra and Prasad 2005), close sales (Lo et al. 2016), generate new business (Kim et al. 2019), reduce customer attrition (Rubel and Prasad 2015), and learn from experience (Misra et al. 2004).

Practice makes perfect. Through learning, salespeople can develop skills and improve productivity over time (Syverson 2011). Indeed, the very act of selling increases experience and reveals opportunities for improvement. For example, in cosmetics sales, a new hire can increase sales by 2%–7% in the first week, through experience- and peer-based learning (Chan et al. 2014).[1] In general, salespeople's current efforts and experience can have persistent impact on their future productivity, through learning firm-specific skills (Lazear 1998).

Salespeople are the eyes and ears of the firms they serve. Through selling, they also learn new information arising over time. For example, because of experience and proximity, they gain superior information on evolving local conditions and sales prospects (Lal and Staelin 1986). If truthfully shared, this information can help improve forecasting, enhance service, and increase productivity. Yet without proper incentives, salespeople may abuse it for personal gain (Kim et al. 2019).

The importance of salesforce learning has long been documented in the empirical literature. Yet the theoretical literature is largely silent on how to incentivize such salespeople. In this paper, we seek to understand how learning affects compensation design. We address three questions: (i) What is the optimal compensation plan? (ii) How does it perform in the long run? (iii) What are the managerial implications of salesforce learning?

We model the employment relationship as a dynamic game of incomplete information. At the outset, the firm offers the agent an incentive plan that pays for performance. Both parties are risk-neutral and forward-looking. They can learn over time: each period the agent observes new information, exerts effort, and learns from experience; from the agent's sales and compensation choice, the firm can infer his private information, update its belief, and revise sales targets. The sales are random and dynamic, driven by agent effort, selling skill, and market shocks. Both effort and skill are the agent's private information, unobservable to the firm.

The problem is challenging for two reasons. First, the relationship suffers both moral hazard and adverse selection problems. Because neither effort nor skill is observable, the agent can shirk effort, misreport information, and game the firm. Hence, the firm must deter misbehaving and ensure truth-telling. Second, learning and shocks change the agent's skill and preference over time. The skill change may either come exogenously from the random environment, or

come *endogenously* from learning by doing. As his skill changes, the agent's preference will change, too. Hence, the firm must dynamically adjust the quota and payment based on the new information, taking into account how current adjustment affects future skills and sales. Technically, the game entails multidimensional private information and dynamic incentive constraints—a nontrivial problem (Rust and Staelin 2011).

Leveraging risk neutrality, we solve the problem and make three contributions. The first is to specify the optimal plan. The optimal plan resolves a dynamic tradeoff between exploiting learning, screening information, and maximizing efficiency. (i) To exploit learning, it sets aggressive quota and pay. This incentivizes the agent to work harder, sell more, and thus expedite skill development. (ii) To screen information, the optimal plan deploys dynamic information rent. The rent is the agency cost for neutralizing agent's misbehaving temptation during the entire relationship; it is generous enough so that the agent finds honesty pays. (iii) To maximize efficiency, the optimal plan sets higher quota and pay for higher skilled, and adjust them dynamically over time. The pay differentiation ensures that each agent is willing to perform to the best of his ability. The dynamic adjustment helps the firm to screen fresh information, tailor sales targets to actual skill level, thereby adapting to changing conditions. (iv) Despite the complex tradeoff involved, the optimal plan has a simple implementation with salary, quota, and commission.

Our second contribution is to characterize the role of learning. We find learning can substantially change incentive structure, skill distribution, and sales performance. Indeed, the optimal plan internalizes information friction and learning potential, by trading the short-term cost overrun for the long-term learning gain. In the short run, it resembles the second-best, because information friction is still severe; in the long run, it gravitates toward the first-best (adjusted for learning), because information friction vanishes but learning persists; the pace of the transition depends on the skill carryover rate. Importantly, learning drives the long-run outcomes: the initial type distribution—the main drive of the classic second-best—is irrelevant in the long run.

Our third contribution is to provide practical guidance. The results help explain when and why firms distort sales, favor incumbents, and prefer long-term plans. More importantly, we quantify when and why ignoring learning can mislead compensation design and inflict substantial losses. A regular employment relationship is usually long-term, involving *two-sided learning* (Waldman 2012). Through repeated interactions, the agent can learn the firm-specific knowledge, while the firm can also learn the agent-specific information (type), thereby reducing information friction.[2] If the compensation is properly structured, the information friction will be short-lived, and the firm only needs to pay information rent for initial private information. When learning is salient, the product is lucrative, and the relationship is durable, ignoring learning can inflict substantial losses.

1.1. Related Literature

The salesforce compensation literature builds on agency theory.[3] The central theme is how to design incentives to motivate efforts and reduce agency costs.[4] The basic idea is simple: to induce unobservable effort or elicit private information, firms should reward good sales outcomes and punish bad ones. This requires a payoff *wedge* between outcomes to ensure incentive compatibility. Enforcing the wedge may require rent payment, beyond selling costs. Three basic frameworks are: (i) risk-aversion (e.g., Basu et al. 1985; Mishra and Prasad 2004, 2005; Caldieraro and Coughlan 2009), (ii) limited liability (e.g., Oyer 2000, Dai and Jerath 2013, Simester and Zhang 2014), and (iii) private information (Lal and Staelin 1986; Rao 1990; Mantrala et al. 1994; Mishra and Prasad 2004, 2005; Simester and Zhang 2014). To control rent, firms can distort sales effort and quota. A main insight from this literature is that, the first-best is generally untenable—firms should pay agency rents and distort output downward. We show this insight may not hold when salespeople can learn over time.

A few recent papers study dynamic incentives. For example, Rubel and Prasad (2015) examine how risk aversion affects optimal incentives; Kräkel and Schöttner (2016) and Schöttner (2016) investigate when commissions dominate bonuses; Jerath and Long (2020) study when two-period contracts dominate spot ones. These studies focus on homogeneous salesforces with symmetric information. By contrast, we study how to compensate heterogenous salesforces with asymmetric information. We show context matters: the optimal compensation differs substantially from the existing ones, in structure and performance.

Extensive empirical studies reveal that learning is critical to salesforce productivity; see, for example, Sujan et al. (1994), Kohli et al. (1998), Wang and Netemeyer (2002), Misra et al. (2004), Ahearne et al. (2010), Bell et al. (2010), Chan et al. (2014), Lu et al. (2015), Lo et al. (2016), and Tan and Netessine (2019). Yet in the theoretical salesforce literature, only Schöttner (2016) considers learning under full information. Our work is the first to study how learning affects salesforce compensation, when private information evolves over time.

2. Model

A firm hires a sales agent-z_0 to sell a product over $T \leq \infty$ periods. At the outset, the firm commits to a

compensation system $\pi = (\pi_t)_{t\geq 0}$, where each $\pi_t = (p_t, q_t)$ specifies sales quota q_t and payment p_t for period t. In period t, the agent privately observes skill z_t and selects quota q_t; he then exerts effort a_t at cost $a_t^2/2$, producing sales

$$s_t = q_t + \epsilon_t = z_t + a_t + \epsilon_t,$$

subject to random shock $\epsilon_t \sim G$. Once sales figure s_t is in, the firm gets revenue $m \cdot s_t$ and pays the agent p_t, based quota q_t and sales s_t. The agent's skill then evolves to z_{t+1}, and the stage game repeats. Both parties are strategic, risk-neutral, and forward-looking, with discount factor $\delta \in (0,1)$. The skill z_t and effort a_t are the agent's private information, *unobservable* to the firm; all other parameters are common knowledge. Let $z^t \equiv (z_0, z_1, \ldots, z_t)$, and $z_s^t \equiv (z_s, z_{s+1}, \ldots, z_t)$. We call the agent at history z^t the agent-z^t.

2.1. Dynamic Learning Process

Salespeople differ in selling skill $z_t \in \mathcal{Z} \equiv [\underline{z}, \bar{z}]$. For the same effort a_t, agents with higher skill z_t generate higher average sales.[5] The skill z_t can change over time, for three reasons. (i) The skill can improve, because of learning from experience: the more time the agent spends selling a particular product, the more skilled he becomes. (ii) The skill can erode, because of random effects; for example, forgetting, technological changes, and market fluctuations. (iii) The skill can persist, because of the carryover effect: a high-skill agent is more likely than his low-skill peers to remain high skilled in the next period.[6]

Formally, the skill evolves according to $z_{t+1} = \alpha z_t + \beta q_t + \omega_{t+1}$. It determines the next period skill z_{t+1} by current skill k_t, endogenous learning (from selling q_t), and exogenous random shock $\omega_{t+1} \sim \mathcal{N}(\mu_\omega, \sigma_\omega^2)$. Parameter α is the *carryover rate*, which measures the marginal effect of current skill z_t on future z_{t+1}. Parameter β is the *learning rate*, which measures the marginal effect of current experience q_t on future skill z_{t+1}. Given (z_t, q_t), the next skill $z_{t+1} \sim \mathcal{N}(\alpha z_t + \beta q_t + \mu_\omega, \sigma_\omega^2)$. Depending on effort and luck, the skill can go either up or down over time. We assume σ_ω is sufficiently small, so that z_t is bounded almost surely. We focus on *regular* skill processes, with $0 \leq \alpha < 1$ and $0 \leq \beta < \infty$. Intuitively, $\alpha \geq 0$ captures skill carryover, $\alpha < 1$ implies skill erosion, $\beta \geq 0$ models learning from experience, and $\beta < \infty$ suggests learning saturation.[7] This model captures a rich set of learning processes; for example, deterministic learning, learning from experience, and outcome-based learning.[8] For general learning processes $z_{t+1} = Z(z_t, q_t, \omega_{t+1})$ with correlated shocks $(\omega_t)_{t\geq 1}$, we can take the stochastic-difference-equation approach (Stachurski 2009).

2.2. Compensation Design

By the revelation principle, we can focus on direct, truth-telling mechanisms: in equilibrium, it is optimal for each agent-z^t to self-select the quota q_t of his caliber, exert the requisite effort $a_t = q_t - z_t$ to hit the quota in expectation, and incur *selling cost* $\psi(q_t, z_t) = \frac{1}{2}(q_t - z_t)^2$. Next, we specify how the agent responds to plan π. First, the agent is forward-looking. Given past choice $\hat{z}^{t-1}$, the agent-z^t takes action $\hat{z}_t$ to maximize his continuation payoff:

$$\tilde{U}_t(\hat{z}^t; z_t) = \mathsf{E}[p_t(\hat{z}^t, q_t(\hat{z}^t) + \epsilon_t)] - \frac{1}{2}(q_t(\hat{z}^t) - z_t)^2 + \delta \mathsf{E}[U_{t+1}(\hat{z}^t, z_{t+1}) \,|\, z_t, q_t(\hat{z}^t)], \tag{1}$$

where $U_t(z^t) \equiv \tilde{U}_t(z^t; z_t)$ is his continuation payoff under the *truth-telling* strategy ($\hat{z}_\tau = z_\tau$, $\forall \tau$).

Second, the agent is individual rational. He would sign the contract only if it is more profitable than his outside option (normalized to 0):

$$U_0(z_0) \geq 0, \quad \forall z_0. \tag{IR}$$

Third, the agent is strategic. Whenever possible, he pursues self-interest, even at the firm's expense. In response, the firm must ensure incentive compatibility: truth-telling must be the agent's best response in each period, that is,

$$U_t(z^t) = \max_{\hat{z}_t \in \mathcal{Z}} \tilde{U}_t(z^{t-1}, \hat{z}_t; z_t), \quad \forall z^t. \tag{IC_t}$$

Given the agent's best responses, the firm must factor them into compensation design:

$$V_0 = \max_{\pi \in \Pi} \mathsf{E}^\pi \left[\sum_{t\geq 0} \delta^t (m \cdot (q_t(z^t) + \epsilon_t) - p_t(z^t, q_t(z^t) + \epsilon_t)) \right], \tag{$\mathcal{P}$}$$

where $\Pi \equiv \{(p_t, q_t)_{t\geq 0} : IR, IC_t, \forall t\}$ is the set of all *feasible plans* that respect individual rationality and incentive compatibility. The expectation E^π is taken with respect to stochastic processes of skill and sales, induced by plan π.

In this model, learning is dynamic and two-sided. As time goes by, both-parties learn new information, update posterior beliefs, and forecast the future with increasing precision. In particular, the firm can learn from interacting with the agent: using the incentive plan, it can screen the new information z_t and update its belief accordingly. To focus on learning, we abstract away from other well-known issues, such as risk aversion, turnover, and ratcheting; we discuss them in §5.

3. The Optimal Compensation

We first characterize the optimal plan. The problem has compounding effects. To build intuition, we isolate each effect through four regimes, with increasing complexity in information structure and learning process. For each *r*egime $\mathcal{P}^r$, we use the same payment

format $p_t^r = A_t^r + B_t^r[s_t - q_t^r]$ to implement the optimal plan: in each period t, the firm asks the agent to pick a *quota* q_t^r before engaging in selling; then after the sales figure s_t is in, it pays the agent with the base *salary* A_t^r, plus the *commission* $B_t^r[s_t - q_t^r]$. The rationale for this implementation choice is the *revenue equivalence theorem*.

3.1. Full Information Benchmarks: $\bar{\mathcal{P}}^n$ and $\bar{\mathcal{P}}$

When the firm is omniscient, it can observe skill, verify effort, and dictate pay. Such tight control preempts misbehaving. The central issue is how to set quota to exploit salesforce learning potential. The main trade-off is intertemporal, between current selling cost and future learning gain. Ignoring $(IC_t)_{t\geq 0}$, the problem reduces to: $V_{T+1}(z_{T+1}) = 0$,

$$V_t(z_t) = \max_{q_t \geq 0}\left\{m \cdot q_t - \frac{1}{2}(q_t - z_t)^2 + \delta \mathsf{E}_t V_{t+1}(\alpha z_t + \beta q_t + \omega_{t+1})\right\}, \quad 0 \leq t \leq T, \tag{2}$$

where E_t is the expectation conditional on the information available at time t. Let $v_t \equiv \partial_{q_t} \mathsf{E}_t V_{t+1}$ be the learning gain. We find $v_t = m\beta \cdot \frac{1-[\delta(\alpha+\beta)]^{T-t}}{1-\delta(\alpha+\beta)}$. Also,

Proposition 1.

(a) *In no learning regime $\bar{\mathcal{P}}^n$, the optimal plan $\bar{\pi}^n$ sets*

$$\bar{A}^n = \frac{1}{2}m^2, \qquad \bar{B}^n = m, \qquad \bar{q}^n(z_t) = m + z_t. \tag{$\bar{\pi}^n$}$$

(b) *In learning regime $\bar{\mathcal{P}}$, the optimal plan $\bar{\pi}$ sets*

$$\bar{A}_t = \frac{1}{2}(m + \delta v_t)^2, \qquad \bar{B}_t = m + \delta v_t,$$
$$\bar{q}_t(z_t) = (m + z_t) + \delta v_t. \tag{$\bar{\pi}$}$$

(c) *The optimal quota $\bar{q}_t$ and pay $\bar{p}_t$ increase in profit margin m, carryover rate α, and learning rate β; the learning gain v_t increases in m, α, β, and T.*

Part (a) is the *myopic first-best* $\bar{\pi}^n$. In regime $\bar{\mathcal{P}}^n$, the agent has exogenous skill $z_{t+1} = \alpha z_t + \omega_{t+1}$; his action q_t today has no bearing on the future. So current quota q_t has only short-term consequence. In each period, the firm sets the quota to equalize marginal revenue m with marginal cost ψ_q; on average, it only pays actual selling cost $m^2/2$, but extracts the entire surplus.

Part (b) characterizes the *dynamic first-best* $\bar{\pi}$. In regime $\bar{\mathcal{P}}$, the agent has endogenous skill $z_{t+1} = \alpha z_t + \beta q_t + \omega_{t+1}$. By selling more today, he suffers short-term cost increase, but more sales also brighten his future: he can improve future skill (at rate β), reduce future cost ($\psi_z \leq 0$), and boost future profit ($v_t = \partial_{q_t} \mathsf{E}_t V_{t+1} \geq 0$). As such, current action q_t has both short- and long-run consequences.

To exploit learning, the firm must balance the short and long run. The resulting quota $\bar{q}_t$ has a simple geometric structure, with upward adjustment $\bar{q}_t - \bar{q}^n = \delta v_t = \delta m\beta \cdot \frac{1-[\delta(\alpha+\beta)]^{T-t}}{1-\delta(\alpha+\beta)}$. This structure arises as the present value of the future learning gains: the longer the relationship, the higher the learning gain. The takeaway is, learning is best cultivated in a durable relationship.

3.2. Asymmetric Information Without Learning: $\mathcal{P}^n$

When the firm can neither observe skill nor monitor effort, the *information friction* arises. The agent may abuse his information superiority for personal gain. To keep a myopic agent honest, the literature prescribes two measures: pay rents and cut quota (Rao 1990). The optimal quota q^c is determined by the trade-off across type, between reducing rents for high skilled and losing sales from low skilled.

These prescriptions, however, may not work for a forward-looking agent. In dynamic regime $\mathcal{P}^n$ ($\beta = 0$), the agent has more leeway to misbehave; his actions have both short- and long-term effects. Motivating him becomes much harder: the firm must ensure *sequential* incentive capability $(IC_t)_{t\geq 0}$, across both type and time. The central issue is how to deter misbehaving over time. Let $\hbar(z_0) \equiv \frac{1-\Lambda(z_0)}{\lambda(z_0)}$, $U_t(z^t) = \kappa_t(z_0)(z_t - \underline{z})$, and $\kappa_t(z_0) = m \cdot \frac{1-(\delta\alpha)^{T-t+1}}{1-(\delta\alpha)} - \hbar(z_0)\alpha^t \cdot \frac{1-(\delta\alpha^2)^{T-t+1}}{1-\delta\alpha^2}$. We find:

Proposition 2.

(a) *In regime $\mathcal{P}^n$, the optimal plan π^n sets*

$$A_t^n(z^t) = \frac{1}{2}(m - \hbar(z_0)\alpha^t)^2 + U_t(z^t) - \delta \mathsf{E}_t U_{t+1}(z^{t+1}),$$
$$B_t^n(z_0) = \kappa_t(z_0) - \delta\alpha\kappa_{t+1}(z_0),$$
$$q_t^n(z^t) = (m + z_t) - \hbar(z_0) \cdot \alpha^t.$$

(b) *In regime $\mathcal{P}^n$, the information rent for agent-z_0 is*

$$U_0(z_0) = \frac{m[1-(\delta\alpha)^{T+1}]}{1-\delta\alpha} \cdot (z_0 - \underline{z}) - \frac{1-(\delta\alpha^2)^{T+1}}{1-\delta\alpha^2} \cdot \int_{\underline{z}}^{z_0} \hbar(z)\, \mathrm{d}z.$$

Under information asymmetry, the firm can pay information rent and distort quota to ensure truth-telling. For a myopic agent, the optimal plan π^c considers only *type*. For a forward-looking agent, the plan must also consider *time*: not only does agent-z_0 enjoy the cost advantage today, but also he is more likely to enjoy it in the future. To keep him honest, the firm must also pay for those future savings, in *every* period t affected by initial z_0. As such, the total rent $U_0(z_0)$ is dynamic: it must *price in* all the misbehaving opportunities across type and time.

The dynamic rent payment has a simple structure. To see why, we observe at each z^t, the skill z_0 has the residual effect α^t that confers cost advantage $(q_t^n - z_t)$, resulting in cost saving $(q_t^n - z_t)\alpha^t\, \mathrm{d}z_0'$ at time t. Taking all such savings together, the firm must pay agent-z_0 with $\mathsf{E}[\sum_t \delta^t (q_t^n - z_t)\alpha^t \,|\, z_0'] \cdot \mathrm{d}z_0'$ more rent—the sum of

carryover α^t-weighted cost savings agent-z_0 can expect from imitating lower agent-z_0' over time. There are $[\underline{z}, z_0)$ such lower skill agents. So the firm must pay

$$U_0(z_0) = \int_{\underline{z}}^{z_0} \mathsf{E}[\textstyle\sum_t \delta^t (q_t^n - z_t)\alpha^t \,|\, z_0'] \, \mathrm{d}z_0'.$$

The rent structure then arises as the present value of future cost savings.

The rent structure suggests how to set quota. Crucially, rent $U_0(z_0)$ increases in the quotas $q_t^n(z_0', z_1^t)$ for $z_0' < z_0$. As such, to reduce rent, the firm should cut the quota for lower agent-(z_0', z_1^t).[9] The cut would reduce the efficiency of agent-(z_0', z_1^t), but the firm is willing to take such a costly measure, because it helps reclaim the rents it would otherwise pay to high skilled. What complicates quota cut (distortion) is the dynamic effects of the initial type z_0: they spill over time. In response, the quota cut should also be dynamic: the firm must cut not only the current quota for his lower peers $z_0' < z_0$, but also their future quotas along each path (z_0', z_1^t). The size of the quota cut is $(\bar{q}^n - q_t^n) = \hbar(z_0)\alpha^t$: the higher the initial heterogeneity $\hbar(z_0)$, the stronger the skill carryover α, the earlier the relationship $(t \downarrow 0)$, the deeper the quota cut. However, both quota cut and information rent are costly: they erode efficiency.

3.3. Asymmetric Information with Learning: $\mathcal{P}$

In regime $\mathcal{P}$, the agent enjoys information advantage and can learn from experience. To compensate him, the firm must price in both information friction and learning potential. The two forces in isolation put competing pressures on sales quota: learning boosts quota while information friction depresses it, $q_t^n(z^t) \le \bar{q}^n(z_t) \le \bar{q}_t(z_t)$, $\forall z^t$. When both forces are at work, how should the firm compensate salespeople? Let $v_t = m\beta \cdot \frac{1-[\delta(\alpha+\beta)]^{T-t}}{1-\delta(\alpha+\beta)}$, $U_t(z^t) = \kappa_t(z_0)(z_t - \underline{z})$, and $\kappa_t(z_0) = m \cdot \frac{1-[\delta(\alpha+\beta)]^{T-t+1}}{1-\delta(\alpha+\beta)} - \hbar(z_0)\alpha^t \cdot \frac{1-(\delta\alpha^2)^{T-t+1}}{1-\delta\alpha^2}$. We find:

Proposition 3.

(a) *In regime $\mathcal{P}$, both parties benefit from higher skill*: $\partial_{z_t} U_t(z^t) \ge 0$, $\partial_{z_t} V_t(z^t) \ge 0$.

(b) *The optimal plan π^* sets*

$$\begin{aligned} A_t^*(z^t) &= \frac{1}{2}(m + \delta v_t - \hbar(z_0)\alpha^t)^2 + U_t(z^t) - \delta \mathsf{E}_t U_{t+1}(z^{t+1}), \\ B_t^*(z_0) &= \kappa_t(z_0) - \delta(\alpha+\beta)\kappa_{t+1}(z_0), \\ q_t^*(z^t) &= (m + z_t) + \delta v_t - \hbar(z_0) \cdot \alpha^t. \end{aligned}$$

(c) *In regime $\mathcal{P}$, the optimal quota for top agent-$\bar{z}$ is always efficient, $q_t^*(\bar{z}, z_1^t) = \bar{q}_t(z_t)$, $\forall z_1^t$. The quotas for other agents are bounded by*

$$q_t^n(z^t) \le q_t^*(z^t) \le \bar{q}_t(z_t).$$

(d) *The sales quota increases in skill z_t, profit margin m, and learning rate β, but it can either increase or decrease in carryover rate α.*

In regime $\mathcal{P}$, the optimal plan π^* reconciles two competing pressures—exploit learning and control rent—in a dynamic way. Relative to the firs-best $\bar{q}^n(z_t) = m + z_t$, it employs two distortions: the upward distortion δv_t is productive, driven by the marginal gain δv_t of learning, for the purpose of skill development; the downward distortion $\hbar(z_0)\alpha^t$ is unproductive, driven by precontract information z_0, for the purpose of rent control. Both distortions decrease over time. Depending on their relative strength, the overall distortion $(q_t^* - \bar{q}^n)$ can be either upward or downward, and the direction can change over time.

Proposition 3 provides guidelines on how to compensate salespeople with private information and learning potential. First, *the firm should set aggressive quota to encourage learning*. The extra output is to expedite skill improvement, reduce selling cost, and increase future revenue. Doing so seems wasteful in the short run—inefficiently higher output than the marginal profit m can justify—but the long-run gain δv_t is more than offsetting the short-term loss. The aggressive quota also improves current performance. In the static setting, to control rent, the firm induces too little output. When the firm increases output to encourage learning, it also brings current selling $q_t^*(z^t)$ closer to the efficient level $\bar{q}_t(z_t)$, thus improving current performance.

Second, *the firm should set higher quota for higher skilled* $(\partial_{z_t} q_t^* \ge 0)$. This is because both parties benefit from higher skill (part (a)). To fully use high performers, the firm should motivate them to sell more once they reach normal targets. This helps explain why in practice, removing quota ceilings can often improve performance (Misra and Nair 2011). Third, *the firm should keep the top agent at the efficient level* $(q_t^*(\bar{z}, z_1^t) = \bar{q}_t(z_t), \ \forall z_1^t)$. Doing so leverages his skill and learning potential, thereby maximizing future surplus; although the firm pays information rent, the increase in total surplus is more than offsetting the rent cost.

Fourth, *the firm should calibrate quota to business specifics*. The higher the profit margin m, the higher the learning rate β, the longer the relationship T, the higher the quota $\bar{q}_t$. Intuitively, all these factors enhance the future gain v_t of learning, thereby pushing up the current quota. However, higher carryover rate α has two countervailing effects: it increases both learning gain δv_t and information friction $\hbar(z_0)\alpha^t$. Depending which effect dominates, the net effect of higher α can go either way. As we shall show next, ignoring these guidelines can produce misleading polices and substantial losses.

4. Managerial Insights

We now address three managerial questions: (i) Is the optimal plan robust to random sales? (ii) How does it

perform in the long run? (iii) What are the incentive implications of salesforce learning?

4.1. Is the Optimal Plan Robust to Random Sales?

In problem $\mathcal{P}$, the firm must address three challenges: adverse selection, moral hazard, and random sales. Such complexity often denies tractability, let alone simple implementation. Yet Proposition 3 reveals the optimal solution π^* has a simple implementation with quota, salary, and commission. Why?

The logic is as follows. First, *the firm can leverage quota adjustment to screen private information*. This is imperative because the agent is better informed of evolving skill z_t. To best use his talent, the firm should allow the agent to adjust quota over time, *within* the system π^* of prespecified terms. When the system is well-designed, the agent-z^t would pick the quota $q_t^*(z^t)$ matching his skill; from the quota choice, the firm can then infer his type. Hence, the firm can use dynamic quota adjustment to screen private information $(z_t)_{t\geq 0}$ arising over time. This use of quota adjustment is common in practice; see, for example, Gonik (1978) and Chen (2005).

Second, *the firm can use base salary to control adverse selection, and commission to control moral hazard*. Given the quota choice q_t^*, the firm sets type-dependent salary A_t^* to ensure that honesty pays. It then sets commission rate B_t^* so that the agent-z^t is motivated to hit his chosen quota $q_t^*(z^t)$ in expectation. When the payment sensitivity to quota is set to $B_t^* = \kappa_t - \delta(\alpha+\beta)\kappa_{t+1}$, the agent equates his marginal cost of output to its expected marginal benefit, thereby maximizing his continuation payoff. Crucially, the firm must set all the instruments jointly, so that the higher the quota, the higher the salary, the higher the pay. This monotone property keeps the incentives alive even after the high performers reach low targets. When these instruments are properly calibrated, it is impossible for the agent to gain from manipulation.

Still, one may wonder, how do random sales affect performance? In practice, random sales are a headache for compensation design: firms want to pay for actual effort, not luck; yet random shocks garble the measurement so that sales do not reveal precise effort. Depending on luck ϵ_t, the same agent with the same effort can produce drastically different sales $s_t = q_t + \epsilon_t$. The limited compensation instruments further worsen the headache. As such, one may predict profit loss, because both random sales and contractual constraints weaken managerial control. Yet we find:

Proposition 4. *In regime $\mathcal{P}$, the optimal plan π^* is robust to sales randomness.*

In regime $\mathcal{P}$, sales randomness does not hurt efficiency. This is reassuring: using π^*, the firm can see through random noise ϵ_t, tell two random sales $q_t^*(z^t)+\epsilon_t$ and $q_t^*(\hat{z}^t)+\hat{\epsilon}_t'$ apart, and reap the same profit as if sales are deterministic. This insight assumes risk neutrality, which allows the firm to freely shift payment over time, thereby filtering out random noises from sales outcomes. When the agent is risk-averse, however, sales randomness can hurt efficiency (§5).

Our robustness result echoes the literature. In the moral hazard literature, Mirrlees (1999) finds that effort unobservability does not matter if random outcomes are stochastically distinguishable. In the auction literature, Maskin and Riley (1980), Crémer and McLean (1988), and McAfee and Reny (1992) find that the auctioneer can extract information rent under correlated valuation and risk neutrality. In the regulation literature, Picard (1987), Melumad et al. (1992), Caillaud et al. (1992), and Laffont (1993) find that outcome randomness does not hurt efficiency, if the agent is risk neutral. Yet all these studies assume static context; we show the same insight extends to our dynamic settings.

4.2. How Does the Optimal Plan Perform in the Long Run?

In static settings, the optimal solution to information asymmetry is the *c*lassical second-best π^c. It pays information rent $U^c(z) \equiv m\cdot(z-\underline{z}) - \int_{\underline{z}}^{z}\hbar(\tilde{z})\,\mathrm{d}\tilde{z}$, and cuts quota $q^c(z) \equiv (m+z) - \hbar(z) \leq \bar{q}^n(z)$. As both measures reduce efficiency, it predicts that the first-best is unattainable.[10]

Is the first-best truly unattainable? The classical prediction takes a static perspective: it assumes one-shot interaction with fixed skill, ignoring two-sided learning in the relationship. It is true that the agent holds real information advantage initially. But a typical employment relationship involves multiple interactions, through which both parties can learn about each other. Given this reality, one may argue, the firm-side learning should weaken the agent's advantage, soften his temptation, and dampen the distortion in the long run. When the relationship is sufficiently long, the firm should be able to achieve the first-best. Indeed, we find:

Proposition 5.

(a) *In regime $\mathcal{P}$ under plan π^*, the skill and sales converge to the first-best under $\bar{\pi}$. Formally, $z_t \to Z \sim \mathcal{N}(\mu_Z, \sigma_Z^2)$, and $s_t \to S \sim \mathcal{N}(\mu_S, \sigma_S^2)$, where $\mu_Z = \frac{\delta m\beta(1-\delta\alpha)+\mu_\omega}{(1-(\alpha+\beta))\cdot(1-\delta(\alpha+\beta))}$, $\sigma_Z = \frac{\sigma_\omega}{\sqrt{1-(\alpha+\beta)^2}}$, $\mu_S = m + \mu_Z + \frac{\delta m\beta}{1-\delta(\alpha+\beta)}$, and $\sigma_S = \sqrt{\sigma_Z^2+\sigma_\epsilon^2}$.*

(b) *In regime $\mathcal{P}$ under plan π^*, the steady-state skill Z and sales S increase stochastically in profit margin m, carryover*

rate α, and learning rate β. Their variances σ_Z^2 and σ_S^2 increase in α and β.

Proposition 5 characterizes the long-run performance of π^*. It is driven by three factors: the higher the skill carryover α, the higher the learning β, the higher the profit margin m, the higher the long-run performance (e.g., higher skill μ_Z and sales μ_S). Tellingly, the prior belief $\lambda(z_0)$—the key driver of plan π^c—is irrelevant in the long run; over time its influence is "washed out" by skill carryover and learning. Hence, the classical solution π^c can be misleading in the long run.

Figure 1 illustrates the long-run performance. It tracks 30 sample paths for the top and bottom agents. We observe, despite huge initial gap, the bottom agent-$\underline{z}$ learns to improve skill and catches up quickly. After just six periods, he is stochastically indistinguishable from the top agent—they are all normal $Z \sim \mathcal{N}(\mu_Z, \sigma_Z^2)$. The convergence is driven by skill carryover and learning: as time goes by, the distortion $\hbar(z_0)\alpha^t$ fades away; skill carryover and learning improve the low skilled, but they are insufficient for keeping the top talent at the peak level. As a result, the salesforce homogenises, gravitating toward $Z \sim \mathcal{N}(\mu_Z, \sigma_Z^2)$.[11]

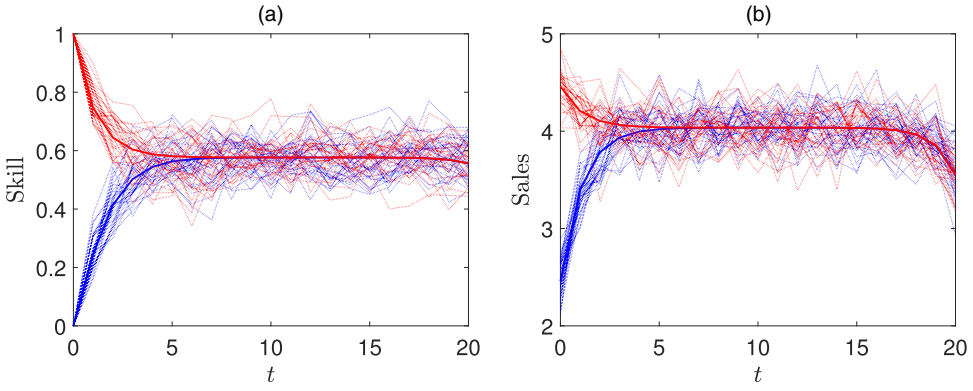
Proposition 5 has three implications. (i) *Optimal sales can either increase or decrease over time*. Observing increasing sales pattern, one may suspect that salespeople have strategically postponed effort, and hence compensation is ill-designed (Jerath and Long 2020). We call for caution: the increasing pattern may not be pathological; it can also arise as the optimal response to learning potential and vanishing information friction (Part (a)).[12] (ii) *The firm should favor incumbents in salesforce staffing*. This strategy has an efficiency justification: the firm-side learning reduces information asymmetry and distortion; for the same skill level, the incumbent demands less rents, sells more, and hence more preferable. (iii) *The firm should take a long-term perspective*. To the extent the employment relationship is stable, the short-run concern of information friction should not dictate its long-term goals of salesforce composition Z and performance S. The long-term goals should focus on enduring fundamentals: when sales are lucrative, knowledge can accumulate, and salespeople can learn, the firm should strive for talented salesforce with high performance (Part (b)).

4.3. What Are the Incentive Implications of Salesforce Learning?

The classical solution π^c has two key prescriptions (Rao 1990): (i) the firm should downward distort the quota for low skilled ($\hbar(z) > 0$, $\forall z < \bar{z}$); (ii) it should pay information rent for high skilled ($U^c(z) > 0$, $\forall z > \underline{z}$) These prescriptions assume static environment, myopic agent, and fixed skill. Yet in reality, selling environments change over time, salespeople are forward-looking, and they can learn from experience. In such a dynamic environment, a manager may ask: do the prescriptions still work? If not, what structural changes are necessary? More importantly, what is the cost of ignoring learning?

To address these questions, we compare the optimal plan π^* with the classical solution π^c. We show that, once learning is considered, the optimal compensation can change substantially in structure, prescriptions, and performance. The structural changes are apparent. Although both plans pay rent and distorts quota, π^* involves far more considerations. For example, It deploys the geometric structure to account for forward-looking, and makes upward adjustment to exploit learning. Neither consideration is present in π^c for the static context, but both are imperative for dynamic settings.

Figure 1. (Color online) Optimal Skill and Sales Processes under π^*: 30 Sample Paths for $z_0 \in \{0,1\}$, $m = 3$, $\alpha = 0.3$, $\beta = 0.1$, $\delta = 0.95$, $T = 20$, $\Lambda(z_0) \sim \mathcal{U}[0,1]$, $\sigma_\omega = 0.06$, $\sigma_\epsilon = 0.2$



Notes. (a) Skill $(z_t)_{t\geq 0}$. (b) Sales $(s_t)_{t\geq 0}$.

These structural changes refine the two prescriptions. Specifically, the downward-distortion prescription assumes fixed skill, ignoring learning. In this case, to control rent, the firm should indeed distort quota downward, $q^c(z_t) < \bar{q}(z_t)$, as π^c recommended. When learning is possible, the firm also has the incentive to exploit that potential with upward adjustment. The learning incentive can overtake rent control concern, producing upward distortion, even for the low skilled.

The rent-payment prescription assumes the agent has permanent advantage, ignoring its temporary nature ($\alpha < 1$). When the agent-z_0 is forward-looking, he seeks to profit from both current and future advantages of possessing z_0. The current advantage grants him immediate rent $\int_{\underline{z}}^{z_0} [-\psi_z(\hat{z}_0, q_0^*(\hat{z}_0))]\,\mathrm{d}\hat{z}_0$, as π^c predicted. But the future advantages, $\alpha^t \cdot \int_{\underline{z}}^{z_0} \mathsf{E}[-\psi_z\ (z_t, q_t^*(z^t))\,|\,\hat{z}_0]\,\mathrm{d}\hat{z}_0$, dissipate at exponentially fast rate α. Therefore, when skill carryover is low, or the relationship is long, the firm may stop paying rents for all but initial periods. Indeed,

Proposition 6. *In regime $\mathcal{P}$, the firm should pay rent for initial private information z_0 only; it can screen future private information $(z_t)_{t\geq 1}$ at no cost.*

Next, we quantify the cost of ignoring forward-looking and learning behaviors. We compare π^* and π^c: the former accounts for both forward-looking and learning, while the latter ignores both. We find:

Proposition 7. *In regime $\mathcal{P}$, the firm prefers long-term plan π^* over short-term π^c.*

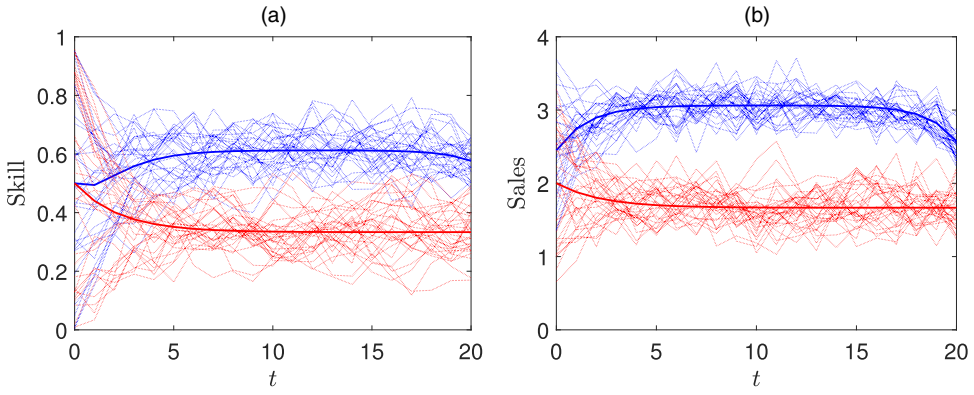
The proposition helps explain the prevalence of long-term contracts.[13] Intuitively, the short-term plan π^c suffers two deficiencies. First, it suffers *learning deficiency*. Focusing on the immediate gain, π^c ignores the impact of current selling activities on the future skill development. Such a myotic perspective sets low targets, reduces sales, and thus discourages learning. For example, the classic second-best π^c sets quota $q^c(z_t) = (m + z_t) - \hbar(z_t)$, way below the optimal level $q^*(z^t)$. The low expectation produces a vicious cycle: the agent generates low initial sales; the low sales reduces learning and future skill; the low future skill depresses sales still further. As the vicious cycle stabilizes, the system is trapped in an inefficiently low equilibrium, wasting the learning potential of the salespeople. See red curves π^c in Figure 2.

Second, the short-term plan π^c also suffers *information deficiency*. By definition, the short-term plan must be renewed every period, *after* the agent has learned new information z_t. The new information is precontract, allowing the agent to demand rent upon each renewal. In response, the firm must distort quota and pay rent in *every period*. The optimal short-term plan is a sequence of π^c, which prolongs the rent and distortion, thereby perpetuating inefficiency.

By contrast, the long-term plan π^* can address both deficiencies. First, it encourages learning. Taking a long-term perspective, it sets aggressive targets $q_t^*(z^t) \geq q^c(z_t)$. Using the upward adjustment δv_t, it internalizes all the future gains from learning. As such, it produces a virtuous cycle: higher targets motivate the agent to sell more; more sales accelerate learning and skill improvement; improved skill boosts sales still further. As the virtuous cycle stabilizes, the system sustains a high level of skill and sales, thereby fulfilling the learning potential of the salesforce. See blue curves π^* in Figure 2.

Second, the long-term plan improves information efficiency. Except the initial z_0, it can eliminate all the information advantage the agent has. At the outset it specifies all foreseeable contingencies for the entire

Figure 2. (Color online) In Regime $\mathcal{P}$: 30 Sample Paths under Plan $\pi \in \{\pi^*, \pi^c\}$, $m = 2$, $\alpha = 0.4$, $\beta = 0.15$, $\delta = 0.95$, $\Lambda(z_0) \sim \mathcal{U}[0,1]$, $\sigma_\omega = 0.06$, $\sigma_\epsilon = 0.2$



Notes. (a) Skill $(z_t)_{t\geq 0}$. (b) Sales $(s_t)_{t\geq 0}$.

Table 1. Performance Comparison

	Improvement %			Long-Term Plan π^*			Short-Term Plan π^c		
Parameter	Rent $\Delta U\%$	Profit $\Delta V\%$	Surplus $\Delta W\%$	Rent U^*	Profit V^*	Surplus W^*	Rent U^c	Profit V^c	Surplus W^c
$T = 0$	0.00	0.00	0.00	0.68	2.15	2.83	0.68	2.15	2.83
3	−22.05	19.54	11.63	1.41	9.23	10.65	1.82	7.72	9.54
5	−35.65	26.41	15.51	1.49	13.73	15.22	2.32	10.86	13.18
10	−53.85	34.26	20.26	1.51	23.27	24.78	3.27	17.33	20.61
$\alpha = 0$	−32.20	25.76	17.65	0.77	8.79	9.57	1.14	6.99	8.13
0.2	−25.95	22.68	14.84	1.04	8.99	10.03	1.41	7.33	8.74
0.4	−22.05	19.54	11.63	1.41	9.23	10.65	1.82	7.72	9.54
0.6	−21.70	17.46	8.45	1.90	9.55	11.45	2.43	8.13	10.56
0.8	−25.74	18.54	5.96	2.48	9.96	12.44	3.34	8.40	11.74
$\beta = 0$	19.13	13.00	13.79	1.19	7.61	8.79	1.00	6.73	7.73
0.05	−4.38	15.69	12.52	1.30	8.37	9.66	1.35	7.23	8.59
0.1	−22.05	19.54	11.63	1.41	9.23	10.65	1.82	7.72	9.54
0.15	−35.82	25.18	11.29	1.55	10.23	11.77	2.41	8.17	10.58
$m = 2$	−22.05	19.54	11.63	1.41	9.23	10.65	1.82	7.72	9.54
2.5	−36.11	17.88	7.41	1.87	14.32	16.19	2.92	12.15	15.07
3.0	−45.42	17.58	5.25	2.32	20.54	22.86	4.25	17.47	21.72
3.5	−52.19	17.76	4.01	2.77	27.89	30.66	5.80	23.69	29.48
4.0	−57.38	18.11	3.23	3.22	36.38	39.60	7.56	30.80	38.36

Note. Base case: $m = 2$, $\alpha = 0.4$, $\beta = 0.1$, $\delta = 0.95$, $T = 3$, $\sigma_\omega = 0.06$, $\sigma_\epsilon = 0.2$, $z_0 \sim \mathcal{U}[0,1]$.

duration T, *before* the agent can learn any new information z_t beyond z_0. This preemptive arrangement denies the agent the new edge from learning $(z_t)_{t\geq 1}$ arising later, thereby limiting his real advantage to z_0 only. After initial period, the agent still learns private information z_t, but the firm can screen z_t, without paying real rent or distorting quota. Indeed, the distortion $\hbar(z_0)\alpha^t$ in quota q_t^* is entirely driven by initial z_0, and it vanishes over time. Hence, except initial period, the long-term plan is information efficient (Proposition 6).

Finally, we quantify *how much* the long-term plan π^* can improve short-term π^c. In Table 1, we measure improvement by percentage changes in rent, profit, and surplus; for example, $\Delta U\% = \frac{U^*-U^c}{U^c} \times 100\%$. We observe: (i) The long-term plan can cut agent rent ($\Delta U\% \leq 0$), because it pays rents only initially. (ii) It can increase total surplus ($\Delta W\% > 0$), because it leverages learning and minimizes distortion. (iii) It can improve firm profit ($\Delta V\% > 0$), because it eliminates the two deficiencies of the short-term plan. The improvement can be substantial, when the relationship T is long, learning β is strong, and profit margin m is high. To the extent these situations prevail, ignoring forward-looking and learning is costly, and the firm should entertain the long-term plan π^*.

5. Limitations and Extensions

We focus on dynamic learning and information efficiency. In practice, other factors—such as reputation, monitoring, enforcement, relational contracts, legal institutions, and industry norms—can also dominate compensation design. In particular, our model $\mathcal{P}$ assumes linear dynamics, full commitment, risk neutral agent, and negligible turnover. As such, the results hold to the extent these assumptions reasonably depict the reality. Next, we discuss the limitations and extensions of our model and results. The linear dynamics assumption is innocuous. It simplifies exposition and helps deliver closed-form solution. We can extend our model to nonlinear dynamics $z_{t+1} = Z(z_t, q_t, \omega_{t+1})$ with correlated random shocks. Although technical details differ, the main insights remain the same.

We assume the firm can commit to a contract for certain periods. Empirically, this assumption respects the legal requirement of full-time employment.[14] Theoretically, it is the starting point of recent dynamic incentive literature; see, for example, Rubel and Prasad (2015), Schöttner (2016), and Jerath and Long (2020). When the firm cannot commit, however, the *ratchet effect* can arise: the agent has stronger incentive to misreport private information, because the firm can use it to set tougher standards in the future. As such, screening becomes much hard. Technically, the revelation principle no long applies, and the problem becomes a dynamic game without commitment. This is a challenging problem for future research.

We assume risk neutrality. Empirically, both risk-neutral and risk-averse salesforce are common.[15] The main implication of risk aversion is the uncertainty-incentive tradeoff; however, the empirical support for this tradeoff is mixed; see Lafontaine and Bhattacharyya (1995), Prendergast (2002), and Lo et al. (2011) for extended reviews. Our model features both adverse selection and moral hazard. It relies on risk neutrality

to control both issues dynamically (§4.1). When the agent is risk averse, the firm must consider additional trade-off between inducing effort and providing insurance, and pay additional rent (risk premium) for the agent to bear risk. All our results still hold for pure adverse selection, that is, when the effort is contractible. In the mixed case, when the insurance concern dominates, our results (in Propositions 2–4) will break down. The reason is that, without risk neutrality, the firm can no long shift compensation freely over time.[16]

To focus on the firm-agent interactions, we assume negligible salesforce turnover. This assumption is standard in the salesforce and agency literature.[17] Empirically, it captures the situations where the skill is firm-specific,[18] and the firm has dominant bargaining power (Lazear 1998). When turnover is substantial, the labor market competition may drive salesforce compensation. Our results no long hold for such situations. For example, instead of developing incumbent agents, firms may favor recruiting talent from the market. In this case, the force of market competition dominates the gain from learning.[19]

Methodologically, one can model labour market competition directly.[20] One can also model the competition indirectly, using type-dependent outside option (reservation utility) $\hat{u}(z_0)$. Since good salespeople have better outside options, $\hat{u}(z_0)$ increases in z_0. Formally, we solve $\max\{\tilde{V}(\pi): IR^o, IC_t, \forall t\}$, where IR^o is the type-dependent participation constraint: $U_0(z_0) \geq \hat{u}(z_0), \forall z_0$. This class of agency problems are notoriously difficult to solve, even for static settings (Jullien 2000). The reason is that, standard contract theory relies on the monotone information rent to derive optimal contracts. When the outside option is type-dependent, the monotone property may fail, so that interior types can have a vanishing rent or be excluded from trade. Using the Lagrangian technique, one may identify specific conditions for the information rent to be monotone (hence outside options have no bite). But the general characterization is still an open question.

6. Conclusion

Salespeople can learn fresh information and develop new skills over time. Both activities are valuable to firm performance. How should a firm incentivize them? We solve this new class of compensation problems. We find the optimal plan differs substantially from existing ones: in the short run, the optimal plan resembles the second-best, because information friction is still acute; in the long run, it converges to the learning-adjusted first-best, because information friction vanishes and learning persists; to adapt to changing conditions, it adjusts quota dynamically for the best use of new private information. Despite these complex tradeoffs, the optimal plan has a simple implementation.

Our study sheds light on when and why firms distort sales, favor incumbents, and prefer long-term plans. We show in a long-term relationship, information friction is often short-lived, but learning is enduring. Hence, ignoring learning can mislead compensation design and inflict substantial losses. By highlighting the critical role of learning in long-run performance, this study advances our understanding of salesforce theory and practice.

Endnotes

[1] Also, in office equipment sales, a new hire can achieve 42% of maximal sales within a year, through learning by doing (Misra et al. 2004); in restaurant business, low-skill salespeople can increase productivity by 2.48%, through learning from high-skill coworkers (Tan and Netessine 2019).

[2] We thank the review team for this insight.

[3] See Coughlan and Joseph (2012) for a review. There is also an extensive economic literature on agency theory; see, for example, Holmstrom and Milgrom (1987), Spear and Srivastava (1987), Phelan and Townsend (1991), Fernandes and Phelan (2000), Courty and Hao (2000), Battaglini (2005), Williams (2011), Pavan et al. (2014). For other applications, see auctions (Eső and Szentes 2007), compensation design (Garrett and Pavan 2015, Chen et al. 2021), procurement (Akcay and Gao 2020), and taxation (Stantcheva 2017). For comprehensive references, see Laffont and Martimort (2001), and Bolton and Dewatripont (2005).

[4] In general, the inefficiencies can arise from information asymmetry (Gao and Mishra 2018), incentive misalignment (Li and Gao 2008, Gao et al. 2020), exogenous uncertainty (Xu et al. 2007; Gao et al. 2012, 2014b; Luo et al. 2016; Gao et al. 2017; Kuzu et al. 2019), strategic uncertainty (Gao et al. 2014a), and restrictive contract structures (Gao 2015a, Gao et al. 2021). They often make the first-best solution unattainable (Bolton and Dewatripont 2005).

[5] We use *skill* as an overarching term to model agent's private information. Our results hold for either market or ability interpretation; see Lal and Staelin (1986), Rao (1990), and Mishra and Prasad (2004).

[6] For example, in pharmaceutical and financial industries, the effect of selling effort on sales persists over time, because of buyer inertia, reputation, goodwill, and entrenchment (Rubel and Prasad 2015, Kim et al. 2019).

[7] These assumptions are widely used in the literature; see, for example, Misra et al. (2004), Rubel and Prasad (2015), and Schöttner (2016). They capture the essential features of common skill processes. For example, Chan et al. (2014) estimate that carryover rate α in cosmetics selling is between 0 and 1; by working 40 hours in the first week, rookies can improve skill by 2%–7%, which suggests a positive but bounded learning rate ($0 \leq \beta < \infty$).

[8] Specifically, (i) *outcome-based learning*: the more the realized sales s_t deviate from the expectation $\mathsf{E}s_t$, the more the agent learns. Hence, $z_{t+1} = \alpha z_t + \beta q_t + \gamma|s_t - \mathsf{E}s_t| + \omega_{t+1}$, where γ is the outcome-based learning rate. It follows that $z_{t+1} = \alpha z_t + \beta q_t + \omega'_{t+1}$, where $\omega'_{t+1} = \gamma|\epsilon_t| + \omega_{t+1}$ is the aggregate shock. (ii) *Learning from experience*: let z_t denote the cumulative experience, and q_t the incremental experience at time t; then the cumulative experience at time $t+1$ is $z_{t+1} = z_t + q_t + \omega_{t+1}$. (iii) *Deterministic learning*, when $\omega_{t+1} = 0$. (iv) *No skill carryover or learning*, when $\alpha = 0$ or $\beta = 0$.

[9] Other instruments can also serve rent control purpose. For example, Simester and Zhang (2014) find that, although lobbying reduces sales in low demand state, it can also reduce the rent paid in high demand state. In this sense, lobbying is similar to downward distortion (the sales in low demand).

[10] Specifically, because hidden information and action allow the agent to manipulate, information rent $U^c(z)$ is imperative for truth-telling and

quota distortion $\hbar(z)$ is necessary for rent control; both measures reduce efficiency; hence the first-best is unattainable.

[11] In Figure 1, the sales drop in the end. This is driven by vanishing learning gain, $v_t \downarrow t$. The skill and sales of the top agent-$\bar{z}$ drop in early periods, because of random shocks and insufficient selling experience to retain skill. In this case, to maintain his initial skill level, the firm would have to assign him unprofitably higher quota. In other cases, all agents may improve skill and sales over time.

[12] This insight is consistent with Jerath and Long (2020), who find a variety of sales patterns can arise in a pure moral hazard setting.

[13] The prevalence of long-term contracts can also be driven by other factors, such as risk management, career concerns and incomplete contracts; see, for example, Gao (2015b), Gibbons and Murphy (1992), and Tirole (1999).

[14] For example, the U.S. labor laws require that the commission salesperson agreement must be in writing and signed by both parties. Due to reputation and litigation concerns, normal firms are reluctant to renege on the employment contracts. See, for example, New York state labor law (https://www.labor.ny.gov/legal/counsel/pdf/payment-of-commissions-frequently-asked-questions.pdf).

[15] For risk neutral salesforce, see, for example, Rao (1990), Mantrala et al. (1994), Oyer (2000), Dai and Jerath (2013), Kräkel and Schöttner (2016), and Schöttner (2016). For the risk averse framework, see Basu et al. (1985); Lal and Staelin (1986); Mishra and Prasad (2004, 2005); Caldieraro and Coughlan (2009); Rubel and Prasad (2015).

[16] Technically, the risk-averse agent has continuation payoff $\tilde{U}_t(\hat{z}^t; z_t) = \mathsf{E}[p_t(\hat{z}^t, q_t(\hat{z}^t) + \epsilon_t)] - r\mathsf{var}[p_t(\hat{z}^t, q_t(\hat{z}^t) + \epsilon_t)] - \frac{1}{2}(q_t(\hat{z}^t) - z_t)^2 + \delta\mathsf{E}[U_{t+1}(\hat{z}^t, z_{t+1}) | z_t, q_t(\hat{z}^t)]$, where r is the risk aversion coefficient (Misra and Nair 2011). The new term $r\mathsf{var}[p_t(\hat{z}^t, q_t(\hat{z}^t) + \epsilon_t)]$ makes the agent much hard to motivate, and it can change our results significantly. For example, risk aversion can reduce learning, because the agent is more cautious to put efforts.

[17] See, for example, Rao (1990), Mishra and Prasad (2004), Simester and Zhang (2014), Tirole (1988), Fudenberg and Tirole (1991), Laffont and Martimort (2001), Bolton and Dewatripont (2005).

[18] For example, firm-specific skills involve (i) *intangible knowledge*, such as idiosyncratic corporate culture, informational networks, and power relationships inside the firm (Pfeffer 1994). This kind of knowledge helps the agent perform better, but it is probably useless elsewhere. (ii) Also, if the agent have developed close working *relationships with clients* and deep knowledge of their particular organizations, that may be primarily firm-specific capital—unless the agent can take the clients with him to a new job; see, for example, Shi et al. (2017), Kim et al. (2019).

[19] We thank the review team for this insight.

[20] For this line of inquiry, see Lucas Jr et al. (1987), Futrell and Parasuraman (1984), Wotruba and Tyagi (1991), Farber (1999), Chand et al. (2002), Sunder et al. (2017), and references therein.

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6. 预约调度：Managing Appointment Booking Under Customer Choices (MS, 2019)

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Managing Appointment Booking Under Customer Choices

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Abstract. Motivated by the increasing use of online appointment booking platforms, we study how to offer appointment slots to customers to maximize the total number of slots booked. We develop two models, *nonsequential* offering and *sequential* offering, to capture different types of interactions between customers and the scheduling system. In these two models, the scheduler offers either a single set of appointment slots for the arriving customer to choose from or multiple sets in sequence, respectively. For the nonsequential model, we identify a static randomized policy, which is asymptotically optimal when the system demand and capacity increase simultaneously, and we further show that offering all available slots at all times has a constant factor of two performance guarantee. For the sequential model, we derive a closed form optimal policy for a large class of instances and develop a simple, effective heuristic for those instances without an explicit optimal policy. By comparing these two models, our study generates useful operational insights for improving the current appointment booking processes. In particular, our analysis reveals an interesting equivalence between the sequential offering model and the nonsequential offering model with perfect customer preference information. This equivalence allows us to apply sequential offering in a wide range of interactive scheduling contexts. Our extensive numerical study shows that sequential offering can significantly improve the slot fill rate (6%–8% on average and up to 18% in our testing cases) compared with nonsequential offering. Given the recent and ongoing growth of online and mobile appointment booking platforms, our research findings can be particularly useful to inform user interface design of these booking platforms.

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1. Introduction

Appointment scheduling is a common tool used by service firms (e.g., tech support, beauty services, and healthcare providers) to match their service capacity with uncertain customer demand. With the widespread use of the internet and smartphones, customers often resort to online channels when searching for information and reserving services. To keep up with customers' preferences and needs, many service organizations have developed online appointment scheduling portals. For instance, Teachers Insurance and Annuity Association of America (TIAA) allows its clients to book appointments with their financial consultants online. There is also a rising number of online service reservation companies that offer online appointment booking software or applications as a service for (small) businesses. Examples include zocdoc.com for medical appointments, opentable.com for dinner reservations, mindbodyonline.com for fitness classes, booker.com for spa services, and salonultimate.com for haircuts.

The interfaces of these online appointment booking systems vary. Some are more toward one-shot offering (i.e., a single list of available appointments is shown on a single screen for customers to choose from). Others offer a small number of options to start, and customers must press "more" or "next" to view additional appointments that are available. This way of scheduling resembles the traditional telephone-based scheduling process, in which the scheduling agent may reveal availability of appointment slots in a sequential manner. Such a sequential way of displaying options is often seen on mobile devices with a small screen as well.

Our research is motivated by these various ways of appointment booking, and we seek to understand how a service provider can best use these (online) appointment booking systems. In scheduling practice, service providers first predetermine for each day an appointment template, which specifies the total number of slots, the length of each slot, and characteristics of customers (e.g., nature of the visit) to be scheduled for each slot. For instance, in a gym setting, one has to determine the

number of classes and their capacity, and in healthcare, the service provider first determines the number of patients that a clinician will see that day and at what times. With an appointment template in place, service providers then decide how to assign incoming customer requests to the available slots—nowadays, this process is often done via online appointment scheduling as mentioned above. The relevant performance metric for this process is the *fill rate* (i.e., the fraction of slots in a template booked before the scheduling process closes). Although the fill rate is not equivalent to the eventual capacity utilization because of various postscheduling factors (e.g., cancellations, no shows, and walk ins), it is the first, and in many cases, the most important step to achieving a high utilization (and thus, a high revenue), and it is the objective of the research presented in this paper.

Our focus is on modeling the scheduling process and developing stochastic dynamic optimization models to inform appointment scheduling decisions in the presence of customer choice behavior. Notwithstanding the surge of interest in service operations management in the past decade, basic single-day, choice-based dynamic decision models are absent for a broad class of real-world scheduling systems. To our knowledge, the existing operations research and management literature on this type of dynamic appointment scheduling is very limited; most, if not all, related research assumes that customers reveal their preferences first and that the scheduler decides to accept or reject (e.g., Gupta and Wang 2008, Wang and Gupta 2011). However, as discussed above, in many real-world scheduling platforms, the system (i.e., the scheduler) offers its availability to customers to choose from either in a one-shot format or in a sequential manner, with no explicit knowledge on customer preferences. Customers interact with the scheduler in ways that have not been fully explored in the literature. This paper fills a gap in the literature by proposing the first choice-based dynamic optimization models for making scheduling decisions in systems where customers are allowed to choose among offered appointment slots from an established appointment template. We show how the current appointment booking processes can be improved by developing optimality results, heuristics, and managerial insights in the context of the proposed models.

We propose and study two models for the interaction between customers and the service provider. The first one is referred to as the *nonsequential offering* model. In this model, the scheduler offers a single set of appointment slots to each customer. If some of the offered slots are acceptable to the customer, she chooses one from them; otherwise, she does not book an appointment. This simple, one-time interaction resembles the mechanism of many online appointment systems that provide one-shot offerings, and our results on this model have direct implications on how to manage these systems. Our second model is a *sequential offering* model, in which the scheduler may offer several sets of appointment choices in a sequential manner. This is motivated by (1) web-based appointment applications designed to reveal only a small number of appointment options one web page at a time (e.g., mobile-based appointment applications) and (2) the traditional telephone-based scheduling process, in which the scheduler offers appointment slots sequentially. This second model is stylized in the sense that it does not incorporate customer recall behavior (i.e., a customer choosing a previously offered slot after viewing more offers), which is allowed in both online and phone-based scheduling. Our goal here is to glean insights on how the fill rate can be improved by "smarter" sequencing when sequential offering is part of the scheduling process.

For both cases, we are interested in which slots to offer to improve and maximize the fill rate. We answer this question by investigating the optimal offering policy using Markov decision processes (MDPs) as well as by discussing heuristics. Intuitively, sequential offering should lead to a higher fill rate than nonsequential offering, because sequential offering gives the scheduler more control over the service capacity. We are also interested in how much improvement a service provider can get by switching from nonsequential scheduling to sequential scheduling. We answer this question by comparing the fill rates resulting from these two models, and the gap in the fill rates represents the "value" of sequential offering.

We make the following main contributions to the literature.

To the best of our knowledge, our paper is the first to study and compare two main scheduling paradigms, nonsequential (online) and sequential (mobile or telephone based), used in the service industries.

For the nonsequential offering model, we characterize the optimal policy for a few special instances and show that the optimal policy can be highly complex in general. We then identify a static randomized policy (arising from solving a single linear program), which is *asymptotically optimal* when the system demand and capacity increase by the same factor. We further show that the offering-all policy (i.e., offering all available capacity throughout) has a *constant factor of two performance guarantee*.

For the sequential offering model, we show that there exists an optimal policy that offers slot types one at a time based on their marginal values. We are able to determine these values for a broad class of model instances, which leads to a *closed form* optimal policy in these cases. For model instances without an explicit optimal policy, we develop a simple, effective heuristic.

We show that a sequential offering model is equivalent to a nonsequential offering model with perfect customer preference information. This equivalence ensures that sequential offering can be optimally applied in

various interactive scheduling contexts, in particular when customer-scheduler interaction can (partially) reveal customer preference information during the appointment booking process.

Via extensive numerical experiments, we show that the offering-all policy and the heuristic developed for sequential offering work remarkably well in their respective settings, and thus, they can serve as effective approximate scheduling policies for practical use. We also show that, by switching from nonsequential to sequential offering, the slot fill rate can be significantly improved (6%–8% on average and up to 18% in our testing cases).

The remainder of the paper is organized as follows. Section 1.1 briefly reviews the relevant literature. Section 2 introduces the common capacity and demand model that will be used in both the nonsequential and sequential settings. Sections 3 and 4 discuss the nonsequential offering case and the sequential offering case, respectively. Section 5 presents an extensive numerical study that complements our analytic work. In Section 6, we make concluding remarks. All proofs of our technical results can be found in the online appendix.

1.1. Literature Review

From an application perspective, our work is related and complementary to the literature on appointment template design, a topic that has been studied extensively (Cayirli and Veral 2003, Gupta and Denton 2008, Ahmadi-Javid et al. 2016). Our work departs from this literature in that we start from an established template and then study how to manage the interaction between the customers and the scheduler to best direct customers to various slots. Among the existing work on dynamic appointment scheduling, Feldman et al. (2014) is the only study, other than the few papers mentioned in the previous section, that explicitly models customer choice behavior. However, Feldman et al. (2014) focus on customer choices across different days and use a newsvendor model to capture the use of daily capacity; this aggregate daily capacity model does not allow them to consider (allocating customers into) detailed appointment time slots within a daily template.

From a modeling perspective, Zhang and Cooper (2005) look at a similar choice model to ours in the context of revenue management for parallel flights. In contrast to this paper, their approach focuses on deriving bounds on the value function of the underlying MDP and using them to construct heuristics. A few recent studies on assortment optimization are particularly relevant to our paper: Golrezaei et al. (2014), Bernstein et al. (2015), Gallego et al. (2016a), and Gallego et al. (2016b). Bernstein et al. (2015) study a dynamic assortment customization problem, which is mathematically similar to our nonsequential appointment offering problem, assuming multiple types of customers, each of which has a multinomial logit choice behavior over all product types. They assume that the customer type is observable to the seller (corresponding to our scheduler), which differs from our setting. Golrezaei et al. (2014) adopt a general choice model and also allow an arbitrary customer arrival process. Gallego et al. (2016b) extend the work by Golrezaei et al. (2014) to allow rewards that depend on both the customer type and the product type. In addition, Gallego et al. (2016a) study assortment optimization in an online retail setting, where each customer picks the number of pages to view according to a fixed distribution and then chooses among all of the products offered on those pages following some choice model. The last three studies also assume that the customer type is known to the seller, and their focus is on developing control policies competitive with respect to an offline optimum, a different type of research question from ours. The other distinguishing feature of our research from all previous work is that we consider sequential offering, an offering paradigm that has not been studied before.

Finally, our work is related to two other branches of literature. The first is on online bipartite matching (Mehta 2013), and the second is on general stochastic dynamic optimization, in particular stochastic depletion problems (e.g., Chan and Farias 2009) and submodular optimization (e.g., Golovin and Krause 2011). These two lines of research mainly aim to obtain performance guarantee results with respect to offline optimums, which is not our research goal.

2. Capacity and Demand Model

We consider a single day in the future that has just opened for appointment booking. The day has an established appointment template, but none of the slots are filled yet. We divide the appointment scheduling window (i.e., the time between when the day is first opened for booking and the end time of this booking process) into N small periods. Specifically, we consider a discrete time N period dynamic optimization model with I customer types (that may come) and J appointment slot types (in the template), where customer types are characterized by their set of *acceptable* slot types. Denote by Ω_{ij} the zero-one indicator of whether slot type j is acceptable by customer type i, and therefore, the $I \times J$ *choice matrix* $\Omega := [\Omega_{ij}]$ consists of distinct row vectors, each representing a unique customer type. Such a customer type structure is similar to those in the literature that model customer segments characterized by different product preferences (e.g., Bernstein et al. 2015).

We now present the details of our customer arrival and choice model. In each period, at most one customer arrives. The customer is type i with probability $\lambda_i > 0$, and with probability $\lambda_0 := 1 - \sum_{i=1}^{I} \lambda_i$, no customer arrives. On a customer arrival, the scheduler offers her a set $S \subseteq \{1, \ldots, J\}$ of slot types without knowledge of the

customer type. When *offer set* S contains one or more acceptable slot types, the customer chooses one uniformly at random. If no type in S is acceptable to this customer, we distinguish two possibilities. Either we use a *nonsequential* model, where the scheduler can only offer a single set and the customer immediately leaves if none of the offered slots are acceptable (Section 3), or we use a *sequential* model, where the scheduler may offer any number of sets sequentially until either the customer encounters an acceptable slot or the customer finds no acceptable slots in any offer set and leaves without booking a slot (Section 4). We start from an initial capacity of b_j slots of type j at the beginning of the reservation process and denote $\mathbf{b} := (b_1, \ldots, b_J)$. Every time a customer selects a slot, the remaining slots of this type are reduced by one. The scheduler aims to maximize the fill rate at the end of the reservation process by deciding on the offer set(s) in each period. This is also equivalent to maximizing the *fill count* (i.e., the total number of slots reserved at the end of the booking process), because the initial capacity $\mathbf{b}$ is fixed.

Our capacity and demand model generalizes that of Wang and Gupta (2011) in the following sense. Our notion of "slot type" can be viewed as an abstraction of the service provider and time block combination in their model, and thus, we allow a generalization of using other attributes of a slot that may affect its acceptability to customers, such as duration. Wang and Gupta (2011) consider distinct customer panels, each characterized by a possibly different acceptance probability distribution over all possible combinations of service providers and time blocks and a set of revenue parameters. In contrast, we define the notion of customer type and identify it with a unique set of acceptable slot types. Their arrival rate (probability) parameters are associated with each customer panel, whereas we directly have the demand rate for each of the I customer types as model primitives.

Our choice model assumes that, for a particular customer type, slot types are either "acceptable" or "unacceptable." This dichotomized classification of slots closely mimics the decision process on whether a time slot works for one's daily schedule. For instance, such a slot-choosing process is seen at the popular polling website www.doodle.com, where each participant responds to a poll by indicating whether a particular time works (i.e., is acceptable) for him or her. This relatively parsimonious choice model enables a tractable analysis of the interplay between appointment booking and customer choice. Its parameters may, for instance, be estimated by conducting a market survey on customers' acceptance on various slot types.

As discussed earlier, the distinction between the nonsequential and sequential customer-scheduler interactions reflects the differences present in various real-life appointment scheduling systems. The nonsequential model is best suited for web-based appointment scheduling systems, such as www.zocdoc.com. In such systems, the customer is presented with a list of time slots to choose from, which corresponds to a single offer set. In contrast, sequential scheduling reflects the iterative nature of, for instance, telephone-based appointment scheduling. Here, the scheduler may propose one or more slots initially and may present more if these are rejected by the customer. Although allowing an unlimited number of offer sets in sequence does not conform with many real-world systems, the sequential model is a valuable object of study, because the scheduler in this setting enjoys the greatest flexibility, and hence, the resulting optimal fill rate serves as an upper bound for that in both the nonsequential model and some intermediate paradigms, such as those allowing a limited number of offer sets or with customer reneging.

The assumption that the customer type is unobservable is unique in our work, and it is present to some degree in all real-world systems that we consider. Users of web- and mobile-based appointment scheduling systems often prefer a simple interface soliciting no or minimal personal information before displaying availabilities, and telephone-based schedulers may only know some basic information about the customers. Even if these collected data are useful in predicting customer preferences, not all service firms have the necessary resources (e.g., human, technology, and software) to make such predictions and then use them in scheduling decisions. Another important motivation for why we choose to assume that the exact customer type is unknown to the scheduler is that this case can be considered a harder problem than the setting with full customer information, because the latter would require only offering a single slot type. We will touch on the intermediate case, in which partial knowledge on customer types is available, in Section 4.6 on interactive scheduling. There, we see that incorporating partial knowledge on customer preference does not necessarily lead to better fill rates than sequential offering (Theorem 4), but it may improve the ease of use for sequential offering (e.g., fewer offers to make).

Our objective is to maximize the fill rate (or equivalently, fill count), thereby assuming that each customer contributes to the objective equally. We choose this objective for a few reasons. First, fill rate is a widely used reporting metric by service firms for their operational and financial performance. The simplicity of this metric also makes it more tractable for analysis. Second, fairness may carry more weight than profitability in the vision of a service firm (e.g., a healthcare delivery organization). Third, although different customers may bring different rewards (e.g., revenues) to the service firm, how to associate such rewards with customer (preference) types is not well understood in the literature.

Finally, our discrete time customer arrival model with, at most, one arrival per period is widely accepted and

used by many operations management studies, including those on healthcare scheduling (e.g., Green et al. 2006) and revenue management (e.g., Talluri and Van Ryzin 2004 and Bernstein et al. 2015). One could set N, the total number of time periods, sufficiently large so that the probability of multiple customers arriving during a single period is negligible (and thus, such as is the probability of more than N customers arriving in total). This demand model can be used to approximate an inhomogeneous Poisson arrival process (Subramanian et al. 1999).

In the following sections, we focus on analyzing the models described above. Note that our models do not explicitly capture the rolling horizon feature of the appointment scheduling practice, in which customers may book appointments in future days and unused capacity in a day is wasted when the day is past. However, the rolling horizon multiday scheduling model is known for its intractability (Liu et al. 2010, Feldman et al. 2014), whereas the single-day model is more tractable and often used in the literature to generate useful managerial insights (e.g., Gupta and Wang 2008 and Wang and Gupta 2011). In Section C of the online appendix, we will introduce a rolling horizon version of our model and argue why it is intractable. Moreover, we numerically show how our single-day models can inform decision making in a rolling horizon multiday setting.

3. Nonsequential Offering

We first consider the nonsequential offering model, in which only one offer set S is presented to each arriving customer. Denote by $\mathbf{m} \le \mathbf{b}$ a J-dimensional, nonnegative integer vector that represents the current number of remaining slots of each type and by $\mathbf{e}_j$ the J-dimensional unit vector with its jth entry being one and all others being zero. Define $\bar{S}(\mathbf{m}) := \{j = 1, \ldots, J : m_j > 0\}$, the set of slot types with positive capacity, and $V_n(\mathbf{m})$ as the maximum expected number of appointment slots that can be booked from period n to period 1 with $\mathbf{m}$ slots available at the beginning of period n. Note that we count time backward.

Furthermore, denote by $q_{ij}(S)$ the probability that slot type j is chosen conditional on a type i customer arrival and an offer set $S \in \bar{S}(\mathbf{m})$. We have, for any j,

$$q_{ij}(S) = \begin{cases} \dfrac{\Omega_{ij}}{\sum_{k\in S}\Omega_{ik}}, & \text{if } \sum_{k\in S}\Omega_{ik} > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Then, the probability that slot type j is chosen when offer set S is given is

$$q_j(S) = \sum_{i=1}^{I} \lambda_i q_{ij}(S), \tag{2}$$

and the no booking probability is $q_0(S) = 1 - \sum_{j=1}^{J} q_j(S)$. The optimality equation is

$$V_n(\mathbf{m}) = \max_{S\subseteq\bar{S}(\mathbf{m})} \left[\sum_{j\in S} q_j(S)\left(1 - \Delta^j_{n-1}(\mathbf{m})\right) \right] + V_{n-1}(\mathbf{m}), \quad \text{for } n = N, N-1, \ldots, 1, \tag{3}$$

where $V_0(\cdot) = 0$ and $\Delta^j_{n-1}(\mathbf{m}) := V_{n-1}(\mathbf{m}) - V_{n-1}(\mathbf{m} - \mathbf{e}_j)$ denotes the marginal benefit owing to the m_jth unit of slot type j at period $n-1$.

We first analyze the nonsequential offering model for a few specific instances and show that, in general, the optimal nonsequential offering policy seems to have no appealing structural properties. Thus, characterizing the optimal policy for general large-scale nonsequential offering models is very challenging, if not impossible. We then focus our efforts on constructing simple scheduling policies that have performance guarantees and may perform well in practice. We first consider a limited class of policies (called static randomized offering policies) and identify one such policy that is asymptotically optimal when we increase the system demand and capacity simultaneously. We further show that a simple policy that offers all available slots at all times has a constant ratio of two performance guarantee, independent of all model parameters. In Section 5, we show via extensive numerical instances that this offering-all policy significantly outperforms its theoretical bound. It may thus serve as a simple, effective heuristic offering rule for many practitioners in the nonsequential offering context.

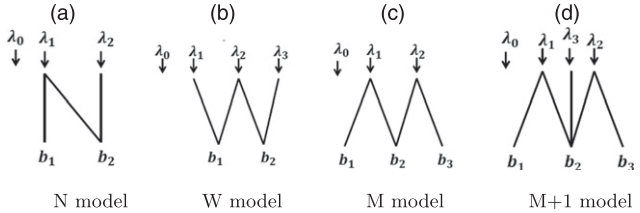
3.1. Results for Specific Model Instances

When there are $J = 2$ slot types, the choice matrix Ω has two possible nontrivial values:

$$\Omega = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \text{and} \quad \Omega = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

These we refer to as the N model instance (Figure 1(a)) and the W model instance (Figure 1(b)), respectively. These two model instances are, for example, applicable to the popular Chinese scheduling system www.guahao.com.cn, which allows customers to book either a morning or an afternoon (medical) appointment for a certain day without providing more granular time

Figure 1. The N, W, M, and M + 1 Model Instances



interval options. In both model instances, we show that it is optimal to offer all available slots at all times (which we call the *offering-all* policy in the rest of this article), because not doing so would unnecessarily risk sending away certain customers. This is formalized in the following result.

Proposition 1. *For the N and W model instances, the offering-all policy is optimal.*

When there are $J = 3$ slot types, the simplest nontrivial choice matrix is the M model instance in Figure 1(c) with

$$\Omega = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

It turns out that, in this case, the offering-all policy is not always optimal; rather, rationing of the versatile type 2 slot is needed. We define policy π_1 according to its offer set:

$$S^{\pi_1}(\mathbf{m}) := \begin{cases} \{1,3\} & \text{if } m_1 > 0 \text{ and } m_3 > 0, \\ \bar{S}(\mathbf{m}) & \text{otherwise.} \end{cases} \quad (4)$$

Therefore, policy π_1 proposes to hold back on offering type 2 slots until either type 1 or type 3 slots are used up. We now formalize that one cannot do better than this.

Proposition 2. *For the M model instance, π_1 is optimal.*

The intuition behind Proposition 2 is that blocking slot type 2 does not lead to any immediate loss of customer demand compared with offering it, while forcing early customers into *less versatile* slot types (types 1 and 3). This preserves the *versatile* slots (type 2) for later arrivals when slots run low. For convenience of discussion, we say a slot type is more versatile if this slot type is accepted by a superset of customer types compared with its counterpart.

Following from Proposition 2, we know that a versatile type 2 slot is at least as valuable as one of the other two less versatile slot types at all times; otherwise, it would be better to offer type 2 slots but not to offer the more valuable, less versatile slot type. To be more specific, we have the following corollary.

Corollary 1. *In the M model instance, for $j = 1$ or 3 or both,*

$$V_n(\mathbf{m} - \mathbf{e}_2) \le V_n(\mathbf{m} - \mathbf{e}_j), \ \forall \mathbf{m} > 0, \ n \in \{1, \ldots, N\}. \quad (5)$$

However, it is important to note that one of two less versatile slot types (1 and 3) may be strictly more valuable than the versatile type 2. For example, for $\lambda_1 = \lambda_2 = 0.5$, it is easy to verify that $V_2(2,1,0) = 1.625 < 1.75 = V_2(2,0,1)$. The reason here is the following. With $m_1 = 2$ and $n = 2$, sufficient capacity is available for all potential type 1 customer demand (i.e., at most two units), but in both cases ($\mathbf{m} = (2,1,0)$ and $\mathbf{m} = (2,0,1)$), at most one slot is available for type 2 customers. If $(m_2, m_3) = (1,0)$, the one unit of type 2 slot has a positive probability of being taken by a type 1 customer (which would essentially be a waste, because type 1 slots are sufficient to satisfy all possible demand from type 1 customers). In contrast, if $(m_2, m_3) = (0,1)$, the one unit of type 3 slot can only be exclusively used by type 2 customers, and thus, this is more efficient. This simple example shows that, because of customers' ability to (randomly) choose from their offer set, less versatile slots may be more valuable than versatile slots because of resource imbalance. This observation implies that the (future) value of keeping a slot type cannot be viewed solely based on the number of accepting customer types, irrespective of the arrival probabilities or slot capacities. This complication renders the optimal policy for a general model instance quite complex as we show now.

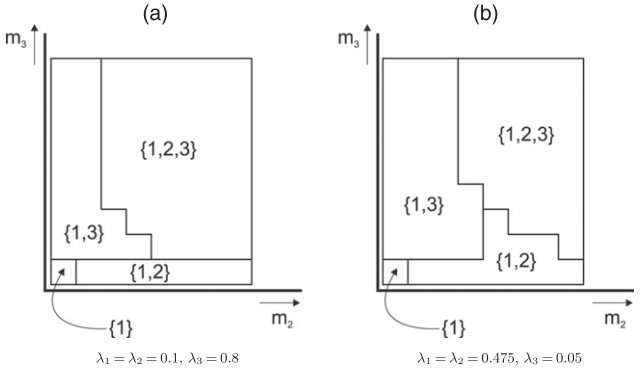
The next model instance that we focus on is the M + 1 model instance shown in Figure 1(d), with choice matrix

$$\Omega = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Note that the only difference between the M + 1 and M model instances is the additional customer type 3 that only accepts type 2 slots. It turns out that the simple, elegant form of the optimal policies in the previous cases does not carry over to the M + 1 model instance.

To illustrate the complexity of the M + 1 model instance, consider the case with $m_1 = 4$ and $n = 5$. Figure 2 shows the *unique* optimal offer set, identified with $S \subset \{1,2,3\}$, as a function of m_2 and m_3. (For instance, if $S = \{1,3\}$, it means offering slot types 1 and 3 but not slot type 2.) Consider $\lambda_1 = \lambda_2 = 0.1$, $\lambda_3 = 0.8$ (Figure 2(a)) or $\lambda_1 = \lambda_2 = 0.475$, $\lambda_3 = 0.05$ (Figure 2(b)). As discussed earlier, resource imbalance can make a less versatile slot more valuable than a more versatile one, which naturally would suggest an action of saving the less versatile slot by only offering the versatile slot.

Figure 2. The Optimal Policy for an M + 1 Model Instance, with $m_1 = 4$, $n = 5$



Indeed, in Figure 2(b), we see that action $\{1,2\}$ can be the unique optimal action, even when $\mathbf{m} > 0$. This is true, because m_3 is relatively small (equal to one or two in this case), whereas $m_1 = 4$ is ample given $n = 5$ and the symmetric arrival rates of type 1 and type 2 customers. Blocking type 3 and offering the versatile type 2 earlier rather than later can help to resolve the resource imbalance by maximizing the total expected amount of type 2 slots taken by type 2 customers (and thus, saving type 3 slots that can only serve type 2 customers for the future).

In addition, we see that the arrival rate now has a strong impact on the optimal policy in contrast to the other cases that we discussed so far: when λ_3 is large, it is often optimal to include type 2 slots in the offer set, whereas when λ_3 is small, this is not the case. The reasoning here is that, for λ_3 small, the model is very close to the M model instance, for which we know it is optimal to save versatile type 2 slots for later in the booking process. However, not offering type 2 slots also implies turning away all type 3 customers, which explains why this slot type should be offered when λ_3 is large.

These observations we make on the M + 1 model instance suggest that the optimal policy depends on the customer preference profiles, arrival rates, and available slot capacity of the specific model instance under consideration. The optimal policy for a general model can be quite complex and may have no straightforward structural properties. Thus, we shall focus our efforts on identifying simple scheduling policies that have performance guarantees and perform well in practice.

3.2. Asymptotically Optimal Policy

In this section, we construct a *static randomized policy* that is asymptotically optimal when we increase the system demand and capacity simultaneously. We first introduce the class of static randomized policies. At any decision epoch, there are altogether 2^J possible actions in terms of which slot types to offer. Here, we use a binary vector to denote the offer set with a one at position j, meaning that slot type j is offered, and zero otherwise. For example, we denote by the action of closing all slots as $\mathbf{w}^1 := (0,\ldots,0)$, the J-dimensional zero vector, and the action of opening all slots as $\mathbf{w}^{2^J} := (1,\ldots,1)$, the J-dimensional one vector. We call the set of all 2^J J-dimensional zero-one vectors as set $\mathcal{W} := \{0,1\}^J$ and name the elements of the set as $\mathbf{w}^1, \mathbf{w}^2 \ldots, \mathbf{w}^{2^J}$. Define $\mathcal{K} := \{1,\ldots,2^J\}$ as the action index set, and therefore, $\mathcal{K}$ and $\mathcal{W}$ have the same cardinality.

A policy π^p is a static randomized policy if π^p offers $\mathbf{w}^k$ with some fixed probability p_k, independent of the system state and the time period.[1] The class of static randomized policies contains all π^p values, such that the vector $\mathbf{p} = \{p_k\}_{k=1}^{2^J}$ is a probability vector. For instance, the offering-all policy is a special case in this class with $p_{2^J} = 1$ and $p_k = 0$ for all $k \neq 2^J$.

We show that there exists a vector $\mathbf{p}^*$, such that π^{p^*} is asymptotically optimal when the demand and capacity are scaled up simultaneously. The choice of p_k^* relies on the fluid model corresponding to the stochastic model (3) considered above, in which we can readily determine the optimal offering policy. We choose p_k^*, such that it represents the fraction of the time in which the action $\mathbf{w}^k$ is used in a fluid model under optimal control. Below, we construct this asymptotically optimal policy π^{p^*} and defer more technical details to the online appendix.

3.2.1. Fluid Model.

We first introduce our fluid model. To differentiate from the notation in the stochastic model formulation (3), we shall put the time index n in parentheses instead of as a subscript. Instead of discrete customers arriving in each slot, we represent a customer by a unit of fluid. In total, one unit of demand arrives in each time period, a fraction λ_i of which corresponds to customer type i. This fluid is distributed evenly among all available slots that are offered and accepted by the corresponding customer type.

For each $n = 1,\ldots,N$, the decision vector in the fluid model is $\mathbf{z}(n) = (z_1(n),\ldots,z_{2^J}(n))$, which is a 2^J-dimensional vector, each component $z_k(n) \in [0,1]$ representing the time during which action $k \in \mathcal{K}$ is being used in period n. Note that each action can be used for any fractional unit of time. Thus, we require that

$$0 \le z_k(n) \le 1, \forall k \in \mathcal{K}, n = 1,\ldots N; \tag{6}$$

$$\sum_{k\in\mathcal{K}} z_k(n) = 1, \forall n = 1,\ldots N. \tag{7}$$

Constraint (7) ensures that the total time spent on all possible actions (including the one that closes all slot types) in one period adds up to one.

Let $\boldsymbol{\tau}(n) = [\tau_{k,j}(n)]$ be a $2^J \times J$ matrix, each row of which corresponds to one of the 2^J possible actions. We use $\tau_{k,j}(n)$ to indicate the amount of time for which type j slots are offered during the time when the kth action is taken in period n. We have that

$$\tau_{k,j}(n) = z_k(n)\mathbf{w}_j^k, \ \forall k \in \mathcal{K}, j = 1,2,\ldots,J, n = 1,\ldots N, \tag{8}$$

where $\mathbf{w}_l^k$ denotes the lth coordinate of vector $\mathbf{w}^k$. Constraint (8) is presented mainly to make the formulation clearer and easier to understand. It ensures that slot type j can be open when action k is chosen only if action k offers slot type j. If action k does not offer slot type j, then $\mathbf{w}_j^k = 0$, and $\tau_{k,j}(n)$ is zero by (8). Let $\mathcal{J}_k = \{j : \mathbf{w}_j^k = 1, j = 1,2,\ldots,J.\}$ be the full set of slot types offered by action k. Note that (8) implies that

$$\tau_{k,j_1}(n) = \tau_{k,j_2}(n), \ \forall k \in \mathcal{K}, n = 1,\ldots N, j_1, j_2 \in \mathcal{J}_k.$$

That is, if an action k offers multiple slot types, the offering durations of these slot types are the same.

Let $y_{i,j}(n)$ denote the amount of type j slot's capacity filled by type i customers during period n and $\mathcal{H}_j = \{s : \mathbf{w}_j^s = 1, s \in \mathcal{H}\}$ be the index set of actions that offer type j slots. If $\Omega_{i,j} = 1$,

$$y_{i,j}(n) = \sum_{k \in \mathcal{H}_j} \tau_{k,j}(n) \cdot \frac{\lambda_i}{\sum_{l=1}^{J} \min\{\Omega_{i,l}, \mathbf{w}_l^k\}}, \; i = 1, 2, \ldots, I, \; j = 1, 2, \ldots, J, \; n = 1, \ldots, N; \tag{9}$$

otherwise, if $\Omega_{i,j} = 0$, then

$$y_{i,j}(n) = 0, \; i = 1, 2, \ldots, I, \; j = 1, 2, \ldots, J, \; n = 1, \ldots, N. \tag{10}$$

Note that all terms in (9), except $\tau_{k,j}(n)$, are constants, and therefore (9), as a set of constraints for the optimization problem, they are linear in the decision variables $\tau_{k,j}(n)$.

Let $M_j(t)$ be the amount of type j slots left with t time periods to go, and let $Z_N(\mathbf{m})$ be the optimal amount of (fluid) customers served with initial capacity vector $\mathbf{m}$ and N periods to go. The goal is to choose $z_k(n)$ (and $\tau_{k,j}(n)$) to solve for

$$Z_N(\mathbf{m}) = \max \sum_{n=1}^{N} \sum_{j=1}^{J} \sum_{i=1}^{I} y_{i,j}(n), \tag{P1}$$

subject to: (6), (7), (8), (9), and (10), and

$$M_j(N) = m_j, \qquad j = 1, 2, \ldots, J, \tag{11}$$

$$M_j(n-1) = M_j(n) - \sum_{i=1}^{I} y_{i,j}(n), \quad j = 1, 2, \ldots, J, \; n = 1, \ldots, N, \tag{12}$$

$$M_j(n) \geq 0, \quad j = 1, 2, \ldots, J, \; n = 0, 1, \ldots, N-1. \tag{13}$$

In (P1), constraint (11) specifies the initial capacity vector, (12) updates the capacity vector for each period, and (13) ensures that all slot types have nonnegative capacity throughout. We remark that, in our formulation, control can be exerted anytime continuously throughout the horizon, but the system is observed only at discrete time epochs $0, 1, 2, \ldots, N$ to match the stochastic model formulation (3).

3.2.2. Choice of $\mathbf{p}^*$. Let p_k^* be the fraction of the time in which the optimal policy chooses action k in the fluid model (P1). That is,

$$p_k^* = \frac{\sum_{n=1}^{N} z_k^*(n)}{N}, \tag{14}$$

where $z_k^*(n)$ is the optimal solution to (P1). We now translate this optimal policy for the fluid model to our original discrete and stochastic setting by defining a policy π^{p^*}, such that, in each period n, this policy offers $\mathbf{w}^k$ with probability p_k^*, independent of everything else.

The intuition behind choosing $\mathbf{p}^*$ as the offering probability vector is that, if we scale up the system demand (i.e., N) and capacity (i.e., $\mathbf{m}$) in the stochastic model, using π^{p^*} makes the proportion of total customer demand going to each slot type in the stochastic model approximately match that in the fluid model. Thus, the total fill counts in the stochastic model are similar to those of the fluid model. Because the fluid model is a deterministic model that provides an upper bound on the objective value of the stochastic model (more on this below), we know that $\mathbf{p}^*$ is (close to) optimum in the stochastic model as the system becomes large. We formalize this intuition in the next section.

3.2.3. Main Result. Consider a sequence of problems indexed by $K = 1, 2, 3, \ldots$. The problems in this sequence are identical except that, for the Kth problem, the number of total periods is NK and the capacity vector is $\mathbf{m}K$. We call the problem instance with $K = 1$ the base problem instance. Let $V_n^{\pi^p}(\cdot)$ be the total expected number of slots filled under a policy π^p with the offering probability vector $\mathbf{p}$ in the stochastic model. The main result is shown in the following theorem.

Theorem 1.

(i) $K^{-1} V_{NK}(\boldsymbol{m}K) \leq K^{-1} Z_{NK}(\mathbf{m}K) = Z_N(\mathbf{m}), \; \forall \mathbf{m} \geq 0, \; K = 1, 2, 3, \ldots;$

(ii) $\lim_{K \to \infty} K^{-1} V_{NK}^{\pi^{p^*}}(\boldsymbol{m}K) = Z_N(\mathbf{m})$.

Recall that $V_n(\cdot)$ is the optimal value of the stochastic model defined in (3). Thus, Theorem 1(i) says that the "normalized" optimal value of the nonsequential offering stochastic model (i.e., the original value divided by K) is bounded from above by that of the corresponding fluid model and that the normalized objective value of the fluid model is the same as the objective value of the base fluid model with $K = 1$. Theorem 1(ii) states that, as the system grows large, the normalized objective value in the stochastic system under policy π^{p^*} converges to this constant upper bound, and thus, π^{p^*} is asymptotically optimal.

The proof of Theorem 1 entails a few key steps, which are outlined below (full details can be found in the online appendix). We first show that the optimal objective value of the fluid model is an upper bound of the optimal value of the stochastic model (i.e., $Z_N(\mathbf{m}) \geq V_N(\mathbf{m})$) for any given set of model parameters. Then, based on any static randomized policy π^p, we construct a lower bound for $V_n^{\pi^p}(\cdot)$, and this lower bound is naturally a lower bound for the optimal value of the stochastic model $V_N(\mathbf{m})$ (because π^p is not necessarily optimal). Finally, we show that, when p is chosen as p^* defined in (14), the normalized lower bound converges to $Z_N(\mathbf{m})$ when the system grows in both demand and capacity. Thus, π^{p^*} is asymptotically optimal.

Our findings build on the early classic results in the revenue management literature, which show that allocation policies arising from a single linear program make the normalized total expected revenue converge to an upper bound on the optimal value (Cooper 2002). Our results are different and new in several important aspects. The model in Cooper (2002) can designate/allocate a particular product (slot) type on a customer arrival (because he assumes that customer preference is known on arrival), whereas our model offers multiple product (slot) types for customers to choose from (because the customer preference is not known). Using the offer set as a decision in the model creates significant new challenges. Our fluid model formulation needs to explicitly take care of customer choice processes and is much more complicated than that in Cooper (2002). Leveraging the fluid model formulation, the asymptotically optimal policy in Cooper (2002) accepts customer requests up to some customer type-specific thresholds, because the optimal solution in the fluid model of Cooper (2002) prescribes such thresholds for each customer type. As a result, the asymptotic policy of Cooper (2002) leads to a closed form expression for each type of the customer demand served, allowing him to show that the normalized demand served converges in distribution to a constant that matches the optimal fluid model decision, by directly using a result in Billingsley (1968). However, because of customer choices, our fluid model cannot give rise to such a simple policy. Our fluid model informs the optimal duration in which a particular offer set is used, and we use this information to construct our policy, which has a completely different form compared with the policy of Cooper (2002). Because we cannot control the exact product (slot) type in an offer set that will be chosen by an arriving customer, we do not have a closed form expression as in Cooper (2002) for the total demand that goes into each product (slot) type and eventually gets served. To deal with this difficulty, we construct a (very) tight lower bound on the objective value and show that this lower bound, after being normalized, converges to the optimal objective value of the fluid model. The idea of our proof may be useful to identify effective approximate policies in other capacity management contexts when the manager cannot directly control the product that a customer may pick.

3.3. Constant Performance Guarantee of the Offering-All Policy

In this section, we focus on a simple scheduling policy: the offering-all policy. Let π_0 represent this policy, and the offer set under π_0 is the full set of all slot types, irrespective of the period n. Note that the effective offer set at state $\mathbf{m}$ is $\bar{S}(\mathbf{m})$. That is, when customers arrive, they only consider those slot types with positive capacity when making a choice. We denote by $V_n^{\pi_0}(\mathbf{m})$ the expected fill count attained by applying the offering-all policy π_0 throughout.

Indeed, this simple policy has a constant performance guarantee that states that, for any set of parameters, the offering-all policy π_0 achieves at least one-half of the optimal fill count.

Theorem 2. *For any* Ω, n, *and* $\mathbf{m}$, $V_n(\mathbf{m}) \le 2V_n^{\pi_0}(\mathbf{m})$.

It is worth noting that Theorem 2 in fact holds more broadly for all so-called *myopic policies*, which at each period, offer a set maximizing the expected number of filled slots for that period. Myopic policies, however, do not have to offer all slot types in all periods. For instance, offering slot types 1 and 3 in the M model instance would constitute a myopic policy.

Performance guarantee results on myopic policies exist in various dynamic optimization settings, and a ratio of two is often the best provable performance bound (e.g., Chan and Farias 2009 and Mehta 2013). Although this performance bound may seem a little loose, we shall see empirically in Section 5.1 that the offering-all policy performs very well and much better than this lower bound; in finite regimes, the offering-all policy also seems to perform better than the asymptotically optimal policy constructed in Section 3.2.

4. Sequential Offering

We now present our second scheduling paradigm, which allows the scheduler to offer multiple sets of slots sequentially. Recall that this way of offering slots may represent, for instance, web-based scheduling where available slots are not revealed simultaneously as well as telephone-based scheduling. Intuitively, having the scheduler offer slots sequentially instead of all at once will be able to steer customers into selecting more favorable slots from the perspective of system optimization. The questions that we address in this section are then how many and what sets of slots to offer to maximize the fill rate. We start by introducing the sequential offering model next.

4.1. Model Outline

On customer arrival, the scheduler chooses a K, $1 \le K \le J$, and sequentially presents the customer with K mutually exclusive subsets $S_1, S_2, \ldots, S_K \subseteq \bar{S}(\mathbf{m})$. We denote this action as $\mathbf{S} := S_1 - S_2 - \cdots - S_K$. Denote by $\mathcal{S}(\mathbf{m})$ the set of all possible such actions at state $\mathbf{m}$ and by $I_k(\mathbf{S}) := \{i : \sum_{j \in S_k} \Omega_{ij} \ge 1, i \notin \cup_{l=1}^{k-1} I_l(\mathbf{S})\}$, $k = 1, \ldots, K$, the set of customer types who do not accept any slot from the first $(k-1)$ offer sets but encounter at least one acceptable slot in S_k. Therefore, $I_k(\mathbf{S})$ represents the set of customers who, given sequential offering $\mathbf{S}$, accept some slot on arrival into the system. Moreover, the slot chosen by these customers belongs to the kth offer set S_k. The probability that slot type j is chosen under action $\mathbf{S}$ may then be written as $q_j(\mathbf{S}) := \sum_{k=1}^{K} \sum_{i \in I_k(\mathbf{S})} \lambda_i q_{ij}(S_k)$, with $q_{ij}(\cdot)$

as in (1). The assumption that the sets $S_1, S_2, \ldots, S_K$ are mutually exclusive is made from a practical rather than mathematical standpoint: there is simply no reason to offer the same slot type in two or more sets, because the customer will book a slot as soon as she is offered a set with at least one acceptable slot. Thus, only the first set in which such a slot is included is relevant.

For ease of presentation, we still use $V_n(\mathbf{m})$ to denote the maximum expected number of slots that can be booked with $\mathbf{m}$ slots available and n periods to go in this section. For an action $\mathbf{S}$, we let $\bigcup \mathbf{S} := \bigcup_{i=1}^{K} S_i$ denote the set of all slot types offered throughout action $\mathbf{S}$. Then, for the sequential offering model, we have

$$V_n(\mathbf{m}) = \max_{\mathbf{S}\in\mathcal{S}(\mathbf{m})}\left[\sum_{j\in\bigcup\mathbf{S}} q_j(\mathbf{S})\left(1-\Delta^j_{n-1}(\mathbf{m})\right)\right] + V_{n-1}(\mathbf{m}), \quad \text{for } n = N, N-1, \ldots, 1, \tag{15}$$

where $V_0(\cdot) = 0$ and $\Delta^j_{n-1}(\mathbf{m}) := V_{n-1}(\mathbf{m}) - V_{n-1}(\mathbf{m}-\mathbf{e}_j)$ denotes the marginal benefit owing to the m_jth unit of slot type j at period $n-1$. We observe that both the transition probability $q_j(\mathbf{S})$ and the set of feasible actions $\mathcal{S}(\mathbf{m})$ are much more complicated than their counterparts in the nonsequential model.

The sequential offering setting can be viewed as a generalization of nonsequential scheduling to any number $K \geq 1$ of offer sets. Consequently, it stands to reason that the offering-all policy will not perform well in the sequential setting, because this would limit the scheduler to a single offer set ($K = 1$). We indeed numerically confirm this conjecture in Section 5.3. Note that, in contrast to the nonsequential case, an offering-all policy is unlikely to be used in a practical setting, such as telephone scheduling (because it would take too much time for the scheduler to go over every possible appointment option). In the online setting, there is a way to take advantage of sequential offerings by redesigning the customer interface that releases information sequentially.

To provide a roadmap of analyzing the sequential model, we summarize our key findings in this section as follows.

1. We first consider a general setting and derive various structural results that provide more insights; in particular, we show that it is optimal to offer slot types one by one.

2. For a large class of problem instances with nested preference structures (to be discussed later), we derive a closed form optimal sequential offering policy.

3. For problem instances not in this class, we develop a simple and highly effective heuristic based on the idea of balanced resource use and fluid models.

4. We prove that the optimal sequential offering does as well as in the nonsequential case where the scheduler has full information of the customer type on arrival; we argue that this equivalence allows us to apply the idea of sequential offering in various interactive scheduling contexts.

4.2. Results for the General Sequential Offering Model

We now present some properties of the sequential model with general choice matrices. We first derive some structural properties of the value function.

Lemma 1. *The value function $V_n(\mathbf{m})$ satisfies*

(i) $0 \leq V_{n+1}(\mathbf{m}) - V_n(\mathbf{m}) \leq 1, \ \forall \mathbf{m} \geq 0, \ \forall n = 1, 2, \ldots, N-1$; *and*

(ii) $0 \leq V_n(\mathbf{m}+\mathbf{e}_j) - V_n(\mathbf{m}) \leq 1, \ \forall \mathbf{m} \geq 0, \ \forall n = 1, 2, \ldots, N$.

Part (ii) of Lemma 1 implies that $V_n(\mathbf{m}+\mathbf{e}_j) \leq V_n(\mathbf{m}) + 1$ (i.e., it is better to have a slot booked now rather than saving it for future). Therefore, in the context of sequential offering, it is better to keep offering slots if none have been taken so far. This is formalized in the following result, which shows that there exists an optimal sequential offering policy that exhausts all available slot types in each period.

Lemma 2. *For any Ω, $\mathbf{m}$, and n, there exists an optimal action $\mathbf{S}^*$, such that $\bigcup \mathbf{S}^* = \bar{S}(\mathbf{m})$.*

Building on Lemma 2, we are able to characterize the structure of an optimal sequential offering policy described in the theorem below.

Theorem 3. *Let $\mathbf{m} > 0$ be the system state at period $n \geq 1$, and let $j_1, j_2, \ldots, j_J$ be a permutation of $1, 2, \ldots, J$, such that $V_{n-1}(\mathbf{m}-\mathbf{e}_{j_k}) \geq V_{n-1}(\mathbf{m}-\mathbf{e}_{j_{k+1}})$, $k = 1, 2, \ldots, J-1$. Then, the action $\{j_1\} - \cdots - \{j_J\}$ is optimal.*

Theorem 3 implies that there exists an optimal policy that offers one slot type at a time. More importantly, this result shows a specific optimal offer sequence based on the value function to go. To understand this, recall that $V_{n-1}(\mathbf{m}) - V_{n-1}(\mathbf{m}-\mathbf{e}_j)$ can be viewed as the value of keeping the m_jth type j slot from period $n-1$ onward. Because all customers bring in the same amount of reward, it benefits the system the most if an arrival customer can be booked for the slot type with the least value to keep (i.e., the slot type with the largest $V_{n-1}(\mathbf{m}-\mathbf{e}_j)$).

Even if the scheduler does not know the exact customer type, following the optimal offer sequence described in Theorem 3 ensures that the arriving customer takes the "least valuable" slot (as long as there is at least one acceptable slot remaining). Indeed, matching customers with slots in this way would be the best choice for the scheduler, even if she had perfect information about customer type (i.e., she knew exactly the customer type on arrival). Following this rationale, our next result shows an interesting and important correspondence between (1) the sequential offering without customer type information and (2) the nonsequential offering with *perfect* customer type information. To distinguish these

two settings, we let $V_n^s(\mathbf{m})$ and $V_n^f(\mathbf{m})$ represent the optimal value functions for settings (1) and (2), respectively, in the next theorem.

Theorem 4. $V_n^s(\mathbf{m}) = V_n^f(\mathbf{m})$, $\forall\, \mathbf{m} \geq 0$, $n = 0, 1, 2, \ldots, N$.

Theorem 4 suggests that the optimal sequential offering can fully exploit the value of customer type information; however, it does not imply that it can fully elicit the customer type. Specifically, optimal sequential offering happens to result in the same system state changes as if the scheduler had full information about customer type but does not let the scheduler know exactly the customer type (Remark 2 in Section 4.4). Theorem 4 suggests that sequential offering is a useful operational mechanism to improve the scheduling efficiency in the absence of customer type information. Our numerical experiments in Section 5 confirm and quantify such efficiency gains.

4.3. Optimal Sequential Offering Policies

In this section, we fully characterize the optimal sequential offering policy for a large class of choice matrix instances, which include the N, M, and M + 1 model instances (Figure 1). To this end, let $I(j)$ be the set of customer types who accept slot type j (i.e., $I(j) = \{i = 1, 2, \ldots, I : \Omega_{ij} = 1\}$, $\forall j = 1, 2, \ldots, J$). It makes intuitive sense that, if $I(j_1) \subset I(j_2)$, then slot type j_2 is more valuable than j_1, and thus, slot type j_1 should be offered first. Combining this observation with Theorem 3 could then help us to design an optimal policy. Let us first introduce a specific class of model instances.

Definition 1. We say that a model instance characterized by Ω is nested if, for all $j_1, j_2 = 1, 2, \ldots, J$ and $j_1 \neq j_2$, one of the following three conditions holds: (i) $I(j_1) \cap I(j_2) = \emptyset$, (ii) $I(j_1) \subset I(j_2)$, or (iii) $I(j_1) \supset I(j_2)$.

Note that not all model instances are nested. One simple example is the W model instance from Figure 1(b), where $I(1) = \{1, 2\}$ and $I(2) = \{2, 3\}$. None of the conditions (i)–(iii) from Definition 1 hold in this case for $j_1 = 1$ and $j_2 = 2$. However, it is readily verified that the N, M, and M + 1 model instances are all nested.

Remark 1. The concept of a nested model instance is related to the *star structure* considered in the previous literature on flexibility design (e.g., Akçay et al. 2010). Consider a system with a certain number of resource types (corresponding to slot types in our context), which can be used to do jobs of certain types (customer types in our context). A star flexibility structure is one such that there are specialized resource types, one for each job type, plus a versatile resource type that can perform all job types. The nested structure generalizes the star structure.

It turns out that we can fully characterize an optimal policy for nested model instances as follows.

Theorem 5. *Suppose Ω is nested; any policy that offers slot type j_1 before offering slot type j_2 for any j_1, j_2 such that $I(j_1) \subset I(j_2)$ is optimal.*

Theorem 5 proposes to offer nested slot types in an increasing order of the accepting customer types. Note that, when two slot types are mutually exclusive (i.e., $I(j_1) \cap I(j_2) = \emptyset$), the order in which they are offered is irrelevant, because customers who would select a slot from one set could never from the other. To give some specific examples, we can fully characterize the optimal policy for the N, M, and M + 1 model instances using Theorem 5.

Corollary 2. *For the N model instance and any n and $\mathbf{m}$, an optimal sequential offering policy is to offer $\mathbf{S} = \{1\} - \{2\}$.*

Corollary 3. *For the M and M + 1 model instances and any n, an optimal sequential offering policy is to offer*

$$\mathbf{S} = \begin{cases} \{1, 3\} - \{2\}, & \text{if } m_1, m_2, m_3 \geq 1, \\ \{1\} - \{2\}, & \text{if } m_3 = 0, \\ \{3\} - \{2\}, & \text{if } m_1 = 0. \end{cases}$$

4.4. Beyond Nested Model Instances

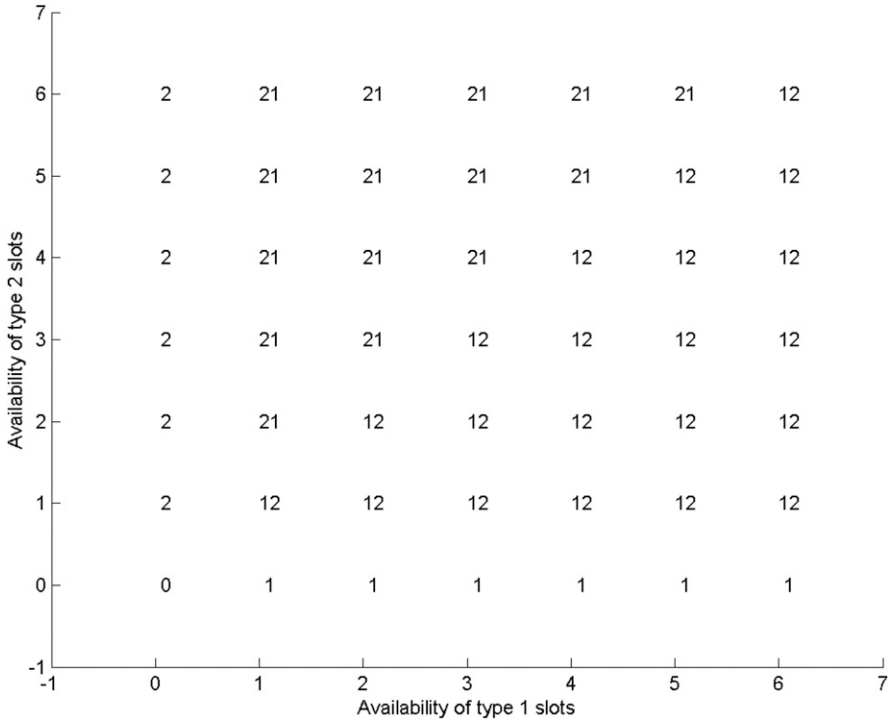
Although Theorem 5 solves a large class of the sequential model instances, not all instances have a nested structure. In this section, we analyze the W model instance (Figure 1) to glean some insights into the instances that are not nested.

To analyze the W model instance, one can formulate an MDP with three possible actions: $\{1, 2\}$, $\{1\} - \{2\}$, and $\{2\} - \{1\}$ (and the corresponding actions at the boundaries). However, there exist no straightforward offering orders for slot types, and the optimal sequential policy turns out to be state dependent. Specifically, we find that the optimal policy is a *switching curve* policy: with the availability of one type of slots held fixed, it is optimal to offer the other type of slots first as long as there is a sufficiently large amount of such slots left.

Figure 3 illustrates the optimal actions for the W model instance at different system states with $\boldsymbol{\lambda} = (0.2, 0.5, 0, 3)$ and $n = 6$. The numerals "0," "1," "2," "12," and "21" correspond to the actions of offering nothing, offering type 1 slots only, offering type 2 slots only, offering type 1 slots and then type 2 slots, and offering type 2 slots and then type 1 slots, respectively. The optimal actions at boundary are obvious. In the interior region of the system states, we can clearly see the switching curve structure. For instance, when the system state is (3, 3), it is optimal to offer $\{1\} - \{2\}$. When the number of type 2 slots increases to four, then it is optimal to offer $\{2\} - \{1\}$.

The intuition behind this is different from that of the model instances considered above where customer preferences are nested (e.g., the N, M, and M + 1 model

Figure 3. Structure of the Optimal Policy Under W Model Instance with Sequential Offers



instances). In the W model instance, type 1 (type 3) customers only accept type 1 (type 2) slots, but type 2 customers accept both types of slots. If there are relatively more type 1 slots than type 2 slots, then it makes more sense to "divert" type 2 customers to choose type 1 slots, thus saving type 2 slots only for type 3 customers. Accordingly, the switching curve policy stipulates that type 1 slots be offered first, ensuring that type 2 customers, if any, will pick type 1 slots. The intuition above is formalized in the proposition below.

Proposition 3. *Consider the W model instance with sequential offers. Given* m_2*, if there exists an* m_1^* *such that the optimal action at state* (m_1^*, m_2) *is* $\{1\}-\{2\}$*, then* $\forall \mathbf{m} \in \{(m_1, m_2), m_1 \geq m_1^*\}$*, and the optimal action is* $\{1\}-\{2\}$*. Similarly, given* m_1*, if there exists an* m_2^* *such that the optimal action at state* (m_1, m_2^*) *is* $\{2\}-\{1\}$*, then* $\forall \mathbf{m} \in \{(m_1, m_2), m_2 \geq m_2^*\}$, and *the optimal action is* $\{2\}-\{1\}$.

Remark 2. In Section 4.3, we state that sequential offering may not fully reveal exact customer types but allows the system to evolve in the same optimal way as if the scheduler knew exactly the customer type. We use the W model instance to illustrate this point. Consider the W model instance with nonsequential offers and the scheduler knowing the exact type of arriving customers. Suppose that the optimal action is to offer {1} when type 1 or type 2 customers arrive and offer {2} when type 3 customer arrives. Now, in a sequential offering model where the scheduler does not know the exact type of arriving customers, the scheduler would have offered $\{1\}-\{2\}$ to any arriving customer. If we encountered type 1 or 2 customers, type 1 slot would be taken, but we do not know the exact type of this customer (we know, however, that she must be either type 1 or type 2); if a type 3 customer arrived, she would reject type 1 slot but take type 2 slot. In this way, the system evolves as if the scheduler had perfect information on customer type. The structural properties of the optimal policy described in Proposition 3 are likely the best that we can obtain for the W model instance; the exact form of the optimal policy depends on model parameters and the system state, much like with the M + 1 model instance in the nonsequential case. If customer preference structures become more complicated, it is very difficult, if not impossible, to develop structural properties for the optimal sequential offering policy. Thus, for model instances that do not satisfy the conditions of Theorem 5, we propose an effective heuristic below.

4.5. The Drain Heuristic

If customer preferences are not nested, the analysis of the W model instance suggests that the optimal policy is to offer slots with more capacity relative to its customer demand. Inspired by this observation and using the idea of fluid models, we propose the following heuristic algorithm, which aims to "drain" the abundant slot type first followed by less abundant ones. This heuristic aims to have all slot types emptied simultaneously, thus maximizing the fill rate. That is, this heuristic tries to "balance" the resource use. Specifically,

the drain algorithm works in the following simple way. At period n and for each slot type $j \in \bar{S}(\mathbf{m})$, we calculate

$$I_j := \frac{m_j}{n\sum_{i=1}^{I}\lambda_i \frac{\Omega_{ij}}{\sum_{k\in\bar{S}(\mathbf{m})}\Omega_{ik}}}. \tag{16}$$

Note that the fraction term in the denominator of (16) represents the share of type i customers who will choose type j slots, assuming that all available slot types are offered simultaneously. Taking expectation with respect to the customer type distribution and multiplying by n, the number of customers to come, the denominator of (16) can be viewed as the expected load on type j slots in the next n periods. As a result, the index I_j can be regarded as the ratio between capacity left and "expected" load.

The drain algorithm is then to calculate all I_j values at the beginning of each period and offer slots in decreasing order of the I_j. The algorithm calls for offering slot types with larger I_j first, because these slot types have relatively more capacity compared with demand. In other words, a slot type with a larger I_j is likely to have a smaller marginal value to keep and thus, can be offered earlier. We could of course safely remove n in the definition of I_j and obtain the exact same order of slots. However, we leave n in the denominator of (16), because this allows us to interpret I_j as the ratio between capacity left and expected number of requests. Based on this interpretation, it is clear that this heuristic aims to have all slot types emptied simultaneously, thus maximizing the fill rate. We will test the performance of this algorithm in Section 5.2.

4.6. Applications to Interactive Scheduling

In Sections 3 and 4, we discuss two different models of customer-scheduler interactions in the appointment booking practice. In one model, the scheduler makes a one-shot offering, and in the other, the scheduler enjoys the full flexibility of sequential offering. The appointment booking process, however, can fall in between these two models in terms of the degree to which the customer preference information is collected and used during the interaction between the scheduler and each customer. Such interactions may be present both in a traditional setting with human interaction (e.g., a customer, after being offered an appointment at 8 a.m. by a receptionist, may indicate that none of the morning slots are acceptable) or fully digitally (e.g., the Partners HealthCare Patient Gateway online booking website allows patients to indicate their acceptable time slots upfront).

When additional customer preference information is gathered during the appointment booking process, the scheduler can still follow the optimal list of slot types $\{j_1\} - \cdots - \{j_J\}$ obtained from Theorem 3 but simply skip all slots known to be unacceptable either up front or dynamically as additional information is collected. This offering strategy is still optimal, because it would end up with the same system state compared with not skipping those slots indicated as unacceptable before or during the booking process (e.g., directly declared by the customer) and thus, give the exact same fill count that can be obtained if the scheduler had full information about the customer type (Theorem 4). Recall that the order of $\{j_1\} - \cdots - \{j_J\}$ can be readily obtained with nested customer preferences (Theorem 5), or otherwise, an approximate order can be easily formed by the drain heuristic (16).

Although outside the scope of this paper, these considerations on interactive scheduling raise various issues related to the tradeoff between obtaining the best fill rate and providing a convenient experience to the customer. For instance, the scheduler may want to limit the number of sets offered to the customer to provide a smooth user experience. In light of Theorem 3, one potential idea for future study is to group slot types based on the order of $\{j_1\} - \cdots - \{j_J\}$.

5. Numerical Results

In the last two sections, we consider nonsequential offering and sequential offering. For each setting, we derive optimal or near-optimal booking policies. In this section, we run extensive numerical experiments to test and compare these policies and the two scheduling paradigms.

We organize this section as follows. Sections 5.1 and 5.2 discuss the performance of the offering-all policy in the nonsequential model and the drain heuristic in the sequential model, respectively. We show that these two algorithms obtain fill rates that are remarkably close to those of the respective optimal policies, and therefore, they can serve as simple, effective heuristics for practical use. Section 5.3 compares the differences in the expected fill rate under the nonsequential and sequential offering models, where this difference represents the value of sequential offering. Specifically, we evaluate the differences between the optimal policies and those between the heuristics. The former represents the "theoretical" value of sequential offering compared with nonsequential offering, whereas the latter can be thought of as the "practical" value if practitioners adopt the heuristics mentioned above for each setting.

5.1. Performance of the Offering-All Policy

We start our evaluation of the offering-all policy in two specific model instances considered above: M and M + 1 model instances. (We need not to evaluate the offering-all policy in the N and W model instances, because the offering-all policy is optimal there.) Here, we use backward induction to determine the expected performance of the optimal policy and compare it through simulation with that of the offering-all policy π_0. To this

Table 1. Optimality Gap of the Offering-All Policy in the M Model Instance

		$(\lambda_1, \lambda_2) = (1/2, 1/2)$, %			$(\lambda_1, \lambda_2) = (1/3, 2/3)$, %			$(\lambda_1, \lambda_2) = (1/4, 3/4)$, %		
N	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
20	45	−4.4	−3.6	−3.6	−4.1	−3.3	−3.3	−3.6	−2.9	−3.0
30	91	−4.8	−3.7	−3.8	−4.5	−3.5	−3.5	−3.8	−3.1	−3.2
40	153	−5.1	−3.8	−3.8	−4.7	−3.6	−3.6	−4.0	−3.2	−3.3
50	231	−5.3	−3.8	−3.8	−4.8	−3.7	−3.7	−4.1	−3.3	−3.4

end, we simulate the offering-all policy for 1,000 days. The performance metric of interest is the percentage optimality gap defined as $(u_g - u_o)/u_o \times 100\%$, where u_o is the expected fill count of the optimal policy and u_g is the average fill count over 1,000 simulated days under the offering-all policy.

Table 1 summarizes the statistics on the optimality gap of the offering-all policy in the M model instance. For each $N = 20, 30, 40, 50$, we evaluate the maximum, average, and median optimality gaps over all possible initial capacity vectors $(b_1, b_2, b_3) \in \mathbb{Z}_+^3$, such that $b_j \geq 0.2N$, $\forall j$ and $b_1 + b_2 + b_3 = N$. The number of initial capacity vectors considered for each N is shown as the number of scenarios in the second column of Table 1. In general, the optimality gap of the offering-all policy in the M model instance is relatively small ($\approx$ 3%–4%) and is not sensitive to model parameters.

Table 2 shows the optimality gap statistics for the M + 1 model instance, and the setup of this table is similar to that of Table 1. When λ_3 is small, the M + 1 model instance is very similar to the M model, and thus, the optimality gaps of the offering-all policy are similar to those observed in Table 1. As λ_3 increases, the performance of the offering-all policy improves, because the offering-all policy becomes more likely to be optimal.

To evaluate the performance of the offering-all policy in settings beyond these two simple instances, we carry out an extensive numerical study using randomly generated customer preference matrices. Fixing the number of slot types J, there are 2^J different possible customer types, including those that accept no slots at all. By allowing any possible combination of these customer types, there could be $2^{2^J} - 1$ possible preference matrices (excluding the empty matrix). To test the performance of the offering-all policy in a robust and yet computationally tractable manner, we compare its performance among many randomly generated such preference matrices.

We also vary the arrival probability vector $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_I)$ for each preference matrix. In particular, we test three possible vectors: $\boldsymbol{\lambda}^{(1)}$, such that $\lambda_i^{(1)} = 1/I$; $\boldsymbol{\lambda}^{(2)}$, such that $\lambda_i^{(2)} = 2(I + i - 2)/(3I^2 - 3I)$; and $\boldsymbol{\lambda}^{(3)}$, with $\lambda_i^{(3)} = 2(I + 3i - 4)/(5I^2 - 5I)$. In all three cases, the λ_i values add up to one. For $\boldsymbol{\lambda}^{(2)}$ and $\boldsymbol{\lambda}^{(3)}$, λ_1 is the largest, and each successive λ_i is smaller by a factor two or four, respectively. Note that the value of I depends on the randomly generated preference matrix and may vary from $I = 1$ (because we exclude the empty matrix) to the maximum number of customer types.

Our results are summarized in Table 3, where we show the optimality gap of the offering-all policy. We compute the performance of the offering-all policy through simulation as before, and the performance of the optimal policy through backward induction. We fix J and N, and then, we generate the number of random instances indicated in Table 3 ("number of instances"). For each instance, we also vary the initial capacity vectors similar to what was done for Tables 1 and 2 ("number of scenarios" in Table 3). Fixing the structure of the arrival rate vector, we then report the maximum, average, and median optimality gaps over all instances and scenarios. It is clear from this table that the offering-all policy continues to do very well, and the average gap with the optimal policy is around 0.5% throughout, independent of the size of the matrix and the arrival rates.

Before proceeding to the next section, we briefly discuss the performance of π^{p^*} (i.e., the static randomized policy arising from the fluid model in Section 3.2). We focus on the M model and vary N, the arrival probabilities, and the initial capacity vectors. We report the optimality gap statistics for π^{p^*} in Table 4. We observe that the average optimality gap decreases from

Table 2. Optimality Gap of the Offering-All Policy in the M + 1 Model Instance

		$(\lambda_1, \lambda_2, \lambda_3) = (9/20, 9/20, 1/10)$, %			$(\lambda_1, \lambda_2, \lambda_3) = (2/5, 2/5, 1/5)$, %			$(\lambda_1, \lambda_2, \lambda_3) = (3/10, 3/10, 2/5)$, %		
N	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
20	45	−3.1	−2.0	−1.9	−2.0	−1.1	−0.9	−0.7	−0.3	−0.2
30	91	−3.4	−2.1	−2.0	−2.3	−1.1	−1.0	−0.8	−0.3	−0.2
40	153	−3.7	−2.1	−2.0	−2.5	−1.2	−1.0	−0.8	−0.2	−0.1
50	231	−3.9	−2.2	−2.0	−2.6	−1.2	−1.0	−0.8	−0.2	−0.1

Table 3. Optimality Gap of the Offering-All Policy for Random Network Instances

				$\lambda = \lambda^{(1)}$, %			$\lambda = \lambda^{(2)}$, %			$\lambda = \lambda^{(3)}$, %		
J	N	No. of instances	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
3	10	100	36	−3.9	−0.2	−0.0	−3.9	−0.2	−0.0	−3.7	−0.3	−0.0
3	20	80	120	−4.6	−0.3	−0.1	−4.6	−0.3	−0.1	−5.1	−0.4	−0.1
3	30	40	253	−5.2	−0.3	−0.1	−5.1	−0.4	−0.1	−3.6	−0.3	−0.1
3	40	10	435	−3.0	−0.3	−0.0	−4.0	−0.5	−0.1	−4.4	−0.5	−0.2
4	10	100	84	−5.0	−0.3	−0.1	−4.7	−0.4	−0.2	−4.0	−0.3	−0.1
4	20	10	455	−3.7	−0.5	−0.3	−2.8	−0.5	−0.3	−4.7	−0.5	−0.3
4	30	10	83	−3.6	−0.6	−0.3	−3.6	−0.7	−0.2	−3.0	−0.6	−0.4
5	10	100	126	−3.9	−0.4	−0.2	−3.7	−0.4	−0.3	−4.2	−0.5	−0.3
5	20	10	126	−2.1	−0.3	−0.2	−3.9	−1.0	−0.6	−4.8	−0.6	−0.3

about 8% to 5% when N increases from 20 to 50. This is consistent with our theory above that π^{p^*} is asymptotically optimal when the demand and capacity increase simultaneously. Because of space constraints, we shall refrain from further exploring the computational issues of π^{p^*} and leave those for future research.

5.2. Performance of the Drain Heuristic

In this section, we evaluate the performance of our drain heuristic developed in Section 4.5. We focus on the N, M, and W model instances. As in Section 5.1, we vary the mix of customer types, the total number of periods, and the initial capacity vectors. The performance of the optimal sequential offering policy is evaluated by backward induction. The performances of the drain heuristic are evaluated by running a discrete event simulation with 1,000 days replication and then computing the average fill count per day. We present the statistics on the percentage optimality gaps of drain in Tables 5, 6, and 7 for the N, M, and W instances, respectively. In particular, for the N and W model instances, the optimality gap statistics are taken over all initial capacity vectors (b_1, b_2), such that $(b_1, b_2) \in \{(x, y) \in \mathbb{Z}_+^2 : x, y \geq 0.2N, x + y = N\}$. The second column of each table shows the number of initial capacity vectors consider for each N.

In the N model, the optimality gap of drain is on average within 0.4% (max 0.8%) in all 264 scenarios that we tested. The performances of drain in the W instance are slightly better than those in the N model. For the M model instance, the optimality gap of drain is on average within 0.7% (max 1.4%) across all 1,560 scenarios that we tested. These observations suggest that the drain heuristic has a remarkable performance. Given its simplicity, it can serve as an effective scheduling rule for practitioners.

5.3. Value of Sequential Offering

5.3.1. Comparison of Optimal Policies. In this section, we investigate the value of sequential scheduling by comparing the optimal sequential policy with the optimal nonsequential policy. We focus on the N, M, and

Table 4. Optimality Gap of the Static Randomized Policy π^{p^*} in the M Model Instance

		$(\lambda_1, \lambda_2) = (1/2, 1/2)$, %			$(\lambda_1, \lambda_2) = (1/3, 2/3)$, %			$(\lambda_1, \lambda_2) = (1/4, 3/4)$, %		
N	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
20	45	−10.7	−7.7	−7.2	−9.8	−7.9	−8.0	−10.8	−8.7	−8.7
30	91	−9.1	−6.4	−5.9	−8.7	−6.7	−6.7	−8.8	−7.2	−7.1
40	153	−8.1	−5.7	−5.2	−7.9	−5.8	−5.8	−7.8	−6.2	−6.2
50	231	−7.5	−5.2	−4.6	−7.1	−5.3	−5.2	−7.0	−5.6	−5.5

Table 5. Optimality Gap of the Drain Heuristic in the N Model Instance

		$(\lambda_1, \lambda_2) = (1/2, 1/2)$, %			$(\lambda_1, \lambda_2) = (1/3, 2/3)$, %			$(\lambda_1, \lambda_2) = (1/4, 3/4)$, %		
N	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
20	13	−0.8	−0.4	−0.4	−0.6	−0.1	−0.2	−0.7	−0.2	−0.3
30	19	−0.6	−0.2	−0.4	−0.8	−0.2	−0.1	−0.5	−0.0	−0.1
40	25	−0.6	−0.2	−0.2	−0.8	−0.1	−0.1	−0.5	−0.0	−0.0
50	31	−0.5	−0.1	−0.2	−0.5	−0.2	−0.2	−0.6	−0.0	−0.1

Table 6. Optimality Gap of the Drain Heuristic in the M Model Instance

		$(\lambda_1, \lambda_2) = (1/2, 1/2)$, %			$(\lambda_1, \lambda_2) = (1/3, 2/3)$, %			$(\lambda_1, \lambda_2) = (1/4, 3/4)$, %		
N	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
20	45	−1.4	−0.7	−0.8	−1.1	−0.4	−0.4	−0.9	−0.2	−0.2
30	91	−0.9	−0.6	−0.6	−0.8	−0.3	−0.3	−0.7	−0.2	−0.2
40	153	−0.7	−0.5	−0.5	−0.7	−0.3	−0.3	−0.9	−0.2	−0.1
50	231	−0.6	−0.4	−0.5	−0.6	−0.2	−0.3	−0.6	−0.1	−0.2

W model instances. To provide a robust performance evaluation, we vary a range of model parameters, including the mix of customer types, the total number of periods, and the initial capacity vectors, like in earlier sections. Table 8 presents the maximum, average, and median percentage improvements in fill count by following an optimal sequential offering policy compared with the optimal nonsequential policy in the N model instance. Tables 9 and 10 present similar information for the M and W model instances, respectively.

We observe that the efficiency gains in the M and W model instances are robust to the initial customer type mix. The efficiency gain in the W model instance is about 6%–7% on average and can be as high as 13%. The efficiency gain in the M model instance is slightly lower. For the N model instance, the gain is relatively more sensitive to customer type mix and ranges between 6% and 11% on average. In certain cases, the efficiency gain in the N model can be as high as 18%. These numerical findings show that sequential offering holds strong potentials to improve the operational efficiency in appointment scheduling systems.

5.3.2. Comparison of Heuristics. In this section, we compare the performances of two heuristic scheduling policies discussed above: the offering-all policy and the drain heuristic developed in Section 4.5. We also consider another policy called the *random sequential offering policy*, which offers available slot types one at a time in a permutation chosen uniformly at random. This policy mimics the existing practice of telephone scheduling, which is often done without careful planning. These three policies are used or can be easily used by practice, and therefore, the comparison results in this section reveal the value of sequential scheduling that may be realized by adopting these policies in practice.

We focus on the N, M, and W model instances and use the combinations of parameters as in earlier sections. The performance of these three policies is evaluated by running a discrete event simulation with 1,000 days of replication and then computing the average fill count per day for each policy. We present the percentage improvement in the fill count of drain over the other two policies. Detailed results are shown in Tables 11, 12, and 13.

For the N model instance, we see an average 9%–11% improvement (max 18%) if using drain compared with using random sequential or the offering-all policy. In the M model, the average improvement is around

Table 7. Optimality Gap of the Drain Heuristic in the W Model Instance

		$(\lambda_1, \lambda_2, \lambda_3) = (1/3, 1/3, 1/3)$, %			$(\lambda_1, \lambda_2, \lambda_3) = (1/5, 1/2, 3/10)$, %			$(\lambda_1, \lambda_2, \lambda_3) = (1/10, 3/10, 3/5)$, %		
N	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
20	13	−0.2	0.0	0.1	−0.1	0.1	0.1	−0.7	−0.2	−0.1
30	19	−0.7	0.0	0.0	−0.4	0.0	0.0	−0.6	−0.1	−0.1
40	25	−0.2	0.0	0.0	−0.2	0.0	0.0	−0.5	0.1	0.0
50	31	−0.2	0.0	0.0	−0.2	0.0	0.0	−0.4	−0.1	−0.1

Table 8. Fill Count Improvement in the N Model Instance (Opt Sequential vs. Opt Nonsequential)

		$(\lambda_1, \lambda_2) = (1/2, 1/2)$, %			$(\lambda_1, \lambda_2) = (1/4, 3/4)$, %			$(\lambda_1, \lambda_2) = (3/4, 1/4)$, %		
N	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
20	13	16.0	10.6	12.4	10.8	9.0	9.6	13.2	6.2	5.5
30	19	16.8	10.9	12.6	11.1	9.3	9.6	14.0	6.3	5.4
40	25	17.2	11.1	12.5	11.2	9.5	9.9	14.5	6.4	5.3
50	31	17.5	11.2	12.9	11.3	9.6	9.8	14.8	6.4	5.3

Table 9. Fill Count Improvement in the M Model Instance (Opt Sequential vs. Opt Nonsequential)

		$(\lambda_1, \lambda_2) = (1/2, 1/2)$, %			$(\lambda_1, \lambda_2) = (1/3, 2/3)$, %			$(\lambda_1, \lambda_2) = (1/4, 3/4)$, %		
N	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
20	45	7.4	4.0	3.4	7.8	4.2	3.8	6.9	4.1	4.0
30	91	7.9	3.7	3.3	8.3	4.1	3.8	7.2	4.2	4.2
40	153	8.3	3.5	2.9	8.5	4.1	3.9	7.4	4.3	4.3
50	231	8.5	3.3	2.6	8.7	4.1	4.0	7.5	4.4	4.4

Table 10. Fill Count Improvement in the W Model Instance (Opt Sequential vs. Opt Nonsequential)

		$(\lambda_1, \lambda_2, \lambda_3) = (1/3, 1/3, 1/3)$, %			$(\lambda_1, \lambda_2, \lambda_3) = (1/5, 1/2, 3/10)$, %			$(\lambda_1, \lambda_2, \lambda_3) = (1/10, 3/10, 3/5)$, %		
N	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
20	13	8.2	6.1	6.5	10.8	6.6	7.0	11.2	7.7	8.3
30	19	9.0	6.6	7.9	11.8	7.0	7.2	11.6	8.1	9.2
40	25	9.5	6.9	7.9	12.3	7.2	7.3	12.0	8.3	9.4
50	31	9.8	7.1	8.3	12.7	7.3	7.3	12.2	8.4	9.4

Table 11. Comparison of the Drain Heuristic with Other Scheduling Policies (the N Model Instance)

		$(\lambda_1, \lambda_2) = (1/2, 1/2)$, %			$(\lambda_1, \lambda_2) = (1/3, 2/3)$, %			$(\lambda_1, \lambda_2) = (1/4, 3/4)$, %		
N	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
% improvement over offering-all policy										
20	13	16.5	10.2	11.6	14.0	10.3	11.6	11.0	9.0	9.6
30	19	17.1	10.8	12.4	14.0	10.6	11.4	11.7	9.1	9.7
40	25	17.8	10.9	12.5	13.9	10.8	11.4	11.1	9.3	9.6
50	31	17.8	11.2	13.3	14.5	10.9	11.5	11.7	9.6	9.8
% improvement over random sequential										
20	13	16.6	10.2	11.9	14.0	10.3	11.2	11.2	8.7	9.0
30	19	16.7	10.8	12.5	13.9	10.5	11.4	11.8	9.3	9.5
40	25	17.4	11.1	12.7	14.0	11.0	11.9	11.9	9.5	9.6
50	31	18.0	11.1	13.2	14.5	10.9	11.2	11.6	9.5	10.0

Table 12. Comparison of the Drain Heuristic with Other Scheduling Policies (the M Model Instance)

		$(\lambda_1, \lambda_2) = (1/2, 1/2)$, %			$(\lambda_1, \lambda_2) = (1/3, 2/3)$, %			$(\lambda_1, \lambda_2) = (1/4, 3/4)$, %		
N	No. of scenarios	Max	Average	Median	Max	Average	Median	Max	Average	Median
% improvement over offering-all policy										
20	45	12.1	7.0	6.8	11.8	7.4	7.1	10.8	6.9	6.9
30	91	13.7	7.1	6.5	13.4	7.6	7.1	11.3	7.4	7.4
40	153	14.0	7.0	6.3	13.8	7.7	7.7	11.5	7.6	7.8
50	231	13.9	7.0	6.4	14.0	7.8	7.9	11.6	7.9	8.0
% improvement over random sequential										
20	45	12.3	7.0	6.9	13.2	7.4	6.8	11.1	6.9	6.7
30	91	13.5	7.1	6.7	13.7	7.6	7.1	11.0	7.4	7.4
40	153	13.8	7.0	6.4	13.7	7.7	7.4	11.4	7.7	7.9
50	231	14.2	7.0	6.4	14.2	7.9	7.7	11.8	7.8	8.0

Table 13. Comparison of the Drain Heuristic with Other Scheduling Policies (the W Model Instance)

N	No. of scenarios	$(\lambda_1,\lambda_2,\lambda_3) = (1/3,1/3,1/3)$, %			$(\lambda_1,\lambda_2,\lambda_3) = (1/5,1/2,3/10)$, %			$(\lambda_1,\lambda_2,\lambda_3) = (1/10,3/10,3/5)$, %		
		Max	Average	Median	Max	Average	Median	Max	Average	Median
% improvement over offering-all policy										
20	13	8.0	6.1	6.9	10.8	6.6	6.7	11.6	7.8	8.5
30	19	9.1	6.5	7.6	11.7	6.9	6.9	11.9	8.1	9.1
40	25	9.7	6.9	7.8	12.5	7.2	7.2	12.3	8.4	9.5
50	31	10.3	7.0	7.9	12.6	7.3	7.3	12.4	8.4	9.3
% improvement over random sequential										
20	13	8.5	6.1	6.8	10.9	6.7	6.9	10.9	7.5	8.1
30	19	9.6	6.5	7.9	11.9	6.9	7.3	11.5	8.0	9.3
40	25	9.8	7.0	8.1	12.5	7.2	7.5	12.3	8.4	9.4
50	31	10.2	7.1	7.9	12.8	7.3	7.4	12.2	8.3	9.4

7%–8%, with max 14%. For the W model, drain makes on average 6%–8% improvement over random sequential or the offering-all policy, with the maximum improvement up to 13%. It is worth remarking on that, in all model instances, the random sequential policy has about the same performance as the offering-all policy. Therefore, although the former is a sequential policy and the latter is not, the potential of sequential offering is not exploited because of the careless choice of the offered slots.

6. Conclusion

Motivated by the increasing popularity of online appointment booking platforms, we study how to offer appointment slots to customers to maximize the total number of slots filled. We consider two models, nonsequential offering and sequential offering, for different customer-scheduler interactions in the appointment booking process. For each model, we develop optimal or near-optimal booking policies.

In our numerical experiments, we find that sequential offering in a proper manner makes a significant improvement over the two benchmark policies: the random sequential offering policy (that mimics the existing practice of telephone scheduling) and the offering-all policy (that resembles many of the current online appointment booking systems). Indeed, carefully designed sequential offerings can achieve the performance of a scheduler with perfect information on each customer but without revealing customer types. These findings suggest substantial potential for improving the current appointment scheduling practice.

Another notable observation from our numerical study is that the two benchmark policies have quite similar performance, which indicates that current online scheduling (that often offers all available slots) and traditional telephone scheduling (without a careful offer sequence) would result in similar fill rates. Thus, one should not expect that implementing an online scheduling system in place of traditional telephone scheduling can automatically lead to more appointments booked. In fact, if the telephone scheduler was (intentionally or unintentionally) offering slots in a "smart" way, moving to online scheduling may hurt the fill rate if the latter is not implemented carefully.

As our research suggests, however, one may improve the performance of online scheduling by designing an interface that uses the idea of sequential offering, collecting information on customer choice behavior, and then, making offers in a smarter way. Given the recent and ongoing growth of online and mobile appointment booking platforms, our research findings can be particularly useful to inform user interface design of these booking platforms.

In summary, our work provides the first analytical framework to model, compare, and improve various

appointment booking processes; we develop optimal or near-optimal booking policies for two representative appointment offering schemes. Our study also suggests many possible directions for future research. To name a few, first, we assume a specific model for customer choice, and future research may consider scheduling decisions under different choice models. Second, it would be interesting to consider other customer behaviors (e.g., cancellations, no shows, recall, and renege after a few trials) in the scheduling models. Third, our numerical study of the multiday scheduling presented in the online appendix is by no means exhaustive, and it would be a fruitful direction to investigate the (optimal) joint offering policy for both day and slot choices. Last but not least, asymptotic regimes with different scaling of model parameters may be interesting objects of study both from a stochastic model theoretical perspective and for informing more efficient operations in practical settings.

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Endnote

[1] Even if some of the slot types are unavailable, π^p would still offer these slot types according to the probability vector $\mathbf{p}$. However, customers only consider those slot types that are available in the booking process.

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